The Law of Fairness Formal Model

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Abstract

Per-Stream Terminal Valence Closure (PTVC) proposes a precise and formally falsifiable candidate boundary hypothesis for conscious experience: within each separately routed maximal qualifying stream, cumulative positive and negative phenomenal magnitudes attain exact balance at the stream's complete strict-pre-endpoint boundary. On the primary finite signed-measure branch, each stream is represented by a finite signed phenomenal measure and the hypothesis is the compact exact-zero condition:

$$Z_i^\star := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\mathrm{phen}}(T_i^\star -), \qquad \mathrm{PTVC}_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^\star = 0.$$

The equality belongs to each stream alone; no stream balances another. Yet it permits immense differences in duration, intensity, temporal order, agency, memory, opportunity, structure, and gross phenomenal variation. This union of exact invariance and experiential diversity gives PTVC its conceptual economy and scientific interest. Rights, welfare, justice, and the direct alleviation of suffering remain indispensable.

The model turns this boundary claim into a typed, mechanism-neutral interdisciplinary research program. Prospectively fixed modules specify representation and scalarization, neutral origin, exposure, bearer and routing, truth-level endpoint, observation and missingness, measured-to-latent recovery, reference class, and inferential modality. Candidate mechanisms—including opponent regulation, exact adaptation, completion-potential dynamics, and endpoint coupling—enter as competing explanations. This common architecture lets psychophysics, affective neuroscience, consciousness science, stochastic-process theory, end-of-life research, and latent-variable inference contribute distinct tests of the same object.

Endpoint timing supplies a particularly informative structural discriminator. On the prospectively declared independent positive-density endpoint branch under the at-risk law P_i^0, the stated measurability and conditional-density premises imply that, when exact closure holds almost surely at the probe, the pre-endpoint ledger vanishes Lebesgue-almost-everywhere on the declared interval for P_i^0-almost every eligible path. The result sharply distinguishes independent termination from state-coupled timing, first hitting or absorption, endpoint atoms, singular carriers, routing change, continued experience, endpoint-inclusive value, and other declared governing branches.

Much of the enabling science already exists. Affective psychophysics supplies instruments; exact biological regulation, opponent processes, and computational mood supply candidate architectures; consciousness, terminal-trajectory, and missing-data research supply endpoint and observation methods. Calibration and open-set recovery establish the measurement foundation, while held-out perturbation and transport studies compare rivals under equal information budgets. On suitable Markov branches, a single frozen completion potential, through its Doob h-transform, links transition, action, endpoint, and terminal-support predictions across held-out modules. As endpoint and latent bridges mature, the same architecture can advance from nested finite-resolution tests toward complete-stream adjudication.

By joining a consequential conjecture to theorem-level discriminators, convergent transport predictions, and a staged confirmatory program, PTVC creates a distinctive scientific opportunity: to determine whether conscious life exhibits a per-stream terminal regularity while building a cumulative science of valence, stream structure, and experiential boundaries.

Keywords: Law of Fairness; valence; signed measures; terminal boundary condition; latent-variable identification; consciousness; endpoint processes; completion potential; affective science; partial identification; model recovery; stochastic processes; measurement theory; random-cut testing

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Confirmatory Core Synopsis

Prospectively fixed target. PTVC is a candidate exact [SET-ALL] hypothesis: on the primary finite signed-measure branch, every separately routed maximal qualifying conscious stream has equal finite positive and negative Jordan masses across its complete strict-pre-endpoint domain, so its terminal signed residual is zero. The claim is intrastream and nontransferable; streams may differ in duration, intensity, history, opportunity, and gross valence variation, and no stream can offset another. Before outcome inspection, the protocol fixes representation and scalarization, sign and neutral origin, exposure, bearer, exhaustive ownership and routing, truth-level domain and endpoint, observation and missingness, recoverability, measured-to-latent transport, reference class, and inferential modality.

Branch-matched adjudication. For each eligible stream, a frozen observation kernel and measured-to-latent bridge generate a branch-typed truth-containing terminal compatibility object. One gate-valid stream whose nonempty object excludes zero decisively rejects exact [SET-ALL] in the registered class. On a finite-band branch, full containment supports compatibility at the declared resolution, disjointness is adverse, and partial overlap remains unresolved. Exact-zero, finite-band, atom-at-zero, [P1], confidence, posterior, probability-one, and [SET-ALL] conclusions retain distinct criteria; [SET-ALL] promotion requires an independent quantifier bridge. If a prerequisite cannot license the next inference, analysis reports the strongest warranted claim or No adequate estimate, preserving a cumulative and interpretable record.

Structural discriminator. Under the declared at-risk conditional law, suppose endpoint timing is independent of the sigma-field generated by the jointly measurable pre-endpoint ledger path and has conditional density strictly positive Lebesgue-almost-everywhere on the declared support interval. Exact closure at the probe almost surely then implies that the ledger is zero Lebesgue-almost-everywhere in time for almost every eligible path under that law. The theorem turns endpoint architecture into a high-value comparison among independent timing, state-coupled timing, hitting or absorption, endpoint atoms, singular carriers, observation shutdown, continued experience, and routing alternatives.

Target-route separation. Finite-resolution compatibility and exact adjudication form a nested target axis; completion potentials, reserve coupling, opponent regulation, action selection, endpoint coupling, and other candidate mechanisms occupy separately declared route axes unless a proved entailment connects them. On the declared action-first route, fixed-policy probability-one completion characterizes policy mass on actions whose frozen continuations close with probability one; it supports a policy statement, not a motivation or endpoint claim. Route evidence can therefore localize mechanisms without substituting for terminal evidence, while target evidence need not privilege a route.

Convergent signature and sequence. Stage 1 establishes representation, origin, exposure, observation, local complementarity, and open-set model-recovery validity. Stage 2 freezes the architecture and tests held-out future-area prediction, compression, signed-dose compensation, endpoint surprise, and transport of one completion potential against equal-information-budget regulatory, observation, selection, memory, bridge, hidden-state, and endpoint-coupling rivals. Replication across strata, sites, perturbations, resolutions, and a distinct held-out module can open the endpoint-bridge stage. Complete-stream adjudication then combines complete-domain coverage, a validated endpoint or terminal-remainder bridge, and the prospectively fixed decision rule. Each passed gate opens a stronger evidential stage; each sound mismatch localizes what to refine, replace, or retire.

Table 1. Terminology and Core Parameters

Symbol / canonical alias	Object and domain / codomain or unit	Definition
Core ontology		
$i,\ s$; $\mathcal{R}^\star$; $\mathcal{R}_\eta^{\text{study}}$; $\mathcal{R}_\eta^{\text{ana}}$	Stream index and reference classes *Domain/type:* i, s in the declared stream-label space; the classes are measurable subsets.	i denotes a fixed stream under a declared law; s enumerates streams within a complete history or routing map. The truth-domain, protocol-study, and analysis classes remain distinct. Observation failure changes evidential status, not truth-domain membership.
PTVC	Per-Stream Terminal Valence Closure *Domain/type:* Predicate/event on maximal qualifying streams; codomain {0,1}	On the finite signed-measure branch, every maximal qualifying conscious stream, separately routed under the declared ownership and routing rule, has equal complete-stream positive and negative phenomenal magnitudes.
GTN	Generalized Terminal Neutrality *Domain/type:* $H_{\text{GTN}}: \mathfrak{M}_{\text{GTN}} \to \mathcal{V}_{\text{GTN}}$, $0_{\text{GTN}} \in \mathcal{V}_{\text{GTN}}$	Jointly measurable functional into a declared measurable codomain with a distinguished zero whose singleton is measurable; distinct from PTVC unless an additive-decomposition bridge is proved.
$(\mathsf{H}, \Sigma_{\mathsf{H}})$; $\mathcal{R}^\star$; $\mathsf{S}^\star(\Gamma)$; $\pi_{\mathsf{S}}^\star(\Gamma)$	Complete-history carrier, qualifying-label carrier, and counting configuration *Domain/type:* $\mathsf{S}^\star(\Gamma) \subseteq \mathsf{S}$; $\pi_{\mathsf{S}}^\star(\Gamma) \in \mathsf{N}_f(\mathsf{S}) \cup \mathsf{N}_c(\mathsf{S})$.	$(\mathsf{H}, \Sigma_{\mathsf{H}})$ is the complete-history measurable space; $\mathcal{R}^\star$ is the truth-domain class; $\mathsf{S}^\star(\Gamma)$ is the finite/countable qualifying-label carrier; and $\pi_{\mathsf{S}}^\star(\Gamma) = \sum_{s\in\mathsf{S}^\star(\Gamma)} \delta_s$ is its counting configuration. Membership uses the carrier; counting and integration use the configuration.
$C_s^{(b)}[\Gamma]$	Closure functional $C_b : \mathfrak{M}_b \to \mathbb{R}$, $C_s^{(b)}[\Gamma] := C_b(\Gamma, s)$. *Domain/type:* Branch-indexed marked histories to signed-value space.	Signed complete-stream residual of stream s in history Γ. On the common finite additive/bipolar domain, Lemma 1 gives the same net residual; the branch label records admissibility and component estimands.
$\mathcal{F}_{\text{PTVC}}^{\text{add}}$; $\mathcal{F}_{\text{PTVC}}^{\text{bip}}$; $\mathcal{F}_{\text{GTN}}$	Admissible history sets *Domain/type:* Measurable subsets of the complete-history carrier	Measurable complete-history sets satisfying additive PTVC, bipolar PTVC, and GTN, respectively.
Q; $\kappa(\cdot \mid \Gamma)$; $\overline{P}$; $Q^{(b)}$; Q^{GTN}	Complete-history and marked-stream population laws *Domain/type:* Probability law on histories; mark-selection kernel; branch-indexed laws	The primary history law is Q. The prospectively fixed kernel κ selects a mark when a marked-stream estimand is required. The corresponding joint law is $\overline{P}(\mathrm{d}\Gamma, \mathrm{d}s) := Q(\mathrm{d}\Gamma)\kappa(\mathrm{d}s \mid \Gamma)$. Typical-history, typical-stream, and exposure-weighted estimands remain distinct and report their weighting law. No uniform law is assumed on a countably infinite mark set.
S_i^θ; $T_i^\star$; $T_{i,\eta}^{\text{adj}}$; J_i; $H_i^K(t)$; $R_i^{\text{risk}}(t)$	Endpoint and occupation objects *Domain/type:* Times in $[0,\infty]$; endpoint marks; indicators in $\{0,1\}$.	Truth-domain origin, truth-level endpoint, protocol endpoint, endpoint-class mark with $J_i = e$ for cause e, predictable off-band indicator, and predictable at-risk process used in Proposition 3.
Measures and ledgers		
$\mathcal{F}_{i,t}^{\text{lat}}$; $\mathcal{H}_{i,t}^{\text{pre}}$; $\mathcal{H}_{i,t}^{\text{joint}}$; $\mathcal{O}_{i,t}^{\text{obs}}$	Information sets *Domain/type:* Filtrations and sigma-fields	Latent-process history; prospective prediction/intervention history; predictable joint-model covariates; and observation filtration.
$D_i^\theta(t)$	Unity duration *Domain/type:* $[0,\infty)$, physical-time units only on the dimensionless time-chart branch; the exposure-clock alternative is reported as accumulated exposure.	Accumulated unity-weighted duration in a declared physical-time chart, typically $D_i^\theta(t) = \int g_{\theta,i}(s)\, ds$ over the stated interval.

Symbol / canonical alias	Object and domain / codomain or unit	Definition
$g_{\theta,i}(t);\ q_i^{\exp}(t)$	Time-chart and exposure-rate factors Domain/type: on the primary branch, $g_{\theta,i}(t)$ is dimensionless and $q_i^{\exp}(t)$ has exposure/time units; on the unity-clock alternative, $q_i^{\exp} \equiv 1$ and $g_{\theta,i}$ carries exposure/time units.	Freeze exactly one convention before analysis. The product $g_{\theta,i}(t)q_i^{\exp}(t)$ is the declared exposure density; neither factor may duplicate ownership or exposure already encoded by μ_i^{phen}.
(Ω_i, Σ_i)	Phenomenal event space *Domain/type:* Measurable phenomenal-event space	Strict pre-endpoint measurable domain and sigma-algebra of stream i.
$\nu_i,\ \mu_i^{\text{phen}},\ f_i$	Signed value, exposure, and density *Domain/type:* Signed/positive measures; density in value per exposure	ν_i is signed phenomenal value; μ_i^{phen} is positive phenomenal exposure; and $f_i = d\nu_i/d\mu_i^{\text{phen}}$ when absolute continuity holds.
$\nu_i = \nu_i^+ - \nu_i^-$; $\nu_i^+ \perp \nu_i^-$; $V_i^\pm = \nu_i^\pm(\Omega_i)$	Jordan measures and magnitudes *Domain/type:* Positive measures and phenomenal-value units	The Jordan positive and negative measures are mutually singular. Complete-stream magnitudes are $V_i^\pm = \nu_i^\pm(\Omega_i)$.
V_i^{gross}	Gross scalar valence variation *Domain/type:* $[0,\infty)$, phenomenal-value units.	$V_i^{\text{gross}} = \lvert\nu_i\rvert(\Omega_i) = V_i^+ + V_i^-$. This is total variation of the declared scalar valence measure, not total consciousness, duration, or every experiential dimension.
$\nu_i = \beta_i^+ - \beta_i^-$; $\beta_i^\pm = \nu_i^\pm + \zeta_i$; $\zeta_i \geq 0$	Bipolar component measures *Domain/type:* Finite positive measures; phenomenal-value units	When the positive and negative component measures are independently identified or structurally specified, they may overlap through ζ_i. Net signed value is unchanged. Primitive gross magnitude is $V_{i,\text{bip}}^{\text{gross}} = V_i^{\text{gross}} + 2\zeta_i(\Omega_i)$; from the net signed measure alone, total variation is only a sharp lower bound unless direct component data or an overlap bound is supplied.
$q_i^{\exp}(t);\ v_i^{\text{real}}(t);$ $m_i(t);\ F_i(t)$	Exposure-rate factor, realized value density, conditional predictor, and contribution rate *Domain/type:* Exposure/time on the primary branch (dimensionless on the unity-clock alternative); value/exposure; conditional value/exposure; and value/time.	$v_i^{\text{real}} = d\nu_i^{\text{time}}/d\mu_i^{\text{time}}$ on the absolutely continuous chart; $m_i = \mathbb{E}[v_i^{\text{real}} \mid \mathcal{F}_{i,t}^{\text{lat}}]$ is a predictor. Pathwise identities use v_i^{real}; substituting m_i requires a validated bridge.
$L_i^{\text{phen}}(t)$	Phenomenal ledger *Domain/type:* Càdlàg signed-value process	Latent cumulative signed phenomenal value.
Endpoint and route		
$U_i;\ U_i^{\text{meas}};\ \widehat{U}_i$	Route-state objects *Domain/type:* Declared state space; measured/estimated counterparts	Latent route state, measured route-state estimand, and estimated proxy.
$r_i^U(t);\ \kappa_i^U(t)$	Reserve-depletion density and proportional contraction rate. Domain/type: reserve/time for r_i^U and 1/time for κ_i^U, on $U_i(t) > 0$.	On the absolutely continuous reference route, $\mathrm{d}U_i(t) = -r_i^U(t)\,\mathrm{d}t$ with $r_i^U(t) \geq 0$, and $\kappa_i^U(t) := r_i^U(t)/U_i(t)$.
$Y_i(t);\ \Psi_i(t)$	Normalized route coordinates *Domain/type:* Dimensionless on the positive-reserve domain	Y_i is signed and $\Psi_i(t) = \lvert L_i^{\text{rc}}(t)\rvert/[c_{\text{LU}}U_i(t)] = \lvert Y_i(t)\rvert$ when $U_i(t) > 0$.
$c_{\text{LU}},\ L_i^{\text{rc}}(t)$	Reserve-route scale and coordinate *Domain/type:* cLU: phenomenal value/reserve; route coordinate: phenomenal value. A dimensionless-reserve branch must say so explicitly	$L_i^{\text{rc}} = c_{\text{LU}}U_iY_i$.
$M_\eta;\ M_{\eta,i};$ $A_\eta;\ \widehat{L}_{i,\eta}$	Measurement pipeline *Domain/type:* Typed maps between latent, measured, and analysis spaces	Measurement map, measured estimand, analysis map, and sampled estimator; each is distinct from structural action availability.

Symbol / canonical alias	Object and domain / codomain or unit	Definition
Observation and inference		
$\mathcal{A}^{\mathrm{do}}_{\eta}(h_t)$	Structural action-availability correspondence *Domain/type:* Nonempty measurable-valued correspondence from histories to a standard-Borel action space; policy kernels are separate objects	Structural action correspondence available at history h_t. A policy satisfies $\pi_\eta\left(\mathcal{A}^{\mathrm{do}}_{\eta}(h_t) \mid h_t\right) = 1$; probability-one admissibility does not imply topological support containment, and policy behavior never defines structural availability.
$\mathcal{A}^{\mathrm{phen}}_{i,\eta}$; $\mathcal{A}^{\mathrm{meas}}_{i,\eta}$	Scientific acceptance region and measured preimage *Domain/type:* Latent phenomenal units; bridge-defined measured preimage	$\mathcal{A}^{\mathrm{phen}}_{i,\eta}$ is calibrated independently in latent units. For each admissible origin z, define $\mathcal{B}_{i,\eta,z}[y] := \{x: (y,x) \in \mathcal{B}_{i,\eta,z}\}$. The origin-specific measured preimage retains exactly those y whose nonempty bridge image lies wholly in $\mathcal{A}^{\mathrm{phen}}_{i,\eta}$; the robust projected rule requires this for every protocol-compatible origin. The set may be asymmetric or disconnected. Measured equivalence, latent finite-band compatibility, and exact PTVC are distinct conclusions.
$\mathcal{C}^{\mathrm{joint}}_{\eta}$; $\mathcal{J}^{\mathrm{meas}}_{\eta}$; Σ_η	Typed joint compatibility construction *Domain/type:* Product of declared component spaces; projected to the named measured-estimand scale	$\mathcal{C}^{\mathrm{joint}}_{\eta}$ combines measurement, endpoint, missingness, representation, calibration, scalarization, route-state, and dependence constraints without scalarizing unlike uncertainties. $\mathcal{J}^{\mathrm{meas}}_{\eta}$ is its one-time projection; Σ_η is the prospectively fixed dependence specification.
N^{term}_{i}	Terminal nuisance state *Domain/type:* Product nuisance-state space	Care context, medication, reporting capacity, device quality, route state, and other observation-relevant variables.
η	Protocol specification *Domain/type:* Versioned protocol record	Declares domain, endpoint, estimand, estimator, observation, missingness, rivals, scoring, and reporting.
$p^{\mathrm{comp}}_{\eta,h_t,g}(a)$	Action-specific completion probability *Domain/type:* $[0,1]$ on the declared risk set.	Probability of the prospectively declared completion event under action a at prefix h_t; explicitly, $p^{\mathrm{comp}}_{\eta,h_t,g}(a) := Q^{a,g}_{\eta,h_t}\left(F^{a,g}_{\eta,\mathrm{comp}}(h_t)\right), \qquad 0 \le p^{\mathrm{comp}}_{\eta,h_t,g}(a) \le 1.$ Endpoint and continuation law are fixed independently.
$D^{\mathrm{obs}}_{i,\eta}$	Protocol-observed data *Domain/type:* Protocol data space	Complete protocol-observed data object after declared observation, retention, missingness, and preprocessing rules.
X_i; $V^{\mathrm{lat}}_{i,c}$; W^{nuis}_{i}; $K^{\mathrm{obs}}_{\eta,c,t}$; $O^{(c)}_{i,k}$	Observation-channel objects *Domain/type:* Typed latent/observed spaces and Markov kernels	Latent state; channel-specific latent target; nuisance state; protocol observation kernel; and observed channel output.
$\mathrm{SNR}_{i,\eta}(t \mid u, x)$	Conditional signal-to-noise ratio *Domain/type:* $[0,\infty)$, dimensionless and finite where used.	Conditional latent route-variance to observation-noise-variance ratio on the domain with a positive finite denominator; a diagnostic, not an identification theorem.
Reserved symbols		
δ_0; $K^{\mathrm{phen}}_{i,\eta}$; δ_{res}; $\delta^{\mathrm{meas}}_{i,\eta}$; d; δ^{TD}_{t}; $\delta_{\log}$; δ_{leak}	Typed margin and role-specific scalar namespace *Domain/type:* Dirac mass, latent half-width, measured-scale margin, residual, delivered dose, TD error, log floor, leakage	Dirac mass; latent scientific half-width on symmetric registered acceptance-set branches; observed-scale planning margin; common-residual rival; assigned or delivered dose d; temporal-difference error; numerical log floor; and leakage tolerance. The latent half-width and measured-scale margin remain distinct unless the bridge condition in §2.3 identifies them.

1. Core hypothesis and layer separation

Registration convention. In this manuscript, registered means fixed in the study protocol before outcome inspection. Publicly preregistered is reserved for a timestamped external registry record with a persistent identifier. When no external record exists, the text uses protocol-fixed or prospectively specified.

Scientific question. PTVC asks whether a precisely typed intrastream terminal-boundary relation is preserved across prospectively fixed representation, bearer, ownership, endpoint, observation, bridge, and rival tests. The model preserves heterogeneity in gross valence variation, duration, trajectory, agency, and circumstance. By declaring support, refinement, counterevidence, and abstention criteria in advance, the model makes every outcome scientifically useful and enables cumulative progress while keeping adjacent evidence in its strongest defensible role.

The Law of Fairness is the ordinary-language statement of the hypothesis; Per-Stream Terminal Valence Closure (PTVC) is its precise scientific formulation. Both names refer to the same proposed per-stream terminal relation.

Universal target scope. PTVC's defining target, formalized here on a declared additive or bipolar scalar branch, is an exact and exceptionless [SET-ALL] proposition: for every separately routed maximal qualifying conscious stream in the declared truth-domain class, whether past, present, or future, $C_s^{(b)}[\Gamma] = 0$, with no cross-stream compensation and no exceptions inside that class. Equivalently on the primary finite signed-measure branch, each complete strict-pre-endpoint stream has equal total positive and negative phenomenal magnitude, so the magnitude ratio is 1 whenever both totals are positive and the terminal signed residual is 0. This concerns the complete cumulative stream, not necessarily a neutral last momentary feeling. Finite bands, probability-one submodels, finite classes, measurement bridges, and abstention states provide the rigorous evidence ladder by which finite observers can test the candidate exact universal law; they do not redefine or weaken the underlying [SET-ALL] target.

Finite-representation scope. The primary [SET-ALL] branch quantifies over every qualifying stream in the declared truth-domain class for which the prospectively registered finite signed-measure representation is admissible. Extension to every possible qualifying stream proceeds through a domain-coverage theorem or a prospectively declared branch-spanning result. An unresolved representation narrows the claim or yields No adequate estimate, while every gate-valid adverse stream remains in the declared test class.

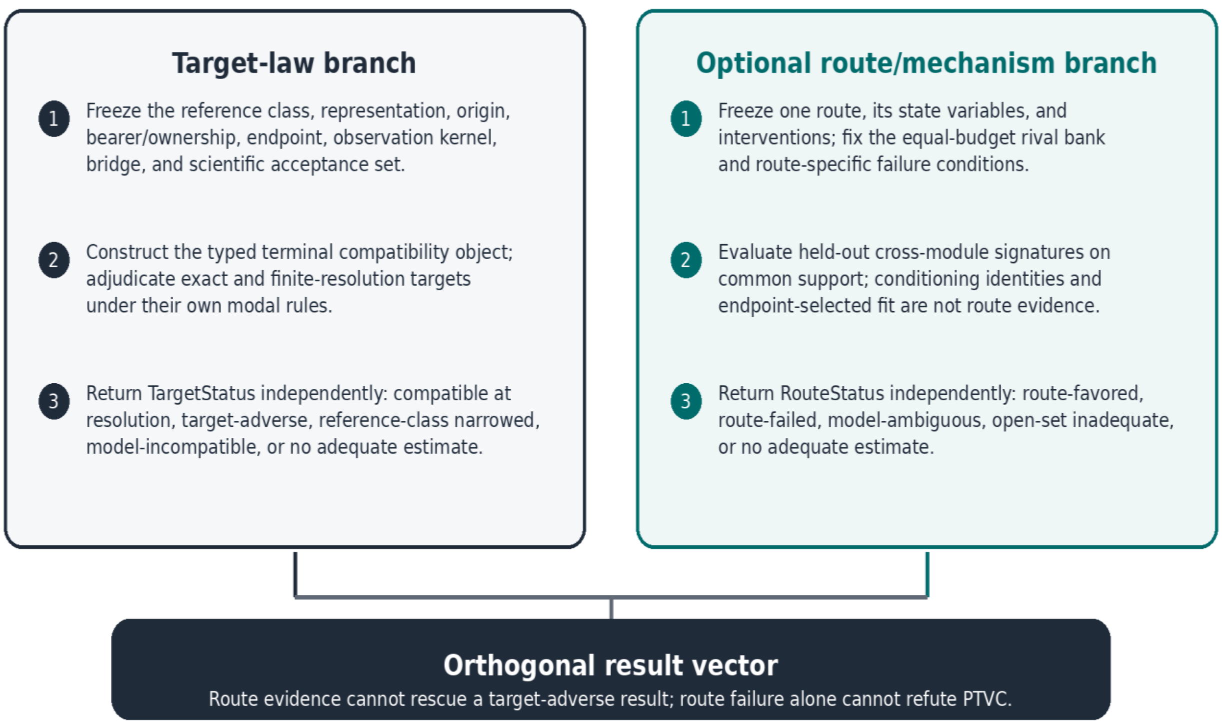
Scientific-role discipline. Definitions, coordinate choices, and algebraic identities establish the framework in which empirical evidence can be interpreted. Representation and identification premises are testable prerequisites; PTVC is the law-level hypothesis; reserve, completion-potential, and other routes are optional mechanism hypotheses; and estimands, estimators, scores, and decisions are protocol objects. Empirical support grows from independently constructed data, validated premises, and preregistered predictions assessed within this architecture.

Distinctive contribution. This manuscript's novelty lies in a typed scientific assembly built around PTVC: the exact intrastream terminal target; its bearer, ownership, endpoint, observation, and modal architecture; the application of the random-endpoint obstruction to that target; its branch-specific route consequences; and a decision system capable of support, principled narrowing, abstention, retirement, or refutation. It brings established tools from signed-measure theory, stochastic processes, constrained reachability, latent-variable identification, equivalence testing, partial identification, and model recovery into a new joint architecture for terminal valence closure.

Artwork by Diego Santos

Figure 1A. Illustrative life-course timeline. The gravestone marks biological death; the truth-level endpoint is defined separately as the last defensible qualifying experience under the declared endpoint branch.

Figure 1B. Dependency and adjudication architecture



The dependency graph contains parallel prerequisites, a target-law branch, and an optional route/mechanism branch. A failed prerequisite narrows the licensed inference; a gate-valid off-band target result remains adverse to the declared target regardless of route performance. Sections 1–9 define the gates, Section 10 supplies empirical foundations and discriminators, Section 11 supplies the staged empirical program, and Appendices A–C preserve methods, mechanism tests, and interpretive scope.

Inference-routing key.

- Representation/origin failure narrows the representation or returns the typed ambiguity or non-evaluable state.
- Bearer/routing failure returns branch-conditional results, Model-ambiguous, or No adequate estimate; it never deletes an adverse valid stream.
- Endpoint failure retains an interval, censoring, unresolved-endpoint, or No adequate estimate state.
- Observation/measurement failure restricts inference to the measured estimand and records the failed bridge.
- Target adjudication returns its own TargetStatus; a route label cannot replace or overwrite it.
- The optional route tournament returns RouteStatus and ComparatorStatus independently of TargetStatus.
- Latent-law adjudication preserves the supported restricted class, evidence-object type, and inferential guarantee; no axis rescues another.

Near-term staged decision path. The program may begin with the listed nonterminal and cohort analyses only after each named resource passes a dataset-specific access, release-version, variable-availability, consent/governance, missingness, preprocessing, and independent-reproduction audit. Stage 1 remains generator recovery, temporal-weighting estimation, local episode-complementarity calibration, and the separately defined residual incremental-value and residual compression-sufficiency tests. Stage R0 of the animal program remains simulation and recovery; animal enrollment is considered only after proxy validation, replacement, welfare, and independent-governance gates pass. The studies exercise different gates and are never pooled as interchangeable confirmation.

Branch-matched decisive-adjudication rule. For [SET-ALL], one independently verified truth-qualifying maximal stream outside the target refutes the registered universal class; for [FINITE-CLASS], one registered member suffices. A [P1] claim is adverse only under a registered procedure whose null, alternative, support conditions, and error guarantee target the declared probability-one statement. An observed or estimated off-target case does not by itself establish positive population probability without that inferential guarantee. [SUPPORT] requires the declared topology and support/closure premises. [POLICY-ALL] and [CONTINUATION-ALL] require a prospectively admissible policy or continuation. Once the branch-matched criterion is met, any post-outcome change to representation, endpoint, scalarization, bearer, routing, or continuation constitutes a new hypothesis requiring fresh prospective evaluation.

1.1 Finite signed phenomenal ledger

Phenomenal-valence representation contract. The primary branch treats phenomenal valence as a signed, neutral-origin-relative quantity assigned to stream-owned phenomenal events. It requires a prospectively fixed sign convention, scalar commensurability on the declared domain, finite total variation, and additive consistency under disjoint event composition. These are independently testable representation premises; terminal fit does not define them. Finite partition experiments test finite-resolution additivity but do not by themselves prove countable additivity. Stable failure routes to a bipolar, vector, ordered, GTN, or No adequate estimate branch.

Additive-conjoint measurement gate. For scalar phenomenal magnitude inferred from joint manipulations, prospectively fix a factorial order and test weak order, factor independence or single cancellation, double cancellation, and, where the design permits, higher-order finite cancellation. Declare rather than infer solvability and Archimedean or continuity premises. The target representation is

$$(a_1, b_1) \succcurlyeq (a_2, b_2) \Leftrightarrow u(a_1) + v(b_1) \geq u(a_2) + v(b_2).$$

Calibrate u and v without terminal outcomes and test held-out orderings and tradeoffs. Stable failure blocks the scalar additive ledger and routes to the bipolar, vector, ordered, or No adequate estimate branch; terminal closure may not select the scale (Luce & Tukey, 1964; Krantz et al., 1971).

Cardinality discipline. Raw Likert categories, ranks, arbitrary factor scores, embeddings, classifier logits, and preference orderings are not integrated as phenomenal magnitudes unless an independently validated interval/cardinal bridge, sign convention, neutral origin, and admissible transformation group have been established. Vector closure requires a prospectively declared point-separating dual class; zero on a convenient subset of projections is not vector zero.

Signed-premeasure extension gate. Before the primary branch is treated as a finite signed measure on Σ_i, the protocol states the generating algebra or semiring $\mathcal{A}_i$, verifies finite additivity of the signed premeasure on $\mathcal{A}_i$, verifies continuity at the empty set or countable additivity on the algebra, establishes finite total variation, names the extension theorem used, and states uniqueness on $\sigma(\mathcal{A}_i)$. Failure preserves only the validated finite-resolution partition model and does not license countable-additive conclusions.

Definition 1 (finite signed phenomenal ledger). Let (Ω_i, Σ_i) be the measurable phenomenal-event space of maximal qualifying stream i. When history dependence matters, write $(\Omega_i(\Gamma), \Sigma_i(\Gamma), \nu_i(\cdot \mid \Gamma))$; the history index is suppressed after a complete history has been fixed. Let $\Omega_i^{\leq}(t) \in \Sigma_i$ be the right-continuous through-time event prefix and let $\Omega_i^{<}(t)$ be the corresponding strict-before prefix. Let ν_i be a finite signed phenomenal measure and μ_i^{phen} a positive sigma-finite exposure measure. The following history–stream–event construction binds these fiberwise objects measurably across histories.

History-stream-event bundle. Let E be a standard-Borel event carrier and define the measurable incidence set

$$\mathcal{E}_{\mathrm{HSE}} := \{(\Gamma, s, e) : \Gamma \in \mathcal{R}^{\star},\ s \in \mathrm{S}^{\star}(\Gamma),\ e \in \Omega(\Gamma, s)\}.$$

On the history–stream–event incidence set, require measurable kernels $\mu^{\mathrm{phen}}(\mathrm{d}e \mid \Gamma, s)$, $\nu^{+}(\mathrm{d}e \mid \Gamma, s)$, and $\nu^{-}(\mathrm{d}e \mid \Gamma, s)$, with

$$\nu(\cdot \mid \Gamma, s) = \nu^{+}(\cdot \mid \Gamma, s) - \nu^{-}(\cdot \mid \Gamma, s), \qquad \nu^{+}(\cdot \mid \Gamma, s) \perp \nu^{-}(\cdot \mid \Gamma, s), \qquad |\nu|(\Omega(\Gamma, s) \mid \Gamma, s) < \infty.$$

and a jointly measurable event-time map $\tau(\Gamma, s, e)$. Define measurable prefix sections by

$$\Omega^{\leq}(\Gamma, s, t) = \{e \in \Omega(\Gamma, s) : \tau(\Gamma, s, e) \leq t\}.$$

The primary implementation scope is finite or countable marked-stream families. A fiberwise-only branch makes no cross-history measure claim and separately proves every probability-bearing scalar measurable.

$$L_i^{\mathrm{phen}}(t) := \nu_i(\Omega_i^{\leq}(t)),$$

$$L_i^{\mathrm{phen}}(t-) := \nu_i(\Omega_i^{<}(t)).$$

Prefix consistency and terminal exhaustion. For every fixed complete history, the through-time prefixes are nested:

$$S_i^\theta \le t_1 < t_2 < T_i^\star \Rightarrow \Omega_i^\le(t_1) \subseteq \Omega_i^\le(t_2).$$

Fix prospectively a deterministic countable dense elapsed-time grid $D \subset [0,\infty)$, such as the nonnegative rationals. For each realized history define

$$\mathfrak{G}_i(\Gamma) := \{S_i^\theta + r : r \in D,\ S_i^\theta + r < T_i^\star\}.$$

The construction rule is fixed independently of terminal balance and is measurable wherever that property is used. Define the strict-before prefix by

$$\Omega_i^<(t) := \bigcup_{\substack{u\in\mathfrak{G}_i(\Gamma)\\ u<t}} \Omega_i^\le(u).$$

Define the covered prefix domain by

$$\Omega_i^{\mathrm{cov}}(\Gamma) := \bigcup_{t\in\mathfrak{G}_i(\Gamma)} \Omega_i^\le(t).$$

Right continuity is imposed in signed-total-variation measure:

$$t_n \downarrow t \Rightarrow |\nu_i|\big(\Omega_i^\le(t_n) \triangle \Omega_i^\le(t)\big) \to 0.$$

Require exhaustion in signed total variation:

$$|\nu_i|(\Omega_i \cap [\Omega_i^{\mathrm{cov}}(\Gamma)]^c) = 0.$$

On every branch that uses phenomenal exposure dynamically or permits a nonzero affine translation, require exposure exhaustion as well:

$$\mu_i^{\mathrm{phen}}(\Omega_i \cap [\Omega_i^{\mathrm{cov}}(\Gamma)]^c) = 0.$$

Continuity from below of the Jordan components gives the complete-stream limit. Assume the through-time ledger admits an adapted càdlàg version under the declared filtration; all subsequent pathwise statements use that version, whose left limit is the strict-before ledger.

Cofinal-limit convention. Every later display written as $t \uparrow T_i^\star$ is shorthand for convergence along the prospectively declared cofinal grid $\mathfrak{G}_i(\Gamma)$, with the branch-specific mode of convergence stated wherever probability, norm, or set convergence is intended.

$$\lim_{\substack{t\uparrow T_i^\star\\ t\in\mathfrak{G}_i(\Gamma)}} L_i^{\mathrm{phen}}(t) = \nu_i(\Omega_i).$$

Any phenomenal contribution assigned exactly at the endpoint is excluded from Ω_i on the primary strict-pre-endpoint branch and belongs only to a separately declared endpoint-inclusive branch.

Endpoint comparison identity. On a matched bearer, unit, origin, and event convention, the endpoint-inclusive residual equals the strict-pre-endpoint residual plus the separately owned endpoint contribution. Unknown endpoint contribution is propagated as a typed set or yields No adequate estimate; endpoint-inclusive compatibility cannot establish strict-pre-endpoint closure.

$$Z_i^{\mathrm{incl}} = Z_i^{\mathrm{pre}} + A_i^\star.$$

When $\nu_i \ll \mu_i^{\mathrm{phen}}$, let $f_i := \mathrm{d}\nu_i/\mathrm{d}\mu_i^{\mathrm{phen}}$.

Exposure-density gauge. For every admissible measurable h satisfying $0 < h(x) < \infty \quad \mu_i^{\mathrm{phen}}$-a.e.,

$$d\mu_i^{(h)} = h\, d\mu_i^{\mathrm{phen}}, \qquad f_i^{(h)} = f_i/h, \qquad f_i^{(h)} d\mu_i^{(h)} = f_i d\mu_i^{\mathrm{phen}} = d\nu_i.$$

The signed measure ν_i, its Jordan decomposition, terminal residual, and additive PTVC are invariant. The factors μ_i^{phen} and f_i, duration/intensity decompositions, and diagnostics normalized by either factor are not invariant. Separate interpretation requires a prospectively external duration/unity anchor. The protocol records ExposureGauge, ExposureAnchor, and AdmissibleGaugeFamily; terminal fit never selects h.

The finite signed-measure branch requires finite total variation:

$$|\nu_i|(\Omega_i) < \infty.$$

Nonvacuity. A qualifying stream has a nonempty phenomenal domain. Unless zero-exposure streams are admitted as a separately declared degenerate class, require

$$\Omega_i \neq \emptyset, \mu_i^{\mathrm{phen}}(\Omega_i) > 0.$$

Ownership convention. Either μ_i^{phen} is already restricted to events owned by stream i, or ownership is supplied by one explicit unity gate; the same relation is not encoded twice. When retained routing hypotheses share candidate tokens, let $\left(\mathsf{E}^{\mathrm{tok}}, \Sigma_{\mathsf{E}}^{\mathrm{tok}}\right)$, $\mathcal{E}^{\mathrm{cand}} \in \Sigma_{\mathsf{H}} \otimes \Sigma_{\mathsf{E}}^{\mathrm{tok}}$, $\mu^{\mathrm{cand}}, \nu_{\mathrm{cand}}^{+}, \nu_{\mathrm{cand}}^{-}$ denote, respectively, the declared candidate-token measurable space, a measurable history-token carrier with its trace sigma-field, and the positive control and finite positive component measures on that carrier. For each routing branch, define jointly measurable weights $w_j^{(r)}: \mathcal{E}^{\mathrm{cand}} \to [0,1]$.

Ownership completeness. For each routing branch r and history-token pair in the declared carrier, apply one frozen allocation to exposure and to both signed components:

Allocation and slack satisfy

$$\begin{aligned} w_j^{(r)}(\Gamma, e) &\geq 0, \qquad w_j^{(r)}(\Gamma, e) = 0 \text{ if } j \notin \mathsf{S}^\star(\Gamma), \\ u_r(\Gamma, e) &:= 1 - \sum_{j \in \mathsf{S}^\star(\Gamma)} w_j^{(r)}(\Gamma, e) \geq 0, \\ \sum_{j \in \mathsf{S}^\star(\Gamma)} w_j^{(r)}(\Gamma, e) + u_r(\Gamma, e) &= 1. \end{aligned}$$

On this carrier, define the allocated exposure and signed-component measures:

$$\begin{aligned} \mathrm{d}\mu_{j,r}^{\mathrm{phen}} := w_j^{(r)}\, \mathrm{d}\mu^{\mathrm{cand}}, &\qquad \mathrm{d}\nu_{j,r}^{\pm} := w_j^{(r)}\, \mathrm{d}\nu_{\mathrm{cand}}^{\pm}. \\ \mathrm{d}\mu_r^{\mathrm{slack}} := u_r\, \mathrm{d}\mu^{\mathrm{cand}}, &\qquad \mathrm{d}\nu_r^{\mathrm{slack},\pm} := u_r\, \mathrm{d}\nu_{\mathrm{cand}}^{\pm}, \\ \sum_{j \in \mathsf{S}^\star(\Gamma)} \mathrm{d}\,\mu_{j,r}^{\mathrm{phen}} + \mathrm{d}\mu_r^{\mathrm{slack}} = \mathrm{d}\mu^{\mathrm{cand}}, &\qquad \sum_{j \in \mathsf{S}^\star(\Gamma)} \mathrm{d}\,\nu_{j,r}^{\pm} + \mathrm{d}\nu_r^{\mathrm{slack},\pm} = \mathrm{d}\nu_{\mathrm{cand}}^{\pm}. \end{aligned}$$

Every nonzero slack token is classified as excluded/nonphenomenal, assigned to a named residual bearer, outside the declared domain, or unresolved. Unresolved positive variation, negative variation, atoms, singular-continuous mass, and exposure are propagated as separate typed slack coordinates rather than collapsed into one net signed slack. Each unresolved coordinate is bounded and propagated or returns Model-ambiguous/No adequate estimate. Fission, fusion, copying, and restoration may redistribute a fixed token or history weight but do not assert conservation of future phenomenal mass.

Maximality rule. Fix prospectively a continuation-and-ownership relation $\preccurlyeq_\eta$ on candidate stream instances. A qualifying instance s is maximal on history Γ when no distinct qualifying instance s' satisfies s $\prec_\eta s'$ while preserving the declared phenomenal-event ownership and continuation rules. Maximality does not mean the largest spatial, anatomical, organismic, legal, social, or computational aggregate. If component and whole candidates overlap or nest, the protocol must provide distinct or explicitly apportioned event ownership satisfying nonduplication; otherwise the bearer verdict is Model-ambiguous or No adequate estimate.

Qualifying-stream truth predicate and protocol bridge. Let $\mathcal{C}$ be the candidate marked-instance carrier. Define the prospectively fixed measurable truth predicate, qualifying-label carrier, and counting configuration by

$$\mathcal{Q}^{\star}: \mathcal{C} \to \{0,1\}, \qquad \mathsf{S}^{\star}(\Gamma) := \{s: (\Gamma, s) \in \mathcal{C},\ \mathcal{Q}^{\star}(\Gamma, s) = 1\}, \qquad \pi_{\mathsf{S}}^{\star}(\Gamma) := \sum_{s \in \mathsf{S}^{\star}(\Gamma)} \delta_s.$$

Thus $\mathsf{S}^{\star}(\Gamma) = \{s \in \mathsf{S}: \pi_{\mathsf{S}}^{\star}(\Gamma)(\{s\}) > 0\}$. The qualifying-label carrier is recoverable from the atoms of its counting configuration on the declared finite or countable measurable-label branch. The label space is standard Borel, so singleton labels are measurable. Membership statements use the carrier; counting and integration use the configuration.

The protocol evidence object is

$$\mathcal{Q}_{\eta}(\Gamma, s) \in \{\emptyset, \{0\}, \{1\}, \{0,1\}\}.$$

Here $\emptyset$ means not adjudicated, $\{0\}$ excluded, $\{1\}$ qualifying, and $\{0,1\}$ unresolved. The truth predicate, not the fallible classifier, defines the universal carrier. Criteria, adjudicator, version, disagreement history, and error calibration are frozen without terminal-residual, endpoint-surprise, or completion-potential output.

Quotient and congruence requirement. If continuation defines a preorder, extension-equivalent labels are quotiented before maximal classes are counted. Exposure, signed measures, residuals, endpoint conventions, and observation maps must be constant on each equivalence class, or the quotient is not admissible. Confirmatory use requires a nonempty measurable maximal-class correspondence and either a justified measurable selector or a rule retaining every incomparable maximal class with history-level accounting. Failure returns Model-ambiguous or No adequate estimate.

Ownership encoding is prospectively fixed in the versioned protocol as either restricted exposure or common exposure plus one unity gate. It is described as preregistered only when an immutable registry identifier, version, and timestamp precede access to the decisive data. On the restricted-exposure branch the ownership gate equals one almost everywhere. On the common-base branch no second multiplicative ownership weight may enter $\hat{F}$, $g_{\theta,j}$, or quadrature weights; the implementation test verifies that ownership enters exactly once.

On the declared physical-time chart, when the signed and exposure push-forwards are absolutely continuous, define the realized density and its conditional predictor separately:

$$v_i^{\mathrm{real}}(t) := \frac{\mathrm{d}\nu_i^{\mathrm{time}}}{\mathrm{d}\mu_i^{\mathrm{time}}}(t), \qquad m_i(t) := \mathbb{E}[v_i^{\mathrm{real}}(t) \mid \mathcal{F}_{i,t}^{\mathrm{lat}}].$$

Pathwise ledger identities use v_i^{real}. The conditional object m_i is a filtering or forecasting target and may replace the realized density only under a separately identified bridge.

$$L_i^{\mathrm{phen}}(t) = L_i^{\mathrm{phen}}(S_i^{\theta} -) + \int_{S_i^{\theta}}^{t} g_{\theta,i}(s) q_i^{\mathrm{exp}}(s) v_i^{\mathrm{real}}(s)\, \mathrm{d}s.$$

On the primary whole-stream branch, S_i^θ is the truth-domain origin and $L_i^{\text{phen}}(S_i^\theta -) = 0$. A later study or measurement entry is represented instead by $S_{i,\eta}^{\text{entry}}$ and the delayed-entry decomposition; any inherited pre-entry residual is identified, bounded, or returns No adequate estimate. Atomic and singular components remain in ν_i and are not absorbed into the density. This chart representation is valid only after the chart, ownership rule, exposure convention, and density bridge have been fixed independently. If μ_i^{phen} is already restricted to stream-owned exposure, $g_{\theta,i}$ may encode only a separately defined unity or exposure-density factor; it may not reapply ownership. If a common exposure base is used, $g_{\theta,i}$ may supply the ownership or apportionment weight. The same ownership effect is never encoded twice.

The exact path bridge is anchored at the pre-origin left limit,

$$L_i^{\text{phen}}(S_i^\theta -) = L_i^{\text{rc}}(S_i^\theta -) = c_{\text{LU}} U_i(S_i^\theta -) Y_i(S_i^\theta -).$$

Any prospectively owned atom at S_i^θ is added once and consistently to both paths. On the named zero-initial absolutely continuous chart, both left limits are zero. The bridge-error process is therefore defined from the same left-limit anchor; no origin-time jump may be assigned to only one path.

Past-fixity axiom (prefix-local branches). If a branch assigns phenomenal value locally to events on the accrued prefix, and two admissible regimes agree on stream identity, event ownership, routing, phenomenal representation, and experienced events through time t, then

$$\nu_i^{(1)}(B) = \nu_i^{(2)}(B) \quad \text{for every } B \in \Sigma_i \text{ with } B \subseteq \Omega_i^{\leq}(t).$$

Later differences may change future experience and routing; they do not revalue phenomenal mass already accrued on the common prefix. A future-contextual or globally nonlocal branch must state its separate consistency rule and may not invoke this prefix-local axiom without a bridge theorem.

Projective prefix consistency. When two complete histories share the same owned phenomenal prefix, the restricted signed and exposure measures, origin, representation, and accrued ledger on that prefix agree, rather than merely their scalar totals. Future-contextual valuation is a separately typed nonlocal branch and cannot borrow prefix-local causal or additivity claims without a bridge theorem.

The construction is conceptually adjacent to experienced-utility aggregation, while adding explicit stream, ownership, endpoint, and measure-theoretic commitments (Kahneman et al., 1997).

Representation additivity/refinement gate. For prospectively defined disjoint event sets $B_1, \dots, B_m$, the latent signed-measure branch requires

$$\nu_i\left(\bigcup_{j=1}^{m} B_j\right) = \sum_{j=1}^{m} \nu_i\left(B_j\right)$$

The measured bridge is tested under randomized partition, concatenation, order, and temporal-resolution manipulations with signs, origins, exposure weights, and bridge parameters frozen outside the test set. Stable nonadditivity routes the claim to GTN, a vector or ordered representation, or No adequate estimate; it is not repaired by refitting ν_i to the terminal result.

1.2 Additive PTVC

Definition 2 (additive PTVC). By the Jordan decomposition theorem for finite signed measures (Folland, 1999), write

$$\nu_i = \nu_i^+ - \nu_i^-, \nu_i^+ \perp \nu_i^-.$$

Define

$$V_i^+ := \nu_i^+(\Omega_i), V_i^- := \nu_i^-(\Omega_i),$$
$$V_i^{\text{gross}} := |\nu_i|(\Omega_i) = V_i^+ + V_i^-,$$
$$C_i[\Gamma] := \nu_i(\Omega_i) = V_i^+ - V_i^-.$$

Additive PTVC (PTVC-1) is

$$Z_i^\star := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\text{phen}}(T_i^\star -), \qquad \text{PTVC}_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^\star = 0.$$

This is an intrastream relation. It permits no cross-stream offset and imposes no equality of V_i^+, V_i^-, or V_i^{gross} across distinct streams. Its one-line controlled summary is equal balance, unequal gross valence variation.

Define the strict pre-endpoint left limit by

$$L_i^{\text{phen}}(T_i^\star -) := \lim_{\substack{t \uparrow T_i^\star \\ t \in \mathfrak{G}_i(\Gamma)}} L_i^{\text{phen}}(t).$$

A useful edge case follows immediately: if ν_i is nonnegative or nonpositive and additive PTVC holds, then ν_i is the zero measure. A nonzero one-sign stream is representable on the finite signed-measure branch but cannot satisfy additive PTVC.

1.3 Bipolar mixed-valence branch

Empirical affect research motivates allowing co-present positive and negative components (Cacioppo & Berntson, 1994; Larsen et al., 2001; Russell, 1980, 2003). It does not by itself establish finite signed phenomenal measures, scalar commensurability, or the representation below.

Definition 3 (bipolar PTVC). As an explicit modeling branch, let β_i^+ and β_i^- be separately identified finite positive measures on the same phenomenal-event space and define

$$\nu_i := \beta_i^+ - \beta_i^-, V_{i,\text{bip}}^+ := \beta_i^+(\Omega_i), V_{i,\text{bip}}^- := \beta_i^-(\Omega_i), C_i^{\text{bip}}[\Gamma] := V_{i,\text{bip}}^+ - V_{i,\text{bip}}^-.$$

Bipolar PTVC is

$$V_{i,\text{bip}}^+ = V_{i,\text{bip}}^- < \infty.$$
$$|\nu_i|(\Omega_i) \le V_{i,\text{bip}}^+ + V_{i,\text{bip}}^-.$$

Define the primitive bipolar gross magnitude by

$$V_{i,\text{bip}}^{\text{gross}} := \beta_i^+(\Omega_i) + \beta_i^-(\Omega_i).$$

The scalar net total variation is a lower bound on this primitive gross magnitude. On the common finite domain, the overlap parametrization below makes the difference explicit.

$$|\nu_i|(\Omega_i) = \beta_i^+(\Omega_i) + \beta_i^-(\Omega_i) \Leftrightarrow \beta_i^+ \perp \beta_i^-.$$

Formal-status ontology. Every formal item is classifiable along four orthogonal dimensions; a displayed bracketed tag is local shorthand and does not by itself instantiate a four-field record:

Field	Allowed values
Role	DEF; ID; AX; HYP; LEMMA; PROP; THEOREM; COR; CM; ED; ST; TT
ProofStatus	not applicable; proved here; proved under cited theorem; conditional derivation; conjectured; open

Field	Allowed values
AssumptionSet	power set of the declared assumption set
Modality	SET-ALL; P1; POLICY-ALL; SUPPORT; CONTINUATION-ALL; FINITE-CLASS; none

$\mathcal{A}_{\text{decl}}$ denotes the explicitly stated or cross-referenced premises governing the item. An empty AssumptionSet is used only when the item is genuinely assumption-free. PTVC-1 has Role = HYP; it is not a proposition. Definitions and identities are not proved claims. Unless a machine-readable four-field registry is deposited, [P]/[CP] markers are local shorthand for the displayed role or proof statement and do not assert an unstated AssumptionSet or Modality.

[P] Lemma 1 (overlap parametrization and residual invariance). On any history where the finite primitive measures β_i^+ and β_i^-, together with the net measure $\nu_i = \beta_i^+ - \beta_i^-$, are defined on the same event space, there is a unique finite positive overlap measure ζ_i such that

$$\beta_i^+ = \nu_i^+ + \zeta_i, \beta_i^- = \nu_i^- + \zeta_i, \zeta_i \geq 0.$$

Consequently,

$$C_i^{\text{bip}}[\Gamma] = C_i[\Gamma], V_{i,\text{bip}}^{\text{gross}} = V_i^{\text{gross}} + 2\zeta_i(\Omega_i).$$

Thus additive and bipolar PTVC have the same net terminal residual wherever both finite representations are defined. The bipolar representation adds independent content for gross-magnitude estimands, mixed-valence measurement, and identification of overlap. If a proposed branch permits σ-finite primitive components, finite bipolar closure additionally requires finite primitive totals.

[P] Proof. Let $\lambda := \beta_i^+ - \nu_i^+ = \beta_i^- - \nu_i^-$ as signed measures. By minimality and mutual singularity of the Jordan pair, λ is nonnegative; set $\zeta_i := \lambda$. Uniqueness follows immediately. Hence every alternative finite positive/negative representation differs from the Jordan pair by one common positive overlap measure.

Primitive-gross identified set. For a fixed net signed measure ν, compatible positive/negative component pairs form

$$\mathfrak{B}(\nu) = \{(\nu^+ + \zeta, \nu^- + \zeta) : \zeta \in \mathcal{M}_+^f(\Omega)\}.$$

For any compatible pair,

$$V_{\text{bip}}^{\text{gross}} = |\nu|(\Omega) + 2\zeta(\Omega), \qquad \inf V_{\text{bip}}^{\text{gross}} = |\nu|(\Omega).$$

The infimum is attained at $\zeta = 0$. There is no finite upper bound without an independently justified overlap bound or direct component data. Net-only analyses therefore report Jordan total variation as a sharp lower bound, not identified primitive gross magnitude.

Representation-domain consequence. Let $\mathcal{D}_{\text{add}}$ and $\mathcal{D}_{\text{bip}}$ denote the complete histories on which the additive and bipolar representations are respectively well defined. On their common domain,

$$C_s^{\text{add}}[\Gamma] = C_s^{\text{bip}}[\Gamma] = C_s[\Gamma].$$

The branch label therefore records representation admissibility and additional component, overlap, and gross-magnitude estimands; it does not create a different net terminal-closure predicate where both finite representations exist.

1.4 Generalized Terminal Neutrality

Definition 4 (Generalized Terminal Neutrality). Let $(\mathsf{H}, \Sigma_{\mathsf{H}})$ be the complete-history measurable space, $(\mathsf{S}, \Sigma_{\mathsf{S}})$ a standard-Borel stream-label space, and $\mathcal{D}_{\mathrm{GTN}} \subseteq \mathsf{H}$ a measurable domain. Let the canonical measurable finite/countable marked-stream map be

$$\pi_{\mathsf{S}}^{\star}: \mathsf{H} \to \mathsf{N}_f(\mathsf{S}) \cup \mathsf{N}_c(\mathsf{S}).$$

Define the GTN marked carrier by

$$\mathfrak{M}_{\mathrm{GTN}} = \{(\Gamma, s): \Gamma \in \mathcal{D}_{\mathrm{GTN}},\ s \in \mathsf{S}^{\star}(\Gamma)\}.$$

Let $H_{\mathrm{GTN}}: (\mathfrak{M}_{\mathrm{GTN}}, \Sigma_{\mathfrak{M}_{\mathrm{GTN}}}) \to (\mathcal{V}_{\mathrm{GTN}}, \Sigma_{\mathrm{GTN}})$ be jointly measurable into the declared measurable codomain. Define $N_{\mathrm{GTN}}(\Gamma)$ as the measurable extended count of marked streams with $H_{\mathrm{GTN}}(\Gamma, s) \neq 0_{\mathrm{GTN}}$, obtained as the monotone limit of finite counts on the countable branch, and define

$$\mathcal{F}_{\mathrm{GTN}} = \{\Gamma \in \mathcal{D}_{\mathrm{GTN}}: N_{\mathrm{GTN}}(\Gamma) = 0\}.$$

GTN counts as literal PTVC only if a prospectively approved bridge constructs an independently grounded finite signed phenomenal measure of finite total variation, or independently grounded finite positive and negative component measures, on a specified event domain and proves equivalence between functional zero and finite positive-negative equality.

On the scalar ordered branch $\mathcal{V}_{\mathrm{GTN}} \subseteq \mathbb{R}$, the following pointwise split is available; if the codomain is not ordered and scalar, omit the split and retain only the declared functional zero.

$$\begin{aligned} H_{\mathrm{GTN}}^{+}(\Gamma, s) &:= \max\{H_{\mathrm{GTN}}(\Gamma, s), 0\}, \\ H_{\mathrm{GTN}}^{-}(\Gamma, s) &:= \max\{-H_{\mathrm{GTN}}(\Gamma, s), 0\}. \end{aligned}$$

This tautological positive/negative split does not satisfy the bridge requirement because it does not construct independently grounded cumulative phenomenal components.

Marked-stream carrier and branch restrictions. Define once the canonical carrier

$$\mathfrak{M}^{\star} = \{(\Gamma, s): \Gamma \in \mathcal{R}^{\star},\ s \in \mathsf{S}^{\star}(\Gamma)\}.$$

Equip the marked-stream carrier with the declared marked-configuration sigma-field. For $b \in \{\mathrm{add}, \mathrm{bip}\}$, let $\mathcal{D}_b \subseteq \mathcal{R}^{\star}$ be the measurable history domain and $\mathfrak{M}_b \subseteq \mathfrak{M}^{\star} \cap (\mathcal{D}_b \times \mathsf{S})$ the measurable restriction on which $C_b(\Gamma, s)$ is defined and jointly measurable. For each history in the branch domain, define the branch fibre $\mathfrak{M}_b(\Gamma) := \{s \in \mathsf{S}: (\Gamma, s) \in \mathfrak{M}_b\}$; let its measurable extended count of nonzero-residual labels be the monotone limit of finite counts. Denote that count and its zero event, respectively, by $N_b(\Gamma)$, $\mathcal{F}_{\mathrm{PTVC}}^{b} := \{\Gamma \in \mathcal{D}_b: N_b(\Gamma) = 0\}$. The GTN branch is the restriction $\mathfrak{M}_{\mathrm{GTN}}$ with functional H_{GTN}. Every probability-one or support statement is made on these measurable restrictions of the one carrier.

1.5 Modal and complete-history formulations

Complete-history carrier. Use the same measurable carrier $(\mathsf{H}, \Sigma_{\mathsf{H}})$, label space $(\mathsf{S}, \Sigma_{\mathsf{S}})$, truth-domain class $\mathcal{R}^{\star}$, and canonical map $\pi_{\mathsf{S}}^{\star}$ throughout. Bearer qualification, routing, representation, and endpoint branches are fixed independently of terminal residuals; §1.4 defines the GTN restriction and the preceding marked-carrier paragraph defines additive and bipolar restrictions.

Truth-carrier semantics. Every target records whether the truth-domain class denotes actual realized histories, histories generated by a declared model class, the topological support of a process law, a finite registered class, or a broader nomologically admissible carrier. The modality field states the quantifier within that carrier and does not change its meaning. A conclusion may not move from an almost-sure process-law carrier to all actual or all nomologically possible histories without a separately registered bridge.

Table 2. Quantifier, attainability, and exactness matrix.

Object	Exact meaning	Does not imply
Fixed actual stream	Whether one completed stream has terminal residual zero	A population law
Fixed-law [P1]	Closure with probability one under one declared law or fixed policy	[SET-ALL]
[SET-ALL]	Every qualifying stream in the registered carrier closes	Finite-sample identifiability
[SUPPORT]	Declared topological support lies in the completion event	Every admissible continuation
[CONTINUATION-ALL]	Every continuation in the declared class closes	Truth outside that class
AS attainability	One admissible policy attains probability one	All policies or universality
LS attainability	A policy sequence approaches probability one	An attaining policy
Atom at zero	One declared law assigns positive mass to zero	Every actual stream closes
Finite-band compatibility	A truth-containing object lies in a registered band	Exact zero
Expected neutrality	A declared expectation is zero	Pathwise or terminal closure

Bearer and routing decisions, inclusions, exclusions, and event ownership are fixed without terminal outcomes. Scientifically live bearer ontologies are reported branch by branch; a favorable union across ontologies is prohibited, and unresolved routing returns No adequate estimate.

For each declared branch, the strict universal reading is restricted to the carrier on which its functional is defined:

$$\begin{aligned} \forall(\Gamma, s) \in \mathfrak{M}_b: &\qquad C_b(\Gamma, s) = 0, \qquad b \in \{\text{add}, \text{bip}\}, \\ \forall(\Gamma, s) \in \mathfrak{M}_{\text{GTN}}: &\qquad H_{\text{GTN}}(\Gamma, s) = 0_{\text{GTN}}. \end{aligned}$$

For a fixed stream under a declared process law P_i, the probability-one branch is

$$P_i\{L_i^{\text{phen}}(T_i^\star -) = 0\} = 1.$$

Modal scope. [SET-ALL] strict universality, [P1] probability-one closure, [POLICY-ALL] all-policy closure, [SUPPORT] topological-support containment, [CONTINUATION-ALL] all-continuation closure, and [FINITE-CLASS] finite-class claims are non-equivalent. Every theorem and empirical verdict names its modal branch.

Ex-ante quantifier order. A protocol version η, reference class, representation, origin, bearer/ownership rule, endpoint branch, observation law, bridge, acceptance set, and decision map are fixed independently of terminal residuals and before the confirmatory histories are evaluated. Universality therefore has the order ‘choose η, then quantify over the declared histories and streams,’ not $\forall\Gamma\ \exists\eta(\Gamma)$. A history-specific or outcome-informed protocol is a new hypothesis version unless a registered transport theorem applies.

Modal evidence rule. Weighted marked-population or sampled-stream summaries may estimate prevalence or a probability-one law under their declared sampling process, but they never establish [SET-ALL] or average away a violating qualifying maximal class. Favorable evidence moves across reference classes only through a registered transport theorem; a valid counterexample remains visible in every containing strict-universal class.

For $b \in \{\mathrm{add}, \mathrm{bip}\}$, let $\mathfrak{M}_b$ be the measurable branch restriction with fibre as defined in §1.4, and let $C_b\colon \mathfrak{M}_b \to \mathbb{R}$ be its jointly measurable residual. By the zero-count definitions in §1.4, the same events have the following equivalent pointwise representations:

$$\mathcal{F}^b_{\mathrm{PTVC}} = \{\Gamma \in \mathcal{D}_b\colon C_b(\Gamma, s) = 0 \text{ for every } s \in \mathfrak{M}_b(\Gamma)\},$$

The GTN event has the corresponding equivalent representation:

$$\mathcal{F}_{\mathrm{GTN}} = \{\Gamma \in \mathcal{D}_{\mathrm{GTN}}\colon H_{\mathrm{GTN}}(\Gamma, s) = 0_{\mathrm{GTN}} \text{ for every } s \in \mathsf{S}^\star(\Gamma)\}.$$

Branch-indexed laws on $(\mathsf{H}, \Sigma_{\mathsf{H}})$ satisfy

$$Q^b\left(\mathcal{F}^b_{\mathrm{PTVC}}\right) = 1, \qquad Q^{\mathrm{GTN}}(\mathcal{F}_{\mathrm{GTN}}) = 1.$$

Variable-stream measurability follows from the declared marked-stream construction; any alternative enumeration must establish the same measurability. Nonvacuity requires a nonempty branch-admissible stream fibre almost surely, or explicit conditioning on its presence:

$$\begin{aligned} Q^b\{\Gamma \in \mathcal{D}_b\colon \mathfrak{M}_b(\Gamma) \neq \emptyset\} &= 1, \\ Q^{\mathrm{GTN}}\{\Gamma \in \mathcal{D}_{\mathrm{GTN}}\colon \mathsf{S}^\star(\Gamma) \neq \emptyset\} &= 1. \end{aligned}$$

The stronger statement $\mathrm{supp}(Q^b) \subseteq \mathcal{F}^b_{\mathrm{PTVC}}$ additionally requires a declared topology, matching support definition, and suitable closedness. The admissible set alone determines neither relative probabilities nor local dynamics, interventions, observations, typicality, or implementation.

Outcome-independent qualification rule. Membership in $\mathcal{R}^\star$, the stream map, representation branch, endpoint branch, and routing ontology are fixed independently of the terminal residual. Excluding an off-boundary stream after observing its result defines a narrower hypothesis.

Primary confirmatory target. Unless otherwise stated, the principal target is the finite signed-measure scalar or explicitly bipolar, strict-pre-endpoint branch over streams satisfying $Q^\star(\Gamma, s) = 1$. GTN, vector closure, endpoint-inclusive closure, extended continuation, infinite-stream targets, and stronger policy or intervention branches remain separately named extensions. Any infinite-stream claim defines its convergence mode and finite-variation or renormalization rule.

1.6 Layer separation

Let $D^{\mathrm{obs}}_{i,\eta}$ denote the complete protocol-observed data object after declared observation, retention, missingness, and preprocessing rules. The analysis map $\mathcal{A}_\eta$ has a prospectively stated measurable domain and codomain; it produces a measured estimator and never changes a truth-level target by definition.

The latent phenomenal object, route-specific consequence coordinate, measurement map, measured estimand, analysis map, and measured estimator are distinct objects.

Primary decision object. Before outcomes, freeze one PrimaryDecisionObject: branch and modality, reference class, terminal compatibility object, acceptance set, and inferential guarantee. Every set or envelope retains its inferential type through unions, projections, and bridges; framework disagreement is Inference-sensitive, not a license for post-inspection selection.

$$M_{\eta,i}(t) := M_\eta\left(L^{\mathrm{phen}}_i, X_i, W^{\mathrm{nuis}}_i, O^{\mathrm{obs}}_i; t\right).$$

$$\hat{L}_{i,\eta} := A_\eta\left(D^{\mathrm{obs}}_{i,\eta}\right).$$

A sampled estimator uses exactly one ownership encoding. On the full-contribution-rate branch, $\hat{F}$ already contains the complete signed rate. On the value-per-exposure branch, the exposure density, value per unit exposure, and quadrature weight remain separate. The protocol records the selected ownership-encoding branch and prohibits multiplying the same ownership gate twice:

$$\widehat{\Delta L}^{(F)}_{i,\eta}[a,b) := \sum_{k:\, a \le t_k < b} \hat{F}_{i,\eta}(t_k) w_k,$$

$$\widehat{\Delta L}^{(\mathrm{dens})}_{i,\eta}[a,b) := \sum_{k:\, a \le t_k < b} g_{\theta,i}(t_k) \hat{q}_i^{\exp}(t_k) \hat{v}^{\mathrm{real}}_{i,\eta}(t_k) w_k.$$

Use one fixed ownership branch; call it preregistered only when its versioned immutable timestamp predates decisive-data access. The estimators are interchangeable only if a validated bridge proves $F = g q^{\exp} v^{\mathrm{real}}$ on the sampled grid.

The measurement map M_η defines $M_{\eta,i}$, and A_η produces $\hat{L}_{i,\eta}$ from sampled observations. Calibration does not identify the estimand, nor does an identified measured estimand identify the latent object without a bridge. L_i^{phen} denotes latent-hypothesis quantities; L_i^{rc} denotes candidate-route quantities.

2. Exact zero, terminal laws, and inference hierarchy

2.1 Exact zero as a terminal-law property

Define the truth-level terminal scalar by

$$Z_i^\star := L_i^{\mathrm{phen}}\,(T_i^\star -).$$

Under a declared baseline or rival law P_i^0, if the law of $Z_i^\star$ has no atom at zero, then

$$P_i^0\,(Z_i^\star = 0) = 0.$$

Absolute continuity is a common sufficient condition; the precise requirement is non-atomicity at the declared neutral point. First hitting, absorption, singular support, endpoint coupling, predictable atoms, exact conditioning, and endpoint-inclusive phenomenal atoms are mathematically different constructions because they alter the terminal law in different ways.

Under the PTVC process law P_i,

$$\mathcal{L}_{P_i}(Z_i^\star) = \delta_0.$$

All spaces on which regular conditional laws or observation kernels are invoked are standard Borel unless existence is established otherwise (Kallenberg, 2002). Let Q be the complete-history law and $\kappa(\cdot \mid \Gamma)$ a prospectively declared probability kernel selecting a mark within history. A complete history already determines its terminal residual; no second within-history random variable is introduced:

$$\overline{P}(\mathrm{d}\Gamma, \mathrm{d}s) := Q(\mathrm{d}\Gamma)\kappa(\mathrm{d}s \mid \Gamma), \qquad Z^\star(\Gamma, s) := Z_s^\star[\Gamma].$$

The marked-population probability-one statement is

$$\mathcal{L}_{\overline{P}}(Z^\star) = \delta_0.$$

This probability-one formulation remains distinct from strict set-theoretic universality. History-weighted, typical-stream (stream-intensity), and exposure-weighted estimands are reported separately. No uniform law is assumed on a countably infinite mark set; history or lineage is the clustering unit, and informative stream count is tested rather than treated as additional independent sample size.

Conditional canonicality. Within the common-residual family, zero is the unique numerical residual preserved under independent positive changes of streamwise unit; within the common-ratio family, one-to-one balance is the fixed point of positive/negative pole exchange. Appendix C states the assumptions and intersection result. These facts give zero and one-to-one balance precise canonical roles within those model families and define mathematical targets for PTVC's empirical tests.

Typed evidence vocabulary. Reserve $\mathcal{J}_{\mathrm{ID}}$ for deterministic identified regions, $\mathcal{J}_{1-\alpha}$ for simultaneous confidence regions, $\mathcal{J}_{\mathrm{Post},q}$ for credible regions, and $\mathcal{J}_{\mathrm{Sens}}$ for sensitivity envelopes. Use respectively: deductive incompatibility under maintained assumptions; frequentist rejection at the registered error level; and posterior incompatibility under the registered prior. Finite-band equivalence is a resolution-indexed positive result; exact-atom adjudication is the nested limiting question under an identified observation model.

2.2 Observation-kernel push-forward

[CP] Proposition 0 (observation-operator identification boundary). Fix a declared latent-law class $\mathcal{P}_Z$ satisfying $\delta_0 \in \mathcal{P}_Z$, and an observation kernel K. If

$$\exists P_1 \in \mathcal{P}_Z: \quad P_1 \neq \delta_0 \text{ and } \mathcal{T}_K(P_1) = \mathcal{T}_K(\delta_0)$$

then the property $P_Z = \delta_0$ (equivalently, unit atom mass $P_Z(\{0\}) = 1$) is not identified over the model class $\mathcal{P}_Z$ from the observed push-forward law $\mathcal{T}_K(P_Z)$. If the operator $P \mapsto \mathcal{T}_K(P)$ is injective on $\mathcal{P}_Z$, the latent law is identified from its observed push-forward. Stable finite-sample recovery is a separate requirement, characterized by a registered inverse modulus, for example

$$\begin{aligned}\omega_{\theta,K}(\delta) \quad &:= \sup\{|\theta(P) - \theta(Q)|: P, Q \in \mathcal{P}_Z,\ d_{\mathrm{obs}}(\mathcal{T}_K P, \mathcal{T}_K Q) \leq \delta\},\\ \omega_{\theta,\mathcal{K}_\eta}(\delta) \quad &:= \sup\{|\theta(P) - \theta(Q)|: P, Q \in \mathcal{P}_Z,\ K, K' \in \mathcal{K}_\eta,\ d_{\mathrm{obs}}(\mathcal{T}_K P, \mathcal{T}_{K'} Q) \leq \delta\}.\end{aligned}$$

The first modulus treats K as known and fixed; the second propagates registered kernel uncertainty. The proposition distinguishes population identification from numerical and statistical stability.

[P] Proof. The first claim follows because two distinct latent laws yield the identical observed law. The second is the definition of injectivity. Protocol use still requires calibration showing that the operative kernel and law class satisfy the asserted premises.

[CP] Proposition 0B (finite-sample nonseparation of a zero atom under locally smoothing observation)

Let K be an observation kernel and define the local total-variation modulus below. Assume the registered latent-law class contains the point mass at zero and atomless laws supported in shrinking neighborhoods of zero.

$$\omega_K(r) := \sup_{|z| \leq r} d_{\mathrm{TV}}\big(K(z,\cdot), K(0,\cdot)\big), \qquad \omega_K(r) \to 0 \text{ as } r \downarrow 0.$$

$$P_m(\{x: |x| \leq r_m\}) = 1, \qquad P_m(\{0\}) = 0, \qquad r_m \downarrow 0.$$

$$Q_0 := \mathcal{T}_K \delta_0 = K(0,\cdot), \qquad Q_m := \mathcal{T}_K P_m.$$

Then the observed laws satisfy the contraction bound

$$d_{\mathrm{TV}}(Q_m, Q_0) \leq \int d_{\mathrm{TV}}\big(K(z,\cdot), K(0,\cdot)\big) P_m(dz) \leq \omega_K(r_m) \to 0.$$

On a branch with n independent observed units, tensorization gives

$$d_{\mathrm{TV}}\big(Q_m^{\otimes n}, Q_0^{\otimes n}\big) \leq 1 - \big(1 - d_{\mathrm{TV}}(Q_m, Q_0)\big)^n \leq n\, \omega_K(r_m).$$

$$Q_0^{\otimes n}(\phi_n = 1) + Q_m^{\otimes n}(\phi_n = 0) \geq 1 - d_{\mathrm{TV}}\big(Q_m^{\otimes n}, Q_0^{\otimes n}\big).$$

Let $\varphi_n = 1$ denote rejection of the zero-atom model in favor of the atomless alternative. Under that convention, the displayed terms are the type-I and type-II errors: false rejection of the zero-atom model under Q_0 and failure to detect the atomless alternative under Q_m.

The tensor-product display applies only to genuinely independent observed units. Repeated prompts within one participant, multiple cuts within one history, streams sharing one lineage, site clusters, household clusters, and correlated endpoint observations require the registered joint law or a dependence-specific bound. Prompt or cut count must not be substituted for the number of independent units.

Interpretation and scope. On this locally smoothing branch, no fixed finite sample uniformly separates a true zero atom from unrestricted atomless laws concentrated arbitrarily near zero. Exactness promotion therefore requires a prospectively registered separation condition, certified inverse modulus, linked resolutions, independent channels, or another structural restriction. Direct noiseless observation does not satisfy the smoothing premise, so the result is branch-specific rather than a universal impossibility theorem.

Atom-functional identified set. For observed law Q^{obs} and admissible kernel class $\mathcal{K}_\eta$, identify the functional actually required for ATOM-1 rather than assuming recovery of the full latent law:

$$\Theta_0\left(Q^{\mathrm{obs}}, \mathcal{K}_\eta\right) := \{P(\{0\}) \colon \mathcal{T}_K P = Q^{\mathrm{obs}} \text{ for some } P \in \mathcal{P}_Z,\ K \in \mathcal{K}_\eta\}.$$

Point identification of the atom mass requires this set to be a singleton. Exact PTVC on this branch is identified only when that singleton contains one and the independently calibrated neutral-origin set contains only zero; if neutral origin remains set-valued, apply §3.4 before interpreting atom location. Finite-sample promotion also requires a certified inverse-stability or recovery analysis over latent-law and kernel pairs. Otherwise report partial identification, instability, or No adequate estimate.

Concrete additive illustration. Under $Y = Z + \varepsilon$, the same observed variation can be assigned entirely to latent residual Z, entirely to observation error ε, or to mixtures of both unless replicate restrictions, external validation, an exclusion restriction, or an injective kernel identifies the decomposition. This example instantiates Proposition 0 without creating a second identification result.

Observation maps the latent terminal law through a protocol-specific kernel. In the additive special case, define

$$P_{\varepsilon,i,T} := \mathcal{L}\left(\varepsilon_{i,T}\right), \quad O_{i,T}^{\mathrm{obs}} := Z_i^\star + \varepsilon_{i,T}.$$

For a general latent law, the observation push-forward identity (OBS-1) is

$$Z_i^\star \perp\!\!\!\perp \varepsilon_{i,T} \quad \Rightarrow \quad \mathcal{L}\left(O_{i,T}^{\mathrm{obs}}\right) = \mathcal{L}(Z_i^\star) * P_{\varepsilon,i,T}.$$

Here * denotes convolution and is distinct from the truth-level star. Under exact PTVC, the latent terminal residual is zero almost surely, so the observed additive-law distribution equals the error law without a separate independence premise. Nuisance-dependent error remains inside the conditional observation kernel.

Let N_i^{term} collect endpoint state, care context, medication, device quality, reporting capacity, route state, and other observation-relevant variables. Then

$$P_{i,T}^{\mathrm{obs}}(A \mid N_i^{\mathrm{term}} = n) = \int K_{\eta,i}^{\mathrm{obs}}(A \mid z, n)\, \mathrm{d}P_{Z_i^\star \mid N_i^{\mathrm{term}} = n}(z),$$

$$P_{i,T}^{\mathrm{obs}}(A) = \iint K_{\eta,i}^{\mathrm{obs}}(A \mid z, n)\, \mathrm{d}P_{Z_i^\star, N_i^{\mathrm{term}}}(z, n).$$

Nuisance-dependent or merely conditionally independent error remains inside the conditional observation kernel rather than an unconditional convolution. The shortcut applies only under the declared additive, endpoint-stable, and nuisance-invariant conditions.

2.3 Scientific equivalence and adjudicability

Let $\mathcal{A}_{i,\eta}^{\mathrm{phen}}$ be the prospectively calibrated scientific acceptance set in declared latent phenomenal units. Estimand guidance and the equivalence-testing literature motivate separation of the target from uncertainty but do not supply this construction (International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use [ICH], 2019; Lakens, 2017).

For each measured value y, define the compatible-origin set $\mathcal{Z}_{i,\eta}(y) := \{z \in \mathcal{Z}_{0,i,\eta}: y \in \mathcal{J}_{i,\eta,T}^{\mathrm{meas}}(z)\}$ and, for every $z \in \mathcal{Z}_{i,\eta}(y)$, the bridge image $\mathcal{B}_{i,\eta,z}[y] := \{x: (y,x) \in \mathcal{B}_{i,\eta,z}\}$. The origin-specific and robust acceptance sets are

$$\begin{aligned}\mathcal{A}_{i,\eta}^{\mathrm{meas}}(z) &:= \{y \in \mathcal{J}_{i,\eta,T}^{\mathrm{meas}}(z): \mathcal{B}_{i,\eta,z}[y] \neq \emptyset,\ \mathcal{B}_{i,\eta,z}[y] \subseteq \mathcal{A}_{i,\eta}^{\mathrm{phen}}\},\\ \mathcal{A}_{i,\eta}^{\mathrm{meas}} &:= \{y: \mathcal{Z}_{i,\eta}(y) \neq \emptyset,\ \forall z \in \mathcal{Z}_{i,\eta}(y),\ \mathcal{B}_{i,\eta,z}[y] \neq \emptyset \text{ and } \mathcal{B}_{i,\eta,z}[y] \subseteq \mathcal{A}_{i,\eta}^{\mathrm{phen}}\}.\end{aligned}$$

The preimage may be asymmetric or disconnected. A symmetric interval $[-K_{i,\eta}^{\mathrm{phen}}, K_{i,\eta}^{\mathrm{phen}}]$ is used only on a registered identity or positive-linear bridge branch. Under a positive unit change, transform the target and any scalar margin together:

$$\tilde{\mathcal{A}}_{i,\eta}^{\mathrm{phen}} = b_i \mathcal{A}_{i,\eta}^{\mathrm{phen}}, \qquad \tilde{K}_{i,\eta}^{\mathrm{phen}} = b_i K_{i,\eta}^{\mathrm{phen}}, \qquad b_i > 0.$$

The observed-scale TOST margin $\delta_{i,\eta}^{\mathrm{meas}}$ is a distinct calibrated object. It is identified with the latent scientific half-width $K_{i,\eta}^{\mathrm{phen}}$ only under a declared identity or known positive-linear measured-to-latent bridge.

Measured equivalence is affirmative only when the complete nonempty measured compatibility set is contained in $\mathcal{A}_{i,\eta}^{\mathrm{meas}}$. Latent finite-band compatibility and exact PTVC remain separately named conclusions.

$$\begin{gathered}\mathcal{C}_{\eta}^{\mathrm{joint}} \subseteq \mathcal{C}_{\eta}^{\mathrm{meas}} \times \mathcal{C}_{\eta}^{\mathrm{end}} \times \mathcal{C}_{\eta}^{\mathrm{miss}} \times \mathcal{C}_{\eta}^{\mathrm{rep}}\\ \times \mathcal{C}_{\eta}^{\mathrm{cal}} \times \mathcal{C}_{\eta}^{\mathrm{scal}} \times \mathcal{C}_{\eta}^{\mathrm{route}},\\ \mathcal{J}_{i,\eta,T}^{\mathrm{meas}} := \phi_{\eta}\left(\mathcal{C}_{\eta}^{\mathrm{joint}}\right).\end{gathered}$$

Each component retains its type: deterministic identified set, simultaneous confidence object, posterior object, sensitivity set, or design tolerance. The declared joint relation Σ_{η} carries dependence; projection through ϕ_{η} occurs once, and bridge or route uncertainty is not added again after projection.

For the nonempty full protocol-adjusted measured compatibility set, measured equivalence is

$$D_{\mathrm{eq},i,\eta} := \mathbf{1}\{\mathcal{J}_{i,\eta,T}^{\mathrm{meas}} \neq \emptyset,\ \mathcal{J}_{i,\eta,T}^{\mathrm{meas}} \subseteq \mathcal{A}_{i,\eta}^{\mathrm{meas}}\}.$$

Equivalence-design planning. For $\mu_0 = 0$, a known observed-scale standard deviation, per-one-sided-test level α, and target joint power $1-\beta$, the exact symmetric-TOST normal-theory solution rounded up is

$$n = \lceil [\frac{\left(z_{1-\alpha} + z_{1-\beta/2}\right)\sigma_{\mathrm{obs}}}{\delta_{i,\eta}^{\mathrm{meas}}}]^2 \rceil.$$

For $\mu_0 \neq 0$, replacing the registered equivalence margin by its distance from $|\mu_0|$ gives a conservative bound; compute exact power from the two normal tails or simulate the registered design. Estimated variance, dependence, clustering, multiplicity, or nonnormality requires the exact registered design model or simulation.

Normalized-imbalance sensitivity. Preserve history and representation-branch indices, impose a positive denominator floor, and infer the numerator and denominator jointly:

$$\begin{aligned} r_{i,b}^{\mathrm{bal}}[\Gamma] &:= \frac{C_i^{(b)}[\Gamma]}{V_{i,b}^{\mathrm{gross}}[\Gamma]}, \qquad V_{i,b}^{\mathrm{gross}}[\Gamma] \geq v_{\min} > 0,\\ \mathcal{R}_{i,\eta}^{\mathrm{bal}} &:= \left\{c/v: (c,v) \in \mathcal{J}_{i,\eta}^{(C,V)},\ v \geq v_{\min}\right\}.\end{aligned}$$

The primary test remains on the absolute calibrated scale. If the joint support contains no denominator at or above $v_{\min}$, return No adequate estimate; report low-gross automatic passage as a false-support diagnostic. The signed ratio is primary and $|r|$ is a separately labeled sensitivity quantity.

Nonvacuity and failure typing. A nonempty compatibility set is adjudicated by the registered containment or adverse rule. An empty set obtained from a valid inversion or deterministic feasibility analysis is Model-incompatible, Bridge-failed, Route-failed, or another named gate rejection at its stated error and assumption level; it cannot pass equivalence by vacuous subset logic. An undefined or uncomputed object caused by zero support, numerical failure, missing prerequisites, or unavailable data is No adequate estimate. Every empty or unavailable object records its cause.

Band calibration. The scientific acceptance set is a substantive, outcome-independent tolerance on a named scale, with unit, transformation law, provenance, and version. Sampling error, instrument resolution, endpoint uncertainty, bridge uncertainty, missingness, and model uncertainty belong in the typed compatibility object, not in the scientific margin. Improved precision narrows that compatibility object. Changing the scientific margin requires a separate substantive calibration and new version. Asymmetric margins are permitted only when independently justified. Normalized imbalance is inferred jointly with its denominator and remains signed; absolute magnitude is a separate sensitivity estimand. A zero-gross denominator is a degenerate branch and supplies no ratio evidence.

Band provenance tiers. A confirmatory band is independently calibrated on the decision-bearing construct and may control the Equivalent verdict. A provisional externally elicited band is frozen before outcomes and permits only “compatible under the provisional margin.” A sensitivity-only band family maps conclusions across plausible margins but cannot control a confirmatory verdict. A later validated margin creates a new protocol version; no result is upgraded retroactively.

2.4 Inference hierarchy

Empirical reports should separate:

1. cohort-mean compatibility;
2. distributional concentration;
3. individual measured compatibility;
4. transfer to a latent-compatible set;
5. route-specific support against declared rivals;
6. strong latent closure in a declared reference class.

Non-ergodic dynamics can separate ensemble and within-stream behavior (Molenaar, 2004). Finite noisy data can support finite-band equivalence, concentration, route-specific cross-equation restrictions, and rejection of declared rivals. A strict universal exact claim ultimately requires a structural theorem connecting independently validated physical and psychophysical postulates to the terminal law.

Strict-universal falsification asymmetry. Fix prospectively the truth-domain class, stream and routing ontology, value-representation branch, neutral origin, sign, scalarization, endpoint branch, observation model, and measured-to-latent bridge. One qualifying stream whose nonempty truth-containing identified set, or deterministic compatibility set guaranteed under the maintained assumptions to contain the latent terminal value, excludes zero refutes strict exact PTVC for that declared class. If the reported object is a confidence set, exclusion of zero supports rejection only at the declared error or coverage level. If it is a Bayesian credible set, exclusion supports the declared posterior-incompatibility statement. Neither latter result is a deductive refutation without an additional truth-containment guarantee. Reclassifying the stream, moving the endpoint, changing the origin or scalarization, or adding continuation after outcome inspection defines a new hypothesis rather than preserving the original universal claim.

Counterexample-search multiplicity. Searching across streams, endpoints, origins, windows, subgroups, bearer ontologies, or preprocessing choices is multiplicity-controlled under the registered search family. A noisy candidate becomes a logical counterexample only after its prerequisites and truth-containing adverse object are independently confirmed; the set-universal consequence then applies to that confirmed instance rather than to an outcome-selected screen.

Finite-resolution exactness program. Rounded, censored, discretized, or finite-precision observations cannot by themselves distinguish a latent atom at zero from sufficiently concentrated continuous, quantized, or observation-floor alternatives. Exactness is therefore evaluated through an identified observation kernel, behavior across resolution levels, replicated channels, and independently predicted structural consequences.

Nested exactness family (ATOM-1). Conditional on covariates x and a fixed reference class, write

$$\begin{aligned} \mathcal{L}(Z_i^\star \mid X = x) &= p_0^{\mathrm{atom}}(x)\delta_0 + [1 - p_0^{\mathrm{atom}}(x)]F_{0^c}(\cdot \mid x), \\ F_{0^c}(\{0\} \mid x) &= 0. \end{aligned}$$

The condition $p_0^{\mathrm{atom}}(x) = 1$ for P_X-almost every x is a [P1] zero-atom closure statement on the declared law and reference class, not [SET-ALL]. Values strictly between zero and one define a mixed zero-atom branch. The condition $p_0^{\mathrm{atom}}(x) = 0$ means only that the conditional law has no atom at zero; it may still contain off-zero atoms, lattice or quantized support, a singular-continuous component, an absolutely continuous component, or a mixture of these. Section 9.3 supplies the decision rules, and §11.7.4 supplies the estimation and recovery program.

At $p_0^{\mathrm{atom}}(x) = 1$, $F_{0^c}(\cdot \mid x)$ has zero mixture weight and is not identified; remove it from the boundary parameterization or fix it by a stated noninferential convention.

ATOM-1 scope, rivals, and stability. Fit a location-free rival that estimates both atom location a(x) and mass $p_a(x)$; exact PTVC promotion requires location zero and mass one on the declared modal branch. Recovery includes off-zero atoms, lattice/quantized laws, singular-continuous laws, highly concentrated continuous laws, censoring, heaping, and observation floors. Under kernel uncertainty, inverse stability ranges jointly over latent-law and kernel pairs K, K'; a shared-known-kernel result is reported separately. The resolution ladder uses a common latent target, linked instruments or a justified garbling order, nested acceptance sets with intersection $\{0\}$, and simultaneous or sequential error control.

3. Representation, neutral origin, scalarity, and partial identification

3.1 Affine transformation and exposure

The sign, origin, unit, scalarity, and interpersonal comparability of phenomenal value are identification problems rather than notation conventions (Krantz et al., 1971). Let

$$E_i^{\mathrm{phen}}(t) := \mu_i^{\mathrm{phen}}\left(\Omega_i^{\leq}(t)\right), L_i^{\mathrm{phen}}(t) := \nu_i\left(\Omega_i^{\leq}(t)\right).$$

When $\nu_i \ll \mu_i^{\mathrm{phen}}$, let $f_i := d\nu_i / d\mu_i^{\mathrm{phen}}$. Under a stream-constant affine transformation,

$$\tilde{f}_i(s) = a_i + b_i f_i(s), b_i > 0.$$

The accrued value becomes

$$\tilde{L}_i^{\mathrm{phen}}(t) = a_i E_i^{\mathrm{phen}}(t) + b_i L_i^{\mathrm{phen}}(t).$$

The transformed measure remains inside the finite signed-measure branch only when the translated exposure term is finite. A sufficient requirement is

$$a_i \neq 0 \Rightarrow E_i^{\mathrm{phen}}(T_i^{\star}-) < \infty.$$

In a physical-time chart,

$$E_i^{\mathrm{phen}}(t) = \int_{S_i^{\theta}}^{t} g_{\theta,i}(s) q_i^{\exp}(s)\, \mathrm{d}s.$$

Additive translation can alter levels, increments, dispersion, covariance, autocorrelation, reserve-normalized coordinates, regression coefficients, endpoint relations, and intervention contrasts whenever exposure varies. The positive and negative Jordan masses also change nonlinearly because the sign regions may move under recentering.

Every primary diagnostic must state its admissible transformation group, exposure assumptions, and whether exposure is fixed, matched, modeled, conditioned upon, or residualized.

Zero-preserving transformation criterion. Let $w_i : \Omega_i \to (0,\infty)$ be a fixed measurable multiplier chosen independently of the ledger. Assume the following implication for every finite signed measure on the declared event space whose weighted total variation is finite:

$$\nu(\Omega_i) = 0 \Rightarrow \int_{\Omega_i} w_i \, \mathrm{d}\nu = 0 \quad \text{for every finite signed } \nu \text{ with } w_i \in L^1(|\nu|).$$

Then the multiplier is constant on the event space: $w_i \equiv c_i > 0$. Indeed, applying the implication to $\delta_x - \delta_y$ yields equality at any two points. Hence a stream-constant positive scale is the maximal fixed pointwise multiplicative change of units that preserves every zero ledger. Event-, state-, or time-dependent positive reweighting is a new representation unless a branch-specific theorem proves invariance. Register its integrability domain and effects on atoms, singular components, exposure, acceptance sets, and route diagnostics.

3.2 Neutral-relative covariance and cross-sign calibration

Event-space-to-time transport. Let $\pi_i^{\mathrm{time}} : \Omega_i \to [S_i^{\theta}, T_i^{\star})$ be the declared measurable time map. Define the push-forward exposure and signed origin measure before introducing a physical-time origin field:

$$\mu_i^{\text{time}} = \left(\pi_i^{\text{time}}\right)_{\#}\mu_i^{\text{phen}}, \qquad m_i^{\alpha}(B) = \int_{\left(\pi_i^{\text{time}}\right)^{-1}(B)} \alpha_i\,(x)\,\mu_i^{\text{phen}}(\mathrm{d}x).$$

$$m_i^{\alpha} \ll \mu_i^{\text{time}} \Rightarrow \alpha_i^{\text{time}} = dm_i^{\alpha}/d\mu_i^{\text{time}}.$$

On the absolutely continuous clock-time branch, $\mathrm{d}\mu_i^{\text{time}}(s) = g_{\theta,i}(s)q_i^{\exp}(s)\,\mathrm{d}s$. If the Radon–Nikodym derivative does not exist, the time representation remains measure-valued; no pointwise origin field is asserted.

$f_i - \alpha_i \in L^1\left(\mu_i^{\text{phen}}\right)$ is required; a simple sufficient branch is $f_i, \alpha_i \in L^1\left(\mu_i^{\text{phen}}\right)$. A nonzero constant translation additionally requires $\mu_i^{\text{phen}}(\Omega_i) < \infty$. If an independently calibrated neutral density $\alpha_i(x)$ is available on the phenomenal-event domain, define

$$L_{i,\alpha}(t) := \int_{S_i^{\theta}}^{t} g_{\theta,i}\,(s)q_i^{\exp}(s)[v_i^{\text{real}}(s) - \alpha_i^{\text{time}}(s)]\,\mathrm{d}s.$$

Replacing v_i^{real} by $m_i(s) := \mathbb{E}[v_i^{\text{real}}(s) \mid \mathcal{F}_{i,s}^{\text{lat}}]$ requires the separately validated predictor-to-realization bridge.

Under the affine transformation

$$\tilde{\alpha}_i(x) = a_i + b_i\alpha_i(x),$$

the neutral-relative object transforms covariantly:

$$\tilde{L}_{i,\tilde{\alpha}}(t) = b_i L_{i,\alpha}(t).$$

Time-map transport. The event-space field induces the clock-time origin through the Radon–Nikodym derivative $\alpha_i^{\text{time}} = dm_i^{\alpha}/d\mu_i^{\text{time}}$. When the time map is many-to-one, this derivative is the μ_i^{phen}-weighted conditional fiber average rather than a pointwise composition. On a one-to-one differentiable chart, the signed and exposure push-forwards carry the same Jacobian, so their density ratio cancels that common factor; the displayed clock-time integrand therefore remains chart-consistent. Without absolute continuity, the transported object remains measure-valued and no pointwise density is asserted.

Time-fiber contraction. For a finite signed measure ρ and measurable time map π,

$$|\pi_{\#}\rho| \le \pi_{\#}|\rho|, \qquad |\pi_{\#}\rho|(\mathcal{T}) \le |\rho|(\Omega), \qquad (\pi_{\#}\rho)(\mathcal{T}) = \rho(\Omega).$$

Equality of gross variation requires an injective map, sign-pure fibers, or another proved no-cancellation condition. Measurable relabeling preserves ledger conclusions; noninjective coarse-graining can preserve net residual while changing gross and mixed-valence estimands. Protocols therefore register temporal-resolution sensitivity and retain the event-space branch when cancellation cannot be bounded.

Under $\tilde{f}_i = a_i + b_i f_i$, transport the neutral origin as $\tilde{\alpha}_i = a_i + b_i\alpha_i$; then $\tilde{L}_{i,\tilde{\alpha}} = b_i L_{i,\alpha}$. This chart representation is valid only when the chart, ownership rule, exposure convention, and neutral-origin transport rule are fixed independently.

This is affine covariance of the neutral-relative object, not blanket origin invariance of every diagnostic.

A common positive rescaling of both poles preserves balance:

$$V_i^{+} \mapsto c_i V_i^{+}, V_i^{-} \mapsto c_i V_i^{-}, c_i > 0.$$

Under independent pole rescaling,

$$V_i^{+} \mapsto c_i^{+} V_i^{+}, V_i^{-} \mapsto c_i^{-} V_i^{-}.$$

This independent pole rescaling preserves one-to-one balance for every nonzero balanced stream only when $c_i^+ = c_i^-$. Neutral-origin identification, cross-sign calibration, and scalarization are therefore distinct requirements.

3.3 Sign and origin identification ladder

A channel is not positive or negative merely because its sign improves terminal fit. Sign requires independent phenomenological, behavioral, clinical, perturbational, or cross-channel support, and construct validity requires a declared bridge between the observed channel and the phenomenal target (Barrett, 2006; Borsboom, 2005). Response shift (Sprangers & Schwartz, 1999), disability adaptation, medication, culture, language, reporting capacity, and expectation can alter measured signs without changing the latent target.

Functional-response nonidentity. Approach, withdrawal, reward pursuit, stress signaling, gene expression, membrane depolarization, homeostatic error, and task performance are behavioral, physiological, or computational variables; none is v_i, f_i, or μ_i^{phen} by definition. Before any such channel contributes to a phenomenal ledger, the declared reference class requires a class-appropriate psychophysical representation and measured-to-latent bridge, equally for neural, aneural, collective, and artificial candidate bearers.

Origin status should be reported as:

- Level 0 - coordinate origin: questionnaire midpoint, sample mean, model coordinate, or analyst recentering.
- Level 1 - within-stream sign-supported origin: local sign evidence and within-stream anchors.
- Level 2 - perturbation-calibrated neutral point: bidirectional perturbations, indifference tasks, no-change tasks, or convergent channels support a bounded neutral region.
- Level 3 - multi-method invariance-supported origin: independent channels, instruments, contexts, and populations support the same neutral interpretation.
- Level 4 - transportable origin: neutral semantics remain stable across the declared reference class.

Level 4 origin transportability is necessary but not sufficient for pooled cardinal comparison; transportable units, scale, scalarization, and measurement invariance are also required.

3.4 Partial identification

When the neutral origin is not point-identified, preserve its index through the complete measured-to-latent construction. The admissible stream-constant domain contains only translations that preserve integrability; it equals all real shifts only on a branch where finite exposure makes every finite translation admissible. Variable origins belong to a prospectively specified admissible subset of $L^1\left(\mu_i^{\mathrm{phen}}\right)$. Every compatibility set, bridge, and sensitivity result uses this same frozen domain.

$$\mathcal{A}_i^{\mathrm{const}} := \{a \in \mathbb{R}: f_i - a \in L^1\left(\mu_i^{\mathrm{phen}}\right)\}, \qquad \mathcal{A}_{i,\eta}^{\mathrm{field}} \subseteq \{\alpha \in L^1\left(\mu_i^{\mathrm{phen}}\right): f_i - \alpha \in L^1\left(\mu_i^{\mathrm{phen}}\right)\}.$$

Define branch-specific admissible sets $\mathcal{Z}_{0,i,\eta}^{\mathrm{const}} \subseteq \mathcal{A}_i^{\mathrm{const}}$ and $\mathcal{Z}_{0,i,\eta}^{\mathrm{field}} \subseteq \mathcal{A}_{i,\eta}^{\mathrm{field}}$. For each branch-tagged origin, $\Theta_{i,\eta}(z)$ is defined only for $z \in \mathcal{Z}_{0,i,\eta}$, and the bridge relation $\mathcal{B}_{i,\eta,z} \subseteq \mathcal{Y}_{i,\eta}^{\mathrm{meas}} \times \mathcal{Y}_i^{\mathrm{phen}}$ is evaluated on the branch-appropriate domain.

The protocol freezes the constant-origin or origin-field branch and its branch-specific admissible set before terminal outcomes. $\mathcal{Z}_{0,i,\eta}$ denotes the tagged union below. The constant-origin component is declared for stream i; the field-origin component lies in $\mathcal{A}_{i,\eta}^{\mathrm{field}}$ and carries its own independently calibrated uncertainty set. The common origin parameter space is the tagged disjoint union

$$\mathcal{Z}_{0,i,\eta} := \left(\{\mathrm{const}\} \times \mathcal{Z}_{0,i,\eta}^{\mathrm{const}}\right) \,\dot{\cup}\, \left(\{\mathrm{field}\} \times \mathcal{Z}_{0,i,\eta}^{\mathrm{field}}\right).$$

For every branch-tagged origin in the admissible origin set, bind the branch and origin before defining the protocol-compatible nuisance and structural parameter set:

$$\mathrm{orig}(z) := \zeta, \qquad \Theta_{i,\eta}(z) := \left\{\vartheta : \begin{matrix} \vartheta \text{ is protocol-compatible with the observed data} \\ \text{when branch } b \text{ and origin } \mathrm{orig}(z) \text{ are imposed} \end{matrix}\right\}, \quad z = (b, \zeta) \in \mathcal{Z}_{0,i,\eta}.$$

Let $m_{i,\eta,T}(z,\vartheta)$ denote the named measured terminal estimand evaluated under the same bound branch-tagged origin. Define the origin-specific measured compatibility set and bridge by

$$\mathcal{I}_{i,\eta,T}^{\mathrm{meas}}(z) := \{m_{i,\eta,T}(z,\vartheta) : \vartheta \in \Theta_{i,\eta}(z)\}.$$

The latent terminal compatibility set is then the union of origin-matched bridge images:

$$\mathcal{I}_{i,\eta,T}^{\mathrm{phen}} := \bigcup_{z \in \mathcal{Z}_{0,i,\eta}} \left\{x : \exists y \in \mathcal{I}_{i,\eta,T}^{\mathrm{meas}}(z) \text{ with } (y,x) \in \mathcal{B}_{i,\eta,z}\right\}.$$

$$\mathcal{B}_{i,\eta,z} \subseteq \mathcal{Y}_{i,\eta}^{\mathrm{meas}} \times \mathcal{Y}_i^{\mathrm{phen}}, \qquad \mathcal{Y}_{i,\eta}^{\mathrm{meas}} \text{ and } \mathcal{Y}_i^{\mathrm{phen}} \text{ are standard Borel.}$$

Here $\mathcal{B}_{i,\eta,z}$ is a measurable compatibility relation. A protocol states whether it is the graph of a function, an identified set, a confidence relation, or another declared object before its image is used for a verdict.

Bridge graph and certified projection. Define

$$\mathcal{G}_{i,\eta} := \{(z,\vartheta,m,\ell) : z \in \mathcal{Z}_{0,i,\eta},\ \vartheta \in \Theta_{i,\eta}(z),\ m = m_{i,\eta,T}(z,\vartheta),\ (m,\ell) \in \mathcal{B}_{i,\eta,z}\}.$$

The protocol states whether this graph is Borel or analytic, requires nonempty closed or compact values wherever containment or optimization is decision-bearing, and states the continuity or hemicontinuity condition used. Computation publishes certified approximations

$$\underline{\mathcal{I}}_{i,\eta,T}^{\mathrm{phen}} \subseteq \mathcal{I}_{i,\eta,T}^{\mathrm{phen}} \subseteq \overline{\mathcal{I}}_{i,\eta,T}^{\mathrm{phen}}.$$

and makes adverse or equivalence decisions from the conservative outer set. Numerical tolerance is part of the typed uncertainty object; a verdict that changes within tolerance is unresolved.

A measured value generated under one origin may be paired only with the bridge calibrated for that same origin. If terminal neutrality holds only for a narrow or data-selected subset of admissible origins, the result is origin-sensitive compatibility. Transfer to the latent set additionally requires sign, scalarization, endpoint, route-state, missingness, observation, and rival-model adequacy. The protocol must state whether each reported set is a deterministic identified region, partial-identification set, confidence set, or Bayesian credible set; these objects do not have interchangeable guarantees (Manski, 2003).

3.5 Origin-relative terminal atoms

Affine neutral-coordinate covariance. For $\tilde{\alpha}_i(x) = a_i + b_i\alpha_i(x)$, the neutral-relative ledger obeys $\tilde{L}_{i,\tilde{\alpha}}(t) = b_i L_{i,\alpha}(t)$. Finite total exposure is imposed only on the affine-translation subbranch that uses a nonzero a_i; scale-only branches inherit no extra finite-exposure premise.

Affine-translation subbranch. When nonzero additive translations are admitted, additionally require finite total phenomenal exposure under the declared scalar representation:

$$E_i^{\mathrm{tot}} := \mu_i^{\mathrm{phen}}(\Omega_i) < \infty.$$

Under a stream-constant affine transformation $\tilde{f}_i = a_i + b_i f_i$ with $b_i > 0$, the terminal residual transforms as

$$\tilde{Z}_i^\star = a_i E_i^{\mathrm{tot}} + b_i Z_i^\star, b_i > 0.$$

Exact terminal zero is invariant under positive scaling but not under additive translation whenever total exposure is positive. The atom-at-zero statement is therefore well posed only relative to an independently fixed neutral origin. Under origin partial identification, the accessible object is origin-indexed compatibility or band concentration rather than an unqualified terminal atom.

Completed-record mean centering is not an origin-identification procedure. For a completed finite-exposure record, define the mean and recentered signed measure once by

$$\overline{f}_{\mathrm{real},i} := \frac{Z_i^\star}{E_i^{\mathrm{tot}}}, \qquad \tilde{\nu}_i := \nu_i - \overline{f}_{\mathrm{real},i}\mu_i^{\mathrm{phen}}, \qquad \tilde{\nu}_i(\Omega_i) = 0.$$

Only on the absolutely continuous branch may the same identity be written as an integral of $f_i - \overline{f}_{\mathrm{real},i}$. This is a Level-0 coordinate operation and is prohibited as confirmatory evidence for PTVC.

Prospective adaptation-gap diagnostic. Freeze an operative neutral origin using nonterminal calibration only. With a preregistered positive exposure floor, define the following residual. Report its sign, scale, and origin-sensitivity envelope; it is a prospective residual diagnostic, not a recentering rule.

$$\Delta_i^{\mathrm{adapt}} := \frac{Z_{i,\alpha_{\mathrm{op}}}^\star}{E_i^{\mathrm{tot}} \vee e_{\min}}, \qquad e_{\min} > 0$$

This diagnostic does not identify the neutral origin; it tests held-out transport of a prospectively fixed origin and is reported with the full origin-uncertainty envelope.

For the neutral-relative coordinate introduced in Section 3.2, transporting the neutral point with the coordinate removes the additive term:

$$\tilde{L}_{i,\tilde{\alpha}}(t) = b_i L_{i,\alpha}(t), \tilde{\alpha} = a_i + b_i\alpha.$$

The same origin dependence applies to the nearest balance-law rivals. Recentering repartitions the Jordan masses, so both V_i^+/V_i^- and C_i are origin-sensitive. A fixed neutral origin is a prerequisite for discriminating $\rho = 1$ from $\rho \neq 1$ and $\delta = 0$ from $\delta \neq 0$, not only for stating exact closure.

Origin-error transport identity (ORI-1). If the analysis subtracts a stream-constant origin shift a from the independently calibrated neutral-relative density, $f_i^{(a)} = f_i - a$, then $Z_{i,a}^\star = Z_i^\star - aE_i^{\mathrm{tot}}$. Hence one common origin correction can restore zero across a reference class only when the residual vector is proportional to total exposure under the registered sign convention.

Prospective cross-sign exchange experiment. Perturb independently validated positive and negative channels on common exposure units; estimate the exchange map outside terminal data; test sign symmetry, state dependence, hysteresis, and transport; then freeze the admissible origin set $\mathcal{Z}_{0,i,\eta}$. A promoted band or route result must retain its sign, equivalence verdict, and cross-equation predictions for every $z \in \mathcal{Z}_{0,i,\eta}$, or report the exact origin subset producing each verdict.

The worst-case statistic or identified set is fixed before terminal fitting, so outcome-selected recentering cannot create closure.

Origin-class no-saturation gate. Freeze the admissible origin class and its complexity on nonterminal calibration data; estimate it by blocked cross-fitting; and report terminal-origin leverage, effective degrees of freedom, and sensitivity to each admissible class. If a saturated or outcome-adaptive origin family can manufacture terminal closure, return Origin-sensitive or No adequate estimate rather than selecting the favorable origin.

4. Endpoint, stream routing, reachability, and terminal windows

4.1 Truth-level and protocol endpoints

Delayed-entry target. Whole-stream inference separates the unobserved prefix, observed segment, and terminal suffix on one scale. A latent-scale branch uses the decomposition below; a measured branch constructs the three components jointly and applies the measured-to-latent bridge exactly once.

Delayed-entry decomposition.

$$Z_i^\star = Z_{i,\mathrm{pre}} + Z_{i,\mathrm{obs}} + Z_{i,\mathrm{suf}}.$$

$$\mathcal{J}_{i,\eta,T}^{\mathrm{entry}} := \left\{ z_{\mathrm{pre}} + z_{\mathrm{obs}} + z_{\mathrm{suf}} : \left(z_{\mathrm{pre}}, z_{\mathrm{obs}}, z_{\mathrm{suf}} \right) \in \mathcal{J}_{i,\eta}^{\mathrm{pre,obs,suf}} \right\}.$$

The joint object carries its simultaneous guarantee and dependence model. Whole-stream language is prohibited unless both unobserved ends are identified or bounded; otherwise the report is entry-conditioned, partially identified, or No adequate estimate.

Endpoint-ontology axiom. The truth-domain endpoint branch, protocol predicate, competing-event convention, and revision rule are fixed without terminal-ledger outcomes. Protocol-indexed bounds $T_{i,\eta}^{\mathrm{lo}} \le T_i^\star \le T_{i,\eta}^{\mathrm{hi}}$ are used only under their declared coverage or identification semantics, and the full set $\{L_i^{\mathrm{phen}}(t-) : t \in [T_{i,\eta}^{\mathrm{lo}}, T_{i,\eta}^{\mathrm{hi}}]\}$ is propagated.

Let $T_i^\star$ be the truth-level terminal boundary of maximal qualifying stream i. Protocol endpoint predicates, adjudicated estimates, intervals, and censoring objects are denoted separately, including $T_{i,\eta}^{\mathrm{cand}}$ and $T_{i,\eta}^{\mathrm{adj}}$. They operationalize the truth-level endpoint; they do not define it by convenience.

A prospective endpoint predicate must be adapted to the declared endpoint-adjudication filtration, with dwell time, hysteresis, return logic, competing events, and reliability fixed before outcome inspection. Retrospective classification may support retrospective analysis but does not automatically license stopping-time formulas. If no endpoint occurs in the study horizon, the observation is censored or unresolved rather than forced to the last available time.

Hitting-time convention. Every first-hitting construction in this model uses $\inf\emptyset = \infty$ unless a different prospectively declared convention is stated.

Study entry is distinct from stream origin. Without an identified or bounded origin, a study estimates an entry-conditioned increment rather than whole-stream closure.

Endpoint-evidence status register. Every protocol endpoint record stores (i) one current status and (ii) an immutable, time-stamped status history. Current status is Candidate crossing, Estimated endpoint, Confirmed under protocol, Revised endpoint, or No adequate estimate; every transition preserves the prior state, evidence, adjudicator, model version, and reason.

1. Candidate crossing - a preregistered indicator crossed its threshold; irreversibility and the truth-level endpoint are not established.

2. Estimated endpoint - a model supplies an endpoint distribution, interval, or classification under declared assumptions.

3. Confirmed under protocol - persistence, signal-quality, competing-event, return, and independent-review criteria are satisfied under the registered rule.

4. Revised endpoint - later return, resuscitation, new channel evidence, or discovered error changes the classification while preserving the original analysis version.

5. No adequate estimate - available evidence cannot support terminal inference.

Protocols predeclare which statuses enter primary, sensitivity, exploratory, and No adequate estimate analyses. Mandatory flow reporting includes crossing, confirmation, revision, return, correction, and unresolved frequencies, with the status available at every decision time reconstructible from the immutable history.

Endpoint-adjudication observation model. “Confirmed under protocol” is a registered evidential status, not an error-free truth label. The primary analysis propagates disagreement, misclassification, interval uncertainty, and later revision through a versioned endpoint kernel:

$$K_\eta^{\mathrm{end}}\big(\mathrm{d}\tilde{t}, \mathrm{d}\tilde{j}, \mathrm{d}h^{\mathrm{status}} \mid T_i^\star, J_i, \mathcal{H}_i^{\mathrm{end}}\big).$$

The kernel conditions on true endpoint time and class only as latent variables and may include care context, signal quality, and observation history. A zero-error branch requires independent justification; otherwise endpoint and ledger uncertainty are propagated jointly.

4.2 Random-endpoint structural restriction

[CP/ED] Conditional zero-atom screen. Let P_i^0 be a declared baseline or rival law, $(\mathcal{F}_t)$ the stated filtration, and T_i^{end} an endpoint time. On the risk set $\{P_i^0\big(T_i^{\mathrm{end}} > t \mid \mathcal{F}_t\big) > 0\}$, assume that a regular conditional law of the jointly measurable $L_i^{\mathrm{pre}}\big(T_i^{\mathrm{end}} -\big)$, given $\mathcal{F}_t$ and continued risk, exists and assigns no atom to zero for P_i^0-almost every conditioning history. Then, under that at-risk regular conditional law,

$$P_i^0\{L_i^{\mathrm{pre}}\big(T_i^{\mathrm{end}} -\big) = 0 \mid \mathcal{F}_t, T_i^{\mathrm{end}} > t\} = 0.$$

Proof and scope. A conditionally non-atomic probability measure assigns zero probability to $\{0\}$ on its almost-sure risk-set domain. Null risk sets receive No adequate estimate rather than a probability claim. This is a conditional screening implication, not an exogeneity result; Proposition 1 below supplies the separate independent positive-density endpoint restriction.

[CP] Proposition 1 (independent positive-density endpoint collapse). Let $\tilde{L}_i(t -)$ be a jointly measurable pre-killing left-limit process on a deterministic interval I. Let T_i^{exo} be supported on I, independent of the sigma-field generated by $\tilde{L}_i$, and have, under the same declared reference law, density f_T positive for Lebesgue-almost every $t \in I$. If exact closure holds at T_i^{exo}, then

$$0 = P_i^0\{\tilde{L}_i(T_i^{\mathrm{exo}} -) \neq 0\} = \int_I P_i^0\,\{\tilde{L}_i(t -) \neq 0\} f_T(t)\,\mathrm{d}t.$$

[P] Proof. The nonnegative joint integral vanishes, so Tonelli’s theorem gives equality for the product of Lebesgue time and the declared reference probability. Fubini’s section conclusion then gives an event of reference probability one on which $\tilde{L}_i(t -) = 0$ for Lebesgue-almost every $t \in I$. If paths are left-continuous, equality extends to every interior point of I. Inclusion of a left endpoint requires corresponding right-continuity or an independently imposed endpoint value there. The result does not apply when timing is state-coupled, the endpoint contributes phenomenally, routing changes the stream, the pre-killing extension does not exist, or independence fails.

[CP] Corollary 1.1 (approximate conditional-dependence occupation bound). Let $X_i = \left(\tilde{L}_i(t-)\right)_{t\in I}$, let C_i be prospectively frozen pre-endpoint covariates, and let T_i^{probe} denote the endpoint probe. This branch permits residual dependence after conditioning on the declared covariates and does not label the probe exogenous. Assume standard Borel path and covariate spaces, joint measurability of $(x,t) \mapsto x(t-)$, and existence of the required regular conditional laws and density $f_{T_{\text{probe}}|C}$. For each ε≥0, define

$$d_{\text{TV}}(P,Q) := \sup_A |P(A) - Q(A)|,$$

$$\begin{aligned}
\rho_i(c) \quad &:= d_{\text{TV}}\left(P^0_{i,X,T_{\text{probe}}|C=c}, P^0_{i,X|C=c} \otimes P^0_{i,T_{\text{probe}}|C=c}\right),\\
\overline{\rho}_i \quad &:= \mathbb{E}_{P_i^0}[\rho_i(C_i)],\\
g_{i,\varepsilon}(t,c) \quad &:= P_i^0\left(\left|\tilde{L}_i(t-)\right| > \varepsilon \mid C_i = c\right),\\
p_{i,\varepsilon} \quad &:= P_i^0\left(\left|\tilde{L}_i\left(T_i^{\text{probe}}-\right)\right| > \varepsilon\right).
\end{aligned}$$

Then

$$\mathbb{E}_{P_i^0}\left[\int_I g_{i,\varepsilon}(t, C_i) f_{T_{\text{probe}}|C}(t \mid C_i)\, \mathrm{d}t\right] \le p_{i,\varepsilon} + \overline{\rho}_i.$$

For a prospectively fixed Borel set $J \subseteq I$ with $0 < \lambda(J) < \infty$ and a constant $m_J > 0$, if $f_{T_{\text{probe}}|C}(t \mid C_i) \ge m_J > 0$ for Lebesgue-almost every $t \in J$, P_i^0-almost surely, then

$$\mathbb{E}_{P_i^0}\left[\int_J g_{i,\varepsilon}(t, C_i)\, \mathrm{d}t\right] \le \frac{p_{i,\varepsilon} + \overline{\rho}_i}{m_J}.$$

If additionally $I_{P_i^0}\left(X_i; T_i^{\text{probe}} \mid C_i\right) \le \iota$, conditional Pinsker and Jensen give

$$\mathbb{E}_{P_i^0}\left[\int_J g_{i,\varepsilon}(t, C_i)\, \mathrm{d}t\right] \le \frac{p_{i,\varepsilon} + \sqrt{\iota/2}}{m_J},$$

equivalently the minimum-dependence requirement

$$I_{P_i^0}\left(X_i; T_i^{\text{probe}} \mid C_i\right) \ge 2\left[m_J\, \mathbb{E}_{P_i^0} \int_J g_{i,\varepsilon}(t, C_i)\, \mathrm{d}t - p_{i,\varepsilon}\right]_+^2.$$

[P] Proof. Apply the total-variation event bound to the jointly measurable event $\{(x,t)\colon |x(t-)| > \varepsilon\}$, compare the actual conditional joint law with its conditional product law, lower-bound the product-law integral on J, and average over C_i. The information form follows from conditional Pinsker and Jensen (Cover & Thomas, 2006). Exact conditional independence and exact closure recover Proposition 1 by exhausting the positive-density support with sets on which the density is bounded below.

Interpretation. Substantial off-band occupation together with rare off-band endpoints forces quantifiable endpoint–ledger dependence under the declared covariate adjustment. This gives a graded falsifier and a minimum-information target; it does not convert a population average into SET-ALL closure.

4.3 Directional reachability

[CP] Proposition 2 (directional reachability necessity). Let h_t be the realized stream-owned prefix through time t, and let $\mathcal{C}_i(t,u \mid h_t)$ be the prospectively declared class of admissible suffixes under which stream i persists to u. If $\mathcal{C}_i(t,u \mid h_t) = \emptyset$, exact completion at u is impossible and no capacity supremum is evaluated. Assume below that this class is nonempty.

For $\gamma \in \mathcal{C}_i(t, u \mid h_t)$, define the completed continuation and future event slice.

$$\Gamma_\gamma := h_t \oplus \gamma, \qquad \mathcal{E}_{i,\Gamma_\gamma}(t,u) := \left(\pi_{i,\Gamma_\gamma}^{\text{time}}\right)^{-1}\left((t,u)\right).$$

The ledger decomposes along the continuation as

$$L_i^{\text{phen}}(u-) = L_i^{\text{phen}}(t) + \nu_{i,\Gamma_\gamma}\left(\mathcal{E}_{i,\Gamma_\gamma}(t,u)\right),$$

with any phenomenal atom assigned exactly at t governed by the separately declared through-time versus strict-before convention. The attainable future signed-contribution set is

$$\mathcal{A}_i^{\text{fut}}(t,u \mid h_t) := \left\{\nu_{i,\Gamma_\gamma}\left(\mathcal{E}_{i,\Gamma_\gamma}(t,u)\right) : \gamma \in \mathcal{C}_i(t,u \mid h_t)\right\}.$$

Exact additive completion at u is possible only if an attainable future contribution cancels the current residual:

$$\exists a \in \mathcal{A}_i^{\text{fut}}(t,u \mid h_t) \text{ such that } L_i^{\text{phen}}(t) + a = 0.$$

For each continuation, suppose the future density obeys the directional envelope

$$-m_{i,\Gamma_\gamma}^{-}(x) \le f_{i,\Gamma_\gamma}(x) \le m_{i,\Gamma_\gamma}^{+}(x) \quad \mu_{i,\Gamma_\gamma}^{\text{phen}} - a.e.$$

Define the directional capacities on the same continuation class and event slice:

$$\text{Cap}_i^{\pm}(t,u \mid h_t) := \sup_{\gamma \in \mathcal{C}_i(t,u \mid h_t)} \int_{\mathcal{E}_{i,\Gamma_\gamma}(t,u)} m_{i,\Gamma_\gamma}^{\pm}(x)\, \mathrm{d}\mu_{i,\Gamma_\gamma}^{\text{phen}}(x)$$

The directional capacity bound (RCH-1), a necessary condition for exact completion, is

$$-\text{Cap}_i^{+}(t,u \mid h_t) \le L_i^{\text{phen}}(t) \le \text{Cap}_i^{-}(t,u \mid h_t).$$

Under a symmetric envelope M_{i,Γ_γ}, define

$$\begin{aligned} \text{Cap}_i^{M}(t,u \mid h_t) \quad &:= \sup_{\gamma \in \mathcal{C}_i(t,u \mid h_t)} \int_{\mathcal{E}_{i,\Gamma_\gamma}(t,u)} M_{i,\Gamma_\gamma}(x)\, \mathrm{d}\mu_{i,\Gamma_\gamma}^{\text{phen}}(x), \\ |L_i^{\text{phen}}(t)| \quad &\le \text{Cap}_i^{M}(t,u \mid h_t). \end{aligned}$$

Future atomic or singular phenomenal contributions require separately justified positive and negative allowances added to the corresponding capacities. In a declared physical-time chart, a complete stream-owned contribution-rate bound $M_i(s)$ yields

$$\begin{aligned} |L_i^{\text{phen}}(t)| \quad &\le \int_t^u M_i(s)\, \mathrm{d}s, \\ M_i(s) \equiv M_i,\ u = T_i^{\star} \quad &\Rightarrow |L_i^{\text{phen}}(t)| \le M_i(T_i^{\star} - t). \end{aligned}$$

This construction is related to viability and constrained reachability, but the object is terminal residual solvency rather than a pathwise invariant viability kernel (Aubin, 1991; Aubin et al., 2011).

[P] Proof. Any completing continuation contributes $a = -L_i^{\text{phen}}(t)$. The directional envelope gives $-\text{Cap}_i^{-} \le a \le \text{Cap}_i^{+}$ on the common continuation class and event slice; substituting $a = -L_i^{\text{phen}}(t)$ yields (RCH-1). The symmetric bound is the same argument with a common envelope.

Reachability interpretation. The attainable future-contribution set is the primary object. Directional capacities are measurable outer envelopes and provide only a necessary screen: finite capacity can still overstate what is jointly attainable, an infinite capacity is uninformative on that side, and a supremum need not be attained. A sufficiency or optimizer claim requires a separate compactness, closure, measurable-selection, and existence argument on the registered continuation class.

4.4 Endpoint-support and hazard-weighted occupation

[CP] Proposition 3 (endpoint-support and hazard-weighted occupation). Work in a completed, right-continuous filtration $(\mathcal{H}_{i,t})_{t\geq 0}$ with an adapted càdlàg ledger. For terminal band $\mathcal{C}_K = [-K, K]$, define the predictable off-band process below. All random measures live on the fixed carrier $(0,\infty] \times \mathrm{E}$; stream entry is represented by the predictable indicator $\mathbf{1}_{\{S_i^\theta < t\}}$.

$$H_i^K(t) = \mathbf{1}_{\{L_i^{\text{phen}}(t-)\notin \mathcal{C}_K\}}.$$

Stopping-time branch. Suppose the truth-level endpoint is a stopping time for the declared completed, right-continuous endpoint filtration and the endpoint-class mark is measurable at that endpoint. Define the all-cause and cause-specific endpoint processes and the predictable at-risk process by

$$N_i^{\text{all}}(t) = \mathbf{1}_{\{T_i^\star \leq t\}}, \quad R_i^{\text{risk}}(t) = \mathbf{1}_{\{T_i^\star \geq t\}}, \quad N_i^e(t) = \mathbf{1}_{\{T_i^\star \leq t, J_i = e\}}.$$

Let Λ_i^e be the predictable compensator of N_i^e. Assuming the displayed integrand is integrable, for every finite horizon τ the endpoint-support identity (END-1) is

$$\mathbb{E}\int_{(0,\tau]} \mathbf{1}_{\{S_i^\theta < t\}} H_i^K(t)\, d\Lambda_i^e(t) = \mathbb{P}\{S_i^\theta < T_i^\star \leq \tau, J_i = e, L_i^{\text{phen}}(T_i^\star -) \notin \mathcal{C}_K\}.$$

Let the probability on the right be denoted $p_{K,e}(\tau)$, and let $\bar{\varepsilon}_{K,e}(\tau)$ be its preregistered upper tolerance. If $\mathrm{d}\Lambda_i^e(t) = R_i^{\text{risk}}(t)\alpha_i^e(t)\,\mathrm{d}t$ and $\alpha_i^e(t) \geq \alpha_{\min} > 0$, then the occupation integral is at most $p_{K,e}(\tau)/\alpha_{\min}$. Under the additional registered constraint $p_{K,e}(\tau) \leq \bar{\varepsilon}_{K,e}(\tau)$, it is also at most $\bar{\varepsilon}_{K,e}(\tau)/\alpha_{\min}$.

Consequently, for every $B \in \mathcal{B}((0,\tau])$,

$$\mathbb{E}\int_{(0,\tau]} \mathbf{1}_B\,(t)\mathbf{1}_{\{S_i^\theta < t\}} R_i^{\text{risk}}(t) H_i^K(t)\,\mathrm{d}t \leq \frac{p_{K,e}(\tau)}{\alpha_{\min}} \leq \frac{\bar{\varepsilon}_{K,e}(\tau)}{\alpha_{\min}}.$$

Non-stopping random-time branch. Define the raw marked endpoint random measure on the fixed carrier by

$$N_i^{e,\text{raw}}(B) := \mathbf{1}\{T_i^\star \in B,\ J_i = e\}, \qquad B \in \mathcal{B}((0,\infty]).$$

Let $\Pi_i^{e,p}$ be its dual predictable projection. Under the full-information filtration in which H_i^K is predictable and under the same integrability condition, for every finite τ,

$$\mathbb{E}\int_{(0,\tau]} \mathbf{1}_{\{S_i^\theta < t\}} H_i^K(t)\, d\Pi_i^{e,p}(t) = \mathbb{P}\{S_i^\theta < T_i^\star \leq \tau, J_i = e, L_i^{\text{phen}}(T_i^\star -) \notin \mathcal{C}_K\}.$$

If the ledger is not adapted to the smaller filtration used for endpoint projection, use the dual predictable projection of the marked random measure $H_i^K(t) N_i^{e,\text{raw}}(\mathrm{d}t)$, or a justified Radon–Nikodym density relative to $\Pi_i^{e,p}$. An ordinary predictable projection of H_i^K multiplied by $\Pi_i^{e,p}$ is not substituted without a theorem. Infinite-horizon forms are corollaries only under separate finite-endpoint and integrability assumptions.

[P] Proof. On the stopping-time branch, predictability and the compensator identity replace integration against the endpoint counting measure by integration against its compensator; the counting-measure integral equals the endpoint indicator in (END-1) (Andersen et al., 1993; Fleming & Harrington, 2005).

The random-time identity follows from the defining duality of the dual predictable projection. Restriction to the declared region and the lower intensity bound gives the occupation bound.

4.5 Boundary accessibility

An endpoint-support equation must be paired with an accessibility mechanism. If the support coordinate reaches zero only at isolated nonsticky times and the applicable endpoint measure has bounded clock-time density, an exact-point route has zero endpoint mass. Positive exact-point mass therefore requires failure of at least one applicable premise of that ordinary zero-mass argument. Nonexhaustive mechanism classes include first-hitting absorption, sticky residence, a predictable endpoint atom, a singular endpoint carrier, a local-time carrier on a separately validated rough semimartingale coordinate, and global or last-exit constructions.

First-hitting anti-tautology gate. A first-hitting, absorption, zero-surface, or last-exit construction contributes route evidence only when the truth-level endpoint is defined and adjudicated independently of the ledger value whose closure is tested and the predicted coincidence is evaluated on held-out data. Defining the endpoint as the first or last time the same ledger equals zero makes closure an identity and supplies no target evidence.

Biological-death-bounded candidate branch. Let the biological-death time be prospectively defined and independently adjudicated. On this optional branch,

$$T_i^\star \le D_i^{\mathrm{bio}}.$$

No qualifying continuation or reassignment of the same stream occurs after biological death. The mapping from biological and clinical events to the phenomenal endpoint remains a separate endpoint model: a cause of biological death is not automatically a cause or observation of the end of qualifying experience. Continued-stream, endpoint-inclusive, and alternative-routing branches remain separate.

Endpoint-inclusive branch. A phenomenal contribution assigned at the endpoint belongs to a separately declared endpoint-inclusive target and cannot establish strict-pre-endpoint singleton closure.

Continuation branch. Independently supported continuation changes the routing or endpoint ontology and is evaluated under a separately declared extended-continuation target.

Finite-band branch. A terminal band $[-K, K]$ defines an approximate closure target, not an accessibility mechanism for exact singleton closure.

If the coordinate is the ordinary continuous finite-variation phenomenal ledger, its semimartingale local time is zero (Karatzas & Shreve, 1991; Revuz & Yor, 1999). A nontrivial local-time route requires a separately declared rough coordinate with nonzero quadratic variation and an independently validated bridge to the ledger.

[CP] Corollary 3.1 (cause-specific zero-occupation identity; ACC-1). Fix endpoint class e and finite horizon τ, and let

$$G_i^0(t) := \mathbf{1}\{L_i^{\mathrm{phen}}(t-) = 0\}.$$

Because the declared ledger is adapted and càdlàg, its left limit is predictable; hence G_i^0 is predictable in the filtration used for the endpoint process. On the stopping-time branch take $A_i^e := \Lambda_i^e$; on the non-stopping branch take $A_i^e := \Pi_i^{e,p}$. Under the stated integrability conditions, the accounting identity is

$$\mathbb{E}\Big[\int_{(0,\tau]} \mathbf{1}_{\{S_i^\theta < t\}}\, G_i^0(t)\, \mathrm{d}A_i^e(t)\Big] = \mathbb{P}\{S_i^\theta < T_i^\star \le \tau,\ J_i = e,\ L_i^{\mathrm{phen}}(T_i^\star -) = 0\}.$$

Therefore

$$\mathbb{E}[\int_{(0,\tau]} \mathbf{1}_{\{S_i^\theta<t\}}\, G_i^0(t)\, \mathrm{d}A_i^e(t)] = 0 \Rightarrow \mathbb{P}\{S_i^\theta < T_i^\star \le \tau,\ J_i = e,\ L_i^{\mathrm{phen}}(T_i^\star -) = 0\} = 0.$$

On an absolutely continuous subbranch, a substantive sufficient condition is $\mathrm{d}A_i^e(t) = \lambda_i^e(t)\,\mathrm{d}t$ with λ_i^e finite and integrably bounded on the named risk interval, together with a ledger zero set of Lebesgue measure zero almost surely. Endpoint atoms, singular or local-time carriers, first-hitting absorption, sticky zero occupation, and last-exit constructions remain separate exception branches rather than consequences of the identity.

[P] Proof. Apply the compensator or dual-predictable-projection identity to the predictable integrand G_i^0. The integral equals the displayed cause-specific endpoint probability; a zero integral therefore forces that probability to zero.

4.6 Routing, terminal windows, and route-state coincidence

Pause, sleep, anesthesia, fragmented unity, fission, fusion, copying, restoration, and rebinding raise personal-identity and bearer questions that must be assigned prospective routing rules (Parfit, 1984). When routing remains ambiguous, terminal claims are conditional on the declared ontology. Maximality and event ownership are evaluated under the same prospectively declared continuation relation used in Section 1.1.

Routing-uncertainty propagation. Let $\mathcal{R}_{i,\eta}^{\mathrm{route}}$ denote the prospectively retained bearer and routing hypotheses. For each $r \in \mathcal{R}_{i,\eta}^{\mathrm{route}}$, let $\mathcal{J}_{i,\eta,T}^{\mathrm{phen}}(r)$ be the corresponding terminal-compatible set. When every hypothesis concerns the same bearer and estimand, define

$$\mathcal{J}_{i,\eta,T}^{\mathrm{route}} := \bigcup_{r\in\mathcal{R}_{i,\eta}^{\mathrm{route}}} \mathcal{J}_{i,\eta,T}^{\mathrm{phen}}(r).$$

Routing-robust exact incompatibility requires $0 \notin \mathcal{J}_{i,\eta,T}^{\mathrm{route}}$; finite-band incompatibility requires $\mathcal{J}_{i,\eta,T}^{\mathrm{route}} \cap [-K,K] = \emptyset$. Mixed gate-valid verdicts yield Model-ambiguous. If live hypotheses define different bearers or estimands, report branch-specific results rather than taking a numerical union.

Bearer and routing nonidentifiability. Let Π_η^{adm} be the prospectively registered class of history-dependent intervention regimes with common structural support, consistency, and observation semantics. Write $P_{\eta,r}^{\mathrm{obs},\pi}$ for the complete protocol-observed law under routing hypothesis r and regime π. If

$$P_{\eta,r}^{\mathrm{obs},\pi} = P_{\eta,r'}^{\mathrm{obs},\pi} \qquad \text{for every } \pi \in \Pi_\eta^{\mathrm{adm}}.$$

then protocol η does not identify r against r' on that regime class. Equality under a smaller or context-free action set licenses only the correspondingly narrower nonidentification conclusion. PTVC compatibility under one unresolved routing hypothesis cannot be transferred to another.

If the intended object is the measured estimand rather than the estimator, use the distinct symbol $\Delta L_{i,\eta,T}^{\mathrm{meas}}(\Delta)$ and its own domain. Do not mix an estimator and estimand in one definition.

For $\Delta > 0$, require $T_{i,\eta}^{\mathrm{adj}} - \Delta \in \mathrm{dom}(\hat{L}_{i,\eta})$ and require the left limit $\hat{L}_{i,\eta}\left(T_{i,\eta}^{\mathrm{adj}} -\right)$ to exist; in particular, $T_{i,\eta}^{\mathrm{adj}} - \Delta \ge S_{i,\eta}^{\mathrm{entry}}$ on the delayed-entry branch. Then define

$$\Delta\hat{L}_{i,\eta,T}^{(\Delta)} = \hat{L}_{i,\eta}\left(T_{i,\eta}^{\mathrm{adj}} -\right) - \hat{L}_{i,\eta}\left(T_{i,\eta}^{\mathrm{adj}} - \Delta\right).$$

Otherwise the protocol reports No adequate estimate for that window.

Terminal-window contraction is a route-specific signature, not whole-stream closure by itself.

For the reserve route, fix $\varepsilon_{i,\eta} > 0$ prospectively in the declared units of U_i, or as a dimensionless fraction of a named baseline, with calibration, tolerance, and uncertainty-propagation rules that do not use terminal outcomes. Define exact latent exhaustion and a threshold crossing separately:

$$T_i^{U,0} = \inf\{t \geq S_i^{\theta} : U_i(t) = 0\}, \quad T_{i,\eta}^{U,\varepsilon} = \inf\{t \geq S_i^{\theta} : U_i(t) \leq \varepsilon_{i,\eta}\}.$$

Define $\tau_i^{\mathrm{rc}} := \inf\{t \geq S_i^{\theta} : (U_i(t), Y_i(t)) \notin \mathcal{D}_i^{\mathrm{rc}}\}$ as the first exit from a prospectively declared route domain $\mathcal{D}_i^{\mathrm{rc}}$, with $\inf\emptyset = \infty$. On a named reserve-exhaustion subbranch one may assume $\tau_i^{\mathrm{rc}} = T_i^{U,0}$; coincidence with the truth endpoint, $\tau_i^{\mathrm{rc}} = T_i^{\star}$, is a separate bridge assumption.

Reserve exhaustion, route termination, truth-level endpoint, and zero ledger are distinct objects. $T_i^{U,0}$ is defined by $U_i = 0$; τ_i^{rc} is a route-boundary time; $T_i^{\star}$ is the truth-level endpoint; and $L_i^{\mathrm{phen}}(T_i^{\star}-) = 0$ is the PTVC consequence. Equality between any pair is a separate route, bridge, or coincidence assumption. A protocol may compare an estimated threshold crossing $\hat{T}_{i,\eta}^{U,\varepsilon}$ with $T_{i,\eta}^{\mathrm{adj}}$ while propagating endpoint and route-state uncertainty.

Delayed entry and left truncation require explicit modeling or bounds before terminal concentration is interpreted.

Table 2A. Clock-and-routing crosswalk

Object	Meaning	Filtration / information set	Observable?	Equality requiring a bridge
S_i^{θ}	Truth-domain stream origin	Truth-domain history	Usually no	Study entry = origin
$S_{i,\eta}^{\mathrm{entry}}$	Protocol entry	Enrollment history	Yes	Entry-conditioned increment = whole stream
$T_i^{\star}$	Truth-level stream endpoint	Truth endpoint branch	No	Protocol endpoint = truth endpoint
$T_{i,\eta}^{\mathrm{adj}}$	Adjudicated protocol endpoint	Adjudication filtration	Estimated	Endpoint bridge
$T_i^{U,0}$ / $T_{i,\eta}^{U,\varepsilon}$	Reserve exhaustion / protocol band crossing	Route-state filtration	Latent/estimated	Reserve endpoint = stream endpoint
τ_i^{rc}	Latent route-domain exit time	Route-state filtration	Latent; protocol estimate uses a distinct hatted symbol	$\tau_i^{\mathrm{rc}} = T_i^{U,0}$ on the reserve-exhaustion branch; $\tau_i^{\mathrm{rc}} = T_i^{\star}$ on the endpoint-coincidence branch.
L_i^{rc}	Reserve-route coordinate	Route-state filtration	Latent or estimated under the registered measurement model	$L_i^{\mathrm{rc}} = L_i^{\mathrm{phen}}$ requires the declared path or terminal bridge.
Prompt / observation time	Recorded sampling event	Observation filtration	Yes	Observation = latent state
Analysis horizon	Registered administrative time	Protocol	Yes	Horizon = terminal boundary

Union rule. Take a numerical union over routing branches only when every retained branch concerns the same bearer, unit, endpoint convention, and estimand; otherwise report branch-specific results and Model-ambiguous or No adequate estimate.

Clock transport. Every dynamic signature names its privileged clock. A deterministic increasing time change transports drift and diffusion with the transformed clock and a newly defined Brownian motion; the original Brownian symbol is not reused as if unchanged. Adapted random clocks require a separate time-change theorem and filtration conditions. Endogenous or anticipative clocks, including retrospective time-to-endpoint alignment, are descriptive/predictive unless a causal design establishes otherwise. Route-boundary risk sets condition on continued route membership; substitution of truth-endpoint risk requires the route–endpoint bridge.

Adapted absolutely continuous branch. If $a_s > 0$ is predictable and the inverse clock is finite on the analyzed domain, the continuous semimartingale and hazard transform as follows under the theorem-matched filtration conditions. Flat, jump, singular, or merely nondecreasing clocks require a general time-change theorem and may not use this smooth formula.

$$\begin{aligned} A_t &:= \int_0^t a_s \, \mathrm{d}s, & \tau_u &:= \inf\{t: A_t > u\}, \\ \mathrm{d}X_{\tau_u} &= \frac{b_{\tau_u}}{a_{\tau_u}} \, \mathrm{d}u + \frac{\sigma_{\tau_u}}{\sqrt{a_{\tau_u}}} \, \mathrm{d}\widetilde{W}_u, & \lambda_u^{(u)} &= \frac{\lambda_{\tau_u}^{(t)}}{a_{\tau_u}}. \end{aligned}$$

5. Reserve-Coupled Route Model and Testable Consequences

The reserve-coupled model is one candidate consequence route. Here U_i is a route-specific nonnegative capacity coordinate, not automatically thermodynamic energy, metabolic free energy, information, time, or moral credit. Its physical dimension is whatever the declared route bridge assigns; c_{LU} is a route-specific scale constant.

5.1 Route coordinate and reference dynamics

Define the route product (RC-1) by

$$L_i^{\mathrm{rc}}(t) := c_{\mathrm{LU}} U_i(t) Y_i(t), c_{\mathrm{LU}} > 0,$$

Here U_i is a nonnegative route state and Y_i is a signed normalized coordinate.

Gauge nonidentifiability. For any $a > 0$, $U_i \mapsto aU_i$ and $Y_i \mapsto Y_i/a$ leave U_iY_i unchanged; corresponding changes can also be absorbed into c_{LU}. RC-1 therefore identifies only the product unless at least one factor's scale and unit are fixed by an external perturbation, calibration, conservation relation, or independently validated measurement model. Terminal fit cannot supply that anchor. Table 1 records units for U_i, Y_i, c_{LU}, r_i^U, κ_i^U, and L_i^{rc}.

Predictable reserve-product gauge. A strictly positive predictable process may rescale the reserve and inversely rescale the normalized coordinate while leaving their product unchanged, subject to the stated semimartingale and integrability conditions. A time-varying gauge introduces derivative, finite-variation, covariation, or jump terms in the factor dynamics; it preserves only product-level claims unless those extra terms are carried explicitly. Dynamics-level claims about either factor therefore require an external anchor or a gauge-invariant formulation.

For $U_i(t) > 0$,

$$Y_i(t) = \frac{L_i^{\mathrm{rc}}(t)}{c_{\mathrm{LU}} U_i(t)}, \Psi_i(t) = \frac{|L_i^{\mathrm{rc}}(t)|}{c_{\mathrm{LU}} U_i(t)} = |Y_i(t)|.$$

A reference case uses a nonincreasing, absolutely continuous finite-variation reserve and arithmetic diffusion: the Itô reference construction uses standard stochastic-calculus conventions (Øksendal, 2003).

$$\mathrm{d}U_i(t) = -r_i^U(t)\,\mathrm{d}t, r_i^U(t) \geq 0,$$
$$\mathrm{d}Y_i(t) = \sigma_i(t)\mathrm{d}W_i(t), L_i^{\mathrm{rc}}(t) = c_{\mathrm{LU}} U_i(t) Y_i(t).$$

On the pre-exhaustion domain $U_i(t) > 0$,

$$\mathrm{d}L_i^{\mathrm{rc}}(t) = -\frac{r_i^U(t)}{U_i(t)} L_i^{\mathrm{rc}}(t)\,\mathrm{d}t + c_{\mathrm{LU}}\sigma_i(t) U_i(t)\,\mathrm{d}W_i(t).$$

A more general finite-variation reserve may contain singular components and must be represented directly through $\mathrm{d}U_i$ rather than solely through a clock-time density.

5.2 Product rule, past-fixity, and bridge hierarchy

For general semimartingales, the integration-by-parts formula is (Protter, 2005; Jacod & Shiryaev, 2003):

$$\mathrm{d}L_i^{\mathrm{rc}}(t) = c_{\mathrm{LU}}[U_i(t-)\,\mathrm{d}Y_i(t) + Y_i(t-)\,\mathrm{d}U_i(t) + \mathrm{d}[U_i, Y_i]_t].$$

Here $[U_i, Y_i]$ is total quadratic covariation. A separate jump-product term would double-count jumps unless the bracket were explicitly redefined as continuous covariation only.

If the route coordinate is claimed to equal the literal phenomenal ledger, the local bridge must satisfy

$$\mathrm{d}L_i^{\mathrm{phen}}(t) = \mathrm{d}L_i^{\mathrm{rc}}(t).$$

This bridge inherits the Section 1.1 pre-origin left-limit convention, including one-time ownership of any origin atom; no route-specific origin reset or one-sided atom assignment is permitted.

On an interval with no new phenomenal accrual, past-fixity then requires

$$\mathrm{d}\big(U_i(t)Y_i(t)\big) = 0.$$

Reserve decline alone cannot shrink value already experienced.

Four bridge strengths must be kept distinct:

1. Exact path bridge: $L_i^{\mathrm{phen}}(t) = L_i^{\mathrm{rc}}(t)$ for every declared preterminal t.
2. Exact terminal bridge: the equality $L_i^{\mathrm{phen}}(T_i^\star -) = L_i^{\mathrm{rc}}(T_i^\star -)$ is required only at the terminal boundary.
3. Approximate latent bridge: a fixed latent norm, tolerance, path class, filtration, calibration procedure, and held-out failure rule.

Approximate path-bridge promotion controls the terminal left-limit trace. L^p, uniform, Skorokhod, and finite-dimensional closeness earn promotion when the terminal jump and endpoint trace are controlled.

4. Protocol bridge: an empirical surrogate evaluated at an adjudicated endpoint or endpoint interval.

5.3 Path class and quadratic variation

For the continuous reference route,

$$[L_i^{\mathrm{rc}}]_t - [L_i^{\mathrm{rc}}]_{S_i^\theta} = c_{\mathrm{LU}}^2 \int_{S_i^\theta}^t \sigma_i^2\,(s) U_i^2(s)\,\mathrm{d}s.$$

An absolutely continuous phenomenal ledger is a continuous finite-variation process and therefore has zero quadratic variation. A finite signed-measure ledger remains finite variation but may jump, in which case its quadratic variation is the sum of squared jumps.

$$[L_i^{\mathrm{phen}}, L_i^{\mathrm{phen}}]_t^c = 0, \qquad [L_i^{\mathrm{phen}}, L_i^{\mathrm{phen}}]_t = \sum_{s \le t} \left(\Delta L_i^{\mathrm{phen}}(s)\right)^2.$$

The route-product and phenomenal-ledger path classes are therefore not identical without a bridge.

[CP] Proposition 4 (finite-variation obstruction to a nondegenerate exact diffusion bridge; QV-1). Under the prospectively declared route-process law, suppose the phenomenal ledger and remaining-reserve process are continuous finite-variation processes on $(S_i^\theta, t]$, the remaining reserve stays positive, the normalized route coordinate is a continuous semimartingale with a Brownian component plus finite variation, and the exact path bridge holds. Then

$$[L_i^{\mathrm{rc}}]_t^c - [L_i^{\mathrm{rc}}]_{S_i^\theta}^c = c_{\mathrm{LU}}^2 \int_{S_i^\theta}^t U_i\,(s)^2 \sigma_i(s)^2\,ds.$$

Exact path identity and the zero continuous quadratic variation of a continuous finite-variation ledger imply

$$\int_{S_i^\theta}^{t} U_i\,(s)^2 \sigma_i(s)^2\, ds = 0.$$

Because $U_i > 0$ on the interval, $\sigma_i = 0$ $\mathrm{d}t \otimes P_i^{\mathrm{rc}}$-almost everywhere there. Thus a nondegenerate Brownian component is incompatible with the exact finite-variation path bridge.

[P] Proof. Continuous quadratic variation is interval-additive. Exact path identity transfers zero continuous quadratic variation from the continuous finite-variation phenomenal ledger to the route coordinate. Substitution in the first display gives the second; on the positive-reserve region this forces the Brownian coefficient to vanish $\mathrm{d}t \otimes P_i^{\mathrm{rc}}$-almost everywhere. The route may still serve as an exact terminal bridge, approximate latent bridge, protocol bridge, or consequence-only coordinate.

5.4 Terminal degeneracy and drift-diffusion signature

Clock consequence. On the positive absolutely continuous reference branch $\mathrm{d}U_i = -\kappa_i^U U_i\, \mathrm{d}t$,

$$\int_{s}^{t} \kappa_i^U\,(u)\, du = \log \frac{U_i(s)}{U_i(t)}.$$

Thus cumulative contraction diverges when $U_i(t) \downarrow 0$. Pointwise strengthening requires an additional rate or monotonicity premise, and innovation amplitude vanishes only when $\sigma_i(t)U_i(t) \to 0$.

The reserve product alone does not imply exact terminal closure. The exact route condition is

$$\lim_{t \uparrow \tau_i^{\mathrm{rc}}} U_i(t) Y_i(t) = 0.$$

A sufficient special case is $U_i(t) \to 0$ with $Y_i(t) = o\big(1/U_i(t)\big)$; other routes may close through absorption, a bridge, or a boundary condition without this rate form.

A literal route-boundary implementation requires

$$\tau_i^{\mathrm{rc}} = T_i^\star, L_i^{\mathrm{phen}}(T_i^\star -) = L_i^{\mathrm{rc}}(T_i^\star -) = 0.$$

This coincidence is a route and bridge claim, not part of the definition of PTVC. It must be evaluated using independently validated reserve-state, endpoint, and measured-to-latent objects.

For the local diffusion route, the route SDE (RC-2) is

$$\mathrm{d}L_i^{\mathrm{rc}}(t) = -\kappa_i^U(t) L_i^{\mathrm{rc}}(t)\, \mathrm{d}t + \sigma_{L,i}(t)\, \mathrm{d}W_i(t),$$
$$\kappa_i^U(t) := \frac{r_i^U(t)}{U_i(t)},$$
$$\sigma_{L,i}(t) := c_{\mathrm{LU}} \sigma_i(t) U_i(t).$$

The route predicts divergence of cumulative contraction as the route boundary is approached. Pointwise strengthening is predicted only on a subbranch with $\kappa_i^U(t) \to \infty$ or another declared rate/monotonicity premise, and innovation amplitude vanishes only on a subbranch with $\sigma_i(t)U_i(t) \to 0$. These signatures are compared against bridges, regulatory rivals, and observation effects under common scoring rules.

5.5 Route-state validation

A useful validation ladder is:

- R0: endpoint clock only;
- R1: descriptive route-state proxy;

- R2: construct-valid route-state channel;
- R3: predictive route-state proxy independent of endpoint labels;
- R4: mechanism-supported proxy transported across sites, instruments, care contexts, and reference classes.

Confirmatory reserve-route support requires at least R3; mechanism and transport claims require R4. A time-to-endpoint coordinate or smoothing artifact cannot become reserve by relabeling.

Scalar-reserve admissibility rule. A scalar route state may support confirmatory reserve claims only when its construction is fixed without terminal neutrality or endpoint labels and its dimensional reduction preserves route-relevant predictions within a preregistered tolerance. It must show discriminant validity beyond frailty, prognosis, disease severity, treatment intensity, care setting, device dropout, and observation availability. Perturbations of the underlying reserve domains must yield transported route-consistent effects, and the scalarization must remain stable or be explicitly state-indexed across sites, treatments, and endpoint classes. Otherwise the route retains a vector-valued state, narrows its interpretation, or is reported as unidentified. Predicting endpoint proximity alone does not identify the reserve required by the route.

Reserve-endpoint nonidentity. Route-state exhaustion and the truth-level stream endpoint are distinct objects. Their coincidence is an empirical bridge proposition, not a definition.

R4 quantitative transport. The protocol preregisters a held-out metric appropriate to the object: predictive log score, calibration error, a simultaneous equivalence region, or an integral probability metric, together with its margin, overlap requirement, and retirement rule. No single R^2 or Wasserstein threshold is imposed across incomparable objects. Promotion additionally requires prediction of future response capacity beyond prognosis, frailty, device dropout, and endpoint proximity.

R4-P/R4-C separation. R4-P freezes the target environment, support, observation process, loss, calibration rule, and practical margin; successful prediction licenses only predictive transport. R4-C additionally freezes the intervention regime, causal estimand, treatment versions, source/target populations, consistency, exchangeability or a justified transport model, positivity, interference rule, endpoint/intercurrent-event strategy, and measurement invariance. If the causal functional is not identified, report R4-P or No adequate estimate; never causal transport by relabeling predictive success.

6. Discriminating signature package

6.1 Route entailment and completion nonidentifiability

No single statistic is route-specific. The reduced reserve route becomes informative through a package of signatures under a common observation model and serious rivals.

Completion nonidentifiability rule. Terminal closure alone does not identify a preterminal drift, volatility law, reserve process, endpoint hazard, bridge kernel, covariance form, perturbation response, terminal-window contraction, or selector. For any finite prefix with residual value ℓ, an unconstrained continuation can be assigned future value $-\ell$ without fixing how that continuation is generated. Therefore every signature below is route-specific: it becomes a prediction only after the mechanism, admissible continuation class, endpoint rule, observation model, and rival class have been declared.

6.2 Reserve-dispersion and boundary-exponent restrictions

Conditional reserve-dispersion restriction. Let $\Xi_i^\circ(t)$ be a declared conditioning state that excludes every component that deterministically fixes current reserve $U_i(t)$. For reserve level $u > 0$ and conditioning-state value x, define

$$V_{Y,i}(u,x,t) := \mathrm{Var}\,(Y_i(t) \mid U_i(t) = u,\ \Xi_i^\circ(t) = x),$$

$$V_{L,i}(u,x,t) := \mathrm{Var}\,(L_i^{\mathrm{rc}}(t) \mid U_i(t) = u,\ \Xi_i^\circ(t) = x) = c_{\mathrm{LU}}^2 u^2 V_{Y,i}(u,x,t).$$

Where the conditional surface has support and both functions are positive and differentiable in u,

$$\frac{\partial \log V_{L,i}}{\partial \log u} = 2 + \frac{\partial \log V_{Y,i}}{\partial \log u}.$$

A slope of two is an overidentifying restriction only where normalized-state dispersion, reserve uncertainty, and innovation-time effects are controlled, stratified, or modeled.

Boundary-exponent discipline. Define the distance to the realized route boundary by

$$d_i^{\mathrm{rc}}(t) := \tau_i^{\mathrm{rc}} - t.$$

Fixed or conditioned-reserve branch. Suppose, conditional on the displayed reserve and state, that

$$\begin{aligned} U_i(t) \quad &= c_U [d_i^{\mathrm{rc}}(t)]^\beta \{1 + o(1)\}, \\ \mathrm{Var}\big(Y_i(t) \mid U_i(t), \Xi_i^\circ(t)\big) \quad &= c_Y [d_i^{\mathrm{rc}}(t)]^\gamma \{1 + o(1)\}. \end{aligned}$$

Here $c_U, c_Y \in (0,\infty)$ and the remainders are conditional-uniform on the preregistered state set (or converge in another explicitly named mode).

$$\mathrm{Var}\big(L_i^{\mathrm{rc}}(t) \mid U_i(t), \Xi_i^\circ(t)\big) = c_{\mathrm{LU}}^2 c_U^2 c_Y [d_i^{\mathrm{rc}}(t)]^{2\beta+\gamma} \{1 + o(1)\}.$$

The special exponent 2β therefore additionally requires $\gamma = 0$ or an independently justified locally constant normalized-dispersion condition.

Random-reserve branch. If reserve is not fixed after conditioning, the claim is governed by the complete conditional total-variance identity

$$\mathrm{Var}\big(L_i^{\mathrm{rc}}(t) \mid \Xi_i(t)\big) = \mathcal{V}_{i,\mathrm{within}}(t) + \mathcal{V}_{i,\mathrm{between}}(t),$$

$$\mathcal{V}_{i,\mathrm{within}}(t) := c_{\mathrm{LU}}^2\, \mathbb{E}[U_i^2(t) \mathrm{Var}\big(Y_i(t) \mid U_i(t), \Xi_i(t)\big) \mid \Xi_i(t)],$$

$$\mathcal{V}_{i,\mathrm{between}}(t) := c_{\mathrm{LU}}^2 \mathrm{Var}\big(U_i(t)\, \mathbb{E}[Y_i(t) \mid U_i(t), \Xi_i(t)] \mid \Xi_i(t)\big).$$

A random-reserve exponent is reported only after both terms receive conditional-uniform, in-probability, almost-sure, or L^2 asymptotics and any claimed negligibility is proved. The realized-boundary branch is retrospective unless the route boundary was known prospectively; prospective diagnostics integrate over the filtered conditional distribution of remaining route time.

Limiting regime for the conditional route diagnostic. Use exactly one registered branch. (i) Deterministic boundary: $t \uparrow \tau_i^{\mathrm{rc}}$ through a declared deterministic sequence. (ii) Announced predictable boundary: choose stopping times $\tau_{i,n} < \tau_i^{\mathrm{rc}}$ with $\tau_{i,n} \uparrow \tau_i^{\mathrm{rc}}$ almost surely and state the mode of convergence. (iii) Retrospective distance-to-boundary analysis: index completed records by $u = \tau_i^{\mathrm{rc}} - t \downarrow 0$ and do not treat u as prospectively observable. Replacing τ_i^{rc} by $T_i^\star$ requires an independently validated route–endpoint bridge. No undeclared limit toward a random boundary is used.

[CP/ED] Conditional Chebyshev route diagnostic. Let Q_t denote the conditional law given continued membership in the declared route domain through t, on the risk set where that conditional probability is positive. Take the displayed second-moment contraction as an explicit route premise:

$$\mathbb{E}_{Q_t}[|L_i^{\mathrm{rc}}(t)|^2] \to 0,$$

then for every $\varepsilon > 0$,

$$Q_t\{|L_i^{\mathrm{rc}}(t)| > \varepsilon\} \le \varepsilon^{-2}\mathbb{E}_{Q_t}[\left(L_i^{\mathrm{rc}}(t)\right)^2] \to 0.$$

This is route-level convergence in probability under the changing at-risk laws. Estimation through a reserve ratio additionally requires $U_i(t) \ge u_{\min} > 0$ on the registered analysis region and the stated uniform integrability/error regime. Transfer to the phenomenal ledger requires $L_i^{\mathrm{phen}}(t) - L_i^{\mathrm{rc}}(t) \to 0$ in probability (or in L^2) under the same conditional laws. A denominator approaching zero with nonvanishing proxy error, or a bridge error that does not contract, defeats the corresponding promotion.

6.3 Two-time dependence and bridge comparators

The following correlations are defined only for $S_i^\theta < s \le t < \tau_i^{\mathrm{rc}}$ with $U_i(s)U_i(t) > 0$ and $0 < \tau_i^Y(s) \le \tau_i^Y(t) < \infty$. In the constant-volatility specialization require $\sigma_i \ne 0$. Outside this domain report covariance rather than correlation.

Two-time covariance and autocorrelation. For $s \le t$, deterministic U_i, origin-aligned $Y_i(S_i^\theta) = 0$, and deterministic square-integrable σ_i (or conditionally on a fixed volatility path), let

$$\tau_i^Y(t) := \int_{S_i^\theta}^{t} \sigma_i^2\,(r)\,dr.$$

For $s \le t$ in the deterministic-reserve reference case,

$$\mathrm{Cov}\left(L_i^{\mathrm{rc}}(s), L_i^{\mathrm{rc}}(t)\right) = c_{\mathrm{LU}}^2 U_i(s)U_i(t)\tau_i^Y(s), \qquad s \le t.$$

and

$$\rho_i^{\mathrm{rc}}(s,t) = \sqrt{\frac{\tau_i^Y(s)}{\tau_i^Y(t)}}.$$

For constant σ_i,

$$\rho_i^{\mathrm{rc}}(s,t) = \sqrt{\frac{s - S_i^\theta}{t - S_i^\theta}}, \qquad S_i^\theta < s \le t, \quad \sigma_i \text{ constant.}$$

Reserve changes marginal scale; normalized two-time dependence is inherited from the innovation clock under the stated deterministic or conditional assumptions.

Brownian bridge comparator. A zero-pinned Brownian bridge on [0, T] satisfies

$$dB_t = -\frac{B_t}{T-t}\,dt + \sigma\,dW_t,$$

with

$$\mathrm{Cov}(B_s, B_t) = \sigma^2 s \frac{T-t}{T}, \quad s \le t,$$

and

$$\rho_{\mathrm{bridge}}(s,t) = \sqrt{\frac{s(T-t)}{t(T-s)}}.$$

A bridge can mimic terminal contraction and terminal-window mean reversion. It can also look martingale-like after its own endpoint-normalizing transform. If B_t is a zero-pinned Brownian bridge and

$$Y_t^{\mathrm{br}} = \frac{B_t}{T-t},$$

then for $t < T$,

$$\mathrm{d}Y_t^{\mathrm{br}} = \frac{\sigma}{T-t}\mathrm{d}W_t.$$

Thus conditional-mean neutrality of a normalized coordinate is not a standalone discriminator. The bridge comparison turns on the full package: operationally separated reserve identity, reserve-squared dispersion, covariance and autocorrelation kernel, singular versus nonsingular normalized volatility, endpoint-generation likelihood, perturbation response, and model recovery under the same observation process.

Integrated-rate bridge rival. A Brownian bridge directly on L is path-class mismatched to an ordinary finite-variation accumulated ledger. A path-class-matched rival can bridge the rate process instead, and finite-band comparison should be explicit:

$$dF_t = b_F(t,F_t)\,dt + \sigma_F(t,F_t)\,dW_t, \qquad L_t = \int_{S_i^\theta}^t F_s\,ds.$$

$$Q_K(A) := P_0(A \mid |L_T| \le K), K > 0, P_0(|L_T| \le K) > 0.$$

Let $(\mathcal{X}, \tau_{\mathcal{X}})$ be the declared topological path space, with Q_K and any limit Q_0 probability laws on $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$. An exact zero-area bridge may be introduced through a regular conditional law, explicit bridge kernel, disintegration, Doob construction, or a shrinking-band limit only after tightness or another existence condition has been established (Chetrite & Touchette, 2015; Billingsley, 1999).

$$Q_K \Rightarrow Q_0 \quad \text{as } K \downarrow 0 \quad \text{in } (\mathcal{X}, \tau_{\mathcal{X}}).$$

If only subsequential limits exist, the result is a family of candidate bridge laws. A unique Q_0 requires a uniqueness theorem or an independently specified conditional law, disintegration, or bridge kernel. Weak convergence, finite-dimensional convergence, total-variation convergence, and convergence of terminal-band probabilities are not interchangeable.

Reciprocal-process discriminator. Let X be the prospectively frozen augmented state used by the bridge rival, including every measured component needed for its state-sufficiency claim, and write $\mathcal{F}_A^X := \sigma(X_t : t \in A)$. A bridge-mixture law predicts the two-sided conditional-independence restriction

$$\mathcal{F}_{(s,u)}^X \perp\!\!\!\perp \mathcal{F}_{[0,s)\cup(u,T]}^X \mid \sigma(X_s, X_u), \qquad 0 \le s < u \le T.$$

Every Markov law is reciprocal, but an arbitrary mixture of bridges of a Markov process generally remains reciprocal while failing to be Markov (Léonard et al., 2014). On held-out trajectories, compare a one-sided Markov test with this two-sided reciprocal test under the same state augmentation, observation kernel, missingness correction, and information budget. Reciprocal-without-Markov structure raises relative support for a mixed two-boundary bridge family over a sufficient-state local Markov controller; failure of reciprocity disfavors that bridge family. Hidden state and observation error can create apparent failures, so this is a mechanism-localization diagnostic, not a standalone PTVC result.

This rival can generate terminal pull while preserving a finite-variation accumulated ledger (Doob, 1957). It is a mechanism if baseline, bridge kernel, and endpoint event are independently fixed; a rival if used as an alternative conditioning explanation; and a descriptive representation if fitted to the same joint law without causal identification.

6.4 Terminal-window and intervention diagnostics

Primary fixed-time branch. Let $(\mathcal{F}_t)$ be the declared filtration, let $t_1 < t_2$ be deterministic, let Y_i be an integrable $(\mathcal{F}_t)$-martingale, and assume $U_i(t_1), U_i(t_2) > 0$. Assert the contraction identity only under $\mathbb{E}[Y_i(t_2) - Y_i(t_1) \mid Y_i(t_1), U_i(t_1), U_i(t_2)] = 0$ almost surely. A sufficient branch is that $U_i(t_1)$ and $U_i(t_2)$ are deterministic; another is that both are $\mathcal{F}_{t_1}$-measurable. With $L_i^{\mathrm{rc}}(t) = c_{\mathrm{LU}} U_i(t) Y_i(t)$, the displayed contraction identity follows. A diagnostic indexed by a realized future endpoint remains retrospective unless its information state is reconstructed without future endpoint information.

$$\mathbb{E}[L_i^{\mathrm{rc}}(t_2) \mid L_i^{\mathrm{rc}}(t_1), U_i(t_1), U_i(t_2)] = \frac{U_i(t_2)}{U_i(t_1)} L_i^{\mathrm{rc}}(t_1).$$

A stopping-time version is used only for bounded stopping times and a uniformly integrable martingale, or under another cited optional-sampling theorem with assumptions matching the displayed conditioning set. The schedule and reserve must not reveal later innovations. Otherwise the diagnostic is restricted to fixed times. The corresponding measured regression remains a route screen because Brownian-bridge and regulatory rivals can generate similar contraction.

[CP] Martingale terminal-pinning no-go. Let the endpoint time be an almost surely finite stopping time in the declared filtration. If the stopped process $(M_{t\wedge T})_{t\ge 0}$ is a uniformly integrable martingale that converges in mean to its stopping-time-measurable terminal variable, and $M_T = 0$ almost surely, then for every $t \ge 0$,

$$M_{t\wedge T} = \mathbb{E}[M_T \mid \mathcal{F}_{t\wedge T}] = 0 \quad \text{a.s.}$$

Thus a nontrivial martingale cannot explain exact pathwise terminal pinning merely through optional stopping; a candidate route must introduce conditioning, drift/control, an absorbing boundary, endpoint coupling, or another explicitly modeled mechanism. This no-go result localizes mechanism classes and does not itself establish PTVC.

Operational route-state intervention estimand. The identity

$$\left.\frac{\partial L_i^{\mathrm{rc}}(t)}{\partial U_i(t)}\right|_{Y_i(t)\ \mathrm{fixed}} = c_{\mathrm{LU}} Y_i(t).$$

This identity is algebraic, not automatically causal. At history h_t, let $a, a' \in \mathcal{A}_\eta^{\mathrm{do}}(h_t)$, where $\mathcal{A}_\eta^{\mathrm{do}}$ is the structural action domain and $\pi_\eta\left(\mathcal{A}_\eta^{\mathrm{do}}(h_t) \mid h_t\right) = 1$. The protocol states whether the intervention targets latent reserve, a measured reserve proxy, or an upstream manipulable cause. The measured total-effect contrast is defined below.

$$\begin{gathered}\Delta_A\left(\Delta; a, a' \mid \mathcal{H}_t^{\mathrm{pre}}\right) := \mathbb{E}[\{L^{\mathrm{meas}}(t+\Delta; a) - L^{\mathrm{meas}}(t; a)\} \\ -\{L^{\mathrm{meas}}(t+\Delta; a') - L^{\mathrm{meas}}(t; a')\} \mid \mathcal{H}_t^{\mathrm{pre}}].\end{gathered}$$

A reported value at history h_t evaluates a prospectively selected regular conditional version on its supported-history domain; unsupported histories return No adequate estimate.

A causal interpretation requires consistency, randomized assignment or sequential exchangeability, positivity on the structural action domain, a declared interference convention, and a specification of whether route-state mediation is included in the total effect. If any premise is unavailable, Δ_A is reported as an associational operational contrast. Errors-in-variables, shared calibration, endpoint uncertainty, and attenuation are handled through the registered validation or partial-identification branch.

Whole-stream intervention contract. Every intervention indexes the entire potential phenomenal stream, bearer/lineage map, ownership rule, routing, endpoint, observation process, and intercurrent events. Cross-world contrasts require an explicit bearer correspondence. Randomized assignment is the primary causal estimand; delivered physical dose may be analyzed when manipulable. Experienced phenomenal dose is ordinarily a latent mediator, not an object of $\mathrm{do}(\cdot)$. Treatment-induced endpoints and truncation by death require a prospectively chosen treatment-policy, composite, survivor-average, principal-stratum, separable-effect, or partial-identification estimand rather than conditioning on postrandomization survival.

6.5 Martingale, variance-ratio, and detectability diagnostics

Conditional signal-to-noise ratio. Where the observation-noise variance is positive and finite, define

$$\mathrm{SNR}_{i,\eta}(t \mid u, x) := \frac{\mathrm{Var}(L_i^{\mathrm{rc}}(t) \mid U_i(t) = u, \Xi_i^\circ(t) = x)}{\mathrm{Var}\left(\varepsilon_{i,\eta}(t) \mid U_i(t) = u, \Xi_i^\circ(t) = x\right)}.$$

This unbounded conditional signal-to-noise ratio is a local detectability diagnostic, not a compression coefficient or identification theorem. Outside its positive finite denominator domain it is undefined.

For calibrated unit-loading channels $O^{(c)} = V + \varepsilon^{(c)}$, $c \in \{1,2\}$, assume $\mathbb{E}[V^2 + \varepsilon_1^2 + \varepsilon_2^2 \mid X] < \infty$ almost surely, $\mathbb{E}[\varepsilon^{(c)} \mid V, X] = 0$, and $\mathbb{E}[\varepsilon_1\varepsilon_2 \mid V, X] = 0$ almost surely. Then $\mathrm{Var}(V \mid X) = \mathrm{Cov}(O_1, O_2 \mid X)$. A stronger sufficient alternative is mutual conditional independence of V, ε_1, and ε_2 given X, with zero conditional error means and the same moment premise.

$$\mathbb{E}(\varepsilon_c \mid V, X) = 0 \quad (c = 1,2), \qquad \mathbb{E}(\varepsilon_1\varepsilon_2 \mid V, X) = 0 \quad \text{a.s.}$$

$$\begin{aligned}\mathrm{Var}(V \mid X) &= \mathrm{Cov}(O_1, O_2 \mid X), \\ \mathrm{Var}(\varepsilon_1 \mid X) &= \mathrm{Var}(O_1 \mid X) - \mathrm{Cov}(O_1, O_2 \mid X).\end{aligned}$$

Analyst separation reduces leakage but does not imply these stochastic restrictions. A confirmatory overidentification branch uses at least three method-diverse channels, a negative-control channel, held-out loading calibration, and preregistered covariance or tetrad constraints. Failure of those constraints invalidates the atom-deconvolution branch.

Martingale and variance-ratio diagnostics are computed on reserve-normalized levels under the declared generative measure. Let

$$\Delta Y_{i,k} := Y_i(t_{k+1}) - Y_i(t_k), \qquad \mathbb{E}\left(\Delta Y_{i,k} \mid \mathcal{H}^{\mathrm{pre}}_{i,t_k}\right) = 0,$$

For square-integrable sampled values and a positive finite denominator, define

$$\mathrm{VR}_i(q;k) := \frac{\mathrm{Var}\big(Y_{i,k+q} - Y_{i,k}\big)}{\sum_{r=0}^{q-1} \mathrm{Var}\left(Y_{i,k+r+1} - Y_{i,k+r}\right)}, \quad q \geq 2.$$

For the reserve-ratio expansion, first declare the proxy equations

$$\hat{L} = c_{\mathrm{LU}} U Y + \varepsilon_L, \qquad \widehat{U} = U + \varepsilon_U, \qquad \hat{Y} = \frac{\hat{L}}{c_{\mathrm{LU}} \widehat{U}}.$$

On the registered region $U \geq u_{\min} > 0$, assume $|\varepsilon_U|/U = o_p(1)$ and $|\varepsilon_L|/(c_{\mathrm{LU}} U) = o_p(1)$, and $Y = O_p(1)$, uniformly whenever a uniform approximation is claimed. Then

$$\hat{Y} = Y + \frac{\varepsilon_L}{c_{\mathrm{LU}} U} - Y \frac{\varepsilon_U}{U} + o_p\left(\frac{|\varepsilon_L|}{c_{\mathrm{LU}} U} + \frac{|\varepsilon_U|}{U}\right).$$

No reserve-normalized inference is made at or below the validated proxy floor. The variance-ratio and serial-dependence diagnostics follow Lo and MacKinlay (1988) and Ljung and Box (1978).

6.6 Conjunctive signature package and ordered route verdict

Conjunctive signature package. Required diagnostics, identification gates, discrimination gates, rival comparisons, adequacy, availability, and open-set checks remain distinct multistate inputs. The finite nonempty required set is fixed before analysis; no binary product score substitutes for the ordered verdict.

Ordered route verdict. (1) A failed required condition or disqualifying gate yields Route-failed. (2) If no decisive failure occurs but a required diagnostic lacks an adequate estimate, report No adequate estimate. (3) If adequate models remain practically indistinguishable, report Model-ambiguous. (4) Route-favored language is permitted only when all required diagnostics are adequately estimated and favorable, every identification gate passes, the discrimination gate passes, and no serious rival is favored.

Controlling decision rule. Table 2C is the deterministic multistate truth table. Its inputs are prerequisite validity, estimate availability, band result, recovery, absolute adequacy, comparative score, and open-set check. A favorable local route never overrides a failed representation, endpoint, observation, or bearer gate.

6.7 Completion-potential overidentification fingerprint

Genericity and transport criterion. For an ordinary Markov reference law and a positive-probability terminal event, the committor and corresponding h-transform describe a conditioned law under their regularity conditions. Their PTVC relevance becomes distinctive when one object is estimated without test-endpoint leakage, frozen, and prospectively transported across held-out transition, action, endpoint, and terminal-support modules with superior performance to equal-budget conditioned, selection, regulation, hidden-state, and observation-artifact rivals.

Completion-potential contract. Freeze the reference law, ensemble weighting, state/history filtration, endpoint event, tolerance family, support, and normalization before scoring. If state sufficiency fails, use the full observed history or withdraw the Markov-generator claim. The ε-family is jointly estimated or projected to be nested, with simultaneous uncertainty; continuity from above licenses the exact-atom limit only under the declared latent law, kernel, and risk set. Off-support actions require exploration, an externally validated structural model, or partial identification. Potentials are coordinate-covariant, and a completion-conditioned selector is not a causal policy.

Positive-probability completion potential. Define $Z_i^\star := L_i^{\mathrm{phen}}(T_i^\star -)$ and $A_{i,\varepsilon} := \{|Z_i^\star| \le \varepsilon\}$.

Before using a state-indexed completion potential, require the following state-sufficiency condition to hold almost surely on the positive-risk set:

$$P_i^0\left(A_{i,\varepsilon} \mid \mathcal{H}_{i,t}, T_i^\star > t\right) = P_i^0\left(A_{i,\varepsilon} \mid \Xi_{i,t}, T_i^\star > t\right).$$

Otherwise augment $\Xi_{i,t}$ or use the full history. On the positive-risk set satisfying this condition, define

$$h_{i,\varepsilon}(t,\xi) := P_i^0\{A_{i,\varepsilon} \mid \Xi_{i,t} = \xi, T_i^\star > t\}, \qquad 0 \le h_{i,\varepsilon}(t,\xi) \le 1.$$

The tolerance family is nested and, when the declared regular conditional version and limit exist,

$$\begin{aligned} 0 \le h_{i,\varepsilon_1}(t,\xi) &\le h_{i,\varepsilon_2}(t,\xi) \le 1 \quad (0 \le \varepsilon_1 \le \varepsilon_2), \\ \lim_{\varepsilon \downarrow 0} h_{i,\varepsilon}(t,\xi) &= P_i^0\{Z_i^\star = 0 \mid \Xi_{i,t} = \xi, T_i^\star > t\}. \end{aligned}$$

A positive-probability terminal-band event uses ordinary conditioning; an exact probability-zero target requires a separately registered bridge or density-limit construction.

Use the authoritative $h_{i,\varepsilon}$ family and nested limit defined immediately above. The cross-module construction below introduces observational, interventional, Doob-transform, and action objects without redefining $h_{i,\varepsilon}$ or changing its limit order.

Observational-to-interventional bridge.

The observational committor and an action-plus-continuation-policy potential are distinct objects. Equality or transport requires a prespecified causal model with consistency, treatment versions, exchangeability or randomization, positivity, interference, endpoint, censoring/missingness, measurement-invariance, and continuation-policy assumptions. Otherwise the object is predictive or descriptive only.

$$\begin{aligned} h_{i,\varepsilon}^{\mathrm{obs}}(t,\xi) &:= P_i^0\left(A_{i,\varepsilon} \mid \Xi_{i,t} = \xi, T_i^\star > t\right), \\ h_{i,\varepsilon}^{a,g}(t,\xi) &:= \mathbb{P}\{A_{i,\varepsilon}(a,g) \mid \Xi_{i,t} = \xi, T_i^\star(a,g) > t\}. \end{aligned}$$

Positive-probability and null-target branches. When the terminal-band event has positive conditional probability, $h_{i,\varepsilon}$ is an ordinary conditional probability. Conditioning on an exact continuous-state endpoint or other probability-zero target instead requires a separately declared regular conditional bridge law or transition-density limit; the notation $\Pr(A \mid B)$ is not used with $\Pr(B) = 0$. The selected bridge version, reference measure, support, and limiting construction are fixed prospectively (Doob, 1957; Chetrite & Touchette, 2015).

Normalized completion-event martingale check. On the positive-probability branch, let $A_{i,\varepsilon}$ be the frozen completion event, require current-risk status $\mathbf{1}_{\{T_i^\star > t\}}$ to be $\mathcal{H}_{i,t}$-measurable for every t, and assume $P_i^0\left(A_{i,\varepsilon} \mid \mathcal{H}_{i,0}\right) > 0$ almost surely. Define

$$M_{i,\varepsilon}(t) := \frac{P_i^0\left(A_{i,\varepsilon} \mid \mathcal{H}_{i,t}\right)}{P_i^0\left(A_{i,\varepsilon} \mid \mathcal{H}_{i,0}\right)}.$$

Then $M_{i,\varepsilon}$ is a nonnegative P_i^0-martingale relative to $\left(\mathcal{H}_{i,t}\right)_{t\geq 0}$, with $M_{i,\varepsilon}(0) = 1$. Under the declared Markov-sufficiency branch, on $\{T_i^\star > t\}$ and with $P_i^0(T_i^\star > 0) = 1$,

$$M_{i,\varepsilon}(t) = \frac{h_{i,\varepsilon}\left(t, \Xi_{i,t}\right)}{h_{i,\varepsilon}\left(0, \Xi_{i,0}\right)}.$$

A frozen estimated potential must therefore pass out-of-fold reliability and conditional-increment tests over prespecified times on common support, with a prospectively fixed denominator floor and simultaneous uncertainty. Failure is adverse to that estimated completion-potential route. Passing is not an independent PTVC vote: the martingale property is generic to a correctly specified conditioned law.

Doob-transform branch. Let $D_{i,\varepsilon} := \{(t,\xi): h_{i,\varepsilon}(t,\xi) > 0\}$. On $D_{i,\varepsilon}$, assume Ξ_i is Markov under P_i^0 with generator $\mathcal{L}_i^0$; assume $h_{i,\varepsilon} \in C^{1,2}$ and $(\partial_t + \mathcal{L}_i^0)h_{i,\varepsilon} = 0$ on the conservative branch with declared terminal and boundary data. A nonharmonic h belongs to a separately specified killed, Feynman–Kac, or eigenfunction branch with its potential or diagonal adjustment.

$$A_i^0 := \sigma_i^0(\sigma_i^0)^\top, \qquad b_i^\varepsilon - b_i^0 = A_i^0 \nabla \log h_{i,\varepsilon}, \qquad q_i^\varepsilon(t,\xi,\mathrm{d}\xi') = q_i^0(t,\xi,\mathrm{d}\xi')\frac{h_{i,\varepsilon}(t,\xi')}{h_{i,\varepsilon}(t,\xi)},\ \xi' \neq \xi$$

Here q_i^0 is the off-diagonal jump-rate kernel, not a normalized transition probability. The diagonal is the negative total outgoing rate on a conservative branch; killed or sub-Markov branches state the added killing term. If A_i^0 is singular, only $A_i^0 \nabla \log h_{i,\varepsilon}$ is identified; components in $\ker(A_i^0)$, disconnected-support normalizations, and zero-rate edges remain unidentified without extra structure.

Endpoint-aware action typing. For horizon $\Delta > 0$, structural action a, and frozen continuation regime g, define the strict-pre-endpoint terminal-band event and feasibility payoff

$$A_{i,\varepsilon}^{a,g} := \left\{\left|L_i^{\mathrm{phen},a,g}(T_i^\star(a,g)-)\right| \leq \varepsilon\right\}.$$

$$G_{i,\varepsilon}^{a,g}(\Delta;t) := \mathbf{1}_{\{T_i^\star(a,g)\leq t+\Delta\}}\mathbf{1}_{A_{i,\varepsilon}^{a,g}}$$

$$+\mathbf{1}_{\{T_i^\star(a,g)>t+\Delta\}}h_{i,\varepsilon}^{a,g}\left(t+\Delta, \Xi_{i,t+\Delta}^{a,g}\right).$$

$$H_{i,\varepsilon}^{a,g}(\Delta;t,\xi) := \mathbb{E}_i^{a,g}[G_{i,\varepsilon}^{a,g}(\Delta;t) \mid \Xi_{i,t}^{a,g} = \xi, T_i^\star(a,g) > t].$$

$$\pi_i^\varepsilon(\mathrm{d}a \mid t,\xi,g,T_i^\star > t) := \frac{H_{i,\varepsilon}^{a,g}(\Delta;t,\xi)\,\pi_i^0(\mathrm{d}a \mid t,\xi,g,T_i^\star > t)}{\int_{\mathcal{A}_{i,\eta}^{\mathrm{do}}(t,\xi)} H_{i,\varepsilon}^{a',g}(\Delta;t,\xi)\,\pi_i^0(\mathrm{d}a' \mid t,\xi,g,T_i^\star > t)}$$

Require the denominator to be strictly positive. Since the registered feasibility payoff lies in [0,1], its conditional expectation also lies in [0,1], and the baseline policy is a probability kernel, finiteness is automatic. If the denominator is zero, the reweighted policy is undefined and the route returns No adequate estimate. Feasible policies are progressively measurable, assign probability one to $\mathcal{A}_{i,\eta}^{\mathrm{do}}(t,\xi)$, and are evaluated under the prospectively frozen continuation regime g, conditionally on $T_i^\star > t$, with declared reward, endpoint, and missingness integrability.

Cross-module discrepancy. Estimate $h_{i,\varepsilon}$ in one transition, action, endpoint, or terminal-support module, freeze it, and score the other modules on common support. Use

$$\Delta_{\mu_{\mathrm{ov}}}(f,g) := \|y \mapsto d_{\mathcal{P}}(f(y,\cdot), g(y,\cdot))\|_{L^\infty(\mu_{\mathrm{ov}})}.$$

Here f and g are identified only as μ_{ov}-almost-everywhere equivalence classes on the registered overlap support; if no adequate overlap law exists, report No adequate estimate.

Alternatively, use the preregistered μ_{ov} -integral analogue, with simultaneous uncertainty. Finite ε adjudicates the band branch; exact-atom evidence advances through the shrinking-band and atom decision ladder. Exactness requires the shrinking-band and atom decision ladder in Sections 2.1–2.4 and 11.7.4.

Controlling definition. The $h_{i,\varepsilon}$, $H_{i,\varepsilon}^{a,g}$, and π_i^{ε} displays above are authoritative. Later modules may add branch-specific assumptions by cross-reference but do not redefine these objects.

Common-support contract. Training and scoring modules share a preregistered overlap law μ_{ov}; discrepancy, normalization, integral versus essential-supremum choice, simultaneous uncertainty, and the no-overlap return are frozen before fitting. For path-valued modules the discrepancy may use signature embeddings with frozen interpolation, augmentation, truncation depth, normalization, and window (Chevyrev & Kormilitzin, 2016). Generator recovery validates the signature choice; it does not create a phenomenal bridge.

Conditioned-model scoring. When a bridge, h-transform, or selector is conditioned on the event used for evaluation, any law-level score includes the selection/event probability and normalizer, or the model is labeled support-by-construction. Model selection, hyperparameter tuning, representation learning, and stacking occur inside the training loop; the outer promotion set is untouched. R4-P predictive transport and R4-C causal/interventional transport are reported separately.

6.8 Counterfactual no-law worlds and unexpected-endpoint stress tests

The no-law counterfactual is operational: fit a serious reference ensemble on nonterminal data, including regulatory control, opponent kernels, flexible set points, mood momentum, selection, endpoint-conditioned sampling, memory reconstruction, censoring, measurement floors, and open-set generators. Let m index a candidate or rival latent law P_m^Z, and let K_η be the frozen observation kernel. For a finite or countable registered dominated bank, let $\lambda_\eta^{\text{dom}}$ dominate every observed-data push-forward on common support and define the following score for protocol-observed datum D. For a nondominated bank, use pairwise or model-specific proper scores only where the requisite Radon–Nikodym derivatives exist.

$$P_m^{\text{obs}}(\mathrm{d}D) = \int K_\eta(\mathrm{d}D \mid z) P_m^Z(\mathrm{d}z), \qquad P_m^{\text{obs}} \ll \lambda_\eta^{\text{dom}}, \qquad \mathcal{S}_m(D) := -\log \frac{\mathrm{d}P_m^{\text{obs}}}{\mathrm{d}\lambda_\eta^{\text{dom}}}(D).$$

Joint-versus-composite score. A module sum is a joint log score only when the terms are nonoverlapping conditional factors of one normalized observed-data law:

$$p_m(D) = \prod_{r=1}^{R} p_{m,r}(D_r \mid D_{<r}), \qquad -\log p_m(D) = \sum_{r=1}^{R} -\log p_{m,r}(D_r \mid D_{<r}).$$

If modules overlap, use incompatible coarsenings, or are separately normalized diagnostics, use prospectively fixed weights with a common score orientation, $w_r \geq 0$ and $\sum_{r=1}^{R} w_r = 1$, and call the result a composite score:

$$S_m^{\text{comp}}(D) = \sum_{r=1}^{R} w_r\, S_{m,r}(D_r).$$

Composite performance may compare models but does not by itself identify the full joint dependence law; selection and uncertainty are calibrated by registered simulation.

Every candidate and rival is scored through its observation-kernel push-forward on the same observed datum and common support. A terminal-band label enters the scored datum only in synthetic recovery or when independently identified and observed; otherwise it is integrated out. The registered factorization may supply transition, action, endpoint, and observation-channel log-score components, but their sum is evaluated as one joint score and its dependence is calibrated by simulation.

Unexpected endpoints create a discriminating fork. Fix landmark t_0, protocol version η, an observed endpoint-record space $\mathcal{D}_{i,\eta}^{\text{end}}$, and a preregistered finite cell set $\mathcal{C}_{i,\eta}$ including time-bin × cause, tail, censoring, observation-loss, and coarsening cells. Freeze the measurable map

$$\begin{aligned} \chi_{i,\eta}^{\text{end}} &: \mathcal{D}_{i,\eta}^{\text{end}} \to \mathcal{C}_{i,\eta}, \qquad c_{i,\eta}^{\text{obs}} := \chi_{i,\eta}^{\text{end}}\left(d_{i,\eta}^{\text{obs}}\right), \\ \text{Surp}_{i,\eta}^{\text{joint}} &:= -\log P_{0,\eta}^{\text{obs}}\left\{\chi_{i,\eta}^{\text{end}}\left(D_{i,\eta}^{\text{end}}\right) = c_{i,\eta}^{\text{obs}} \mid R_{i,\eta}^{\text{obs}}(t_0) = 1, \mathcal{O}_{i,t_0}^{\text{obs}}\right\}. \end{aligned}$$

Here $D_{i,\eta}^{\text{end}}$ is a generic random protocol-observed endpoint record and $d_{i,\eta}^{\text{obs}}$ is the realized record; $P_{0,\eta}^{\text{obs}}$ is the observation-kernel push-forward of the frozen base law. The base law, landmark, cell map, coarsening rule, common support, and zero-support rule are frozen before endpoint reveal. A separate truth-endpoint score may be used only on synthetic or identified branches and cannot contain censoring or observation-loss cells. Repeated-landmark predictions are generated out of fold; each stream receives a prospectively normalized total weight, and dependence across landmarks, estimation, competing risks, censoring, and generated-predictor uncertainty is propagated jointly.

Test whether terminal-residual dispersion, completion-potential calibration, and cross-equation transport deteriorate with endpoint surprise after cause, acuity, care, availability, and measurement precision are controlled. Persistence across highly surprising endpoints weighs against purely local horizon-aware mechanisms but does not by itself identify a global law.

6.9 Signed-dose compensation and ledger-residual sufficiency

Safe nonterminal interventions can test whether a randomized, brief, reversible physical stimulus recruits a quantitatively matched future response. Distinguish assignment, delivered physical dose, an independently calibrated direct phenomenal-dose map $a_{\text{dir}}(d,x)$, and the later ledger response. The do-operator applies to assignment or a manipulable delivered dose; experienced dose is a measured or latent mediator unless a separate intervention makes it directly settable. Intention-to-treat is primary. Let d denote the registered manipulable dose and let $\Delta C_{t:t+H}^{\text{later}}$ denote the later cumulative signed response, excluding the direct delivered contribution but retaining all protocol-induced anticipation, effort, frustration, relief, and downstream experience. On a declared differentiability region with $\partial_d a_{\text{dir}}(d,x) \neq 0$, define

$$\rho(H,d,x) := -\frac{\partial_d \mathbb{E}\{\Delta C_{t:t+H}^{\text{later}} \mid \text{do}(d), X_t = x\}}{\partial_d a_{\text{dir}}(d,x)}.$$

If a local coefficient is needed, set $\rho_0(H,x) := \rho(H,0,x)$. Exact local compensation is the registered dimensionless claim $\rho(H,d,x) = 1$ on a named dose region; it is not an unqualified global result. If the direct-dose calibration is unavailable or its derivative is zero, report the dimensional later-response surface and do not test it against one. Ordinary adaptation, opponent, and allostatic models predict latency, incomplete plateaus, sign asymmetry, nonlinearity, and state dependence, so the response surface across horizons, signs, modalities, controllability, predictability, repetition, sleep, and reserve is the target.

Let S_t contain the registered current state and let R_t^{ledger} be an out-of-sample cumulative residual not mathematically reducible to that state. For a future-outcome vector $G_{i,t:t+H}^{\text{fut}}$, test

$$H_0\colon \mathcal{L}\left(G_{i,t:t+H}^{\text{fut}} \mid S_t, R_t^{\text{ledger}}\right) = \mathcal{L}\left(G_{i,t:t+H}^{\text{fut}} \mid S_t\right),$$

against inequality. The future vector may include signed exposure, action selection, recovery dynamics, and endpoint intensity; $Y_i(t)$ remains reserved for the normalized route coordinate. Hidden-state confounding, autocorrelated environments, measurement error, and outcome-trained residual construction are explicit rivals.

Signed-dose estimand. Use d for the registered manipulable dose level throughout. Reserve $\delta_{\text{dose}} := d_1 - d_0$ for a prespecified contrast between two dose levels. Define a derivative only where it exists and is identified; otherwise use the registered finite difference with $\delta_{\text{dose}} \neq 0$ and $a_{\text{dir}}(d_0 + \delta_{\text{dose}}, x) \neq a_{\text{dir}}(d_0, x)$:

$$\rho_{\delta_{\text{dose}}}(H; d_0, x) := -\frac{\mathbb{E}\{\Delta C_{t:t+H}^{\text{later}} \mid \text{do}(d_0 + \delta_{\text{dose}}), X_t = x\} - \mathbb{E}\{\Delta C_{t:t+H}^{\text{later}} \mid \text{do}(d_0), X_t = x\}}{a_{\text{dir}}(d_0 + \delta_{\text{dose}}, x) - a_{\text{dir}}(d_0, x)}$$

Residual incremental value. Let $\mathcal{L}_S$ be held-out loss for the frozen state-only model and $\mathcal{L}_{S+R}$ the loss after adding the prospectively estimated residual. Define

$$\Delta_{\text{inc}} = \mathcal{L}_S - \mathcal{L}_{S+R}.$$

Evidence for incremental value requires the registered lower confidence bound for Δ_{inc} to exceed the practical margin $m_{\text{inc}} > 0$.

Residual compression sufficiency. Let $\mathcal{L}_R$ be held-out loss for the residual-only summary and $\mathcal{L}_H$ the loss for the registered richer-history model. Define

$$\Delta_{\text{comp}} = \mathcal{L}_R - \mathcal{L}_H.$$

Compression sufficiency is promoted only when the simultaneous upper confidence bound for Δ_{comp} is below the preregistered margin $m_{\text{comp}} \geq 0$ and the residual-only model also passes frozen absolute held-out loss, calibration, and coverage thresholds on the same data. Relative noninferiority between two inadequate predictors yields No adequate estimate, not sufficiency.

6.10 Pathwise ledger complementarity and terminal-residual adjudication

For stream i and registered cut k, prospectively partition the complete strict-pre-endpoint domain into disjoint measurable prefix and suffix sets that assign any cut-time atom exactly once. Let P_{ik} and R_{ik} denote the scalar total signed masses of the restrictions of the signed phenomenal measure to the prefix and suffix sets, respectively, under the same representation, origin, exposure, unit, ownership, and endpoint convention. Their shared-residual identity is

$$P_{ik} + R_{ik} = Z_i^\star.$$

[CP] Proposition 5 (cut-point additivity and terminal-residual adjudication; CUT-1). Finite signed-measure additivity gives the displayed identity. Across varying cuts, the regression slope -1 is an additivity diagnostic shared by every conserved residual $Z_i^\star = c$, including $c \neq 0$; it is not independent

PTVC evidence. The PTVC-bearing estimand is the independently adjudicated terminal residual $D_i(\tau) := P_i(\tau) + R_i(\tau) = Z_i^\star$ and its fixed zero target.

[P] Proof. The prefix and suffix sets are disjoint measurable sets whose union is the complete strict-pre-endpoint domain; finite signed-measure additivity therefore gives the displayed identity. PTVC adds $Z_i^\star = 0$. Conversely, an independently constructed complete-domain partition with $D_i(\tau) = 0$ implies terminal closure for that stream.

Observed hierarchical errors-in-variables model. Use a stream-specific intercept for the latent terminal residual $Z_i^\star$ and first fit the unconstrained diagnostic model. Observation-level non-tautology gate: the measured suffix must come from post-cut observations or a separately validated terminal-remainder bridge; it may not be defined by subtracting the measured prefix from the terminal target under test, because that construction guarantees complementarity algebraically.

$$\hat{P}_{ik} = P_{ik} + u_{ik}, \qquad \hat{R}_{ik} = \alpha_i - \beta P_{ik} + v_{ik}.$$

This implies

$$\hat{P}_{ik} + \hat{R}_{ik} = \alpha_i + (1 - \beta) P_{ik} + u_{ik} + v_{ik}.$$

The additivity target is $H_{\text{add}}: \beta = 1$, evaluated with the prospectively registered simultaneous compatibility or equivalence rule. Do not fix β=1 in the model used to test that target.

Declare the joint law of (u_{ik}, v_{ik}) across cuts and channels, with history or lineage as the resampling unit; cuts k are nested within stream i. First evaluate $H_{\text{add}}: \beta = 1$ in the unconstrained errors-in-variables model. If that target is compatible at the registered resolution, fit a second-stage constrained model with β=1 and estimate the stream-specific intercept α_i. The identity between the fitted intercept and $Z_i^\star$ is licensed only when the location of the combined measurement error is fixed at zero by design or independent calibration. Otherwise propagate the joint compatible region for (α_i, β) and every admissible combined bias into a simultaneous truth-containing residual set for all included streams; an unbounded set, or a set whose empirical status remains unresolved, yields No adequate estimate. Replicate cuts improve within-stream precision but do not identify a shared bias or increase the number of independent streams. A conventional point-null failure to reject $\alpha_i = 0$ is not favorable PTVC evidence. Finite-resolution support requires full containment of the truth-containing residual set in the registered acceptance region; exact support requires the separately registered exactness route. The implied slope -1 remains an additivity diagnostic and supplies no independent PTVC vote.

Errors-in-variables identification gate. Before testing the additivity restriction β = 1 (equivalently, regression slope −1), the protocol must preregister at least one valid identification route: externally calibrated error covariance or reliability; theorem-matched method-diverse replicate channels; a valid instrument with exclusion and relevance; a structural measurement model validated in sealed recovery studies; or a bounded error class yielding a truth-containing identified set. Declaring an error distribution is not identification. Shared-method, common-smoothing, and correlated-origin errors can manufacture a slope near minus one; without a valid route, report an identified set or No adequate estimate.

[CP] Corollary 5.1 (randomized arm-level branch). Let assignment occur at τ_{assign} and let $\mathcal{G}_\eta$ be the registered arm/regime family. Require the same cut to be admissible under every registered regime: $\mathbb{P}\{\tau_{\text{assign}} < \tau_{\text{cut}} < T_i^\star(g)\} = 1$ for every $g \in \mathcal{G}_\eta$ on the target class. If an endpoint can precede τ_{cut}, treat it as a declared intercurrent event with a prespecified estimand/strategy; do not condition on postrandomization survival. Arm-level contrasts include every protocol-induced contribution and are not individual cross-world effects.

For a bounded laboratory window $W = [a, b)$ that is not assumed to end at the independently adjudicated endpoint $T_i^\star$, define the omitted prefix and suffix contributions on the same representation, origin, unit, ownership, and endpoint branch:

$$\begin{aligned} Z_{i,\mathrm{pre}}(a) &:= L_i^{\mathrm{phen}}(a-), \\ Z_{i,\mathrm{suf}}(b) &:= L_i^{\mathrm{phen}}(T_i^\star -) - L_i^{\mathrm{phen}}(b-), \\ Z_i^\star &= Z_{i,\mathrm{pre}}(a) + \Delta L_i([a,b)) + Z_{i,\mathrm{suf}}(b). \end{aligned}$$

A bounded laboratory window supplies only the middle term and therefore tests local episode complementarity. Promotion to complete-stream PTVC requires the prefix term to be zero under the declared origin and atom convention or identified by a validated prefix bridge, and the suffix term to be zero under independent endpoint adjudication or identified by a validated terminal-remainder bridge. Without both components, the result remains a local mechanism screen. The confirmatory package rotates cuts, instruments, channels, cohorts, and interventions and requires the terminal-residual target and completion potential to transport under equal rival scoring.

7. Observation, estimands, missingness, and recoverability

7.1 Estimands, intercurrent events, and causal roles

The treatment-policy, composite, while-on-treatment, and hypothetical intercurrent-event strategies follow ICH E9(R1) (ICH, 2019). Principal-stratum targets follow Frangakis and Rubin (2002); any additional manuscript categories are separately defined rather than attributed to ICH.

A terminal study must declare the measured estimand before modeling. The estimand must specify target population, variable, endpoint rule, time coordinate, terminal window, summary measure, observation process, and intercurrent-event strategy. Analgesia, sedation, ventilation status, monitoring withdrawal, hospice enrollment, family consent, acute infection, delirium treatment, care transitions, device removal, caregiver substitution, language change, disability state change, reporting-capacity change, observation shutdown, endpoint timing, and endpoint adjudication may be confounders, mediators, selection mechanisms, competing events, endpoint components, intercurrent events, or parts of the estimand. A protocol must state which role each item plays and must declare the estimand strategy: treatment-policy, hypothetical, composite, while-on-treatment, principal-stratum, descriptive, predictive, mechanistic, causal, partial-identification, or No adequate estimate (Frangakis & Rubin, 2002).

The same variable may not be treated as a nuisance in one paragraph, a mechanism in another, and part of the endpoint in a third without a declared causal or estimand framework. If treatment, sedation, analgesia, monitoring, hospice access, family consent, device tolerance, language access, reporting capacity, or care setting determines who remains observable, the study must state the positivity assumptions needed for any causal interpretation. If one person's care environment, family script, clinical staffing, or institutional policy affects another participant's observation or treatment path, interference must be modeled or the causal claim withdrawn.

7.2 Observation kernels and garbling

Observation should be modeled as a channel system.

Observation-garbling rule. Let candidate latent laws be pushed through the same observation kernel K_t:

$$P_t^{\mathrm{obs}} = P_t^{\mathrm{lat}} K_t, Q_t^{\mathrm{obs}} = Q_t^{\mathrm{lat}} K_t.$$

For a Markov kernel K, the displayed f-divergence contraction is the data-processing inequality for f-divergences (Csiszár, 1967). Blackwell (1951, 1953) supplies the broader comparison-of-experiments interpretation: a garbled experiment cannot be more informative for every decision problem.

$$D_f\left(P_t^{\mathrm{obs}} \| Q_t^{\mathrm{obs}}\right) \leq D_f\left(P_t^{\mathrm{lat}} \| Q_t^{\mathrm{lat}}\right).$$

Observed compression can reflect information loss, censoring, clipping, channel shutdown, device intolerance, family reporting, sedation, or care-context change without latent contraction. Confirmatory protocols therefore require calibration signals, negative-control channels, or independent channel-quality measures that distinguish the two.

Table 2B. Variable-role and observation-kernel matrix

Variable / object	Ontic or observed	Timing	Causal role	Missingness status	Permitted use
Assignment A_i	Assigned/randomized	Before exposure	ITT or instrument	Recorded by protocol	Primary assignment estimand

Variable / object	Ontic or observed	Timing	Causal role	Missingness status	Permitted use
Delivered physical dose d_i^{del}	Delivered/received	Exposure time	Manipulable exposure or adherence mediator	Measured with error	Per-protocol/mediation only under stated assumptions
Experienced/direct phenomenal dose $a_{\mathrm{dir}}(d,x)$	Latent or measured mediator	During/after exposure	Phenomenal mediator	Bridge-dependent	Not a do-object without a separate intervention
Truth endpoint $T^\star$	Ontic	Terminal	Outcome boundary	Latent	Only through endpoint bridge
Protocol endpoint	Observed/estimated	Adjudication time	Measurement of boundary	Interval/revision allowed	Registered endpoint model
Intercurrent event	Observed/latent	Post assignment	Changes interpretation/path	May be incomplete	Prespecified strategy
Latent channel target	Ontic target	State indexed	Measurement target	Never coded as zero	Bridge model only
Nuisance state	Latent/observed	Time varying	Confounding/measurement	May be missing	Adjustment/sensitivity
Retention	Observed process	Longitudinal	Selection	Not a value	Joint/sensitivity model
Availability	Observed indicator	Each occasion	Observation process	Defines †	Kernel/missingness model
Recorded value	Observed	When available	Channel output	Undefined when unavailable	Frozen preprocessing
Protocol-observed datum	Observed pair	Each occasion	Analysis input	Includes availability	Common raw-data manifest
Analysis estimator	Derived	After freeze	Decision statistic	NAE possible	Registered analysis only

Every candidate and rival is pushed through the same frozen raw-data manifest and observation kernel unless a model-specific kernel is itself the registered object under comparison; model-specific preprocessing or channel access counts against its information/tuning budget.

7.3 Ontic-epistemic decomposition

For $Z_t \in L^2$ and observation sigma-field $\mathcal{O}_t$, $\mathrm{Var}(Z_t) = \mathrm{Var}(\mathbb{E}[Z_t \mid \mathcal{O}_t]) + \mathbb{E}[\mathrm{Var}(Z_t \mid \mathcal{O}_t)]$. This controlling identity is VAR-1.

A shrinking posterior mean variance can reflect observation loss, smoothing, clipping, or changing channel quality while posterior uncertainty grows. Reports should therefore separate latent dispersion, posterior uncertainty, observation-garbling, and estimator shrinkage.

Assume $L_t \in L^2$. Let $m_t := \mathbb{E}[L_t \mid \mathcal{O}_t]$ and $v_t := \mathrm{Var}(L_t \mid \mathcal{O}_t)$, where $\mathcal{O}_t$ contains only prospectively available information and excludes future endpoint labels, future observation availability, and retrospectively supplied time-to-endpoint information. Then

$$\mathbb{E}[L_t^2] = \mathbb{E}[m_t^2] + \mathbb{E}[v_t].$$

Consequently, mean-square convergence $L_t \to 0$ forces both the predictable component $\mathbb{E}[m_t^2] \to 0$ and the mean unresolved conditional variance $\mathbb{E}[v_t] \to 0$. A posterior mean approaching zero while unresolved uncertainty remains large or increases is epistemic compression, not identified latent contraction.

For nested same-time information sets $\mathcal{G}_t \subseteq \mathcal{O}_t$ and square-integrable $L(t)$,

$$\begin{aligned}\mathbb{E}[\mathrm{Var}(L(t) \mid \mathcal{G}_t)] &= \mathbb{E}[\mathrm{Var}(L(t) \mid \mathcal{O}_t)] \\ &+\mathbb{E}[\mathrm{Var}(\mathbb{E}[L(t) \mid \mathcal{O}_t] \mid \mathcal{G}_t)] \\ &\geq \mathbb{E}[\mathrm{Var}(L(t) \mid \mathcal{O}_t)].\end{aligned}$$

This is an expectation statement; realized posterior variances need not be ordered record by record. The ex-ante sigma-field excludes terminal outcomes, future observation, and outcome-trained representations.

Prospective-information status rule. For every displayed trajectory or diagnostic, label the object as observed, filtered, forecast, smoothed, endpoint-aligned, retrospectively adjudicated, imputed, or simulated. Only filtered quantities based on information available at time t and genuine forecasts based on that same information may support prospective preterminal prediction. Smoothed, endpoint-aligned, or retrospectively adjudicated trajectories may support description, sensitivity analysis, or model checking, but they may not be presented as signals available before the endpoint. If the historical information set cannot be reconstructed, the prospective claim receives No adequate estimate (Doucet et al., 2001; Durbin & Koopman, 2012; Särkkä, 2013).

Reverse-time causal discipline. Time-to-endpoint plots, backward alignment, and smoothed trajectories are descriptive or predictive unless endpoint timing is exogenous or a causal design identifies the relevant intervention. They may not be narrated as prospective control signals, and conditioning on survival to a backward-time landmark must not create immortal-time or collider bias.

7.4 Channel targets and measurement invariance

For channel c at time t_k, define the observation-conditioning state with the Ξ namespace; $Z_i^\star$ remains reserved for terminal residuals:

$$\Xi_{i,k}^{(c),\mathrm{obs}} := \left(X_i(t_k), V_{i,c}^{\mathrm{lat}}(t_k), W_i^{\mathrm{nuis}}(t_k), O_{i,t_k-}^{\mathrm{obs}}, H_{i,t_k-}^{\mathrm{joint}}\right).$$

Conditional on availability, the channel law is

$$\mathcal{L}\left(O_{i,k}^{(c)} \mid O_{i,c}^{\mathrm{avail}}(t_k) = 1, \Xi_{i,k}^{(c),\mathrm{obs}}\right) = K_{\eta,c,t_k}^{\mathrm{obs}}\left(\cdot \mid \Xi_{i,k}^{(c),\mathrm{obs}}\right).$$

Let $\dagger \notin \mathcal{O}_c$ denote unavailability and define

$$O_{i,k}^{\mathrm{rec},c} := \begin{cases} O_{i,k}^{(c)}, & O_{i,c}^{\mathrm{avail}}(t_k) = 1, \\ \dagger, & O_{i,c}^{\mathrm{avail}}(t_k) = 0, \end{cases} \qquad D_{i,k}^{(c)} := \left(O_{i,c}^{\mathrm{avail}}(t_k), O_{i,k}^{\mathrm{rec},c}\right).$$

The observed-data codomain is $\mathcal{O}_c \cup \{\dagger\}$; missingness is never encoded as scientific zero. The full likelihood includes availability and every state-dependent loading, threshold, sign, unit, and error process.

Terminal measurement-invariance rule. The terminal window cannot assume that a channel keeps the same sign, unit, loading, threshold, or error structure when consciousness, language, pain control, sedation, delirium, family presence, clinical access, or device tolerance changes. Dynamic measurement invariance, response-shift, convergent/discriminant validity, and channel-quality checks must be reported before comparing terminal and nonterminal estimands (American Educational Research Association et al., 2014; Bollen, 1989; Campbell & Fiske, 1959; Cronbach & Meehl, 1955; Lord & Novick, 1968; Millsap, 2011).

Within-stream transport rule. A between-stream association does not establish a within-stream sign, loading, or dynamic relation. Multilevel measurement models must distinguish between-stream and within-stream effects; only a held-out, state-appropriate within-stream relation may contribute to the sign or loading of a channel used to reconstruct a stream trajectory, and only under the independently validated measured-to-latent bridge. Sign reversal, material loading divergence, or failure across medication, sedation, disability, culture, endpoint proximity, or device state requires narrowing, state indexing, or exclusion of the channel from confirmatory ledger construction.

Instrument-version rule. An instrument version includes its active channels, wording, language, placement, firmware, preprocessing, calibration, lag rules, scalarization, channel weights, and missingness treatment. Different versions define different measured objects unless an overlap-calibration or measurement-invariance bridge establishes comparability. Channel removal or down-weighting follows prospectively declared signal-quality, burden, calibration, or negative-control criteria rather than its effect on terminal fit. Material leave-one-channel-out changes in sign, equivalence status, or terminal verdict are reported as channel-sensitive.

Unavailable observations use † outside every scientific-value codomain. Each protocol datum $D_{i,k}^{(c)}$ is the pair (availability, recorded value), and every channel-specific target carries a $\mathtt{lat}$ qualifier. MeasurementVersionGate records active channels, wording/language, firmware, preprocessing, sign/origin, loadings, lag/exposure rules, scalarization, missingness treatment, and calibration bridge. Promotion requires within-stream dynamic invariance or explicit state indexing across terminal proximity, medication, sedation, disability, culture, care, and device state.

When channel-availability indicators are dependent, model the joint vector $\mathbf{O}_i^{\text{avail}}(t) = \left(O_{i,c}^{\text{avail}}(t)\right)_{c\in C_i^{\text{obs}}}$ rather than multiplying marginal probabilities. A scalar availability indicator is used only when all registered channels are attempted and lost jointly by design.

Proxy and memory carrier rule. A report about another stream is generated by an explicit cross-stream observation kernel. The reported person's inferred experience remains assigned to that person's stream; distress, relief, memory, or narrative experienced by the reporter belongs to the reporter's stream and is not counted again as the subject's phenomenal mass. Remembered and experienced quantities are linked by a registered memory kernel rather than summed as independent evidence or value.

7.5 Missingness, retention, and denominator flow

Missingness near the endpoint is potentially informative whenever retention or observation availability may depend on latent or observed state, care, burden, communication, device, or endpoint proximity. The protocol therefore models or sensitivity-bounds these processes rather than assuming ignorability (Scharfstein et al., 1999; Tsiatis, 2006).

For a declared joint-model history, identify or sensitivity-bound the retention and observation-availability processes:

$$\mathbb{P}\left(R_i^{\text{ret}}(t) = 1 \mid \mathcal{H}_{i,t-}^{\text{joint}}\right) = \pi_R\left(\mathcal{H}_{i,t-}^{\text{joint}}\right).$$

$$\mathbb{P}\left(O_{i,c}^{\text{avail}}(t) = 1 \mid \mathcal{H}_{i,t-}^{\text{joint}}, R_i^{\text{ret}}(t) = 1\right) = \pi_{O,c}\left(\mathcal{H}_{i,t-}^{\text{joint}}\right).$$

If these processes are not scientifically identifiable, use a prospectively declared missing-not-at-random sensitivity analysis, partial-identification or tipping-point analysis, or No adequate estimate (Rubin, 1976; Diggle & Kenward, 1994; Little & Rubin, 2019). Terminal missingness is potentially informative whenever retention or availability may depend on distress, sedation, delirium, communication loss, device tolerance, family consent, clinician access, medication, endpoint proximity, reserve decline, care transition, or observation burden.

Long-horizon ecological momentary assessment (EMA) shows that observation availability is itself a dynamic outcome. The TIME study enrolled 246 young adults; 207 contributed analyzable data and 123 completed the full year (50% of those enrolled). During biweekly four-day bursts, participants received a mean of 12.1 prompts per day; mean completion in the analyzable sample was 77% (Prochnow et al., 2025).

Completion odds rose when the phone screen was active and fell with time in study, travel, short sleep, away-from-home contexts, and higher stress at the preceding prompt (Prochnow et al., 2025).

The WARN-D tutorial examined item-performance patterns across 599 participants and approximately 360 scheduled occasions, showing why context, time trends, and within-person variation require direct inspection (Siepe et al., 2025).

Denominator flow is mandatory. Reports state the number screened, eligible, included, stream-qualified, endpoint-adjudicated, route-state-valid, measured-ledger-valid, observed in the terminal window, excluded, downgraded to No adequate estimate, and analyzed; each exclusion category reports its relation to distress, volatility, reserve, communication, medication, delirium, and endpoint proximity.

On the strict-pre-endpoint branch, stream-level phenomenal quantities are undefined for $t \geq T_i^{\star}$. They are not missing zeroes, neutral observations, carried-forward values, or ordinary imputation targets. A post-endpoint quantity requires a separately declared endpoint-inclusive or continuation branch.

7.6 Recoverability, no-lookahead, and measurement reactivity

Finite-observation recoverability is a design requirement. A signature is not confirmatory merely because it is true in the latent reference model. The protocol must show, through simulation or analytic power and identifiability calculation, that the signature is recoverable under the planned cadence, sample size, measurement error, route-state-proxy uncertainty, including reserve-proxy uncertainty where applicable, missingness, endpoint uncertainty, and rival family. Otherwise the study reports design non-recoverability.

Discrete-integral recoverability rule. A finite sequence of measurements does not identify a continuous path integral unless the protocol declares the signal timescale, sampling design, quadrature rule, smoothness or bandwidth assumptions, interpolation rule, aliasing risk, missingness model, and worst-case quadrature error. Sparse, irregular, reactive, event-triggered, terminally missing, smoothed, interpolated, or selectively retained samples may estimate a coarse increment, a bound, or a partial-identification interval, but they cannot be presented as recovery of the continuous ledger path without the declared regularity bridge.

No-neutrality-by-construction and telescoping rule. Every cumulative measured object declares whether its primitive input is a state level, signed rate, interval contribution, derivative, or change score; its units, temporal kernel, quadrature rule, channel signs, scalarization weights, and weight-learning procedure are fixed independently of the terminal result.

If interval contributions are constant-weight first differences, their sum telescopes to a boundary contrast. Its PTVC interpretation requires a validated accrual bridge linking the transformed quantity to cumulative phenomenal value. Variable or learned weights introduce interior cross-terms whose interpretation depends on the weight process. Centering, derivative transforms, endpoint fitting, outcome-selected signs or weights, and outcome-dependent exclusions enter confirmatory neutrality only through an independently validated bridge. A route or endpoint defined to close by construction belongs to the model specification; held-out predictions and observations validated through an independent bridge provide empirical support for the protocol-fixed accrual bridge and candidate route, which remain subject to complete endpoint and transport adjudication.

Joint path-uncertainty rule. On a declared grid, let $\mathbf{F}$ be the vector of latent or measured signed rates or interval contributions and let $\mathbf{w}$ be fixed or D-measurable and contain unity, exposure, duration, and quadrature weights.

$$\mathrm{Var}(\mathbf{w}^\top \mathbf{F} \mid D) = \mathbf{w}^\top \mathrm{Cov}(\mathbf{F} \mid D)\mathbf{w}.$$

If integration weights are estimated or random, propagate their joint law with the rate vector through (OBS-2):

$$\mathrm{Var}(\mathbf{W}^\top \mathbf{F} \mid D) = \mathbb{E}[\mathbf{W}^\top \mathrm{Cov}(\mathbf{F} \mid \mathbf{W}, D)\mathbf{W} \mid D] \\ + \mathrm{Var}(\mathbf{W}^\top \mathbb{E}[\mathbf{F} \mid \mathbf{W}, D] \mid D).$$

A diagonal-only approximation is valid when the relevant conditional covariances are zero or negligibly bounded. Confirmatory cumulative inference propagates joint uncertainty in latent state, serial dependence, calibration, origin, sign, scalarization, missing intervals, endpoint time, instrument linking, and model class. A posterior mean path inside the terminal band earns band support when the joint terminal-compatible distribution or identified set is correspondingly concentrated within the band.

A terminal study also requires no-lookahead rules. Endpoint, reserve, filtering, forecasting, smoothing, imputation, terminal-window width, equivalence-band provenance, measurement channel selection, and exclusion rules must be fixed before outcome inspection or clearly labeled exploratory.

Learned-proxy leakage rule. A reserve proxy, endpoint classifier, measured-ledger estimator, channel weighting rule, embedding, imputation model, smoothing rule, or learned representation may not use future endpoint information, terminal-window labels, ledger neutrality, terminal concentration, outcome-driven exclusion decisions, treatment-response labels, observation-availability futures, or features derived from the target quantity unless the protocol is explicitly exploratory and the leakage path is named.

A leakage audit must report, for every learned object, which features have access to endpoint labels, future observation availability, reserve proxies, treatment events, terminal-window position, measured-ledger summaries, imputation targets, and exclusion decisions. If any feature path can encode the endpoint, the ledger target, observation availability, or terminal status, the analysis must be downgraded, redesigned, or evaluated against leakage-matched rivals.

Measurement reactivity and intercurrent-event discipline. Observation can change the trajectory being measured. Diaries, sensors, prompts, interviews, family reporting, clinical attention, device presence, repeated reporting, or terminal-study enrollment may alter distress, relief, attention, privacy, care, sleep, communication, device tolerance, endpoint behavior, or observation probability. If measurement changes the target trajectory, the estimand must include that intervention, route it through a declared intercurrent-event strategy, or report a reactivity-bounded, exploratory, or No adequate estimate result. Sedation changes, analgesic escalation, ventilation status, transfer, infection, delirium treatment, communication loss, terminal agitation, device intolerance, or family nonconsent must be assigned to a declared treatment-policy, hypothetical, composite, while-on-treatment, principal-stratum, partial-identification, descriptive, or No adequate estimate strategy.

Participant, family, and staff burden is part of the observation process. Protocols must report where study presence changes care time, communication, family contact, device tolerance, clinician attention, privacy, fatigue, distress, relief, or refusal. A low-burden passive channel, an active diary, and an end-of-life interview do not observe the same target unless the difference is modeled or explicitly bounded.

Temporal-coverage rule. Let N_i^{samp} count scheduled and triggered observation attempts and let $\Lambda_i^{\mathrm{samp}}$ be its predictable compensator. Represent

$$d\Lambda_i^{\mathrm{samp}}(t) = \sum_m p_{i,m}\ \delta_{s_{i,m}}(dt) + \lambda_i^{\mathrm{samp}}(t)\, dt + d\Lambda_i^{\mathrm{sing}}(t),$$

where the first term contains scheduled predictable atoms, the second is the absolutely continuous component, and the third is any declared singular component. The familiar small-increment limit for $\lambda_i^{\mathrm{samp}}$ is used only on the absolutely continuous branch of a simple point process with no simultaneous attempts. Simultaneous or multitype attempts use a marked counting process. Scheduled, randomized, participant-triggered, symptom-triggered, clinician-triggered, safety-triggered, endpoint-triggered, and device-failure-related observations retain separate marks.

Crisis-enriched, event-triggered, or terminally enriched observations alone do not identify a time- or exposure-weighted complete-stream integral. Confirmatory designs require an outcome-independent scheduled or randomized coverage component, an identified sampling-process correction with adequate positivity, scientifically justified bounds for unsampled intervals, or a partial-identification or No adequate estimate verdict. If a region of positive phenomenal exposure has zero observation probability and its signed contribution is not otherwise bounded, the complete ledger is not point identified.

Quantitative recovery contract. Bound discrete-integration error using a declared path-regularity, variation, spectral-decay, or multiresolution condition and validate it under the actual cadence and missingness. Publish a no-neutrality-by-construction test for every cumulative proxy; a structured reproducibility record of learned objects, training data, targets, features, horizons, and leakage exclusions; and a measurement-reactivity estimand or bound. Recovery includes quantization, censoring, smoothing, endpoint alignment, and observation-floor generators.

8. Rival models and rigorous comparators

8.1 Common comparison rules

Common observation architecture. Every latent-dynamics candidate and rival is evaluated through one frozen observation architecture, preprocessing rule, missingness/retention process, and information budget. A prospectively declared observation-artifact rival may vary only the named kernel component that constitutes the artifact while keeping latent-dynamics information, tuning budget, and scoring rules equal. Those results are labeled observation-artifact comparisons and do not license a different favorable kernel for each latent model.

Duration-neglect and peak–end observation rival. Retrospective global reports are compared with a prospectively sampled integral branch and a peak–end memory branch (Fredrickson & Kahneman, 1993; Redelmeier & Kahneman, 1996). The rival is an observation or memory kernel: success can invalidate retrospective-report evidence for an additive ledger but does not by itself refute a separately identified latent signed measure.

Equal-budget comparator contract. Freeze for every model the raw variables, observation kernel, train/validation/test split, missingness inputs, tuning calls, parameter/representation complexity, computation, and score. Ordinary regulation, homeostatic control, flexible-state, hidden-state, memory, endpoint-selection, observation-floor, and balance-law rivals receive their strongest scientifically credible forms. Absolute adequacy and open-set checks are fixed in addition to relative rank.

8.2 Regulatory, set-point, heavy-tail, and diffusion rivals

Homeostasis, allostasis, adaptation, and opponent processes. These models can generate return toward set points, rebound, affect regulation, and reserve-dependent changes without implying terminal neutrality. They are serious rivals and mechanism neighbors.

Non-neutral set points and flexible baselines. A rival may converge toward a nonzero terminal set point, a learned subject-specific baseline, or a flexible terminal attractor. Flexible baselines may include spline or Gaussian-process terminal curves, switching state-space models, subject-specific random effects, heteroskedastic terminal states, frailty-linked baselines, and care-context baselines. Such models defeat the empirical reading of universal neutrality if they fit terminal data and out-of-sample structure as well as the PTVC route.

Heavy-tail and stochastic-volatility rivals. Apparent concentration or late volatility change may arise from mixture structure, heteroskedasticity, volatility regime switching, tail truncation, or robust-estimator behavior rather than terminal neutrality (Kim, Shephard, & Chib, 1998; McLachlan & Peel, 2000). Recovery studies must include heavy-tail and stochastic-volatility baselines when observations are sparse, noisy, or clinically heterogeneous.

Integrated OU rival. For $\kappa > 0$ and $\eta \geq 0$, let

$$dX_t = -\kappa X_t\, dt + \eta\, dW_t, \quad X_0 = 0, \quad I_t = \int_0^t X_s\, ds,$$

the zero-start integrated variance (OU-1) is

$$\mathrm{Var}(I_t) = \frac{\eta^2}{2\kappa^3}(2\kappa t - 3 + 4e^{-\kappa t} - e^{-2\kappa t}).$$

For the stationary-start integrated-OU benchmark, take X_0 from the stationary Gaussian law. Then

$$X_0 \sim \mathcal{N}\left(0, \frac{\eta^2}{2\kappa}\right) \Rightarrow \mathrm{Var}(I_t) = \frac{\eta^2}{\kappa^2}[t - \frac{1 - e^{-\kappa t}}{\kappa}].$$

This comparator can create apparent stabilization while its integrated variance grows rather than terminally contracts, so it must remain distinct from the zero-start integrated-OU benchmark.

Integrated-OU audit. The zero-start branch uses (OU-1) above; the stationary-start covariance and variance are separate objects. Symbolic limits as $t \to 0$ and for large t, and Monte Carlo recovery under the actual sampling design, must agree with the analytic moments.

8.3 Bridge, observation-floor, and constrained-control rivals

Endpoint-conditioned bridges and Doob transforms. A bridge or Doob-transformed process can create terminal pinning by conditioning on an endpoint. Random-endpoint bridges can mimic terminal phenomena when endpoint times depend on latent state, care processes, frailty, selection, or observation shutdown. The comparator must include fixed-endpoint and random-endpoint bridge families.

Measurement-floor and censoring-floor rivals. Terminal concentration can arise when instruments lose resolution, values are clipped, reporting channels shut down, devices fail, or censoring preferentially removes high-variance states. A floor artifact is not reserve-coupled contraction.

Predictive-processing, free-energy, and constrained-control rivals. Viability, predictive control, active inference, safe reinforcement learning, and constrained control can produce state-dependent narrowing, avoidance of high-cost states, and terminal-like stabilization without pathwise neutral closure. The PTVC route must add value beyond these accounts (Sutton & Barto, 2018).

8.4 Endpoint selection and terminal-sampling rivals

Global endpoint-process convention. For cause e, $\alpha_i^e(t \mid \mathcal{H}_{i,t-})$ denotes the cause-specific hazard, $\lambda_i^e(t) := R_i^{\mathrm{risk}}(t)\alpha_i^e(t \mid \mathcal{H}_{i,t-})$ the counting-process intensity, and $d\Lambda_i^e(t) = \lambda_i^e(t)\,dt$ on the absolutely continuous branch. Singular or atomic compensator branches are named separately.

Protocol-endpoint selection branch. Begin at protocol risk-set entry $S_{i,\eta}^{\mathrm{entry}}$ and let $N_{i,\eta}^{e,\mathrm{adj}}$ be the cause-e adjudication counting process with predictable compensator $\Lambda_{i,\eta}^{e,\mathrm{adj}}$ and, on its absolutely continuous branch, intensity $\lambda_{i,\eta}^{e,\mathrm{adj}}$. Every numerator, denominator, risk set, and conditioning event in this subsection uses $T_{i,\eta}^{\mathrm{adj}}$ and that protocol process. A separate truth-endpoint analysis uses $T_i^\star$, S_i^θ, and the truth-level endpoint law. Transfer between branches requires an explicit endpoint-error kernel. If a conditioning event has probability zero or a denominator is nonfinite, the conditional distribution is undefined and the result is No adequate estimate.

Define the neutral-centered measured proxy and its balance distance once:

$$\tilde{L}_{i,\eta}^{\mathrm{proxy}}(t-) = L_{i,\eta}^{\mathrm{proxy}}(t-) - z_{i,\eta}^{\mathrm{neutral}}, \quad d_{i,\eta}^{\mathrm{bal}}(t-) = \| \tilde{L}_{i,\eta}^{\mathrm{proxy}}(t-) \|.$$

Compare three cause-specific branches: omission of the proxy; a preregistered constrained spline in $d_{i,\eta}^{\mathrm{bal}}$; and an unrestricted sensitivity spline in the signed proxy $\tilde{L}_{i,\eta}^{\mathrm{proxy}}$. Units, origin, uncertainty, knots, penalties, and monotonicity are frozen without terminal outcomes.

With competing causes, endpoint atoms, singular compensators, or non-stopping ontic endpoints, use the corresponding marked counting-process, compensator, or random-time formulation rather than this density special case.

Ordinary endpoint selection can generate near-neutral concentration. A finite absolutely continuous hazard cannot create a new exact atom at zero when the hazard-weighted occupation law is atomless there. Exact atomic closure requires pre-existing atomic occupation, absorption or stickiness, first-hitting support, a predictable endpoint atom, a singular compensator, or a corresponding limiting construction. This rival must be explicitly modeled in terminal cohorts.

Quasi-stationary survival-conditioning rival. On a declared absorbed-Markov branch with absorption time τ_∂, test whether

$$\mathcal{L}(X_t \mid \tau_\partial > t) \overset{\mathrm{TV}}{\to} \alpha$$

for a quasi-stationary distribution α, together with the associated Q-process when its existence conditions hold. Fit this rival without terminal-residual labels and score its survival-conditioned state law, absorption-time tail, perturbation recovery, endpoint observation kernel, and terminal-band predictions. Near-neutral concentration under α is a survival-selection explanation; it does not create an exact zero atom unless α itself assigns positive mass to the zero-ledger set or an additional absorbing, sticky, or singular mechanism supplies one (Champagnat & Villemonais, 2016).

Each endpoint-selection rival follows the common kernel policy in Section 8.1 and is scored jointly with endpoint, observation, and terminal-band components.

Endpoint-selection typing. Every confirmatory endpoint covariate is predictable on the protocol risk set; future-derived prognosis appears only in a labeled retrospective sensitivity analysis. Proxy centering uses independently calibrated nonterminal rules.

8.5 Recovery, mechanism localization, and action invariance

Model recovery and ambiguity. Recovery studies must include the PTVC route, bridges, OU and homeostatic baselines, non-neutral set points, flexible baselines, selection-conditioned data, observation-floor mechanisms, constrained-control baselines, and open-set alternatives. Verdicts follow the lexicon in Section 9.1.

Mechanism-localization rule. Every proposed route or experiment must first pass the representation gate and then identify the earliest local object whose conditional law differs from its nearest adequate rival: assignment; information; policy; structural action availability; physical transition; outcome or randomizer; precommitment eligibility; postcommitment retention or analysis inclusion; endpoint or stream routing; or observation. An anomaly at one locus may not be narrated as evidence for another. Downstream consequences of one altered conditional law form one evidence family and may not be counted as independent confirmations.

Mechanism-equivalence ceiling. If two candidate mechanisms induce the same distribution of protocol-observable variables under every admissible intervention in the study, that study does not identify their locus or ontology. The report must use mechanism-equivalent, Model-ambiguous, or No adequate estimate. Breaking equivalence requires a new intervention or measurement that produces divergent predictions.

Conditional calibration and action invariance are governed by B.3.1–B.3.3. The exact conditional-uniformity result is stated once in B.3.1; any observed excess first challenges the declared information, calibration, intervention, eligibility, retention, or observation model rather than identifying PTVC.

Modal action-mass discipline. Existence of a completing continuation, positive-probability completion, fixed-policy probability-one completion, almost-sure attainability, limit-sure attainability, topological-support containment, and closure of every admissible continuation are distinct. On a registered action-first route, probability-one completion under the actual policy requires probability-one policy mass on actions whose complete frozen continuations close with probability one. A topological-support claim additionally requires a declared topology and direct support or closedness premises; an all-continuation claim quantifies over the declared admissible class. Existential or positive-probability reachability cannot be promoted to any stronger modality.

Experimental-trajectory rule. The study procedure is part of the trajectory. Reward, stress, frustration, uncertainty, relief, effort, fatigue, social evaluation, later opportunity changes, and measurement burden induced by the protocol must enter the measured estimand or intercurrent-event strategy. Treatment assignment, task label, reward history, task success, or task absorption is not a ledger sign, ledger magnitude, or true conscious-stream endpoint without an independently validated bridge.

Protocol-field additions for mechanism tests. When relevant, a confirmatory protocol must include the primary causal locus and altered kernel, commitment information set and private policy randomization, quantitative cue-leakage budget, modal action-mass semantics, precommitment eligibility, postcommitment retention or analysis inclusion, and observation denominator flow, selected-sample and observation-garbling negative controls, and common-pipeline blinded generator recovery.

8.6 Balance-law rivals

The nearest weaker and orthogonal balance-law comparators are typed on the same declared representation branch b and law $Q^{(b)}$. They require the same origin, sign, scalarization, exposure, bearer, routing, endpoint, and observation discipline as PTVC.

Common-ratio rival:

$$H_\rho^{(b)}: \qquad V_i^{+,(b)} = \rho\, V_i^{-,(b)}, \qquad \rho > 0.$$

Common-residual rival:

For a fixed $\delta_{\mathrm{res}} \in \mathbb{R}$ in the same calibrated residual unit and neutral origin, $H_{\delta_{\mathrm{res}}}^{(b)}$ requires $C_i^{(b)}[\Gamma] = \delta_{\mathrm{res}}$ throughout the declared class. A fitted or uncertain δ_{res} defines a separate hierarchical or interval rival.

$$H_{\delta_{\mathrm{res}}}^{(b)}: \quad C_i^{(b)}[\Gamma] = \delta_{\mathrm{res}}, \qquad \delta_{\mathrm{res}} \text{ on the declared calibrated scale.}$$

Expected-neutrality rival:

$$\mathbb{E}_{Q^{(b)}}[C_i^{(b)}[\Gamma] \mid \mathcal{G}_{i,0}^{\mathrm{ex}}] = 0.$$

The ex-ante information field excludes realized terminal values, future endpoint classifications, later retention, later observation availability, and outcome-trained bearer, origin, sign, scalarization, routing, or representation choices. If $C_i^{(b)}[\Gamma] \in L^2(Q^{(b)})$, then $C_i^{(b)}[\Gamma] = 0$ $Q^{(b)}$-a.s. is equivalent to $\mathbb{E}_{Q^{(b)}}[C_i^{(b)}[\Gamma] \mid \mathcal{G}_{i,0}^{\mathrm{ex}}] = 0$ and $\mathrm{Var}_{Q^{(b)}}\left(C_i^{(b)}[\Gamma] \mid \mathcal{G}_{i,0}^{\mathrm{ex}}\right) = 0$, $Q^{(b)}$-a.s.; expected neutrality alone is weaker.

Common-gross rival:

$$H_g^{(b)}: \quad V_i^{\mathrm{gross},(b)} = g, \qquad g \in [0, \infty).$$

These rivals are compared under the same evidence and claim ceilings. Origin misspecification affects the ratio and residual families as well as PTVC.

Branch-matching rule. Proportional, common-residual, expected-neutrality, population-transfer, and other balance rivals use the same origin, unit, scalarization, exposure, bearer, endpoint, observation kernel, and data budget as PTVC; extra assumptions are visible. Undefined denominators return No adequate estimate rather than a favorable value.

8.7 Computational-mood and ledger-free control rivals

Computational mood models make the comparator set substantially stronger. Momentary happiness can depend on recent reward expectations and prediction errors; mood can summarize momentum in outcomes; early experiences can receive disproportionate weight; and affect can alter the choice of subsequent activities (Rutledge et al., 2014; Eldar et al., 2016; Keren et al., 2021; Taquet et al., 2016). These mechanisms create history dependence, apparent compensation, residual-sensitive choice, and multiscale carryover without a complete-stream terminal law.

Homeostatic reinforcement learning likewise couples internal physiological state, reward, and action so that learned policies defend viable regions (Keramati & Gutkin, 2014). Interoceptive prediction and active inference supply additional ordinary routes by which bodily error, anticipated demand, and action can jointly organize affect (Barrett & Simmons, 2015; Seth & Friston, 2016). A PTVC analysis that omits these comparators risks mistaking adaptive control for completion control.

The discriminating requirement is structural. Latent-dynamics candidates and rivals share the frozen observation architecture specified in §7.2. Only predeclared observation-artifact rivals may vary the named kernel component under the same latent information, tuning budget, and score. PTVC earns differential route support only when the same completion potential or residual geometry transports across transition, action, endpoint, and resolution modules beyond equal-budget rivals. Persistent long-run effects remain mandatory comparators rather than exceptions.

All variants in this subsection inherit the frozen observation-kernel and equal-treatment rule in Section 8.1; no model receives an outcome-selected channel or preprocessing pipeline.

9. Confirmatory decision structure and model recovery

9.1 Modular confirmatory architecture

Orthogonal public result vector. The sole public decision object has four axes: TargetStatus, RouteStatus, ComparatorStatus, and InferenceStatus. Each axis has a closed vocabulary and precedence rule. Target-adverse is recorded even when no route is proposed; route failure alone does not refute PTVC; route favoring cannot rescue target adversity; comparator inadequacy blocks a route-favored claim without erasing a deterministic target counterexample; and InferenceStatus preserves whether the evidence object is identified-set, frequentist, Bayesian, or sensitivity-based.

Verdict lexicon. The following labels govern every module and are not redefined elsewhere:

Table 2C. Controlled Verdict Lexicon

Prerequisite / estimate	Band region	Recovery / adequacy	Comparative / open-set	Primary result	Interpretation
Valid; estimate available	Entire simultaneous region inside band	Recovered; adequate	Any adequate ranking	Equivalent	Resolution-indexed positive result; exact-atom question remains nested
Valid; estimate available	Entire region outside band	Recovered; adequate	Direction registered	Different / off-band	Adverse to the tested band branch
Valid; estimate available	Region overlaps boundary	Recovered; adequate	—	Informative but inconclusive	Report interval/set and next resolution
Failed prerequisite or unavailable estimate	Not evaluated	Not evaluable	—	No adequate estimate / invalid gate	No supportive vote; preserve the failure route
Valid local route	Registered local target passes	Recovered; absolute fit adequate	PTVC route wins equal-budget comparison	Route-favored	Local mechanism only; cannot override representation, endpoint, observation, or bearer gates
Valid; estimate available	Entire simultaneous target region outside the registered band	Recovered; adequate	Any	Target-adverse	Adverse to the tested finite-band target; route status is reported separately.
Valid local route	Target not independently adverse	At least one adequate rival	Rival superior	Rival-favored / Route-failed	Retire the registered mechanism route; do not label the target adverse unless its own compatibility object is adverse.
Valid estimates	Any	Two or more adequate models	Indistinguishable	Model-ambiguous	Report branch-specific results
Valid estimates	Any	No declared model adequate	Open-set check fails	Open-set inadequate	Expand the model bank without relabeling the prior outcome

The controlling confirmatory object is not a single p-value or terminal coefficient. It is a prospectively specified decision structure combining target status, measurement status, route status, rival status, and recoverability.

Noncompensatory evidence-dependency rule. Confirmatory eligibility is conjunctive. Required components include, where applicable, phenomenal representation; neutral origin and sign; cross-sign calibration; bearer and routing; endpoint; measurement; measured-to-latent bridge; candidate realization;

route signatures; rival comparison; recoverability; replication; and transport. Every downstream result inherits the weakest required predecessor, so a failed component restricts the verdict to the corresponding abstention or narrower branch. Replacing that component creates a new versioned branch, with prior evidence and failures retained under the version in which they arose.

Universality-aware reporting. Every prespecified site, subgroup, endpoint class, care context, and instrument version receives its own measurement, endpoint, routing, compatibility, and transport verdict. A pooled or hierarchical mean cannot establish strict per-stream universality. A gate-valid incompatible stream or transported stratum blocks universal-support language for the stated class; a gate-invalid stratum receives No adequate estimate, with 'unresolved prerequisite' recorded as the reason. Reports include site-specific results, prediction intervals, invariance tests, and leave-site-out or leave-stratum-out sensitivity analyses.

Reference-class inclusion lattice. Every claim version locates its reference class in a prospectively declared inclusion lattice. Evidence inherits upward or downward only when the target, representation, bearer, endpoint, observation, bridge, and sampling laws justify that transport. A result in a narrower class cannot be advertised for a broader class by default; a gate-valid counterexample in a subclass remains visible in every containing strict-universal claim.

A confirmatory report is modular.

Core PTVC module - required for every confirmatory claim. Report the reference class; stream origin and routing; endpoint branch, adjudication, and filtration; modal scope; value representation; neutral-origin level; sign and scalarity; phenomenal-exposure rule; measured estimand and estimator; observation kernel; measured-to-latent bridge; missingness, delayed entry, retention, and intercurrent-event strategy; rival family; common scoring rules; power and discrimination analysis; model recovery; false-support calibration; ambiguity and No adequate estimate rules; and separate verdicts for target, bridge, route, and controlled summary.

Route-specific module - required for every claimed implementation route. Report the independently identified route state or structural object, intervention semantics, transported cross-equation predictions, nearest adequate rivals, failure conditions, and generator-recovery requirements.

Reserve-route module - required only when the reserve-coupled route is claimed. Report the reserve proxy and validation level, reserve floor, latent and measured reserve-exhaustion objects, reserve-endpoint coincidence tolerance, reserve-squared and boundary-exponent assumptions, reserve-normalized diagnostics, denominator-error analysis, conditional signal-to-noise detectability, and reserve-specific perturbation and recovery tests.

A nonreserve route substitutes its own preregistered state variables, bridge, cross-equation predictions, intervention semantics, and recovery tests. The core PTVC module must not silently make the reserve route mandatory.

9.2 Model recovery and evidence-theory maturity

Generator recovery is specified but has not been executed for the present theoretical analysis. Future confirmatory or Route-favored language requires code, preprocessing, model versions, random seeds, synthetic generators, scoring rules, calibration data, exclusion routes, and analysis containers to be frozen before analysis or explicitly labeled exploratory.

Benchmark governance. Separate development, public-validation, and sequestered-promotion suites. An independently authored sealed challenge bank is controlled by a clean-room team; a material change to the hypothesis, score, features, kernel, gate, or rival bank after disclosure requires a fresh suite or an adaptive-valid guarantee. Query counts, exposed summaries, model versions, failures, and retirement decisions are archived. A finite registered bank supports only a bank-conditional guarantee unless a covering, complexity, or uniform-convergence theorem justifies a class-wide statement.

Recovery selector. Let s(D) be the registered score vector and g(D) the admissibility-gate vector. Let $\mathcal{G}$ denote the true synthetic generator, $\widehat{M}$ the fitted or selected model, and $\widehat{V}$ the controlled verdict returned by the frozen pipeline. The model-recovery matrix records $\mathcal{G}$ versus $\widehat{M}$; a separate verdict table records $\widehat{V}$, including Model-ambiguous, Open-set inadequate, No adequate estimate, and invalid/failed run.

A model-recovery matrix must show whether each synthetic generator is recovered under the frozen pipeline:

$$R_{ab} := \mathbb{P}(\widehat{M} = M_b \mid \mathcal{G} = G_a).$$

Design-specific operating characteristics must be reported under the planned cadence, dependence, attrition, endpoint uncertainty, observation kernel, open-set generators, and invalid-run rules. Required outputs include multiclass confusion and abstention matrices, noncompensatory per-class promotion thresholds, calibration, false-support rates, power against nearest rivals, and recovery failure by generator. Report pooled, per-stratum, worst-stratum, and simultaneous all-strata performance; time-indexed decisions require simultaneous or time-uniform coverage. Until executed, these are protocol requirements rather than results.

A pipeline that fails blinded generator recovery, or an equally stringent analytic or formally verified validation appropriate to the declared model class, cannot produce a strong confirmatory verdict on real terminal data.

Generator recovery and verdict calibration are related but distinct performance targets. The selector domain is the registered score and gate vectors; ties within practical equivalence return Model-ambiguous, and failure of all candidates yields Open-set inadequate or No adequate estimate according to the preregistered cause.

Two ladders must be reported separately. Empirical evidence: E0 - logical compatibility only; E1 - validated measurement representation; E2 - one route-specific nonterminal signature defeating serious rivals under the same observation model; E3 - a transported cross-equation mechanism package; E4 - coherent routing and endpoint prediction with held-out recovery. Structural theory maturity: T0 - compatibility with PTVC; T1 - an exactness-compatible mechanism with independently fixed carrier, bridge, and failure conditions; T2 - an independently successful physical-psychophysical theory entailing PTVC in a declared reference class. A speculative exactness construction is not automatically stronger empirical evidence than a replicated E4 result.

End-to-end recovery matrix. Publish truth generator/family, selected generator/family, verdict, parameter/band recovery, route/falsifier status, and NAE/open-set state. Run the frozen pipeline on ordinary, challenging, near-boundary, quantized, missing, endpoint-selected, recentered, flexible-state, and deliberately deceptive generators. Calibrate family-wise false Route-favored support and power across all gates. Point-null Bayes factors additionally require predictive calibration, prior sensitivity, and generator recovery.

9.3 Ambiguity, discrimination, and false-support control

Error-family control. Freeze three noninterchangeable events: favorable target classification under a target-false generator; Route-favored classification under a route-false generator; and confirmed-counterexample classification under a target-compatible generator. The display denotes these events by E with subscripts T, R, and C. For each exact finite registered bank, calibrate the corresponding worst-generator probability with simultaneous one-sided uncertainty:

$$\begin{aligned} p_{\mathrm{FS},T}^{\mathrm{bank}} &:= \max_{1\le j\le M_T} P_{G_j}(E_T), \\ p_{\mathrm{FS},R}^{\mathrm{bank}} &:= \max_{1\le j\le M_R} P_{G_j}(E_R), \\ p_{\mathrm{FC}}^{\mathrm{bank}} &:= \max_{1\le j\le M_C} P_{G_j}(E_C). \end{aligned}$$

Each upper confidence bound must meet its own preregistered tolerance; no error family substitutes for another. Report generator counts, dependence, interval construction, multiplicity allocation, seeds, stopping, and invalid runs. Replace max by a class-wide supremum only when a declared covering, continuity/robustness theorem, or verified robustness certificate transfers the finite evaluation to that class; otherwise the guarantee is bank-conditional and out-of-bank generators are stress tests.

Model-selection ambiguity rule. If two or more adequate candidate or rival models remain practically indistinguishable under common scoring, calibration, recovery, controls, sensitivity analysis, and held-out evaluation, the verdict is Model-ambiguous rather than Route-favored. A model that wins among inadequate models has not won the scientific comparison.

Decision-engine metamorphic tests. The implementation must verify that enlarging uncertainty cannot strengthen a conclusion; garbling or dropping an observation channel cannot shrink a latent truth-containing set without added assumptions; adding a serious rival cannot improve route support; weakening a prerequisite cannot promote a verdict; splitting a stratum cannot erase its worst-stratum failure; duplicating a stream token cannot change the all-stream event; and an empty set cannot pass equivalence.

Discrimination-power rule. A confirmatory claim reports the design's ability to separate the declared route from its nearest adequate rivals under the common observation model. If the primary predictions are practically indistinguishable, the preregistered verdict is Model-ambiguous, No adequate estimate, or Reference-class narrowed.

Joint verdict-stability surface. Evaluate the frozen multistate decision jointly across origin error, endpoint misclassification and revision, response shift, missingness, route ambiguity, and rival flexibility. Report interaction regions and worst-case verdict changes; one-factor-at-a-time sweeps cannot establish robustness when these mechanisms interact.

Joint-signature calibration. Correlated diagnostics are inputs to the one frozen multivariate decision map above, not separate evidential votes.

For $H_0: \sigma_Z^2 = 0$, use theorem-matched boundary calibration or a justified bootstrap. Failure to reject H_0 is not affirmative evidence for an exact atom. Report the one-sided upper confidence bound for σ_Z^2, achieved family-wise false support, and power against preregistered positive-variance alternatives.

Local-Gaussian planning branch. On an identified, homoskedastic, conditionally independent branch with known measurement-error variance, use the collision-free variance ratio ρ_{var}.

$$H_0: O_j \sim \mathcal{N}(0, \sigma_\varepsilon^2), \qquad H_1: O_j \sim \mathcal{N}(0, \sigma_\varepsilon^2 + \sigma_Z^2).$$

$$D_{\mathrm{KL}}(P_1 \parallel P_0) = \frac{1}{2}[\rho_{\mathrm{var}} - \log(1+\rho_{\mathrm{var}})] = \frac{\rho_{\mathrm{var}}^2}{4} + O(\rho_{\mathrm{var}}^3).$$

The reverse direction has the same leading fourth-order signal-to-noise rate:

$$D_{\mathrm{KL}}(P_0 \parallel P_1) = \frac{1}{2}[\log(1+\rho_{\mathrm{var}}) - \frac{\rho_{\mathrm{var}}}{1+\rho_{\mathrm{var}}}] = \frac{\rho_{\mathrm{var}}^2}{4} + O(\rho_{\mathrm{var}}^3).$$

$$n \approx \frac{4\Lambda}{\rho_{\mathrm{var}}^2} = \frac{4\Lambda\sigma_\varepsilon^4}{\sigma_Z^4}, \qquad \rho_{\mathrm{var}} := \frac{\sigma_Z^2}{\sigma_\varepsilon^2}.$$

for target expected log-likelihood evidence Λ at leading order. This is a planning approximation, not an evidence law. Unknown or heteroskedastic error, dependence, mixtures, atoms, censoring, kernel uncertainty, or nuisance estimation requires exact branch-specific divergence or simulation.

Boundary-null calibration. Atom weights, zero variances, mixture components, and nuisance parameters unidentified under the null are nonregular problems. Each implementation records its null geometry and nuisance-identification status and uses a theorem-matched cone/mixture law, exact or one-sided procedure, validated parametric bootstrap, profile calibration, subsampling, or another demonstrated method. Default symmetric Wald intervals, ordinary χ^2 calibration, and a generic bootstrap without branch-specific validation are not used on these branches. Failure to reject is not affirmative atom evidence; report one-sided bounds, finite-bank size, power, coverage, and Type-S/Type-M behavior (Crainiceanu & Ruppert, 2004; Self & Liang, 1987).

Bayes-factor prior-sensitivity control. When a point-null analysis is used, define

$$\mathrm{BF}_{01}(y) := \frac{p(y \mid \sigma_Z^2 = 0)}{\int_0^\infty p\,(y \mid \sigma_Z^2 = u)\, g_\psi(u)\,\mathrm{d}u}.$$

Use a proper alternative prior; preregister its family, scale, prior odds, nuisance priors, kernel assumptions, and parameterization. Report near-zero and scale sensitivity and predictive calibration under the point null, local alternatives, and rival generators. Improper alternative priors are prohibited in marginal-likelihood Bayes factors. A Bayes factor cannot identify a latent atom, establish universality, or repair an unidentified observation kernel (Lindley, 1957; Pauler et al., 1999; Sinharay & Stern, 2002).

9.4 Protocol register and clinical-ethical gates

Protocol field register. Every confirmatory protocol must contain the core PTVC module fields: reference class, truth-domain stream origin, stream routing, endpoint branch and adjudication filtration, modal scope, value representation, measured estimand, estimator version, observation filtration, missingness and retention model, intercurrent-event strategy, delayed-entry mechanism, treatment and care context, scalarization rule, neutral-origin bridge, equivalence band, rival matrix, negative and positive controls, generator-recovery matrix, scoring rules, result-status label, and downgrade route. Each claimed implementation adds its own route-specific fields. The reserve proxy, reserve floor, reserve-exhaustion estimand, reserve-normalized diagnostics, and reserve-endpoint coincidence are mandatory only for reserve-route studies. Appendix B mechanism tests additionally declare causal locus, altered kernel, commitment information, cue-leakage budget, intervention-indexed action mass, assignment-level eligibility and postcommitment retention flow, selected-sample controls, observation-garbling controls, and blinded generator recovery.

Canonical protocol declaration. Protocol PTVC-P1 freezes a noncompensatory sequence: (1) calibrate sign, origin, cross-sign units, and the representation branch; (2) validate bearer/routing, endpoint, route-state, observation, retention, and measured-to-latent objects; (3) pass blinded generator recovery, open-set abstention, and mandatory controls; (4) run safe nonterminal mechanism-separation and perturbation studies; and (5) consider terminal observational work only after the preceding gates pass. A failed gate narrows, redesigns, or pauses the corresponding branch; later success cannot erase it.

Claim-and-decision registry. Each prespecified claim records a Claim ID, version, reference class, target object, prerequisites, predicted observable, nearest adequate rivals, quantitative gate, failure or downgrade condition, data split, and final controlled verdict. A substantive change creates a new claim version rather than overwriting the original result.

Prospective independent review. Before outcome access, the flagship confirmatory protocol, simulators, and interpretation rules undergo independent review by methodologists charged with representing PTVC and its strongest registered rivals. Alternative predictions, disagreements, and retirement triggers are archived in advance; this governance strengthens error detection and the credibility of every later evidence claim.

Frozen rival roster. Before outcome access, the confirmatory comparison registers: ordinary adaptation/opponent regulation (event- and horizon-specific incomplete compensation); flexible nonneutral state-space baselines (transportable nonzero terminal location); fixed- and random-endpoint bridges (pinning explained by conditioning); endpoint selection and censoring (concentration tracks risk and observation); computational mood/memory kernels (history-weighted reports without a lifetime ledger); constrained or homeostatic control (viability without terminal-area closure); observation-floor models (concentration follows lost resolution); and open-set generators. Each receives its strongest equal-budget nuisance structure and one held-out discriminating prediction. The roster is co-signed through a preregistered collaboration designed for balanced scrutiny, and any later addition creates a new version.

Controlling safety, non-coercion, and clinical-governance gate. No protocol may induce or accelerate a terminal state, withhold or delay indicated care, analgesia, rescue, nutrition, ventilation, communication support, or comfort, manipulate endpoint classification for fit, or pressure participants, families, clinicians, or institutions to remain observable. Safety events and treatment changes override data collection. Human research requires prospective institutional review board/research ethics committee (IRB/REC) review, consent/assent and surrogate protections appropriate to capacity, privacy and burden controls, adverse-event governance, clinical independence, and transparent withdrawal rights. Human research follows the World Medical Association Declaration of Helsinki, 2024 revision (World Medical Association, 2024); an interventional clinical trial additionally follows the consolidated ICH E6(R3) Good Clinical Practice guideline, Step 4 version dated 16 June 2026 (ICH, 2026). Palliative and terminal care are never altered for measurement or model fit. Later sections add only route-specific safeguards and cross-reference this gate.

Clinical consciousness-adjudication rule. Where human disorders of consciousness, coma, minimally conscious states, anesthesia, sedation, delirium, or neurologic end-of-stream questions enter endpoint or unity-gate adjudication, the protocol must cite current clinical guidance and declare how clinical categories are translated into model categories. Clinical categories do not automatically equal stream identity, unity, or endpoint status. If that translation cannot be justified, the endpoint verdict must narrow or become No adequate estimate (Giacino et al., 2004, 2018; Greer et al., 2023; Schiff, 2015).

Behavioral nonresponse alone cannot establish absence of command-responsive cognition, consciousness, stream continuation, or phenomenal exposure. In a 2024 multicenter study, cognitive motor dissociation was detected in 60 of 241 behaviorally nonresponsive participants; such EEG/fMRI command-following still requires separate bridges to valence, magnitude, unity, and stream identity (Bodien et al., 2024).

Theory-indexed integration, complexity, or perturbational measures may inform sensitivity analyses, but they do not independently establish stream ownership, phenomenal valence, neutral origin, cardinal exposure, or the truth-level endpoint (Casali et al., 2013; Seth et al., 2011; Tononi, 2004).

Palliative-care and sedation ethics gate. When palliative sedation, analgesia, delirium management, withdrawal of monitoring, hospice transition, or end-of-life care transition enters a terminal study, the protocol must cite current palliative-care and medical-ethics guidance, including proportionality, refractoriness, autonomy, and informed decision-making (Surges et al., 2024). It must identify whether the study is observational or interventional and state how symptom relief, consent, capacity, privacy, and non-burdening measurement are protected.

No research inference may compete with comfort care. If the measurement plan would increase distress, delay care, alter treatment, pressure family consent, or burden the patient, the study must be redesigned or abandoned.

Human-participant research follows the fixed governance editions and protections stated in §9.4; specialist palliative-sedation guidance is version-checked again at protocol freeze.

A terminal, palliative, intensive-longitudinal, or otherwise high-burden protocol may not enroll under a confirmatory label until its frozen pipeline passes preregistered generator-recovery, discrimination-power, and false-support thresholds under the planned cadence, missingness, endpoint jitter, observation shutdown, treatment process, and measurement reactivity. Failure permits only simulation, low-burden feasibility work, redesign, or abandonment.

False-support certification follows the definitions, upper-bound construction, threshold, multiplicity rule, and reporting requirements in §9.3.

Governance and retirement registry. The controlling gate covers welfare, noninterference with care, burden, consent/capacity, proxy consent, privacy, stopping, abandonment, independent monitoring, humane endpoints, replacement/refinement/reduction, and applicable IRB/REC/legal review; each source is frozen by edition/date. The versioned retirement record contains branch, trigger, evidence required, effective date, permissible reformulation, and inheritance rule. All adverse and NAE outcomes remain in the registered record; a reformulated branch receives a new ID.

9.5 Controls and challenging null models

Proposed first-study protocol template. A primary design would combine a low-burden nonterminal cohort, independent neutral-origin and scale calibration, route-state measurement, a predictable observation-attempt model, endpoint-negative controls, blinded rival recovery, absolute and normalized equivalence gates, and a stopping rule that cannot use terminal outcomes to select the route; the complete protocol is designed for registration before confirmatory outcome access. Terminal-stage measurement remains observational and subordinate to care.

Let $\mathcal{J}^{\mathrm{PC}}_{j,\eta}$ be the nonempty simultaneous compatibility set for positive-control generator j, distinct from every negative-control set, and let $G^{\mathrm{PC}}_{j,\eta}$ apply its registered recovery criterion. Define a separate verdict-recovery matrix $Q_{av} := \mathbb{P}(\hat{V} = v \mid \mathcal{G} = G_a)$. The global recovery gate is the noncompensatory conjunction below.

$$G^{\mathrm{rec}}_{\eta} := \mathbf{1}\left\{\begin{matrix} \left(p^{\mathrm{bank}}_{\mathrm{FS},T}\right)^{U} \le \alpha_T,\ \left(p^{\mathrm{bank}}_{\mathrm{FS},R}\right)^{U} \le \alpha_R,\ \left(p^{\mathrm{bank}}_{\mathrm{FC}}\right)^{U} \le \alpha_C, \\ Q^{L}_{a,v(a)} \ge q_{a,\min} \text{ for every registered generator } a, \\ \text{open-set abstention, positive-control, and invalid-run gates all pass} \end{matrix}\right\}.$$

Here the three upper bounds and every $Q^L_{a,v(a)}$ lower bound use their separately preregistered tolerances. Multiplicity, dependence, invalid runs, empty sets, abstention, and the worst-generator rule are frozen before simulation. No pooled sensitivity/specificity summary may compensate for failure in any target, route, counterexample, generator, or open-set class; invalid or empty positive-control sets never pass.

Negative-control failure: if pseudo-endpoints, shifted endpoint windows, shuffled endpoint labels, negative-control channels, administratively convenient cutoffs, discharge events, device-removal times, or endpoint-normalized displays produce the same terminal concentration pattern as the declared endpoint, the route-specific interpretation fails or is downgraded to exploratory artifact detection (Lipsitch et al., 2010).

Candidate-counterexample audit. Any stream-level result capable of refuting a strict-universal branch undergoes independent review of bearer, routing, neutral origin, sign, scalarization, endpoint, observation model, missingness, and measured-to-latent bridge, with reviewers blinded to residual direction and magnitude whenever feasible. An unfavorable case is not removed from the reference class. If all prerequisites remain gate-valid and a truth-containing set excludes zero, the case remains a counterexample; if a prerequisite fails independently, report No adequate estimate and record the failed prerequisite as the reason.

Finite counterexample-prevalence summary. If zero gate-valid counterexamples are observed among n independent representative streams, the exact one-sided $(1-\alpha)$ Clopper–Pearson upper bound on sampled-population counterexample prevalence is

$$1-\alpha^{1/n}.$$

This bound licenses a finite registered class. Promotion toward strict universality requires a design-matched bound that addresses selection, dependence, and gate validity; clustered, dependent, and hierarchical samples therefore use their corresponding design-matched analysis.

Unresolved-case prevalence accounting. Let W_A be representative weight with gate-valid adverse results, W_U unresolved or gate-invalid weight, and W_E total eligible weight. Without favorable assumptions about unresolved cases, report the lower and upper violation-prevalence bounds below. Judgment-bearing gates, including qualification, bearer, origin, sign, routing, endpoint, observation, and bridge adjudication, receive blinded inter-rater reliability, disagreement preservation, and sensitivity analysis. Gate validation is a selection process and is reported jointly with target status.

$$\frac{W_A}{W_E} \le p_{\text{viol}} \le \frac{W_A+W_U}{W_E}, \qquad W_E > 0.$$

Inferential qualification. The displayed interval is a sharp deterministic bookkeeping bound only when favorable and adverse classifications are truth-containing for the same target, modality, frame, weighting rule, and gate system. Confidence-set versions require a simultaneous coverage statement; posterior and sensitivity versions retain their registered prior or sensitivity interpretation. Without a joint guarantee, report classification bookkeeping rather than a bound on the true violation fraction.

Route-favored language requires challenging null models. The simulator bank must include reserve-coupled candidates, fixed- and random-endpoint bridges, Doob-transformed processes, OU/homeostatic baselines, non-neutral set points, flexible baselines, stochastic-volatility baselines, observation-floor mechanisms, selection-conditioned terminal data, treatment-process rivals, and open-set alternatives.

Low-gross streams are included in the stress-test bank. Report how often they pass the absolute terminal band while failing the normalized imbalance-sensitivity gate.

Program-level retirement and reformulation rule. For a strict exact [SET-ALL] branch, one independently audited gate-valid stream is adverse when its nonempty truth-containing latent compatibility set excludes zero. For a registered finite-resolution branch, adversity requires the complete set to be disjoint from the scientific acceptance region. Simultaneous confidence and posterior conclusions retain their own error guarantee or prior, and [P1] retirement requires a positive-probability violation or theorem-matched rejection. Each mechanism route is retired under its separately registered transported-fingerprint rule; route failure alone does not refute PTVC.

The overall empirical program enters a registered sunset review after completion of at least one representation-valid study, one route-signature study, and one endpoint-aligned study. If no PTVC-specific result clears its prespecified recovery, false-support, transport, and band gates, the strict empirical program is retired pending a genuinely independent bridge, theorem, or observation technology. All adverse and No adequate estimate outcomes remain in the registered record. A reformulated representation or modal claim receives a new identifier and cannot inherit the evidential record of the retired branch without fresh model recovery.

Global control suite. Every promoted module preregisters a positive control, a direct mechanism-negative control, an observation/leakage control, and a challenging near-miss. Report control similarity, effective sample size, sensitivity, specificity, multiclass confusion, calibration, Type-S/Type-M behavior, and NAE rate. A failed positive control or underpowered negative control blocks promotion rather than counting as a reassuring null.

10. Empirical foundations, decisive rivals, and progressive tests

10.1 Evidence-synthesis method and role coding

This section presents empirical foundations, decisive rivals, and progressive tests. The cited literatures contribute prerequisites, methods, mechanism classes, boundary conditions, rivals, or direct tests; PTVC-specific support arises when distinctive predictions transport across equations, resolutions, cohorts, and interventions. Each subsection therefore follows contribution, discriminator, and next-test order.

Evidence-synthesis method. This manuscript uses a targeted narrative synthesis focused on sources with a direct role in representation, mechanism, endpoint, observation, rival construction, or adjudication. Bibliographic checking used PubMed, DOI or publisher records, official guideline repositories, cited reference chains, and named dataset documentation. Primary studies carry quantitative claims when available, while reviews organize mature fields. Convergence is evaluated by triangulation across methods with materially different bias structures rather than by citation count, and dependency edges among cohorts, instruments, preprocessing pipelines, research groups, and reused datasets are recorded so correlated analyses are not treated as independent evidence votes (Lawlor et al., 2016; Munafò & Davey Smith, 2018). Each quantitative claim identifies its source and evidential role at the controlling location. The review is explicitly scoped to these functions; a future systematic or meta-analytic extension would add a rerunnable search, screening record, and prospectively registered synthesis protocol.

Reference-integrity gate. Each public version records DOI or persistent-link resolution; bibliographic concordance; version-of-record status; correction, retraction, or expression-of-concern checks; claim-to-source fit; primary-versus-secondary role; cross-document concordance; duplicate detection; and currentness review where the field has materially changed. A citation supports only the claim actually established by its source. Reference-style convention. Consortium and international-guideline statements may use the publisher's official recommended short citation; all other records with more than twenty authors follow the manuscript's long-author rule.

Evidence posture. Established patterns, program contributions, and next discriminating tests are kept distinct so that every positive result carries its strongest defensible meaning. The assembled evidence makes PTVC sufficiently well specified, plausible, and scientifically consequential to merit a progressive program of rigorous testing. The reviewed fields already supply convergent foundations relevant to valence representation, compensatory dynamics, predictive control, terminal trajectories, consciousness assessment, cultural recurrence, and exact biological regulation; PTVC-specific support can grow as its distinctive terminal and cross-module signatures transport across held-out studies. This manuscript supplies the formal and protocol foundation for that next stage.

10.2 Valence representation and measurement

10.2.1 Valence is richer than reported happiness

Valence refers to the pleasant-unpleasant character of experience. It is related to, but distinct from, arousal, motivation, appraisal, desire, meaning, life satisfaction, pain intensity, mood, and remembered evaluation. A lifetime ledger requires momentary or interval-specific phenomenal exposure, not a retrospective global judgment alone.

Russell's circumplex model places valence and arousal as central dimensions of affective description (Russell, 1980), while work on bodily feeling and consciousness supplies complementary mechanism and measurement targets (Damasio, 1999). Cacioppo and Berntson's evaluative-space analysis and mixed-emotion studies show that positive and negative activation can be partly separable and can co-occur (Cacioppo & Berntson, 1994; Larsen et al., 2001). Thus, a single bipolar line may be adequate in some tasks and inadequate in others.

A flexible observation model uses two latent channels:

$$F_i^+(t) \geq 0, \qquad F_i^-(t) \geq 0.$$

With a calibrated scalarization, define

$$f_i(t) = a_i^+(t)F_i^+(t) - a_i^-(t)F_i^-(t).$$

The weights $a_i^{\pm}(t)$ cannot be chosen after terminal outcomes are known. They must arise from preregistered psychophysical calibration, revealed-preference constraints, or a partially identified set.

A longitudinal scalar ledger requires every term to be expressed in one fixed cardinal ledger unit. Time-varying weights are admissible only as independently validated transport maps into that common unit and must be estimated without terminal outcomes. If longitudinal unit transport or cross-sign comparability fails, retain the bipolar, vector-valued, partially ordered, or partially identified result.

10.2.2 Experienced, remembered, and decision utility

Kahneman et al. (1997) distinguished experienced utility from decision utility. Later work showed that retrospective evaluations of episodes can overweight peaks and endings and neglect duration (Fredrickson & Kahneman, 1993; Redelmeier & Kahneman, 1996). A participant's answer to "How was your life?" is therefore not a neutral substitute for the integral of momentary experience.

For PTVC, this is not merely measurement noise. Remembered utility is an observation rival with its own generative mechanism. If terminal reports appear unusually neutral, the pattern may reflect memory compression, narrative integration, peak-end weighting, limited recall, social desirability, or changing standards. Protocols should measure experienced and remembered channels separately and test whether their divergence changes near endpoints.

10.2.3 Experience sampling and day reconstruction

Ecological momentary assessment (EMA) collects repeated reports in natural environments and reduces reliance on long recall windows (Shiffman et al., 2008). The Day Reconstruction Method can support larger-scale reconstruction of affective episodes, though it remains more memory-dependent than real-time sampling (Stone et al., 2006). A practical ledger estimator is

$$\hat{L}_i(t) = \sum_{k:t_{ik} \leq t} \hat{f}_{ik}\ \Delta\hat{\mu}_{ik},$$

where $\hat{f}_{ik}$ is a calibrated signed state estimate and $\Delta\hat{\mu}_{ik}$ is estimated phenomenal exposure between samples.

This estimator needs more than frequent prompts. Sampling intensity may depend on distress, sleep, hospitalization, device access, caregiver burden, or the study's own adaptive algorithm. The schedule is part of the observation kernel and should be modeled as a stochastic process. Where observation times are continuous, use a conditional sampling kernel or intensity, not a point probability.

10.2.4 Exposure weighting

Clock time supplies a useful starting coordinate, while a validated exposure model can refine it into the phenomenal exposure measure. Sleep, anesthesia, delirium, seizures, fluctuating arousal, split attention, and altered time perception may change the amount or organization of experience per physical-time interval. A general physical-time representation is

$$d\mu_i^{\mathrm{phen}}(t) = w_i(t)\,dt, \qquad w_i(t) \geq 0,$$

where $w_i(t)$ is an exposure weight, not a moral importance weight. Current consciousness indices should not be treated as cardinal measures of phenomenal mass. They may support sensitivity analyses or exclusion bounds only after their ordering, temporal resolution, state transportability, and relation to conscious unity are validated.

Emotion also distorts subjective duration. A 2023 meta-analysis of 31 studies (95 effect sizes; 3,776 participants) found that valence, arousal, stimulus type, and timing paradigm moderated emotional temporal distortion, with negative and higher-arousal stimuli generally producing greater overestimation (Cui et al., 2023). That result does not dictate $w_i(t)$, but it demonstrates why naive clock-time integration and retrospective duration judgments should be compared rather than presumed equivalent.

10.2.5 Measurement invariance and response shift

Cross-time and cross-state comparisons require measurement invariance. In a factor model

$$Y_{it} = \tau_t + \Lambda_t \eta_{it} + \varepsilon_{it}.$$

Changes in Y_{it} cannot be attributed to latent valence η_{it} if intercepts τ_t, loadings Λ_t, or residual properties shift with age, illness, culture, medication, mode of communication, or proximity to death. Configural, metric, scalar, and, where appropriate, strict invariance should be evaluated rather than assumed (Meredith, 1993; Millsap, 2011).

Response shift is especially relevant in serious illness: people may recalibrate internal standards, reprioritize domains, or reconceptualize quality of life (Sprangers & Schwartz, 1999). Such change may be psychologically adaptive and scientifically meaningful. It also means that stable observed ratings can hide latent change and observed change can reflect scale change. Anchoring vignettes, item-response models, individualized anchors, longitudinal factor models, and multimethod measurements should be built into terminal studies.

Recent digital-measurement evidence illustrates both feasibility and origin risk. In a thirty-day ecological momentary assessment study of 3,761 adults, participants showed high tactile precision and substantial agreement about expected neutral points on unipolar and bipolar visual-analogue scales, but agreement was not universal; deviations were strongly associated with person-specific average ratings (Cloos et al., 2025). The result supports digital EMA as a potentially precise observation channel within its tested context while showing why a visual midpoint cannot be treated as a universally identified phenomenal zero. Prompt burden, habituation, reporting fatigue, response shift, and time-varying item functioning remain part of the observation model.

Prompt schedule, response format, and repeated-assessment exposure belong inside the observation process. Randomized EMA evidence shows that fixed versus random prompting can change measured affect variability, while Likert and visual-analogue formats can differ in means, floor behavior, skewness, and criterion correlations even when several time-series properties are broadly similar (Langford et al., 2026; Haslbeck et al., 2025). Stone et al. (2026) synthesize evidence that completion time, missingness, careless responding, and reactivity may change across intensive repeated assessment, creating a longitudinal-invariance and data-quality threat rather than a universal causal effect. These findings directly strengthen PTVC study design by identifying observation features that must be frozen, calibrated, or transported before terminal patterns are interpreted. Protocols must freeze or calibrate prompt cadence, response format, anchors, initial cursor position, interface defaults, burden, and time-on-study. If more than one format is used, transport must be estimated and held out by format, person or declared subgroup, and state range. A common numerical range does not establish a common latent unit, neutral origin, floor behavior, or cross-sign cardinality. Held-out transport is one necessary test of the proposed mapping; it is not sufficient without origin, sign, scalarity, and measurement-invariance evidence.

10.2.6 Multimethod bridge

Current physiological channels provide informative but indirect and channel-specific access to phenomenal valence. Heart-rate variability, electrodermal activity, endocrine measures, facial action, vocal features, movement, sleep, neural signals, and clinical observations are channel-specific indicators affected by arousal, motor capacity, medication, disease, and context. The correct model is multimethod:

$$Y_{i,c}(t) \sim K_c\left(\cdot \mid V_{i,c}^{\mathrm{lat}}(t), X_i(t), \theta_c\right),$$

where channel c has its own latent target and observation kernel. Only a validated bridge combines channels into a ledger. Campbell and Fiske's multitrait-multimethod logic remains relevant: convergence is persuasive when channels share the intended target but not their dominant method variance (Campbell & Fiske, 1959).

10.2.7 Measurement-science contribution

Measurement science makes PTVC progressively accessible. Nonterminal calibration studies can identify sign, scale, mixed valence, duration weighting, response shift, and channel reactivity before terminal cohorts carry the full inferential burden. This staged design reduces inferential burden: it builds trustworthy bridges with low-burden studies and reserves the most demanding protocols for branches that have earned promotion.

Scalarization validation rule. Scalarization weights, sign, origin, and exposure are learned from nonterminal calibration or transported validation data and frozen before terminal analysis; failure retains bipolar, vector, ordered, or No adequate estimate branches.

10.3 Hedonic adaptation: empirical foundation and discriminating tests

10.3.1 What adaptation contributes

Hedonic adaptation is the attenuation of affective response or movement toward a prior level after changed circumstances, extending the classic hedonic-relativism and treadmill research program (Brickman & Campbell, 1971). The classic lottery-winner and accident-victim study helped popularize the idea that dramatic events need not permanently determine well-being (Brickman et al., 1978). Its small samples, cross-sectional elements, and historical measurement limits mean it should be treated as an important origin point, not a decisive demonstration of universal return.

The modern literature is more informative and more heterogeneous. Diener et al. (2006) argued that adaptation theory required major revision: set points need not be hedonically neutral, differ across people and components, and can change. Lucas (2007) reviewed large panel evidence showing that adaptation is not inevitable. Divorce, bereavement, unemployment, and disability can be associated with lasting changes. Luhmann and colleagues' meta-analysis integrated 188 publications, 313 samples, and 65,911 participants across family and work events; reactions and adaptation differed by event and by affective versus cognitive well-being (Luhmann et al., 2012).

Affective-forecasting error and causal 'explaining away' provide a further rival: apparent recovery can arise because attention, interpretation, or the explanatory status of an event changes, without a compensating signed-area mechanism. This rival predicts appraisal and report dynamics that need not satisfy transported kernel-area or complete-residual constraints (Wilson & Gilbert, 2008).

This literature establishes that valence trajectories can contain compensatory, history-dependent structure and supplies cohorts and perturbations for estimating it. The decisive promotion test for a registered completion-related compensation route is whether estimated signed response areas exhibit held-out transported closure rather than merely attenuating toward baseline. A favorable result supports that route under the tested bridge; complete-stream PTVC adjudication retains the separately defined gates.

10.3.2 Why heterogeneity is informative

A theory that merely says "people bounce back" can absorb many patterns after the fact. PTVC must be more specific. Suppose an event at time s produces an affective impulse response $k(t-s)$ with amplitude A_s. A shock-and-repair representation is

$$f_i(t) = b_i(t) + \sum_{j:t_{ij}\le t} A_{ij}\, k_i\big(t - t_{ij};\theta_i\big).$$

Ordinary adaptation requires $k_i(u) \to 0$ as $u \to \infty$. Integral neutrality of the event response requires the stronger condition

$$k_i \in L^1([0,\infty)), \qquad \int_0^\infty k_i\,(u;\theta_i)\,\mathrm{d}u = 0.$$

These conditions are not equivalent. A monotone exponential decay $k(u) = e^{-\lambda u}$ adapts but has positive integral $1/\lambda$. A biphasic kernel can adapt and have zero integral. A permanently shifted baseline neither fully adapts nor closes locally. Estimating kernel areas rather than only return times is therefore a direct way to convert the adaptation literature into a PTVC-relevant test.

The full-area restriction is defined only for $k_i \in L^1([0,\infty))$, or under a separately declared improper-integral convergence rule. Nonintegrable long-memory kernels use a finite-horizon or registered regularized area with its truncation and limit reported; they are not assigned a nonexistent infinite-horizon zero area.

10.3.3 Event-specific predictions

The meta-analysis by Luhmann et al. (2012) indicates that marriage, divorce, bereavement, childbirth, unemployment, reemployment, retirement, and relocation should not be pooled as interchangeable "life events." A rigorous program estimates event-specific kernels and asks:

- Does the net signed kernel area differ from zero?
- Does the compensatory lobe scale with the initial affective displacement?
- Does the kernel depend on controllability, predictability, chronicity, social support, or meaning?

- Are cognitive life satisfaction and momentary affect governed by different kernels?
- Does repeated exposure change the kernel, as opponent-process theory predicts?
- Do residual event areas predict later compensatory dynamics or terminal behavior?

A primary prospective design combines dense EMA around events with long-run panel follow-up. Large panels establish external validity and persistence; intensive cohorts estimate within-person dynamics and measurement invariance.

10.3.4 From adaptation to terminal closure: the additional identification requirements

The adaptation literature becomes maximally informative when three progressively stronger targets are kept distinct.

First, mean reversion is not path closure. A stationary process can accumulate a nonzero random integral. Second, population averages are not individual closure. Positive and negative residuals can cancel across people while every individual violates PTVC. Third, return to a personal set point is not return to zero. A positive or negative set point has no privileged relationship to the neutral origin of the cumulative ledger.

These distinctions make a successful closure result substantially more informative than an ordinary return-to-baseline finding.

10.3.5 Inference supported by current adaptation evidence

The adaptation literature supplies a mature body of natural perturbations, long-term cohorts, event taxonomies, and alternative mechanisms. PTVC can now be tested against real heterogeneity rather than an idealized treadmill. If a closure-relevant kernel property survives across events, people, countries, measurement channels, and preprocessing choices while ordinary set-point models do not, that would be substantially more persuasive than a simple return-to-baseline plot.

Local adaptation contract. Every claim names its event class, horizon, integrability conditions, heterogeneity model, and held-out compensation criterion. Local return or attenuation is a mechanism result, not a complete-stream terminal conclusion.

10.4 Opponent processes, relief, homeostasis, and allostasis

10.4.1 Opponent-process theory

Solomon and Corbit (1974) proposed that affective reactions recruit opponent processes with different onset and decay dynamics. Repetition can weaken the primary process and strengthen or prolong the opponent response. This architecture is naturally relevant to PTVC because it generates signed, temporally structured compensation.

For $\lambda_a > 0$, $\lambda_b > 0$, and $\gamma \geq 0$, define a normalized unit primary response and a linear opponent state by

$$\begin{aligned} a(t) &= \mathbf{1}_{\{t \geq 0\}} e^{-\lambda_a t}, & \partial_t b(t) &= -\lambda_b b(t) + \gamma a(t), \\ b(0) &= 0, & f(t) &= a(t) - b(t). \end{aligned}$$

Here λ_a, λ_b, and γ have inverse-time units; the primary and opponent responses share the same normalized amplitude unit. Integrating the stable opponent equation gives

$$\int_0^{\infty} f(t)\, dt = \frac{1}{\lambda_a}\left(1 - \frac{\gamma}{\lambda_b}\right).$$

Exact local area cancellation therefore holds if and only if $\gamma = \lambda_b$. This is a model-specific opponent-kernel restriction, not a whole-stream PTVC result.

10.4.2 Pain-offset relief

Pain-offset experiments show that removing an aversive state can simultaneously increase positive affect and reduce negative affect. Franklin et al. (2013) observed both effects for several seconds after painful stimulation ended, supporting bipolar or bivariate dynamics and an opponent-like transition. The decisive promotion test is one preregistered high-temporal-resolution kernel-area study that varies intensity, duration, predictability, controllability, and repetition; estimates positive and negative channels separately; and compares integrated primary and offset areas rather than peak ratings.

10.4.3 Addiction and allostatic counteradaptation

Koob and Le Moal (2001) described addiction as dysregulation of reward systems with counteradaptation and allostatic shifts. Chronic drug exposure can reduce reward function and recruit stress systems, producing a negative emotional state that persists beyond acute intoxication. This is important because it demonstrates that opponent processes can alter baselines and become pathological, providing a stringent real-world test of when compensation is benign, symmetric, transported, or exact.

At the same time, addiction research supplies strong perturbations, repeated exposures, and identifiable transitions for testing dynamic balance models. PTVC-specific predictions must outperform allostatic and incentive-sensitization rivals rather than relabel them.

10.4.4 Homeostasis and allostasis

Homeostatic feedback corrects deviations from regulated variables. Allostasis emphasizes predictive adjustment of internal conditions to anticipated demand (McEwen, 1998; Sterling, 2012; Ramsay & Woods, 2014). These frameworks make long-horizon regulation scientifically familiar. PTVC-specific leverage arises when a validated phenomenal bridge links the regulated state or objective to the signed ledger.

A generic regulated state $x(t)$ may follow

$$dx_t = [Ax_t + Bu_t]dt + \Sigma(x_t, t)dW_t,$$

with control u_t minimizing expected physiological cost. PTVC enters only if the control objective or viability constraint contains the phenomenal ledger or a validated proxy. Otherwise homeostatic recovery is a rival explanation.

10.4.5 Physiological reserve, energetic cost, and loss of complexity

The reserve variable needed by some PTVC mechanisms should not be confused with a single biomarker or a metaphor for “health.” Biology offers several partially overlapping constructs: organ reserve, frailty, resilience, allostatic load, metabolic flexibility, autonomic complexity, and the capacity to recover after perturbation. The energetic model of allostatic load emphasizes that anticipatory regulation consumes finite energetic resources and can trade against maintenance and repair (Bobba-Alves et al., 2022). This supplies a biologically grounded reason to study feasible compensation as a function of changing capacity; a validated phenomenal bridge is the corresponding promotion gate.

Physiological aging has also been associated with reduced multiscale complexity. Lipsitz and Goldberger (1992) proposed that aging can reduce the complex variability that supports adaptive responses. In a cross-sectional sample of 389 community-dwelling older women with moderate to severe disability, lower approximate entropy of heart-rate dynamics was associated with frailty, and conventional heart-rate-variability measures were also lower in frail participants (Chaves et al., 2008). These results motivate reserve and flexibility measurements; they do not show that low variability is universally beneficial. Healthy adaptive systems can require rich variability, whereas terminal failure, sedation, measurement shutdown, and constrained dynamics can all reduce variance.

Accordingly, a reserve construct should be estimated as a latent, preregistered, multichannel state rather than post hoc from whatever variable predicts the endpoint. Candidate indicators may include physical function, frailty phenotype, autonomic complexity, metabolic and inflammatory measures, cognition, recovery slopes after perturbation, and treatment burden. Each indicator requires its own observation equation, lag structure, medication adjustment, and transport test. A valid reserve proxy must predict *future feasible response* out of sample, not merely correlate with age or mortality.

The PTVC-relevant discriminating question is conditional: after reserve and observation quality are modeled separately, does shrinking reserve narrow the attainable future-ledger set or compress residual dynamics in the specific form predicted by a declared branch? This is stronger than showing that frail people have lower variability. It requires a joint state-space model and rival explanations for censoring, immobility, treatment, and physiological collapse.

10.4.6 Sleep and overnight affect processing

Sleep is a recurrent state transition that alters consciousness, memory, autonomic regulation, and next-day affect. Cartwright et al. (1998) linked selected dream-affect variables to overnight mood change in 60 healthy volunteers. Reviews support roles for sleep in emotional memory and regulation while finding stage- and task-dependent, sometimes inconsistent effects (Palmer & Alfano, 2017; Tempesta et al., 2018). Sleep therefore offers a repeated nonterminal laboratory for state-dependent kernels and experience-versus-memory divergence without treating REM architecture or a morning report as a validated valence meter.

Sleep's principal experimental contribution is mechanistic and methodological. By jointly modeling selected dream reports, sleep-stage physiology, circadian change, homeostasis, expectancy, and waking affect, protocols can estimate signed event kernels across wake, non-REM, and REM states; model unobserved exposure during sleep; and test whether overnight changes predict subsequent experience independently of remembered dream content.

An informative protocol would combine evening EMA, polysomnography, prespecified awakenings, immediate affect reports, morning ratings, and next-day sampling. It would distinguish (i) direct experienced affect during sampled dreams, (ii) remembered episode evaluation, (iii) waking baseline shift, and (iv) autonomic or neural correlates. A balance-like pattern that survives those separations would justify further study; a pattern visible only in morning recollection would favor a memory-construction rival.

10.4.7 Reserve-coupled route as a testable branch

One candidate branch uses remaining reserve $U_i(t) \geq 0$ and normalized coordinate $Y_i(t)$. A phenomenal-route bridge is a separately validated empirical relation, not an identity:

$$\| L_i^{\text{phen}} - L_i^{\text{rc}} \|_{\mathcal{B}} \leq \delta_{\text{bridge}}.$$

The norm, tolerance, common-support domain, calibration sample, and held-out failure rule are fixed before terminal outcomes. If $U_i(t) \to 0$ at the same boundary and $Y_i(t)$ remains controlled, the route coordinate closes; transfer to the phenomenal ledger requires the validated bridge, and c_{LU} remains in every normalization and unit check.

10.4.8 A balanced scientific interpretation

Opponent and regulatory science establishes biologically realized examples of delayed, biphasic, predictive, and history-dependent regulation. The same science supplies serious rivals: adaptation can be partial, set points can move, compensation can overshoot, and pathology can accumulate. Treating these patterns as discriminating constraints, including their counterexamples, is therefore central to the research program.

10.4.9 Physiological hysteresis as a route-state diagnostic

Section 10.4.5 defines the multichannel reserve construct and its validation burden. The additional opportunity is geometric: repeated perturbation–recovery cycles can reveal path dependence even when their starting and ending physiological levels match. Hysteresis can therefore test whether regulatory history changes future feasible compensation without repeating the biomarker-to-valence discussion.

For the confirmatory geometric branch, $B_i: [s, t] \to \mathbb{R}^p$ is the preregistered path. Let ω be a fitted one-form and define $\mathcal{A}_{i;s,t} := \int_s^t \omega\left(B_i(u)\right) \cdot \mathrm{d}B_i(u)$.

The one-form is conservative on the relevant domain when $\omega = \nabla\Phi$. Nonzero circulation around a preregistered repeated perturbation–recovery cycle γ_i,

$$\oint_{\gamma_i} \omega\left(B_i(u)\right) \cdot \mathrm{d}B_i(u) \neq 0,$$

quantifies hysteresis or allostatic path dependence. This is an empirical route-state diagnostic: it can test whether regulatory history changes future capacity even when current levels match. A separately calibrated valence channel is still required before any association with the phenomenal ledger is interpreted.

Sleep is especially valuable because it supplies repeated, safe state transitions. Daily protocols can estimate whether pre-sleep residual, REM architecture, autonomic recovery, dream affect, and next-day event kernels form a transported compensation pattern. Trauma-related sleep disruption and noradrenergic abnormalities provide a natural failure mode: a law-specific route should predict precisely how compensation geometry changes when the proposed overnight mechanism is impaired, while allowing other routes to remain possible.

10.4.10 Robust perfect adaptation and integral feedback

Some biomolecular systems implement robust exact set-point restoration and provide valuable mechanistic comparators for exact control. Bacterial chemotaxis, integral-feedback analyses, and antithetic integral feedback supply concrete architectures for testing whether an independently validated phenomenal variable exhibits a corresponding exact-control structure (Barkai & Leibler, 1997; Yi et al., 2000; Aoki et al., 2019).

A canonical controller has an integral state z:

$$\dot{z}(t) = y^\star - y(t), \qquad u(t) = kz(t),$$

Persistent output error accumulates until the specified controller removes the state error; this property remains distinct from cancellation of a signed phenomenal path area.

RPA supplies a direct comparator because returning a present output to $y^\star$ does not imply cancellation of the complete signed phenomenal area $\int f_i \, d\mu_i$. Measure state error and path area under the same perturbations. CUT-1's slope -1 is a partition/additivity identity shared by every fixed terminal residual, including a nonzero residual; it is not a PTVC-specific prediction. The PTVC-bearing contrast is whether the independently adjudicated $D_i(\tau)$ lies in the registered zero or finite acceptance region across cuts and interventions after an equally flexible RPA rival receives the same state information, tuning budget, and observation model.

Integral feedback supplies a concrete proof of principle for exact biological regulation in a specified state variable. It motivates carrier, information, and comparator tests and establishes a powerful foundation for the next PTVC promotion step: pairing exact control architecture with a validated phenomenal bridge and CUT-1 terminal evidence.

10.5 Reward, motivation, predictive control, and affective value

10.5.1 Reward contains dissociable components

Neuroscience distinguishes hedonic impact ("liking"), incentive salience ("wanting"), and learned prediction (Berridge et al., 2009). Dopamine signals are closely related to learning, prediction errors, vigor, and incentive salience rather than a simple meter of experienced pleasure (Berridge & Robinson, 1998; Schultz et al., 1998). This distinction is indispensable for PTVC. A neural signal that predicts reward, motivates pursuit, or updates a policy cannot be integrated as phenomenal valence without an independently validated bridge.

The separation also creates useful experiments. States in which wanting diverges from liking (addiction, cue sensitization, salt appetite, compulsive behavior, and some mood disorders) can test whether the proposed ledger tracks experienced valence, action policy, learned value, or motivational salience. A serious ledger should remain correctly typed when these quantities dissociate.

10.5.2 Reward-prediction errors and temporal-difference learning

In temporal-difference learning, a prediction error may be written

$$\delta_t^{\mathrm{TD}} := r_t + \gamma V(s_{t+1}) - V(s_t).$$

The algebra resembles a telescoping difference, but that does not make the cumulative reward or phenomenal ledger close. If an empirical proxy is itself defined as a conservative state difference, summing it may manufacture a boundary term. PTVC protocols should therefore prohibit proxies whose apparent closure follows from their mathematical construction.

Reward-prediction errors remain valuable mechanism-plausibility evidence. They show that nervous systems compute temporally extended deviations between expectation and outcome. PTVC-relevant work can test whether long-horizon residual estimates modulate learning or policy selection after controlling for standard prediction errors.

10.5.3 Predictive processing and active inference

Predictive-processing and active-inference frameworks describe perception and action through generative models, belief updating, and policy selection (Clark, 2013; Friston, 2010). In one influential process theory, policies minimize expected free energy, combining pragmatic and epistemic terms (Friston et al., 2017). These frameworks offer a language for hidden states, preferences, uncertainty, and horizon-dependent action. PTVC-specific leverage begins with a validated bridge from those computational quantities to phenomenal valence and the signed ledger.

A declared bridge might relate an affective signal to prediction-error or expected-free-energy components, as proposed in computational accounts of emotional valence (Joffily & Coricelli, 2013). That bridge should be judged by predictive performance and dissociation tests. It must handle cases in which surprise is pleasurable, certainty is aversive, wanting diverges from liking, and epistemic actions are pursued despite immediate negative affect.

10.5.4 Constrained-control interpretation

Let $\Pi_{\mathrm{adm}}(t,x,\ell)$ be the progressively measurable policies assigning probability one to the structural action domain and satisfying the declared reward, endpoint, and integrability conditions. For finite strict-pre-endpoint target L_{T-}, define

$$V_K(t,x,\ell) := \sup_{\substack{\pi \in \Pi_{\mathrm{adm}}(t,x,\ell) \\ P^{\pi}(|L_{T}-| \le K | X_t = x, L_t = \ell) \ge 1-\alpha}} \mathbb{E}^{\pi}\left[\int_t^T r_s \, \mathrm{d}s \,|\, X_t = x, L_t = \ell\right].$$

If the feasible policy set is empty, report No adequate estimate. A Lagrange multiplier or shadow price is interpreted only under a stated constraint qualification and strong-duality theorem. The formulation defines a model class; it does not assert that an organism explicitly computes the ledger or endpoint.

10.5.5 Cognitive control, effort, and policy selection

The expected-value-of-control framework treats cognitive control allocation as a decision that weighs prospective benefits against control costs (Shenhav et al., 2013). This provides another ordinary route by which horizon, uncertainty, and available resources can change behavior. It also creates a crucial rival: residual-sensitive choices may arise because imbalance proxies correlate with expected task value, threat, fatigue, or opportunity cost rather than because a terminal constraint is represented.

PTVC-focused experiments can exploit this overlap. After estimating an individual event ledger prospectively, participants can choose among actions that differ in immediate affect, expected future affect, effort, and uncertainty. A standard control model and a residual-aware model should be fit to held-out choices. The PTVC branch earns support only if the residual term improves calibrated prediction across tasks and remains distinguishable from reward prediction error, fatigue, and generic negative urgency.

Policy-to-motivation bridge. A completion-dependent policy becomes a psychological motivation hypothesis only after a prospectively validated bridge connects features of that policy to attention allocation, wanting, avoidance, effort, persistence, goal revision, or stopping. The bridge must add held-out prediction beyond immediate and anticipated reward, habit, learned value, prediction error, fatigue, prognosis, threat, opportunity cost, and ordinary affect regulation. Reflex, bodily transition, structural action availability, endpoint coupling, routing, and global selection remain separate causal loci; failure of the psychological bridge does not by itself refute an action-mediated route implemented elsewhere.

10.5.6 Why predictive-control science strengthens the program

Predictive-control science strengthens PTVC by supplying concrete mechanism candidates, quantitative baselines, and demanding comparative tests. It shows how future-oriented control, latent-state estimation, and information seeking can generate nonlocal behavior. A control-mediated PTVC route becomes especially credible when ordinary observable dynamics remain consistent with known biology while the candidate constraint adds distinctive held-out signatures through physical marginal transparency.

10.5.7 Computational mood, prediction errors, and temporal weighting

Computational affect supplies quantitative detail that the broader reward literature leaves implicit. Rutledge et al. (2014) modeled momentary subjective well-being as a function of recent certain rewards, expectations, and reward-prediction errors. The laboratory sample was small ($n = 26$), but the behavioral form generalized to a smartphone sample of 18,420 participants. This result makes a central point for PTVC measurement: reported mood is dynamically referenced to recent expectations and outcomes, not a direct readout of accumulated phenomenal mass.

Eldar et al. (2016) proposed that mood represents momentum in rewards, allowing recent prediction errors to bias how subsequent outcomes are perceived and learned. Keren et al. (2021) found robust primacy weighting in subjective mood: earlier events within an experience influenced later mood reports more strongly across multiple environments and populations, with corresponding neural analyses in frontal mood-regulation regions. These findings motivate explicit kernels

$$m_t = \alpha_i + \int_0^t k_i\,(t,s)\,\delta_s\,ds + \varepsilon_t,$$

where δ_s denotes event or prediction-error input and k_i is estimated rather than assumed exponential or time-homogeneous. A ledger analysis must distinguish the phenomenally experienced process from the reporting kernel that maps history into mood.

The program gains evidential specificity when it uses computational-mood models as mature ordinary rivals, calibratable temporal kernels, and high-frequency outcomes for the completion-potential program. If a ledger residual predicts future mood, action, or endpoint behavior after these kernels, expectations, and recent outcomes are controlled, the resulting evidence is much more specific.

10.5.8 Mood homeostasis and hedonic flexibility at population scale

Taquet et al. (2016) followed more than 28,000 smartphone users for an average of about 27 days and found a hedonic-flexibility pattern: lower mood increased the probability of choosing subsequently mood-improving activities, whereas higher mood permitted useful activities associated with lower immediate mood. Taquet et al. (2020) then analyzed two large population samples totaling 58,328 participants. Estimated mood homeostasis was lower in people with low mood and in those with a history of depression; their simulations linked impaired homeostasis to greater incidence and longer duration of depressive episodes.

These studies demonstrate that people use behavior to regulate affect in ordinary life, with clinically meaningful heterogeneity. They also give PTVC a concrete rival and experimental platform. A generic mood-homeostasis model predicts action from current mood and accessible activities. A completion-oriented model adds prospectively calibrated residual, horizon, and reachable-set terms. The decisive comparison asks whether those additions transport across tasks and forecast held-out choices after current mood, depression history, opportunity, and expected utility are controlled.

Emotional inertia provides another dimension. Kuppens et al. (2010) associated greater persistence of emotional states with low self-esteem and depression across two naturalistic studies. Inertia, flexibility, rebound, and long-run residual correction should therefore be modeled jointly. A system can return toward a local mood baseline while its lifetime signed integral continues to drift; conversely, a low-inertia process need not approach terminal balance.

10.5.9 Durable shocks as discriminating evidence

Durable shocks provide an especially valuable discriminating test. Lindqvist et al. (2020) surveyed 3,362 Swedish lottery players 5–22 years after randomized prize events totaling approximately US$277 million. An after-tax prize of US$100,000 was associated with about 0.037 standard deviations higher life satisfaction, with estimated persistence across years since winning; effects on happiness and mental health were smaller. This result reveals that adaptation is construct-specific and can remain incomplete for decades, creating long, identifiable perturbations through which later event exposure, choice, memory, reserve use, and terminal dynamics can be tested against one frozen completion geometry.

Accordingly, PTVC's strongest form explicitly accommodates persistent shifts, unequal gross trajectories, and incomplete local compensation rather than relying on a universal 'everything returns to baseline' premise. Durable-shock cohorts are especially valuable because they create long, identifiable residual perturbations. They can test whether later event exposure, choice, memory, reserve use, and terminal dynamics depend on the persistent component in ways predicted by the same completion geometry.

10.5.10 Neural valence channels and typed bipolar representation

Neuroscience provides convergent support for representing positive and negative experience as partly distinct, interacting channels rather than forcing them into a single uncalibrated report. In mice, Kim et al. (2016) identified largely distinct basolateral-amygdala populations associated with positive and negative valence and showed antagonistic interactions between them. In humans, Seymour et al. (2007) found differential encoding of gains and losses along an anterior–posterior striatal axis. Together with behavioral mixed-affect findings, these results support models with separate appetitive and aversive channels, cross-channel inhibition, and state-dependent scalarization.

The PTVC-relevant role is representational and experimental. Simultaneous positive and negative channels can be estimated separately, perturbed separately, and tested for a stable cross-sign map before a scalar ledger is formed. Neural population activity is not itself a phenomenal unit, but independently decoded channel dynamics can serve as a validation and negative-control layer: a proposed scalar ledger should predict held-out choice and report while preserving sign, mixed states, and temporal integration better than arousal, salience, motor preparation, and generic reward-prediction-error models.

10.6 Subjective time, future horizons, and lifespan motivation

10.6.1 Time horizon reorganizes motivation

Socioemotional selectivity theory proposes that perceived future time, not chronological age alone, shifts motivational priorities. When time is experienced as open-ended, information acquisition and future preparation are emphasized; when it is limited, emotionally meaningful goals and present-oriented regulation gain priority (Carstensen et al., 1999; Carstensen, 2006). Similar shifts appear in contexts other than aging, including illness, relocation, and threat.

This is an important mechanism-localization foothold: horizon information demonstrably alters affective policy. The promotion experiment tests whether that policy tracks a validated residual proxy beyond salience, health prognosis, and ordinary emotion regulation.

10.6.2 Positivity and emotional regulation in aging

Experience-sampling and lifespan studies often find reduced frequency or persistence of negative emotion and improved regulation in older adulthood, with substantial heterogeneity and context dependence (Carstensen et al., 2000; Gross et al., 1997). These results are compatible with emotionally meaningful goal selection, accumulated regulation skill, cohort effects, selective survival, and changing environments.

PTVC should predict something more specific than “older adults are more positive.” Candidate predictions include stronger coupling between validated residual proxies and goal selection when perceived time narrows, different compensation kernels under limited versus expansive horizons, and terminal-window effects that track time-to-event more closely than chronological age after health and selection are modeled.

10.6.3 Subjective duration as part of the observation problem

Emotion changes perceived duration, and arousal, stimulus modality, and task alter the direction and magnitude of the effect (Cui et al., 2023). Severe illness, pain, medication, delirium, and sleep disruption may add further distortions. Consequently:

- physical time, remembered duration, and phenomenal exposure should be analyzed separately;
- duration-weighted retrospective scores are not substitutes for dense sampling;
- the exposure measure should be subjected to sensitivity analysis;
- terminal changes in perceived time are potential mechanism variables and potential measurement confounders.

Anticipatory valence is accrued when experienced. Use prospectively declared, pairwise-disjoint measurable event-domain windows contained in the stream event space for anticipation, the focal event, and the post-event interval. A report or physiological response may not be counted in more than one window without an explicit joint model:

$$\begin{aligned} W_{i,\eta}^{\text{ant}} \,\dot{\cup}\, W_{i,\eta}^{\text{event}} \,\dot{\cup}\, W_{i,\eta}^{\text{post}} &= W_{i,\eta}^{\text{episode}}, \\ Z_{i,\eta}^{w} &:= \nu_i\left(W_{i,\eta}^{w}\right), \qquad w \in \{\text{ant}, \text{event}, \text{post}\}, \\ Z_{i,\eta}^{\text{episode}} &= Z_{i,\eta}^{\text{ant}} + Z_{i,\eta}^{\text{event}} + Z_{i,\eta}^{\text{post}}. \end{aligned}$$

Dread and savoring therefore enter the ledger before the focal outcome rather than being reassigned retrospectively to it. They are genuine value components and mechanism or observation rivals in horizon experiments.

10.6.4 Horizon manipulation as a safe experiment

Time-horizon effects can be studied without endangering participants or invoking actual terminal states. Established experimental manipulations ask people to imagine constrained or expansive futures or use naturally occurring relocations and transitions. A PTVC-focused study can cross horizon manipulation with independently calibrated signed events, then test whether horizon changes the estimated compensatory kernel or policy sensitivity to residual imbalance.

The design must distinguish motivational framing from demand characteristics. Blinded hypotheses, active control narratives, indirect behavioral outcomes, and preregistered manipulation checks can reduce false support.

10.6.5 Promotion path

Horizon sensitivity establishes that future-time beliefs reorganize affective regulation. A route mechanism earns promotion when its held-out predictions additionally separate terminal-boundary sensitivity from ordinary salience, health-prognosis, and emotion-regulation explanations.

10.7 Terminal decline and end-of-life trajectories

10.7.1 Time to death is an empirically meaningful axis

Longitudinal aging research has repeatedly found that some domains change more sharply as death approaches. Gerstorf and colleagues analyzed national longitudinal data from Germany, the United Kingdom, and the United States and reported that well-being was relatively stable across much of adulthood but declined more rapidly beginning roughly three to five years before death; rates of decline steepened by a factor of three or more after the estimated transition (Gerstorf et al., 2010b). Broader multivariate work found terminal changes across cognitive, sensory, physical, health, social, and self-related functions (Gerstorf et al., 2013).

Vogel and colleagues followed 1,671 older adults for up to 15 years and modeled positive and negative affect by distance to death. Time to death organized change better than chronological age; positive affect declined and negative affect increased, with estimated transitions several years before death (Vogel et al., 2013). These findings strongly justify terminal alignment as a scientific coordinate.

Their positive- and negative-affect patterns sharply separate terminal affect level from the completed cumulative integral. PTVC concerns the latter, so the next discriminating analysis reconstructs calibrated signed trajectories and tests the derived ledger implications instead of expecting raw mood itself to converge.

10.7.2 Terminal decline is heterogeneous

Terminal trajectories differ across people, causes of death, health states, social environments, and domains. Social participation and valuing social goals have been associated with higher late-life well-being, later onset, and less pronounced terminal decline in SOEP data (Gerstorf et al., 2016). Geography and individual characteristics also predict terminal progression (Gerstorf et al., 2010a). Proxy-reported final-year well-being declines vary with morbidity, care needs, place and cause of death, and baseline function.

Retrospective SOEP proxy reports indicated a 0.57-SD decline during the last year of life, with morbidity-related vulnerabilities and substantial between-person heterogeneity shaping the trajectory (Gerlach et al., 2017).

Recent endpoint-aligned studies broaden this foundation. In the Rush Memory and Aging Project, 1,971 older adults received annual assessment for as long as 22 years and 1,119 died during follow-up. Backward-time and change-point models placed the onset of terminal decline at approximately 11 years before death for purpose in life, 10 years for cognitive activity, 9 years for social activity and depressive symptoms, and 6 years for loneliness; social-network size did not show the same terminal change-point pattern (Wang et al., 2025).

In the multinational IGEMS consortium, 2,411 participants contributed 12,574 depressive-symptom assessments. Accelerated increases emerged about 4.3 years before death, and within 98 mortality-discordant twin pairs the deceased twin showed a steeper terminal increase than the co-twin who survived at least four years longer (Petkus et al., 2025).

The domain-specific timing, co-twin contrast, and between-person variability strengthen the case for endpoint-aligned multivariate models with individual change points. They also provide candidate datasets, subject to resource-specific audits, and design templates for asking whether a common signed-ledger geometry adds transported predictive structure beyond disease, function, social loss, care, and ordinary terminal decline.

10.7.3 Clinical symptom trajectories

Seow et al. analyzed 10,752 cancer decedents with Edmonton Symptom Assessment System (ESAS) symptom assessments and 7,882 with Palliative Performance Scale (PPS) assessments during the final six months of life. Several symptoms and PPS worsened, whereas pain, nausea, anxiety, and depression were comparatively flat (Seow et al., 2011). End-of-life experience is therefore multidimensional; clinical deterioration is not a single monotone valence trajectory.

Symptom control, sedation, opioids, oxygen, delirium, communication capacity, and care setting all affect observation. Clinical measures target symptoms or function, not the latent lifetime ledger. They are nonetheless valuable covariates, negative controls, and indicators of state-dependent missingness.

10.7.4 Palliative care changes the path

Palliative interventions are ethically required care, not nuisances to be removed. They can reduce suffering, alter awareness, change communication, and shift the timing and setting of observation. A PTVC study must specify intercurrent-event strategies: treatment-policy, hypothetical, composite, principal-stratum, or while-on-treatment estimands, as appropriate. It must never withhold symptom relief to create a cleaner test. Palliative care improves quality of life and remains globally underprovided; access and care context therefore belong in the endpoint and observation model (World Health Organization, 2020).

This ethical constraint enhances rather than destroys scientific value. Variation in treatment timing, setting, and response can support target-trial emulation or an observational study after explicit confounding, positivity, and selection assumptions are stated (Hernán & Robins, 2016, 2020). The term natural experiment is reserved for settings with a defensible quasi-random assignment mechanism. The primary goal is to understand paths under actual care; mechanism localization is secondary to participant welfare.

For irregular treatment and monitoring times, a continuous-time marginal-structural-model branch can encode the observational law and a hypothetical randomized-trial law as distinct martingale measures, with their likelihood-ratio process supplying the weights (Røysland, 2011). The protocol prospectively defines treatment, observation, censoring, and endpoint counting processes; requires the target-trial law to be absolutely continuous with respect to the observed law; diagnoses positivity and unstable weights; and returns No adequate estimate when the target law is unsupported.

10.7.5 Endpoint selection and missingness

For each registered observation channel c, let $O_{i,c}^{\text{avail}}(t)$ indicate availability. Let $H_i^{\text{rec}}(t-)$ denote a preregistered predictable summary of the prior recorded pairs $D_{i,k}^{(c)}$ from §7.4, including measured values and unavailability symbols. Model channel availability conditionally. When indicators may be dependent, use the joint vector

$$\mathbf{O}_i^{\text{avail}}(t) = (O_{i,c}^{\text{avail}}(t))_{c \in \mathcal{C}_i^{\text{obs}}}.$$

This preserves cross-channel availability dependence in the observation model.

$$\mathbb{P}\big(O_{i,c}^{\text{avail}}(t) = 1 \mid \mathcal{H}_{i,t-}^{\text{joint}}, R_i^{\text{ret}}(t) = 1\big) = g_c(X_i(t-), H_i^{\text{rec}}(t-), U_i(t-), \text{care setting}, \text{burden}).$$

$O_{i,c}^{\text{avail}}$ denotes channel-specific availability, and $\mathbf{O}_i^{\text{avail}}$ denotes the joint availability vector. Neither notation collides with the route, retention, or reference-class R families.

10.7.6 What current endpoint evidence supports

Existing end-of-life science already supports three constructive statements:

1. terminal alignment is empirically meaningful;
2. the relevant trajectories are heterogeneous and multidimensional;
3. large, harmonizable datasets and clinical infrastructures exist.

That foundation justifies immediate reflection analyses, endpoint-selection audits, and prospective measurement pilots. The measurement platform is no longer merely hypothetical: a mixed-method content-validation study with 43 people living with advanced breast or lung cancer and 8 health-care professionals produced the ESM-AC digital questionnaire, including 31 momentary, 2 morning, and 7 evening core items; the 10 participants in its usability round rated ease of use at a mean of 4.5 (SD 0.5) (Geeraerts et al., 2024). Separately, a five-week palliative-care feasibility study in 15 advanced-cancer patient–caregiver dyads achieved 73% daytime wearable adherence while pairing sensor-triggered ecological momentary assessment with electronic patient-reported outcomes (Schuler et al., 2023). These studies establish concrete burden, trigger, adherence, and missingness parameters for intensive prospective pilots. Developing and validating the cardinal completed-ledger and endpoint bridges remains the next defined promotion task for these terminal-science resources.

10.7.7 Unexpected endpoint timing as an observational discriminator

Unexpected terminal timing is an empirically meaningful observational discriminator. In a prospective cohort of 1,896 patients across 23 Japanese inpatient hospice or palliative-care units, Ito et al. (2021)

reported 30-day cumulative incidence estimates ranging from 6.4% to 16.8% across four definitions of sudden unexpected death. The same event could be classified differently depending on rapid decline, clinician surprise, earlier-than-anticipated death, or recent performance status. This definitional variation is itself scientifically useful: it permits preregistered sensitivity analyses rather than one outcome-trained "unexpected" category.

Such cases provide a high-contrast stress test of local horizon-aware closure, subject to the same endpoint, care, observation, and selection gates. A protocol should freeze the filtered endpoint distribution using information available before the final decline, compute an endpoint-surprise score, and compare terminal residual compatibility across expected and unexpected deaths. Cause-specific hazards, acute complications, palliative treatment, missingness, reporting capacity, and stream-endpoint uncertainty must be jointly modeled. The key prediction is not that every unexpected biological death must display a measured zero; it is whether the same completion-potential model preserves calibration as terminal timing becomes less predictable.

10.7.8 Terminal neurodynamics and consciousness-aware endpoint intervals

Terminal neurophysiology can narrow the interval in which qualifying experience remains possible. Xu et al. (2023) analyzed electroencephalography and electrocardiography from four comatose dying patients after withdrawal of ventilatory support. Two showed marked increases in gamma power, cross-frequency coupling, and directed connectivity, including activity in a posterior cortical 'hot zone.' In this four-patient sample, the signal appeared in two patients and contemporaneous report was unavailable. Its strongest contribution is therefore specific and important: neural activity near death need not decline monotonically, a coarse behavioral endpoint can miss structured terminal dynamics, and conscious experience and valence remain targets for appropriately designed follow-up studies.

The multicenter AWARE-II study likewise examined consciousness and electrocortical activity during cardiopulmonary resuscitation (Parnia et al., 2023). Its value for PTVC is methodological: multimodal resuscitation records, survivor interviews, event timing, and electroencephalographic monitoring can improve endpoint ascertainment and reveal intervals that ordinary chart data collapse. Neither gamma activity nor recalled experience is a valence meter. Their proper role is to refine the observation window, test retained processing, and prevent premature truncation of the candidate stream.

A prospectively registered terminal protocol should therefore synchronize clinical events, medication, perfusion, oxygenation, EEG quality, behavioral responsiveness, and patient-selected low-burden reports where possible. It should classify each interval by evidence for possible experience rather than assign a single unqualified time of "consciousness loss." The output is an endpoint interval and observation-availability model suitable for the ledger analysis.

10.7.9 Paradoxical lucidity and nonmonotone terminal observability

Reports of paradoxical lucidity—unexpected, apparently meaningful communication or recognition in people with severe dementia—matter because observation availability near death need not decline monotonically. In a pilot study, health-care professionals described 29 witnessed episodes with heterogeneous onset and duration (Teresi et al., 2023). Retrospective reports cannot determine the underlying neural state or establish terminal valence. They identify a concrete failure mode for observation models that permanently replace self-report with proxy report after a single incapacity threshold. Prospective multimodal work may investigate neural and observational implications and, only through a separately validated valence bridge, any relation to terminal valence. The retrospective reports themselves are not endpoint or PTVC evidence.

Prospective palliative and dementia protocols should therefore permit time-varying return of behavioral responsiveness, record the exact relation among communication, medication, sleep–wake state, delirium, physiology, and death, and preserve brief participant-led reports when ethically appropriate. Continuous or event-triggered audio, video, electroencephalography, and bedside assessment require consent and independent governance. Their immediate contribution is a more accurate stream-endpoint interval and observation kernel; the methodological opportunity is to recover otherwise missing late-window transitions without assuming that an apparent return of communication is either meaningless noise or direct evidence of closure.

Terminal-alignment role. Terminal alignment is an empirical coordinate that can improve prediction, reveal regime change, and stress-test endpoint and observation models. Paired with a validated signed-ledger bridge and terminal residual, it can contribute evidence about neutral terminal valence. Perturbational complexity, terminal gamma or coupling reports, and paradoxical lucidity add complementary endpoint and observation information.

10.8 Experienced life, remembered life, and narrative integration

10.8.1 The peak-end problem

Retrospective evaluations of episodes can overweight the most intense moment and the ending while underweighting duration (Fredrickson & Kahneman, 1993). In clinical procedures, remembered unpleasantness can favor an objectively longer episode if its ending is less painful (Redelmeier & Kahneman, 1996). These findings motivate simultaneous modeling of terminal life evaluation and the experienced trajectory rather than collapsing them into one channel.

Model the prospectively sampled experienced path and remembered evaluation separately:

$$L_i^{\text{online}}(t) := \int_{\Omega_i^{\leq}(t)} f_i\,(x)\,\mu_i^{\text{phen}}(dx) = L_i^{\text{phen}}(t)$$

$$M_i(t) := \mathcal{R}(\{f_i(s) : s \leq t\}, \text{peaks}, \text{ending}, \text{narrative}, \text{memory}).$$

Dense prospective sampling estimates L_i^{online} imperfectly; retrospective interviews estimate M_i. Their divergence is a scientific outcome rather than a shared superscript that could be misread as exponentiation.

10.8.2 Life review, meaning, and reconciliation

Late-life and palliative contexts often include life review, meaning-making, reconciliation, spiritual practice, and changing priorities. These processes can improve peace or narrative coherence even when symptoms worsen. They may influence experienced valence, remembered utility, both, or neither. Qualitative interviews are therefore useful for mechanism discovery and construct validation, but narrative themes should not be coded as signed ledger entries without demonstrated measurement validity.

10.8.3 Duration neglect as a strong rival

A strong experiment gives the remembered-utility rival a fair chance. Candidate models should include peak, end, duration, recency, and narrative-coherence terms. PTVC earns support only if a calibrated experienced-utility model predicts held-out outcomes better than these rivals and if apparent terminal closure is not confined to retrospective channels.

10.8.4 Measurement contribution

This literature strengthens the project by identifying one of its most important measurement distinctions. A program that simultaneously tracks lived affect and remembered life could make independent contributions to welfare measurement, palliative care, decision science, and the philosophy of well-being even before PTVC is resolved.

10.8.5 Fading affect bias as a memory-layer compensator

Autobiographical memory has its own signed dynamics. Walker et al. (2003) reviewed evidence for a *fading affect bias*: affect attached to negative autobiographical events often fades faster than affect attached to positive events, while dysphoria and related conditions can disrupt the pattern. This supplies a psychologically grounded mechanism by which remembered life can become more positive or apparently balanced even when the lived phenomenal trajectory has not changed.

The bias is therefore both useful science and a strong rival. A prospectively specified design should collect immediate experience, repeated memory ratings, confidence, rehearsal, social disclosure, current mood, and narrative meaning for the same events. A hierarchical model can estimate separate experienced-event and memory-reconstruction kernels. PTVC-relevant evidence strengthens when past–future complementarity and the prospectively specified completion-potential fingerprint appear in prospectively sampled experience and independent channels, rather than only after autobiographical affect has faded or been narratively reorganized.

10.9 Consciousness, stream individuation, and terminal ascertainment

10.9.1 Behavior is an imperfect consciousness channel

Clinical consciousness assessment is vulnerable to motor, language, sensory, sedation, arousal, and examiner limitations. The Coma Recovery Scale–Revised improved standardized behavioral assessment, and clinical guidelines recommend repeated standardized evaluations, treatment of confounds, and optimization of arousal (Giacino et al., 2004, 2018).

Multimodal assessment materially improves endpoint ascertainment. In a multicenter 2024 study, task-based fMRI or EEG detected cognitive motor dissociation in 60 of 241 participants (25%) who showed no observable response to verbal commands (Bodien et al., 2024). The result supports the use of endpoint intervals and serial multimodal adjudication beyond the last overt response, while subsequent bridge work connects command following to valence and unity.

A 2026 multicentre multimodal investigation combining behavioral assessment with neuroimaging, electrophysiology, and machine-learning markers further supports a plural endpoint-assessment strategy in disorders of consciousness (Manasova et al., 2026). The study strengthens that strategy and shows why the final qualifying conscious boundary should be estimated from serial, multimodal evidence. Phenomenal valence and PTVC remain separate targets for measurement.

10.9.2 Perturbational and complexity measures

The Perturbational Complexity Index was developed to measure complex causal responses to transcranial magnetic stimulation independently of sensory processing and overt behavior (Casali et al., 2013). Related EEG and complexity measures can help discriminate conscious level in some contexts. Their validated role is endpoint and conscious-level ascertainment; cardinal phenomenal-exposure weighting remains a separate promotion target.

In PTVC studies, consciousness measures have bounded roles:

- adjudicate whether a stream may continue beyond behavioral silence;
- define endpoint intervals;
- stratify confidence in self-report or command-following channels;
- serve as sensitivity variables for exposure weighting;
- detect transitions that require separate routing rules.

10.9.3 Serial assessment and endpoint intervals

Because arousal fluctuates, one negative examination should not determine the endpoint. A mature protocol combines serial behavior, task EEG or fMRI where feasible, spontaneous EEG, medication and metabolic context, clinician assessment, and timing of cardiorespiratory or neurologic events. The output should be an endpoint posterior or interval, not false precision.

For the truth-level strict-pre-endpoint ledger, report the endpoint-uncertainty image

$$\left\{L_i^{\mathrm{phen}}(t-): t \in [T_{i,\eta}^{\mathrm{lo}}, T_{i,\eta}^{\mathrm{hi}}]\right\}.$$

Alternatively, integrate over an endpoint distribution. If conclusions change materially across plausible endpoint times, the result is endpoint-sensitive and should not be promoted as strong closure evidence.

10.9.4 Stream unity and difficult cases

Ordinary wakeful continuity is already scientifically challenging; fission, fusion, copying, split-brain phenomena, severe dissociation, and restoration add ownership problems. The initial reference class should be narrow and transparent. Frontier cases should be used to test the theory's coherence, not to rescue or destroy its first empirical branch.

An event-token provenance rule can prevent duplication: every phenomenal exposure token is owned once, transferred under declared continuation rules, and never canceled against an independently owned predecessor merely because the algebra permits it.

10.9.5 Ethical priority

All studies in this section are governed by the controlling safety, non-coercion, and clinical-governance gate in §9.4. Cognitive-motor-dissociation findings heighten the duty to presume uncertainty, communicate respectfully, and begin with low-burden observation and safe nonterminal calibration.

10.9.6 Comparative and prelinguistic affect program

If terminal balance is a substrate-independent law of qualifying conscious streams, its prerequisite architecture should not depend on adult language, explicit justice concepts, or one species' cultural narratives. Comparative affect therefore supplies a demanding transport program. Unexpected sucrose induced dopamine-dependent positive emotion-like judgment bias in bumblebees (Solvi et al., 2016). Bumblebees also made motivational trade-offs involving noxious heat and reward, with behavior consistent with modulation rather than a rigid reflex (Gibbons et al., 2022). Octopuses showed conditioned place avoidance after noxious stimulation, preference for a context associated with local anesthetic relief, and directed wound grooming (Crook, 2021).

A bearer gate should therefore be multidimensional rather than species-labeled. Birch et al. (2020) distinguish perceptual richness, evaluative richness, integration at a time, integration across time, and self-consciousness; PTVC particularly requires evidence relevant to evaluative richness and the two forms of integration, because a temporally extended signed ledger cannot be assigned from nociception or task success alone.

In carrion crows, pallial neuronal activity tracked reported perception after stimulus offset in a design separating subjective report from physical input (Nieder et al., 2020). A systematic review of decapod crustaceans organized evidence under eight neural and cognitive-behavioral criteria, illustrating how convergent profiles can replace a single anthropocentric threshold (Crump et al., 2022).

For each proposed bearer, the protocol should preregister the dimensions assessed, channel-specific evidence, temporal-integration interval, false-negative risks, and the rule for qualification, abstention, or exclusion. No single assay assigns stream status; agreement across behavior, learning, relief, neural dynamics, and temporal integration progressively narrows the eligible reference class.

These results justify careful study of valenced state, relief, choice, and compensation in nonhuman systems; they do not, alone, settle consciousness, commensurability, stream boundaries, or lifetime closure. The program should therefore preregister a graded qualifying-stream criterion, use species-appropriate positive and negative channels, and test event kernels, cross-sign calibration, multiscale persistence, and residual-sensitive choice without anthropomorphic language. Prelinguistic human research can provide an additional bridge using preference, soothing, nociception, autonomic recovery, and neural measures under stringent ethical limits.

The comparative discriminator is conservation of *structure*, not equality of coefficients. A shared completion-potential relation, sign-symmetric causal compensation, or residual-sufficiency pattern across distantly related systems would be harder to explain by common religious belief or explicit just-world cognition. Species-specific homeostasis, ecology, and experimental demand remain strong rivals. The research value lies in separating culturally learned narratives from more deeply conserved regulation.

Graded bearer criterion. Record evaluative valence discrimination, temporal integration, learning/generalization, unified access or integration, persistence/continuation, ownership/routing evidence, and measurement reliability. For each dimension report evidence, false-negative control, transport status, and uncertainty; predeclare qualification, branch-conditional inclusion, Model-ambiguous, and NAE thresholds. Conserved relations, rather than identical coefficients or anthropomorphic interpretation, are the transport target.

10.10 Cultural recurrence, religion, fairness cognition, and personal experience

Operational scope. Cultural and religious recurrence, fairness cognition, and personal narratives document persistent human intuitions about balance, restoration, and consequence. They establish human salience, generate comparative and mechanism hypotheses, and supply a globally diverse rival-model program. Their PTVC-specific scientific value comes from testing whether prospectively specified relations persist in nonverbal, cross-cultural, and method-diverse evidence after belief, framing, transmission, motivated reasoning, memory, comfort, cooperation, just-world belief, and selection are measured.

10.10.1 Why recurrence matters

Ideas of moral balance, cosmic justice, karmic consequence, final judgment, reciprocity, and restoration recur across societies and religious traditions. This recurrence matters in three constructive ways. It shows that human communities repeatedly encounter balance as an organizing intuition about life; it supplies cultural variation for studying how justice concepts influence reports, expectations, coping, and cooperation; and it makes possible strong comparative tests that distinguish culturally transmitted interpretation from a deeper, substrate-independent signature.

10.10.2 Fairness behavior across cultures

Economic games show that narrow self-interest fails as a complete description across diverse small-scale societies, but the form and degree of prosociality vary. In studies of 15 societies, market integration and local returns to cooperation explained substantial group-level variation; later work associated market integration, community size, and participation in world religions with aspects of fairness and punishment (Henrich et al., 2005, 2010).

Developmental research likewise shows both recurrence and variation. In seven societies, disadvantageous inequity aversion emerged broadly by middle childhood, whereas advantageous inequity aversion appeared later and in fewer populations (Blake et al., 2015). This pattern argues against both extremes: fairness cognition is neither culturally absent nor a single invariant rule.

10.10.2.1 Developmental and comparative boundary

Fairness expectations appear before mature religious or legal instruction. Fifteen-month-old infants have shown sensitivity to unequal distributions together with individual differences related to sharing (Schmidt & Sommerville, 2011), and 19- to 21-month-old infants have looked longer at unequal allocations in paradigms designed to test fairness expectations (Sloane et al., 2012). These findings make early-developing social expectations a serious part of the recurrence map and motivate nonverbal designs.

Comparative boundary evidence sharpens the research program. An individual-participant-data meta-analysis of accept/reject inequity paradigms reported little reliable inequity aversion across the tested nonhuman animals and tasks (Ritov et al., 2024). That result identifies a productive boundary question: which components—expectation, sharing, disadvantageous inequity response, partner choice, reciprocity, punishment, or terminal balance—transport across development, culture, and species? Early human recurrence may arise through evolved social cognition, learning preparedness, cultural input, or a combination of them; a separately measured completion signature can determine whether any deeper relation accompanies those pathways.

10.10.3 Moralizing gods and supernatural punishment

Cross-cultural work has linked beliefs in knowledgeable, punitive moralizing gods with generosity toward distant co-religionists (Purzycki et al., 2016). A study across 15 field sites found a broad tendency to attribute moral concern to local deities (Purzycki et al., 2022). Such findings support theories in which supernatural monitoring and punishment contribute to cooperation.

These findings permit a direct comparison between two live possibilities: societies may preserve balance concepts because they regulate behavior, coordinate groups, reduce uncertainty, and provide meaning, while a deeper terminal relation could produce additional signatures that persist beyond those social functions. Prospective cross-cultural tests can distinguish the explanatory reach of each account.

10.10.4 Karma and ethical causation

Belief in karma explicitly represents moral actions as producing congruent outcomes across time and, in many traditions, across lives. In culturally diverse samples totaling 8,996 participants, karmic belief was associated with but not reducible to belief in a just world, moralizing-god belief, religious participation, or culture (White et al., 2019b). Experimental and cross-cultural work suggests that karmic beliefs can shift prosocial behavior and revenge, but can also increase blame or derogation of victims (White et al., 2019a; Goyal & Miller, 2023; White & Willard, 2024).

This is scientifically important because it yields measurable moderators. If PTVC reports or interpretations vary with prior justice belief, karma belief, religiosity, or exposure to fairness framing, those effects may reflect observation and expectation rather than the latent law.

10.10.5 Just-world cognition as a mandatory rival

Just-world theory proposes that people are motivated to see outcomes as deserved and use strategies to manage threats posed by innocent suffering (Hafer & Bègue, 2005). Adults can infer moral causation between unrelated conduct and outcomes, especially when the pairing fits deservingness expectations (Callan et al., 2006). Including just-world cognition as a first-class rival makes the cultural-evidence program more informative and credible.

The appropriate response is not to suppress cultural evidence. It is to test it. A preregistered cross-cultural corpus and experiment program can measure:

- prevalence and semantics of terminal-balance motifs;
- historical dependence and borrowing among traditions;
- correlation with social institutions, market integration, mortality salience, and uncertainty;
- effects of justice framing on affect reports and memory;
- whether PTVC-specific predictions persist after belief and demand characteristics are controlled.

10.10.6 Personal experience and anecdote

People often report that hardship was followed by relief, that suffering deepened later joy, that a life “balanced out,” or that endings brought peace. Such reports can generate hypotheses, identify constructs, and guide qualitative work. They are vulnerable to selection, memory reconstruction, peak-end weighting, narrative coherence, survivorship, and confirmation bias.

A high-quality project invites personal testimony as cultural and motivational evidence and then tests its proposed moderators prospectively. It preregisters coding, includes disconfirming narratives, separates prospective from retrospective reports, and tests whether narrative balance predicts independently measured trajectories.

10.10.7 Testable cultural and cognitive evidence family

Society, religion, and personal experience repeatedly organize meaning around fairness, compensation, restoration, and balance. That recurrence makes PTVC culturally consequential and supplies testable psychological moderators, but it does not independently support the terminal relation. Independent phenomenal and endpoint studies must determine whether any registered PTVC relation survives cultural transmission, memory, framing, comfort, cooperation, and just-world rivals.

10.11 What existing evidence supports and what discriminates next

10.11.1 The convergent pattern

The adjacent literatures define the following testable research program:

Table 3. Convergent Evidence Families and Promotion Gates

Evidence family	Established pattern	PTVC-relevant contribution	Leading comparator / dependency cluster	Promotion bridge	Next discriminating test
Affective representation	Valence/arousal differ; positive and negative affect can co-occur	Supports signed, bipolar, or vector candidate ledgers	Measurement model; DC-MEAS	Cardinal units, neutral origin, exposure	Cross-sign calibration with held-out transport
Adaptation	Responses often attenuate, with event/person heterogeneity	Makes compensatory kernels estimable	Ordinary accommodation; DC-REG	Local kernel to complete ledger	Frozen horizon-area and heterogeneity tests
Opponent regulation	Relief, after-reaction, tolerance, and history dependence occur	Provides a signed-compensation mechanism class	Opponent/homeostatic regulation; DC-REG	Exact area and cross-scale transport	Estimate full response areas under equal-budget rivals
Allostasis/control	Organisms regulate prospectively under constraints	Motivates horizon- and reserve-sensitive routes	Predictive control; DC-REG	Control signal to phenomenal value	Held-out residual contribution beyond prognosis/frailty
Reward science	Liking, wanting, learning, and prediction error dissociate	Types proxies and rigorous comparators	Computational mood/RL; DC-REWARD	Phenomenal sign and magnitude	Perturb channels separately; freeze scalarization
Lifespan motivation	Perceived horizon changes emotional goals	Offers safe horizon manipulations	Salience and goal regulation; DC-HORIZON	Terminal-boundary-specific sensitivity	Cross horizon with calibrated residual and rivals
Terminal decline	Multiple functions change with time to death	Validates endpoint-aligned coordinates	Disease/frailty/selection; DC-TERM	Complete signed-ledger bridge	Multivariate endpoint-aligned residual analysis
Palliative trajectories	Symptoms, function, care, and affect are heterogeneous	Supplies clinical covariates and endpoint infrastructure	Treatment/care dynamics; DC-TERM	Low-burden latent valence/exposure	Joint care–missingness–endpoint model
Consciousness science	Behavior can miss covert cognition; serial multimodal tests help	Supports endpoint intervals and NAE rules	Observation failure; DC-OBS	Bearer, valence, exposure, routing	Prospective multimethod endpoint validation

Evidence family	Established pattern	PTVC-relevant contribution	Leading comparator / dependency cluster	Promotion bridge	Next discriminating test
Cultural/moral recurrence	Fairness and balance narratives recur with variation	Generates moderators and cross-cultural predictions	Just-world belief, norms, transmission; DC-CULT	Physical-law differential	Preregister cognitive-rival and nonverbal transport tests
Statistical methods	Equivalence, mixtures, proper scores, and partial identification are mature	Makes band and atom tests executable	Model misspecification; DC-INF	Validated observation model	End-to-end false-support and recovery suite
Robust perfect adaptation	Specified biological variables can return exactly to set points	Provides an exactness comparator architecture	Integral feedback; DC-REG	Phenomenal carrier and terminal-area target	Differentiate output restoration from CUT-1
Neural valence channels	Positive/negative signals can be partly distinct and interactive	Supports typed bipolar calibration	Salience/arousal/learning; DC-NEURAL	Phenomenal sign, unit, and origin	Cross-sign perturbation and multimethod transport
Memory reconstruction	Remembered affect can differ systematically from experience	Provides estimable reconstruction kernels	Peak-end/fading-affect bias; DC-MEM	Prospective experience coverage	Test residual fingerprints after memory adjustment
Terminal observability	Responsiveness can fluctuate or reappear	Motivates nonmonotone availability models	Delirium/device/care; DC-OBS	Latent state and endpoint interval	Event-triggered multimodal ascertainment
Comparative affect	Behavioral/physiological channels partially dissociate	Enables graded nonverbal bearer testing	Homeostasis/ecology; DC-COMP	Qualification, ownership, commensurability	Conserved-relation transport with welfare controls
Direct PTVC adjudication	The complete conjunction is now formally specified	Converts a broad conjecture into an executable target	All dependency clusters	Validated complete-stream ledger plus endpoint bridge	Two-stage generator recovery, then gate-valid terminal study

Dependency-cluster key. DC-MEAS = shared measurement and instrument dependencies; DC-REG = adaptation and regulatory-control dependencies; DC-REWARD = reward-learning and computational-mood dependencies; DC-HORIZON = time-horizon and salience dependencies; DC-TERM = disease, care, frailty, and terminal-process dependencies; DC-OBS = observation, availability, and ascertainment dependencies; DC-CULT = cultural transmission and normative-cognition dependencies; DC-INF = inferential and model-specification dependencies; DC-NEURAL = neural salience, arousal, and learning dependencies; DC-MEM = memory-reconstruction dependencies; DC-COMP = comparative-affect, ecology, and welfare-proxy dependencies.

Each literature contributes a distinct piece with a named promotion gate. This structure avoids treating unlike findings as interchangeable votes and identifies how the program can accumulate dependency-aware evidence.

10.11.2 Current evidence-supported conclusion

The evidence-supported synthesis is:

> PTVC is a formally specified and conditionally testable candidate boundary hypothesis. Existing science establishes relevant representational, regulatory, horizon, terminal, and measurement phenomena; supplies multiple datasets and experimental paradigms; and makes

a progressive test program feasible. The next stage integrates these assets into calibrated terminal-band, atom-versus-diffuse, and route-discrimination studies.

Nontrivial consistency class (formal coherence). Let $I(f) := \int f \, \mathrm{d}\mu$ on $L^1(\mu)$, with I a nonzero continuous functional. Then $\ker I$ is a closed codimension-one subspace. If the carrier contains disjoint measurable A and B with $0 < \mu(A), \mu(B) < \infty$, the following c-indexed family lies in $\ker I$ and has nonzero positive and negative Jordan masses; the mixed-sign family is a subset of the kernel, not itself a hyperplane.

$$\begin{aligned} f_{A,B,c} &:= c\mathbf{1}_A - c\frac{\mu(A)}{\mu(B)}\mathbf{1}_B, \qquad c > 0, \\ \int f_{A,B,c} \, \mathrm{d}\mu &= 0, \\ \int \left(f_{A,B,c}\right)_+ \mathrm{d}\mu &= \int \left(f_{A,B,c}\right)_- \mathrm{d}\mu = c\mu(A) > 0. \end{aligned}$$

This infinite family witnesses joint satisfiability of the finite additive axioms under the stated carrier assumptions and exhibits a nondegenerate mixed-sign subclass. It is neither a mechanism, a probability model, nor empirical evidence. Mechanism, probability-law, and empirical promotion are addressed only by the separately defined scientific gates.

Exact-atom satisfiability witness. On a two-token finite carrier, assign a positive atom of magnitude a to one owned event and a negative atom of the same magnitude to the other, with $a > 0$. The terminal residual is then deterministically zero, so its terminal law has an atom of mass one at zero while both Jordan masses are nonzero. This witness establishes logical satisfiability; the separately defined scientific gates evaluate mechanism and empirical support.

This synthesis warrants a coordinated preregistered research program because every possible outcome—support, principled refinement, or clear counterevidence—would produce reusable advances in measurement, endpoint science, and model discrimination.

10.11.3 How independent convergence accumulates

Evidence families share dependencies. HRS, ELSA, SOEP, and SHARE are related aging-study infrastructures. Several hedonic-adaptation analyses reuse the same panels. Fairness and religious-belief measures share cultural history. Psychophysiological channels share arousal and medication confounding. Counting papers would exaggerate independence.

A formal synthesis should therefore use a dependency graph. Let nodes be datasets, measures, mechanisms, and studies; let edges represent shared samples, instruments, analysts, preprocessing, or cultural ancestry. Confidence should grow most from results that traverse different paths through this graph and have biases predicted in different directions.

10.11.4 Results that would narrow or reject a registered PTVC claim

The program gains credibility by specifying results that would count against branches:

- failure to establish a stable sign or neutral origin weakens the additive ledger branch;
- persistent cross-sign noncomparability weakens scalar PTVC but may leave vector-valued alternatives;
- event kernels with consistently nonzero net areas weaken local balance mechanisms;
- model recovery showing observational indistinguishability blocks confirmatory claims;

- instability of terminal concentration under missing-not-at-random or endpoint-sensitivity analyses narrows the corresponding support claim;
- fixed-kernel rivals that predict as well as PTVC remove differential evidence;
- a diffuse residual model that outpredicts an atom model weakens exactness;
- closure confined to retrospective reports favors remembered-utility rivals;
- dependence on justice framing favors cognitive-demand explanations;
- outcome-driven exclusion of nonclosing streams invalidates universality tests.

10.11.5 Primary differential-evidence target

The primary near-term differential-evidence target is a transported completion-geometry fingerprint rather than another correlation with "balance." Its components are:

1. a prospectively calibrated signed residual that is not reducible to current mood, health, or recent events;
2. an achieved cancellation precision compatible with the prospectively bounded information available to any local controller;
3. one completion potential estimated without endpoint-outcome training;
4. predicted drift, action odds, and endpoint support derived from that same potential;
5. preserved calibration across resolution, dataset, culture, endpoint cause, endpoint predictability, and measurement channel;
6. superior held-out joint score against computational mood, homeostatic reinforcement learning, opponent-process, conditioned-bridge, selection, memory, censoring, and observation-floor rivals;
7. complete-history cut-point contrasts in which an independently measured suffix through the adjudicated endpoint cancels accrued past area, together with bounded-episode analogues used only for local mechanism calibration.

The shared parameterization creates a cross-module restriction. Every candidate and rival must recover the same frozen completion geometry across transitions, actions, complete-history cuts, local calibration episodes, interventions, endpoints, resolutions, and channels without module-specific refitting. A model that satisfies this restriction is an empirically adequate candidate implementation of the shared structure; a model that succeeds only after module-specific refitting does not explain the joint fingerprint.

10.11.6 Candidate-law invariance and decision battery

Quantitative invariance and retirement rules. The preregistration matrix records each environment family, transported parameter or function, equivalence/discrepancy margin, dependence model, family-wise false-support target, minimum detectable violation, leave-environment-out score, and route-retirement threshold. A gate-valid incompatible environment refutes the strict-universal transported claim for the registered class; it may support only a prospectively declared narrower class. Post hoc pooling cannot erase that failure.

A substrate-general candidate boundary hypothesis predicts invariance under changes that alter surface implementation without altering the qualifying stream. The primary prospectively specified transport tests therefore cross cultural and justice beliefs, demographic groups, endpoint causes, anticipated and unexpected timing, valence modalities, care contexts, sampling resolutions, and, where ethical and scientifically justified, prelinguistic or comparative systems. Effect sizes may vary because observation and mechanism vary; the predicted relation among residual, reachability, action, and endpoint must remain coherent.

This battery gives cultural and religious recurrence a constructive role. If a signature appears only after justice framing, belief priming, retrospective life review, or culturally specific narrative prompts, a cognitive-demand or memory rival is favored. If the same formal relation appears in indirect behavior, prospective experience sampling, physiology-typed reserve channels, and nonverbal systems, explicit belief becomes less adequate as a complete explanation. Cross-domain invariance is thus the route from a suggestive cultural pattern to a serious test of substrate-independent structure.

Candidate-law differential. The strongest prospectively specified fingerprint is conjunctive: a residual that shows prospectively specified incremental value beyond current state and, in a separate test, adequate compression relative to the registered richer-history model; one completion potential transported across modules; invariance across endpoint surprise, resolution, substrate, and culture; perturbation-compatible compensation; and failure of mechanism-specific rivals under equal information and tuning budgets. This bundle concentrates the paper's most informative cross-domain predictions and is designed to yield results less expected under ordinary regulation alone.

11. Conclusion and research agenda

11.1 Scientific synthesis

PTVC is formally specified and progressively testable. The present model converts a broad fairness conjecture into linked representation, origin, endpoint, reachability, observation, route, and rival tests. Finite-resolution results are positive when their full typed compatibility objects lie within independently calibrated scientific acceptance sets; sharper measurement and bridges increase resolving power without redefining success. The adjacent literatures supply a broad multidisciplinary base of measurement resources, comparator mechanisms, candidate datasets, near-term discriminators, and informative adverse outcomes. Together these assets make direct terminal-law adjudication a concrete research objective and support immediate, testable lines of inquiry.

Why the hypothesis merits serious attention. The reviewed sciences reveal a striking combination: valence has signed, partly separable channels; affective systems adapt, oppose, anticipate, and regulate across time; exact regulation is biologically realizable in specified variables; perceived horizons reorganize action; terminal trajectories possess measurable structure; and cultures repeatedly return to ideas of balance and restoration. These findings contribute different things: premises, measurement tools, candidate mechanisms, comparators, contextual plausibility, and human salience. PTVC asks whether one compact terminal relation adds a transportable structure beyond them. Its conceptual economy, multidisciplinary reach, and decisive testability make the program a strong candidate for sustained academic discussion, collaborative study, and staged research investment.

Existing datasets can immediately advance PTVC's representation, route, measurement, endpoint, and rival programs, while direct complete-stream adjudication is the culminating objective enabled by a validated terminal ledger and endpoint bridge. The model's immediate scientific value is its disciplined separation of these claims and its ability to produce favorable, adverse, indeterminate, model-ambiguous, or No adequate estimate outcomes without redefining the hypothesis after inspection.

What the combined evidence supports as premises and test opportunities. First, affective value is multidimensional, dynamically regulated, and measurable with improving but imperfect longitudinal methods. Second, adaptation, opponent responses, predictive regulation, memory reconstruction, and observation change supply well-developed ordinary comparators rather than a single generic baseline. Third, some biological systems implement opponent, predictive, and integral-feedback regulation, and particular molecular models achieve robust set-point adaptation. These findings demonstrate that exact and compensatory control are biologically realizable and supply concrete architectures for testing phenomenal-area compensation. Fourth, completion-potential transport and past–future ledger complementarity supply differential finite-class tests that those comparators must reproduce under the same observation process. Fifth, several human cohorts and measured domains show time-to-death changes over months to years, with substantial outcome, cohort, and observation heterogeneity. Sixth, progressively tighter bands, method-diverse channels, endpoint adjudication, and recovery of the zero-atom functional across laws with a positive atom at zero, laws with no atom at zero that are absolutely continuous near zero, and no-zero-atom laws that are locally discrete, singular-continuous, or mixed provide a disciplined path from empirical compatibility to stronger structural adjudication.

Implementation-pressure dashboard. The conditional zero-atom screen shows that exact zero has conditional probability zero whenever the declared at-risk law assigns no atom to zero; Proposition 1 gives the substantive independent positive-density endpoint restriction, while Corollary 1.1 supplies its graded dependence bound; Proposition 2 (RCH-1) places the residual inside the future directional-capacity envelope; Proposition 3 localizes structure to endpoint support or hazard-weighted occupation; Proposition 4 restricts nondegenerate diffusion components; Proposition 5 (CUT-1) requires independently measured future area to complement accrued past area at every admissible cut; Proposition C.4 identifies the joint ratio-residual signature of one-to-one balance; and A2.22 (INF-1) lower-bounds the information available to any local compensator. These linked results turn one broad conjecture into distinct reachability, endpoint, path, conservation, and information pressure tests under declared representation and observation assumptions.

At the program's center, the primary finite signed-measure branch asks whether every qualifying complete conscious stream has equal cumulative positive and negative phenomenal magnitude at its truth-level pre-endpoint boundary, while permitting unequal gross valence variation and radically different lives.

Current empirical position. PTVC enters testing with methodological resources, candidate mechanisms, serious rivals, and relevant datasets from affect measurement, adaptation, opponent regulation, robust biological control, computational mood, horizon-sensitive behavior, terminal trajectories, consciousness assessment, and fairness cognition. After each named resource passes a dataset-specific access, release-version, variable, calibration, consent/governance, missingness, preprocessing, and independent-reproduction audit, high-volume task data and longitudinal cohorts can support model-recovery and terminal-regime studies. Preregistered nonterminal experiments can then test local compensation, residual incremental value, residual compression sufficiency, causal locus, and information-budget fingerprints. These bounded studies calibrate route mechanisms; they adjudicate complete-stream closure only when the suffix reaches the independently adjudicated stream endpoint or is identified through a validated terminal-remainder bridge.

The framework's central contribution is its separation of objects and burdens: latent phenomenal value, stream routing, endpoint ontology, candidate implementation, measured estimand, estimator, observation process, and rival model. Mechanism-neutral results involving non-atomic endpoints, directional reachability, and compensator-weighted occupation provide strong pressure tests. The reserve-coupled process contributes a detailed, falsifiable route signature, but no unique implementation follows from PTVC alone.

11.2 Foundational targets and research priorities

Research priorities. The program now has a clear sequence of tractable targets: establish or bound the phenomenal-value representation, neutral origin, sign, cross-sign calibration, scalar or bipolar structure, stream ownership and routing, truth-level endpoint, finite-ledger domain, measured-to-latent bridge, observation process, and implementation route. Each solved component increases the precision and power of the remaining tests.

Mind-source research horizon. If PTVC is established for a declared reference class, exactness would provide a powerful structural fingerprint capable of constraining theories of which systems are conscious, where experience is located, why qualifying dynamics are phenomenal, and how a substrate relates to experience. Identifying the anatomical or computational source remains a complementary research program; PTVC would narrow that search without prejudging which candidate ultimately succeeds.

Maturity map. Every output is labeled formal identity, conditional theorem, simulation-validated method, measured local mechanism, endpoint-aligned result, or complete-stream result. Promotion requires the next gate; no category is silently upgraded.

11.3 Ordered research sequence

A productive research sequence is now available:

1. formal verification of the core definitions and conditional propositions;
2. psychophysical work on sign, neutral origin, cross-sign calibration, scalarity, and mixed valence;
3. simulator recovery and open-set countermodel tournaments;
4. safe causal-locus and information-separation experiments;
5. route-state measurement and nonterminal perturbation studies;
6. prospective endpoint studies after the measurement and recovery gates mature;
7. independent replication, citation audit, and specialist review across probability, measurement, neuroscience, control, ethics, and philosophy.

Proposed first-study protocol template. No dataset or protocol is called precommitted until a timestamped registration identifier exists. Select the cohort only after a documented access and variable audit. The registration must name the dataset release; endpoint and cause handling; measured estimand; origin/scalarity status; observation and missingness model; primary target and serious rival; one multivariate decision rule; smallest meaningful margin; transport target; adverse-result and abstention wording; code environment; independent reproduction role; and claim ceiling.

Use a two-stage design. Stage 1 applies safe randomized signed doses and random cut points to estimate local episode complementarity and one completion potential on training modules. Stage 2 freezes that potential and predicts a held-out future-area channel and its interaction with prospectively frozen endpoint or horizon surprise. Score PTVC, integral feedback, opponent kernels, flexible state-space models, conditioned bridges, and selection through the same observation model and tuning budget. The recommended first claim ceiling is successful generator recovery plus transported local-mechanism calibration, not strict complete-stream PTVC. The flagship endpoint study should be preregistered as a balanced collaboration, with at least one investigator on record as skeptical of PTVC co-signing the design, analysis, and interpretation.

Proposed first study (registration required before execution). Stage 1 uses an access-audited nonterminal dataset to calibrate measurement and recover candidate/rival generators; Stage 2 freezes the representation, observation kernel, completion potential, rival bank, primary held-out transport estimand, and go/no-go criteria before scoring a disjoint test set. Registry ID, timestamp, data/access state, and final thresholds enter the public record at preregistration.

11.4 Integrated research infrastructure

Reproducibility rule. Apply §10.1's source-location and evidence-posture requirements to every analysis in this section. Executed empirical or computational outputs are claimed only when the corresponding data, code, and protocol record are actually available.

Document architecture. Appendix A organizes operational methods and frontier theorem targets; Appendix B supplies safe mechanism-localization experiments; Appendix C develops the philosophical and scope analysis; Section 11 contains the progressive empirical program. Evidence sources and formal dependencies are stated at their controlling locations rather than repeated in additional appendices.

The structure creates publishable paths in representation, measurement, endpoints, simulation, and mechanism tests, all advancing one formal target with explicit claim ceilings.

11.5 Immediate analyses in existing data

11.5.1 Data resources

Several harmonizable longitudinal studies can support near-term work.

Health and Retirement Study (HRS). A nationally representative U.S. panel of adults over 50, fielded biennially since 1992, with health, cognition, economics, family, psychosocial measures, biomarkers, and administrative mortality linkages. The HRS network harmonizes with international sister studies (Sonnega et al., 2014).

English Longitudinal Study of Ageing (ELSA). A representative English cohort aged 50 and older with repeated interviews, health visits, biomarkers, psychosocial measures, and end-of-life interviews with informants (Steptoe et al., 2013).

German Socio-Economic Panel (SOEP). An annual household panel beginning in 1984 with life satisfaction, health, work, income, family, and mortality information; it underlies influential terminal-decline analyses (Wagner et al., 2007).

Survey of Health, Ageing and Retirement in Europe (SHARE). A cross-national panel covering health, socioeconomic conditions, family networks, life history, and end-of-life interviews across Europe and Israel (Börsch-Supan et al., 2013).

These datasets do not contain a validated lifetime phenomenal ledger. They can still test reflection, timing, selection, response shift, horizon, and competing terminal shapes.

Great Brain Experiment / open computational-mood resource. The Dryad release associated with the Great Brain Experiment contains 91,058 task completions from 47,067 adults, approximately 2.7 million choices, 1.1 million momentary happiness ratings, and more than 6,000 participant-hours; a depression-questionnaire subset includes 1,858 participants (Rutledge, 2023; 2023 repository version, originally published 2021). The short gamified task is not a terminal cohort, but it is immediately useful for event-kernel estimation, expectation and prediction-error rivals, sampling-resolution stress tests, cross-participant heterogeneity, and blinded model recovery at a scale unavailable in most laboratory studies.

This resource should be used first as an open benchmark. Freeze train, validation, and held-out participant splits; reproduce the published computational-mood model; then ask whether a prospectively constructed accumulated residual improves prediction without outcome-trained recentering. Release the exact split, preprocessing, abstention rule, and negative-control residuals so any apparent gain can be independently audited.

11.5.2 Retrospective terminal-cohort reflection screen

Estimand and sampling frame. Define the target population, risk-set entry, delayed-entry mechanism, endpoint categories, observation process, and estimand before alignment by time to death. Entry uses left-truncated risk sets. Death, competing events, dropout, proxy substitution, and observation cessation remain distinct. Informative visit and retention processes require a joint longitudinal–endpoint model, inverse-intensity weighting under stated assumptions, or partial-identification bounds; the decedent sample is not treated as a random sample of complete streams.

Align repeated observations by the terminal-distance coordinate and fit the mixed model

$$\begin{aligned} d_{it} &:= T_i^{\text{death}} - t, \\ O_{it} &= \beta_0 + s_a(\text{age}_{it}) + s_d(d_{it}) + X_{it}^{\top}\beta + b_{0i} + b_{1i}d_{it} + \varepsilon_{it}. \end{aligned}$$

Replacing T_i^{death} by $T_i^{\star}$ requires independent adjudication that death time equals the endpoint of the qualifying stream under the registered endpoint bridge.

Because $\text{age}_{it} + d_{it} = A_i^{\text{death}}$, unrestricted age and time-to-death smooths are not separately identified. Preregister constrained/centered smooths and penalties with explicit age-at-death and cohort structure, or fit age-only and time-to-death models before a constrained joint model. Verify parameter recovery under the actual visit cadence, mortality selection, and missingness by simulation and report sensitivity to alternative constraints.

Compare models indexed by chronological age, time to death, disease stage, functional reserve, and the observation clock. Use random change points where data support them. Model positive affect, negative affect, life satisfaction, depressive symptoms, pain, social connection, and cognitive function separately before any scalar composite.

Primary questions:

1. Does time to death improve held-out prediction beyond age, health, and care setting?
2. Are trajectory shapes reproducible across datasets and countries?
3. Which domains show change-point, smooth, abrupt, or no terminal shift?
4. Does response-shift evidence increase near death?
5. How much apparent terminal concentration is explained by attrition, proxy substitution, or scale floor/ceiling?

Interpretation ceiling: this screen identifies terminal-regime structure, not a completed phenomenal ledger.

11.5.3 Ledger-proxy endpoint-hazard study

Use the global convention: $\alpha_i^e(t \mid \mathcal{H}_{i,t-})$ is the cause-specific hazard, $\lambda_i^e(t) := R_i^{\text{risk}}(t)\alpha_i^e(t \mid \mathcal{H}_{i,t-})$ is the predictable intensity, and $\mathrm{d}\Lambda_i^e(t) = \lambda_i^e(t)\,\mathrm{d}t$ only on the absolutely continuous branch.

Cause-specific hazard, cumulative incidence or subdistribution estimands, and interventional endpoint risk are distinct objects. Each analysis names one target, its risk set and competing-event treatment, and the assumptions required to move among them; no hazard coefficient is interpreted as a causal risk contrast without an interventional identification argument.

After a nonterminal calibration produces the signed proxy $\tilde{L}_i^{\text{proxy}}(t)$, fit a cause-specific hazard and derive its counting-process intensity using the same endpoint class e and filtration:

$$\lambda_i^e(t) = R_i^{\text{risk}}(t)\,\alpha_i^e(t \mid \mathcal{H}_{i,t-}).$$

Intensity-calibration gate. Define the fitted total terminal-event intensity and cumulative intensity by

$$\hat{\lambda}_i^{\bullet}(t \mid \mathcal{H}_{i,t-}) := \sum_e \hat{\lambda}_i^e(t \mid \mathcal{H}_{i,t-}), \qquad \hat{\Lambda}_i^{\bullet}(t) := \int_0^t \hat{\lambda}_i^{\bullet}(s \mid \mathcal{H}_{i,s-})\,\mathrm{d}s.$$

On the fully observed continuous single-terminal-event branch, a correctly specified fitted law implies unit-exponential reference behavior for $Z_i := \hat{\Lambda}_i^{\bullet}(T_i)$, while endpoint class is calibrated against

$$\hat{\pi}_i^e(T_i) := \frac{\hat{\lambda}_i^e\left(T_i \mid \mathcal{H}_{i,T_i-}\right)}{\hat{\lambda}_i^{\bullet}\left(T_i \mid \mathcal{H}_{i,T_i-}\right)}.$$

Evaluate both diagnostics out of fold and propagate parameter-estimation uncertainty. Use marked rescaling for endpoint class and a theorem-matched terminating or right-censored construction when observation terminates or is censored; do not treat a censored residual as an observed event residual. Failed calibration blocks hazard-residual and endpoint-surprise interpretation under the fitted law, rather than the target hypothesis itself (Brown et al., 2002; Tao et al., 2018; Zhang et al., 2025).

The absolute-value spline tests the sharp route prediction that endpoint hazard is suppressed when the candidate residual is far from the feasible completion region. A signed spline can be added to detect asymmetry. This is a branch-specific test, not a universal consequence of PTVC: a model may close through an endpoint jump, unobserved compensation, or a mechanism unrelated to hazard.

Controls should include age, diagnosis, function, frailty, medication, hospitalization, socioeconomic status, and care setting. Negative controls should include pseudo-endpoints and future-proxy leakage checks. For delayed-entry analyses, define the at-risk indicator to be zero before the registered entry time, so the displayed integral from the origin equals the integral over the observed risk interval. If entry is not encoded in the at-risk indicator, replace the lower integration limit with the registered entry time. Use the registered competing-risk and causal-sensitivity analyses.

11.5.4 Endpoint-level reflection and residual-history analysis

The primary model is endpoint-level ANCOVA or a latent state-space analysis, not raw change on a noisy baseline. Let the registered measured-ledger estimand be $M_{\eta,i}(t)$, linked to observed channels by the frozen observation model. Fit

$$M_{\eta,i}\left(T_{i,\eta}^{\mathrm{adj}}\right) = \beta_0 + \beta_1 M_{\eta,i}(t_0) + \gamma^\top X_i + \varepsilon_i.$$

Latent-ledger interpretation requires the separately validated measured-to-latent bridge.

Measurement error in both latent states, endpoint uncertainty, and terminal missingness are propagated jointly. A raw change score is sensitivity analysis only; its null distribution is calibrated by simulation under the estimated measurement model to expose regression-to-the-mean. Past/future ledger analyses use $P_i^{\mathrm{ledger}}(t_0)$ and $C_i^{\mathrm{future}}(t_0, T)$, with protocol and endpoint indices where measured, rather than overloaded R_i and F_i.

11.5.5 Terminal dispersion

For observed channel value O_{it}, latent state $V_{i,t}^{\mathrm{lat}}$, and distance/status covariates d_{it}, use the exact conditional law of total variance:

$$\mathrm{Var}(O_{it} \mid d_{it}) = \mathrm{Var}\left(\mathbb{E}[O_{it} \mid V_{i,t}^{\mathrm{lat}}, d_{it}] \mid d_{it}\right) + \mathbb{E}[\mathrm{Var}\left(O_{it} \mid V_{i,t}^{\mathrm{lat}}, d_{it}\right) \mid d_{it}].$$

If $O_{it} = V_{i,t}^{\mathrm{lat}} + \varepsilon_{it}$ and $\mathbb{E}\left(\varepsilon_{it} \mid V_{i,t}^{\mathrm{lat}}, d_{it}\right) = 0$, this reduces to

$$\mathrm{Var}(O_{it} \mid d_{it}) = \mathrm{Var}\left(V_{i,t}^{\mathrm{lat}} \mid d_{it}\right) + \mathbb{E}[\mathrm{Var}\left(\varepsilon_{it} \mid V_{i,t}^{\mathrm{lat}}, d_{it}\right) \mid d_{it}].$$

When conditional mean zero fails, report the full decomposition:

$$\mathrm{Var}(O_{it} \mid d_{it}) = \mathrm{Var}\left(V_{i,t}^{\mathrm{lat}} \mid d_{it}\right) + \mathrm{Var}(\varepsilon_{it} \mid d_{it}) + 2\mathrm{Cov}\left(V_{i,t}^{\mathrm{lat}}, \varepsilon_{it} \mid d_{it}\right).$$

Model every term.

11.5.6 Experience-versus-memory analysis

In cohorts with daily affect and retrospective well-being, compare predictive models for later global evaluations:

$$M_i = \alpha + \beta_{\text{mean}}\overline{f}_i + \beta_{\text{peak}} f_i^{\text{peak}} + \beta_{\text{end}} f_i^{\text{end}} + \beta_{\text{dur}} D_i + \varepsilon_i.$$

Mean, peak, end, and duration features arise from the same cadence-dependent sampled path and are therefore treated as one dependent evidence family. The confirmatory analysis uses the held-out tournament or training-only orthogonalization in A1.29, propagates generated-feature uncertainty, and stress-tests cadence, noise, and downsampling.

Then test whether terminal or horizon changes modify the held-out predictive contributions of the experience-derived, peak, end, and duration features. This discriminates among experience-weighting, memory-reconstruction, and narrative accounts under the registered measurement model; it does not by itself identify the latent phenomenal ledger or establish that any one channel is phenomenally privileged.

11.5.7 Cross-dataset invariance

Results should be replicated across HRS, ELSA, SOEP, and SHARE using harmonized variables but dataset-specific measurement models. Cross-national agreement is not automatically independent because studies share designs and harmonization. Use leave-one-country-out and leave-one-dataset-out prediction, and report cultural, policy, and healthcare heterogeneity rather than forcing a common coefficient.

11.5.8 Anticipated-versus-unexpected endpoint study

Construct a prospective endpoint forecast from diagnosis, function, symptoms, care setting, treatment, frailty, and longitudinal physiology without access to terminal-ledger outcomes. At a frozen landmark, partition time and cause into disjoint registered cells and use the same canonical joint surprise score as §6.8:

Use the canonical observed-record SURP-1 definition in §6.8 without modification: the frozen $\chi_{i,\eta}^{\text{end}}$ maps a generic protocol-observed endpoint record $D_{i,\eta}^{\text{end}}$ and the realized record $d_{i,\eta}^{\text{obs}}$ into the same registered cell space, and scoring uses $P_{0,\eta}^{\text{obs}}$.

The cell partition, endpoint cause, competing-risk treatment, tail state, censoring and observation-loss states, common-support rule, coarsened or revised-endpoint probability, and forecast model are fixed before endpoint reveal. A zero assigned probability is model inadequacy rather than an arbitrary clipped score. Repeated landmarks receive prospectively normalized per-stream weights with dependence propagated.

Primary outcomes are: terminal compatibility-set width; residual dispersion; completion-potential calibration; observation availability; and the held-out score of local-horizon, global-conditioned-path, selection, and stream-endpoint models as a function of the frozen surprise score. The key interaction is model-by-surprise. A local controller predicts deterioration when realized time is shorter than the actionable horizon; a global path model predicts better transport; a selection model predicts changes explained by the endpoint intensity; and a stream-endpoint model predicts that independently estimated loss of qualifying experience is more informative than cardiopulmonary time.

The design must not convert unexpected death into an experiment. It uses naturally occurring timing variation, protects care, and keeps all clinical decisions independent of the model. A successful retrospective pilot justifies a prospective multisite study with harmonized endpoint and observation records.

Section 11.5 measurement semantics. Use O for measured outcomes and a separately named latent state. Joint age/time-to-death models retain $\text{age}_{it} + d_{it} = A_i^{\text{death}}$ with centered or constrained smooths, cohort and age-at-death structure, recovery simulation, and separate age-only and time-to-death benchmarks. Variance and memory regressions state their measurement-error premises. VAR-1 is the aligned total-variance identity, and §11.5.8 uses the canonical SURP-1 score with endpoint coarsening/revision rules.

11.6 Prospective experimental program

11.6.1 Program logic

Prospective work should advance through gates. Each gate has a success criterion, failure criterion, and promotion rule. No terminal experiment should be asked to solve representation, measurement, mechanism, and endpoint identification simultaneously.

Gate 1: valence and neutral-origin calibration

Use individually titrated appetitive, aversive, and mixed stimuli. Collect continuous or dense ratings on separate positive and negative channels, bipolar judgments, choice tradeoffs, physiological channels, and confidence. Estimate whether a stable neutral origin exists across tasks and days. Test sign consistency, cross-sign comparability, mixed affect, and reactivity.

Add a balanced factorial conjoint arm in which independently manipulated appetitive and aversive attributes are ordered by choice, indifference, and report. Before either dimension enters a scalar ledger, test weak order, factor independence or single cancellation, double cancellation, and held-out additive representations. Prospectively freeze the factorial design, admissible order reversals, exclusion rules, and test statistic before terminal or compensation outcomes are observed. Finite success supports only the declared design; stable cancellation failure triggers the non-scalar branch.

Success means a preregistered measurement model predicts held-out choices and reports with acceptable invariance. Failure redirects the project to vector or partially ordered valence without claiming scalar PTVC. Promotion requires held-out transport to a second task/day and method family with origin, sign, scalarization, and transformation class frozen.

Gate 2: temporal exposure calibration

Manipulate objective duration, attention, arousal, memory delay, and subjective-time judgments. Compare clock-time, subjective-duration, and state-weighted integration. Use interval reproduction and discrimination only as validation tasks, not direct phenomenal-mass meters.

Success means the declared exposure rule predicts held-out experienced and retrospective evaluations better than reasonable alternatives. Failure means no registered exposure rule transports, or a simpler duration/attention/memory rival matches held-out performance. Promotion requires a second timing manipulation and an independent duration channel with units and quadrature frozen.

Gate 3: finite-window kernel-area restriction

$$\int_0^{\tau_{\max}} f_{ij}\,(u)\,du \in [-K_{\text{event}}, K_{\text{event}}],$$

where K_{event} has integrated-valence units. This is $\tau_{\max}$-window balance. A full-area claim requires compact support or a registered tail bound K_{tail}, giving total tolerance $K_{\text{event}} + K_{\text{tail}}$.

For the normalized linear opponent benchmark in §10.4.1, with $\lambda_a, \lambda_b > 0$ and unit impulse, exact full-area cancellation holds if and only if $\gamma = \lambda_b$. This local mechanism restriction is not whole-stream PTVC.

Success requires the simultaneous finite-window area region and registered tail bound to lie within the preregistered total tolerance on held-out event and sign classes. A gate-valid exceedance is failure; unidentified tails or aliasing yield No adequate estimate. Promotion requires transport to a second modality and defeat of equal-budget kernel rivals.

Gate 4: multiscale invariance

Estimate the ledger at multiple sampling resolutions Δ:

$$\hat{L}_i^{(\Delta)}(T) = \sum_k \hat{f}_{ik}^{(\Delta)}\ \Delta\hat{\mu}_{ik}^{(\Delta)}.$$

Here $\Delta\hat{\mu}_{ik}^{(\Delta)}$ contains calibrated duration, exposure, or quadrature weights. The original constant-Δ sum is the complete regular-grid special case. Compare one-minute, hourly, daily, and reconstructed integrations where feasible while propagating irregular timing, missing intervals, interpolation, and aliasing uncertainty. A genuine path property should be stable after modeled aggregation error; a quantization artifact will change with resolution.

Success requires every registered cross-resolution discrepancy to lie inside the frozen aggregation-error envelope. A gate-valid exceedance is failure; unresolved missingness or aliasing yields No adequate estimate. Promotion requires transport to a second cadence or channel with interpolation and weights frozen.

Gate 5: horizon manipulation

Pair independently validated horizon manipulations with controlled event sequences.

Success means the preregistered horizon manipulation changes future-action or ledger signatures in the predicted direction while sign, exposure, and current-state calibration remain stable. Failure means demand, expectancy, current state, or ordinary planning explains the effect under equal scoring. Promotion requires held-out transport to a second horizon and method family.

Gate 6: natural reserve perturbations

Study clinically occurring, noninduced changes such as acute illness, recovery, sleep loss, rehabilitation, or treatment transitions. Estimate whether a declared reserve variable changes the feasible residual envelope and recovery dynamics. Use target-trial emulation where treatment timing is observational.

Success means prospectively measured reserve changes predict the reserve-normalized residual envelope and recovery beyond current state, care, treatment, and observation intensity. Failure means endpoint, selection, observation, or ordinary regulatory rivals match the held-out data. Promotion requires target-trial sensitivity, negative controls, and transport across at least one independent reserve proxy.

Gate 7: prospective intensive longitudinal cohorts

Recruit participants for long follow-up with EMA, wearables, periodic laboratory calibration, major-event bursts, and consent for data linkage and end-of-life follow-up. The purpose is not to wait decades for one terminal number. The cohort yields event kernels, measurement stability, health transitions, subjective-time changes, and increasingly informative terminal windows.

Before confirmatory terminal inference, run a blinded multisite adjudication-reliability study on synthetic and historical records. Freeze the manual, training set, blind set, time-tolerance metric, cause/status confusion matrix, agreement target, disagreement resolution, revision rule, and endpoint-uncertainty propagation. Report time disagreement, class agreement, return/revision rates, and sensitivity of substantive conclusions to adjudicator uncertainty. Adjudicators and endpoint algorithms are blinded to ledger residuals, band verdicts, and completion-potential outputs. Event classes, interval rules, return logic, and disagreement procedures are frozen before outcome analysis; unavoidable shared inputs are declared and included in endpoint-leakage sensitivity analyses.

Success means one frozen measurement-and-dynamics model predicts held-out event kernels, completion-potential outputs, and transport across sites, channels, and resolutions. Failure means recovery is retrospective, leakage-dependent, or confined to one instrument or site. Promotion to terminal inference requires the registered endpoint, missingness, observation, and adjudication gates to pass.

Gate 8: low-burden terminal observational studies

Only after earlier gates mature, add brief patient-selected reports to existing palliative care workflows, integrate clinical and consciousness assessments, and use caregiver or proxy channels without replacing self-report. Independent ethics and palliative teams govern burden, stopping, and communication.

The preterminal bridge-validation program uses method-diverse positive and negative stimuli, individual neutral anchors, scalar and bipolar branches, item-response and longitudinal-invariance models, at least two independent method families, response-shift perturbations, and held-out sign/origin transport across sites. Calibration samples and sites never see terminal outcomes. Failure to transport within preregistered tolerances narrows the terminal program before enrollment.

Success is limited to endpoint-adjudicated, bridge-valid, low-burden observational evidence whose registered terminal targets and rivals recover in held-out data. Failure or any unresolved bridge, availability, care, or stream-endpoint gate yields No adequate estimate. This gate never authorizes manipulation of terminal timing, care, or burden.

11.6.2 Factorial mechanism-localization experiment

A discriminating nonterminal design crosses five factors:

- valence history: positive-heavy, negative-heavy, balanced;
- actual safe episode-end schedule or prospectively randomized stopping-hazard schedule;
- endpoint information: truthful, withheld, uncertain, or an ethically permissible sham cue;
- action access: choices available versus yoked;
- observability: full state feedback versus partial feedback.

The critical orthogonalization crosses actual schedule with disclosed information. A cue effect with schedule fixed identifies anticipation, expectancy, demand, or information; a schedule effect with information fixed identifies exposure, timing, or route sensitivity. A schedule-by-information interaction is diagnostic rather than automatically PTVC-supportive. Before disclosure, assigned future schedule should not affect current outcomes except through a separately modeled leakage or physical channel.

Predictions distinguish mechanisms. A constrained controller should alter action when residual and horizon information are jointly available. A conditioned-bridge account may show trajectory selection without online residual-sensitive action. An observation-demand account should weaken under indirect outcomes and blinded framing. An ordinary regulator should respond to current state but not the preregistered residual after state is controlled.

Use a simulated or virtual environment with safe affective stimuli; do not imply actual death or moral judgment. Fit the full factorial model, preregister interactions, and validate with model recovery.

11.6.3 Replicate-channel terminal-atom experiment

For a validated nonterminal proxy or eventually a terminal measure, collect two or more conditionally independent replicate channels:

$$O_{i1} = Z_i + \varepsilon_{i1}, \qquad O_{i2} = Z_i + \varepsilon_{i2}.$$

Under a Kotlarski-type identification branch, require the theorem-matched package: conditional mutual independence of the latent target and both replicate errors given frozen covariates (or a precisely stated weaker theorem), calibrated loadings, location or zero-mean normalization, required moments, and nonvanishing relevant characteristic functions. Shared-method, state-dependent, correlated, or endpoint-dependent errors invalidate the branch. Method diversity and analyst separation improve design but do not establish stochastic independence; the experiment must test and sensitivity-analyze the required error structure rather than duplicate questionnaire items alone.

11.6.4 Intervention ethics and information value

For design d, research decision a, state θ, and decision utility u_{dec} in common utility units, define

$$\text{EVI}(d) = \mathbb{E}_{Y\sim p(\cdot|d)}[\sup_{a\in\mathcal{A}(d,Y)} \mathbb{E}_{\theta|Y,d}\{u_{\text{dec}}(a,\theta)\}] - \sup_{a\in\mathcal{A}_0} \mathbb{E}_{\theta}\{u_{\text{dec}}(a,\theta)\} - C(d).$$

The prior/design distribution, predictive law, measurable action correspondences, and integrability are declared; any exchange of expectation and supremum is justified. $C(d)$ includes participant burden and ethical risk in the same units. Use the $\sup$ unless attainment is proved. Here d indexes a design, and a indexes a research decision.

11.6.5 Open simulation challenge

Release a simulator containing PTVC and non-PTVC generators: adaptation, set point, opponent process, random walk, heavy tail, observation floor, censoring, selection, conditioned bridge, absorbing boundary, response shift, and remembered utility. Teams receive training datasets with labels and blinded challenge datasets without labels. Promotion requires prespecified performance on open-set detection, not only classification among familiar models.

This early deliverable tests whether the proposed signatures recover their generating mechanism before real-world claims are made.

11.6.6 Completion-potential overidentification benchmark

Build a simulated and then empirical benchmark with four modules: state transitions, discrete action, event or jump intensity, and endpoint support. Estimate the completion potential from one module, freeze it, and predict the other three. Rotate the source module and compare the resulting potential estimates in a common function space.

The benchmark includes at least: ordinary homeostasis; homeostatic reinforcement learning; mood momentum; heterogeneous opponent kernels; conditioned bridges; absorbing boundaries; flexible endpoint selection; measurement floors; censoring; hidden-state confounding; and an open-set generator. Models receive equal tuning budgets and share the frozen raw-data, preprocessing, observation, missingness, retention, and information architecture. Only separately registered observation-artifact rivals may vary one named kernel component. Primary metrics are joint log score, calibration, cross-module potential discrepancy, transport across resolution, and open-set abstention.

A promotion result requires more than correct generator classification. The PTVC family must recover a stable shared potential under realistic measurement error and remain superior when the rival family is expanded. This benchmark operationalizes the paper's most distinctive theoretical proposal before any terminal interpretation is attempted.

11.6.7 Randomized signed-dose and compensation-surface experiment

Use safe, brief, reversible positive and negative stimuli calibrated separately for each participant. Before choosing a continuous-dose derivative, require an open dose region, consistency, randomization or conditional exchangeability, local positivity, differentiability, a declared interference model, explicit censoring, endpoint, observation, and missingness assumptions, and an independently calibrated direct phenomenal-dose map $a_{\mathrm{dir}}(d,x)$. The later response $\Delta C^{\mathrm{later}}$ excludes that direct contribution but includes all task-induced anticipation, effort, frustration, relief, and downstream experience.

$$\rho(H,d,x) := -\frac{\partial_d \mathbb{E}\{\Delta C^{\mathrm{later}}_{t:t+H} \mid \mathrm{do}(d), X_t = x\}}{\partial_d a_{\mathrm{dir}}(d,x)}$$

On discrete support, use the prespecified finite contrast with $\delta_{\mathrm{dose}} \neq 0$ and $a_{\mathrm{dir}}(d_0 + \delta_{\mathrm{dose}}, x) \neq a_{\mathrm{dir}}(d_0, x)$:

$$\rho_{\delta_{\mathrm{dose}}}(H; d_0, x) := -\frac{\mathbb{E}\{\Delta C^{\mathrm{later}}_{t:t+H} \mid \mathrm{do}(d_0 + \delta_{\mathrm{dose}}), X_t = x\} - \mathbb{E}\{\Delta C^{\mathrm{later}}_{t:t+H} \mid \mathrm{do}(d_0), X_t = x\}}{a_{\mathrm{dir}}(d_0 + \delta_{\mathrm{dose}}, x) - a_{\mathrm{dir}}(d_0, x)}$$

Do not report a derivative that the design cannot identify.

Estimate this response surface over preregistered horizons, dose signs, modalities, and state strata. Compare exact local compensation, incomplete opponent response, flexible adaptation, homeostatic choice, and residual-aware constrained control under the same observation model.

The most informative nonterminal result would be a response surface predicted prospectively by the independently estimated completion potential, replicated across modalities and laboratories. Null or asymmetric results remain useful because they retire specific local-balance routes without deciding the terminal law.

11.6.8 Comparative and prelinguistic structure program

Begin with paradigms already capable of separating reflex, learning, preference, relief, and motivational trade-off. In each candidate system, estimate positive and negative event kernels, recovery dynamics, choice under conflict, and multiscale persistence. Avoid verbal justice framing and use independent welfare and sentience oversight.

The confirmatory target is conserved relation, not a shared numerical scale. Test whether the same formal coupling between current residual proxy, reachable actions, and later signed behavior appears after species-specific calibration. Cross-species noninvariance may identify implementation boundaries; positive transport motivates deeper stream-individuation work. No organism is classified as a qualifying conscious stream merely because it passes one behavioral task.

11.6.9 Physiological hysteresis and sleep-transition laboratory

Repeated laboratory and ambulatory cycles can test whether regulatory burden is path-dependent. Estimate a multivariate burden state from autonomic, sleep, metabolic, inflammatory, and functional channels; fit conservative and nonconservative dynamics; and test whether loop integrals predict later recovery beyond current state. Cross pre-sleep residual and stress exposure with natural sleep variation or ethically acceptable sleep manipulations.

Primary discriminators are: overnight change in signed event kernels; next-day residual sufficiency; route-state recovery; memory-versus-experience divergence; and degradation in participants with trauma-related sleep disruption or other independently characterized failures of overnight regulation. Physiology remains a route-state channel; valence is calibrated separately.

11.6.10 Local episode complementarity and channel-rotation mechanism test

For a bounded, safe episode $W = [a, b)$ and an externally randomized cut point $\tau \in (a, b)$, define the prospectively typed past and suffix quantities

$$P_i^W(\tau) := L_i(\tau) - L_i(a-), \qquad R_i^W(\tau) := L_i(b-) - L_i(\tau).$$

Finite additivity gives

$$P_i^W(\tau) + R_i^W(\tau) = \Delta L_i(W).$$

This identity tests a separately declared episode-closure or local-compensation mechanism only when that mechanism predicts $\Delta L_i(W) = 0$. Apply the window-to-whole-stream decomposition in §6.10. The result remains episode-level unless $Z_{i,\mathrm{pre}}(a)$ and $Z_{i,\mathrm{suf}}(b)$ are both zero under the declared origin and endpoint conventions or are identified by prospectively fixed, validated prefix and terminal-remainder bridges. Choosing the truth-level endpoint as the window terminus removes only the terminal suffix; it does not recover the pre-window history. Estimate past and suffix with separated temporal blocks, blinded analyst teams, and at least one method-diverse replicate channel; freeze sign, neutral origin, cross-sign calibration, exposure weights, missingness model, and cuts before outcome access.

The episode-level diagnostic estimates the unconstrained working relation

$$R_i^W(\tau) = \alpha - \beta P_i^W(\tau) + \varepsilon_{i,\tau}.$$

When either quantity is measured with nonnegligible error, use the hierarchical errors-in-variables model and bias-propagation procedure in §6.10 rather than treating the observed quantities as error-free. The point (0,1) is the registered local target, not a restriction in the fitted relation. Declare compatibility only when the joint confidence region lies within the preregistered rectangle around (0,1), with power, within-stream dependence, and false support calibrated by simulation. Include shuffled-cut, sign-reversed, future-leakage, shared-method, recentered-total, and observation-floor controls. Compare ordinary mean reversion, integral feedback, mood momentum, opponent kernels, conditioned bridges, selection, and flexible state-space models under one held-out score.

The joint compatibility region for (α, β) is a population local-episode mechanism and measurement screen. It does not adjudicate per-stream PTVC unless each stream's terminal residual is separately bridged and its simultaneous compatibility set is evaluated. A favorable population intercept cannot replace the SET-ALL or finite-class residual rule.

Factorial signed-dose branch. For parallel randomized arms, prospectively fix a baseline delivered physical dose and a nonzero randomized signed delivered-dose contrast, with both assigned levels in the admissible support. Define arm-level causal contrasts rather than individual cross-world differences:

$$\Delta_P(\tau;\delta_{\mathrm{dose}}) := \mathbb{E}[P_i^W(\tau) \mid \mathrm{do}(d_0+\delta_{\mathrm{dose}})] - \mathbb{E}[P_i^W(\tau) \mid \mathrm{do}(d_0)],$$
$$\Delta_R(\tau;\delta_{\mathrm{dose}}) := \mathbb{E}[R_i^W(\tau) \mid \mathrm{do}(d_0+\delta_{\mathrm{dose}})] - \mathbb{E}[R_i^W(\tau) \mid \mathrm{do}(d_0)],$$
$$D(\tau;\delta_{\mathrm{dose}}) = \Delta_P(\tau;\delta_{\mathrm{dose}}) + \Delta_R(\tau;\delta_{\mathrm{dose}}).$$

The primary equivalence decision is

$$H_0\colon\ |D(\tau;\delta_{\mathrm{dose}})| \geq \varepsilon_D, \qquad H_1\colon\ |D(\tau;\delta_{\mathrm{dose}})| < \varepsilon_D.$$

Use no flexible nuisance term in the primary analysis; any adjustment is frozen in independent training data or cross-fitted without confirmatory outcomes. Individual Δ notation is reserved for an identified paired or crossover design. Hold out one channel and one horizon for transport, and control the preregistered family across cuts, doses, channels, and horizons through §9.3's multivariate decision map or an anytime-valid construction proved for the declared composite null.

Before enrollment, release a simulation grid over stream count, cut points per stream, measurement reliability, within-stream correlation, attrition, endpoint uncertainty, and rival effect sizes. Select a design only when the registered joint-equivalence rule attains its target power while the family-wise false-support bound remains below the preregistered tolerance; no numerical feasibility claim is made until that calibration is executed.

11.6.11 Integral-feedback comparator benchmark

Implement robust perfect adaptation as a first-class rival rather than a verbal control, including deterministic integral feedback, stochastic antithetic-control architectures, and the known structural requirements for maximal robust perfect adaptation (Briat et al., 2016; Aoki et al., 2019; Gupta & Khammash, 2022). Simulate and experimentally fit controllers with parameter heterogeneity and saturation; cross sustained and transient perturbations with positive and negative signed doses; and measure both endpoint state error and complete response area.

The primary contrast is state recovery versus terminal-residual adjudication. An RPA model may return $y(t)$ to $y^\star$ while leaving nonzero signed area. CUT-1's slope -1 checks partition/additivity and is shared by every fixed terminal residual; it is not a PTVC route prediction. The PTVC-bearing result is joint compatibility of the independently adjudicated $D_i(\tau)$ with the registered zero or finite acceptance region across cuts and interventions. Promotion requires that the complete PTVC-bearing model improve held-out joint prediction after the RPA rival receives equal state information, tuning, and observation flexibility.

11.6.12 Prospective multimethod rat welfare cohort and benign completion-choice benchmark

Aim and prospective confirmatory claim. This program specifies a staged ethical animal test of a PTVC measured projection. It asks whether a prospectively frozen bipolar multimethod welfare measurement model is terminally compatible with zero. After proxy and scalarization gates pass, it also asks whether its projected ledger is terminally compatible with zero and whether independently estimated residual history predicts future measured area, benign choices, and endpoint-aligned trajectories. Its claim ceiling is species- and measurement-conditional: no single proxy receives a privileged phenomenal interpretation, and one-species results do not identify unrestricted PTVC.

Stage R0: Simulation and recovery. Before animal enrollment, simulate the planned cadence, channel noise, missed observations, illness, treatment, endpoint intervals, and informative observation process under PTVC-band, diffuse near-zero, nonzero set-point, ordinary adaptation, integral-feedback, selection, memory-bias, and open-set generators. Freeze the latent dimension, sign and origin rules, missingness model, scientific band, held-out score, false-support tolerance, and minimum discrimination power. Failure to recover these generators blocks the animal study or restricts it to feasibility work.

Stage R1: Benign completion-choice benchmark. Use only reversible, welfare-approved, nonterminal choices among perceptually matched routes leading to ordinary enrichment, neutral continuation, or species-appropriate positive outcomes. Residual sign and magnitude come from the independently calibrated multimethod welfare-state model or naturally occurring variation; no negative-affect arm is manufactured. An external randomizer assigns the concealed target, automated apparatus and blinded coding eliminate handler cueing, route identity is remapped across trials, and locomotion, side bias, odor, learning, satiety, and apparatus history are measured. This stage tests the selector subclass of Appendix B, not the terminal law as a whole.

Stage R2: Prospective life-course natural-endpoint cohort. Enroll litters before the earliest stable, welfare-compatible channel calibration and follow through ordinary aging. The primary projection begins at the registered earliest qualifying observation. A complete-lifetime interpretation additionally requires an independently justified stream-origin bridge and a bound on preobservation contribution; without them the result remains a post-calibration segment-to-endpoint projection. Randomization, if used, is limited to standard-care versus welfare-positive enrichment; both arms retain social housing, nutrition, rescue, analgesia, and veterinary care. No endpoint is induced or delayed. Veterinary staff may be blinded to the inferential PTVC score and research hypothesis only when this cannot affect care. All clinically actionable observations, adverse events, pain indicators, and treatment information remain immediately available, and the attending veterinarian retains unilateral authority to intervene or stop data collection. The protocol endpoint remains an interval around the last defensible qualifying observation rather than an assumed exact instant.

Measurement battery. Positive and negative channels are modeled separately before scalarization. Candidate channels include ambiguous-cue judgment, preference and approach, facial action, 50- and 22-kHz vocalization subclasses, play and affiliative behavior, grimace during naturally occurring or clinically documented illness, activity, sleep, autonomic state, treatment, appetite, body condition, and blinded veterinary assessment. Mendl et al. (2009), Harding et al. (2004), Finlayson et al. (2016), and Sotocinal et al. (2011) support channel development; Wright et al. (2012) supplies an explicit nonuniversality control for ultrasonic vocalization. Calibration, measurement invariance, channel failure, medication effects, and missingness are tested prospectively, and no proxy receives a privileged phenomenal interpretation by name.

Primary estimand and fingerprints. The primary object is the nonempty compatibility set for the post-calibration projected welfare ledger on the frozen scalarization branch; it is complete only relative to the declared protocol origin, and bipolar results are retained whenever scalarization fails. Secondary targets are: cut-point complementarity whose suffix runs from the randomized cut to the independently adjudicated segment endpoint; incremental prediction from whole-history residual within the registered segment beyond current health and recent state; sign-appropriate response to naturally occurring events and positive enrichment; stability across sampling resolutions and method-diverse channels; and endpoint-aligned variance behavior under anticipated and unexpectedly rapid natural decline. The cut-point result calibrates the measured segment mechanism, not complete-lifetime PTVC. A single completion potential is trained on one module and scored without retuning on the others.

Decision rules. A nonempty, gate-valid terminal compatibility set wholly outside the registered band rejects the registered post-calibration rat-cohort segment-to-endpoint projection for the declared reference class. Concealed-target performance rejects the registered minimum-effect selector subclass only when its simultaneous confidence region lies wholly inside the preregistered chance-equivalence region and the recovery, calibration, multiplicity, and power gates pass; failure to detect above-chance performance alone is inconclusive. Directionless, apparatus-bound, or finite-capacity effects favor ordinary disturbance or leakage. Within-band terminal results are reported as operational compatibility; promotion requires replicated cross-channel and cross-module fingerprints, and complete-lifetime or latent phenomenal interpretation awaits the corresponding independently validated bridges.

Design and welfare controls. Planning follows PREPARE and the NC3Rs Experimental Design Assistant (EDA; RRID:SCR_017019; Percie du Sert et al., 2017); reporting follows ARRIVE 2.0 (Smith et al., 2018; Percie du Sert et al., 2020). The protocol must justify rats, strain, sex composition, and life stage against the documented replacement review and the specific validation target; species choice may not rest only on apparatus convenience or channel availability. Independent veterinary and welfare oversight governs enrollment, monitoring, and stopping. Preregister site, transport, litter, cage, handler, social composition, enrichment history, and experimental unit; account for clustering and social-housing interference. Power is simulated under the planned clustered longitudinal model across reliability, attrition, endpoint uncertainty, and plausible effects. Raw allocation, exclusion, treatment, endpoint, adverse-event, and observation-flow records are retained.

Welfare-coverage framework. Candidate channels and monitoring procedures are prospectively mapped to the Five Domains framework as a welfare-coverage and harm-monitoring aid (Mellor et al., 2020). The framework is qualitative and does not supply a cardinal cross-sign valence scale.

Proxy-validation go/no-go gate. Promotion to a scalar ledger requires prespecified sign validity, within-animal sensitivity, temporal reliability, cross-context transport, cross-sign comparability, synthetic-generator recovery, and correct negative-control behavior. Failure retains a bipolar or partially identified result and blocks scalar-ledger interpretation.

Independent scientific value and replacement gate. Animal enrollment proceeds only if the protocol answers a preregistered welfare-science question with value independent of PTVC and a documented replacement review shows why simulation, existing data, or nonanimal methods cannot answer it adequately.

Operational stopping and endpoint rules. Before enrollment, preregister individual humane endpoints, cumulative-severity and handling or sensor-burden limits, adverse-event triggers, device-removal criteria, cage and cohort suspension boundaries, and proxy-invalidity stops. The attending veterinarian has unilateral authority to stop or modify data collection and care. Natural death, veterinary euthanasia, welfare removal, observation cessation, and technical loss remain distinct endpoint states and are analyzed with cause-specific competing-risk or multistate models under blinded adjudication; pooling requires an independently justified endpoint bridge.

Scientific yield under every verdict. Positive transport would justify a stronger comparative bridge and a multisite replication. Off-band or rival-favored results would retire a stated branch while leaving a reusable longitudinal welfare dataset, validated observation models, and comparative affect measures. Gate failure would identify the exact measurement obstacle to solve next. The design therefore moves the research program regardless of outcome without exposing animals to terminal manipulation or preventable suffering.

Replicate and suffix contract. Replicate channels require a shared latent target, conditional orthogonality or independence, method-diverse errors, nuisance conditioning, and the stated Kotlarski/completeness premises; replicates do not identify an atom automatically. The bounded suffix is $R_i^{\text{local}}(\tau, H)$, with arm-valid cuts and measured EIV. This protocol calibrates local episode complementarity; it promotes to CUT-1 only with an independently adjudicated terminal suffix or validated terminal-remainder bridge.

11.6.13 Randomized observation-schedule calibration

Within each nonterminal participant, randomize otherwise identical assessment blocks to fixed and random prompts while balancing time of day, item content, burden, and expected compliance. Estimate schedule-specific observation kernels and compare latent variance, serial covariance, signed area, compliance, reactivity, and missingness. Pool schedules only after a held-out transport map or invariance test succeeds. A schedule effect is an observation-process result unless the measured-to-latent bridge transports.

Preregister block order, period, carryover, washout, block length, device and interface version, prompt-delivery success, response latency, and the compliance estimand. Analyze observed-scale and bridge-adjusted latent outcomes separately. A schedule effect on latent variance is identified only under a measured-to-latent bridge frozen independently of the randomized comparison; otherwise report the schedule effect on the observed measurement process.

11.6.14 Serial multimodal endpoint-bridge study

In clinically indicated observational cohorts, collect repeated standardized behavioral assessments, EEG, and available structural, functional, metabolic, and diffusion imaging without altering treatment or care. Construct separate diagnostic and prognostic models, retain modality-specific disagreement, and return a registered confidence, identified, posterior, or sensitivity interval or compatibility set together with its explicit inferential guarantee rather than forcing one timestamp. No modality, clinical label, or behavioral nonresponse alone defines stream identity or the truth-level endpoint.

The output must not be called truth-containing unless the procedure supplying that guarantee has been established under the declared assumptions. Endpoint adjudicators remain blinded to later outcomes and terminal-residual estimates. Missing-modality rules are frozen prospectively. Future prognosis, later recovery, or outcome information may enter only in an explicitly retrospective endpoint analysis and may not be used to validate a contemporaneous classifier on the same cases.

11.6.15 Linked-resolution atom challenge

In simulations, generate one known latent process through at least two independently calibrated observation kernels. In empirical data, posit one frozen latent-law family and calibrate each observation channel independently on nonterminal or external data before terminal adjudication. The empirical analysis therefore tests cross-channel compatibility with the frozen latent family; it does not inherit the simulation's knowledge of the true latent process.

Model shared errors and conditional dependence between channels. Agreement between two instruments that share calibration data, preprocessing, raters, or physiological confounding does not constitute independent replication. Freeze calibration and testing partitions before comparing atom, near-zero, and observation-floor explanations.

11.7 Statistical adjudication of bands, exactness, and false support

11.7.1 Equivalence, difference, and indeterminacy

Equivalence receives affirmative support through a prospectively specified interval within registered bounds. The two one-sided tests procedure evaluates whether an interval lies within preregistered bounds (Schuirmann, 1987; Lakens, 2017). For an estimand θ and symmetric bound K:

$$H_{01}: \theta \leq -K, \qquad H_{02}: \theta \geq K.$$

Both nulls must be rejected to establish equivalence. Report four states: (1) difference established / equivalence not established; (2) equivalence established / difference not established; (3) both established; or (4) neither established / indeterminate. Reserve 'nonequivalent' for a separately specified test supporting an effect outside the equivalence region. For individual PTVC, group-mean equivalence is a preliminary result; hierarchical individual bounds and population-coverage targets complete the registered inference.

11.7.2 Scientific band versus sampling uncertainty

The scientific band K_{sci} follows calibrated phenomenology, decision relevance, or an explicit smallest effect of interest rather than affordable sample size. Equivalence requires the entire confidence interval or joint confidence region to lie inside the preregistered equivalence region; width alone is never sufficient. A wide interval crossing zero and the band is indeterminate.

11.7.3 Boundary-null variance testing

On a square-integrable branch,

$$\Pr(Z_i^\star = 0 \mid X_i) = 1 \Leftrightarrow \begin{cases} \mathbb{E}[Z_i^\star \mid X_i] = 0, \\ \mathrm{Var}(Z_i^\star \mid X_i) = 0. \end{cases}$$

Expected neutrality tests only the first condition. On a registered variance-component implementation of exact closure, zero latent variance is a boundary null; atom-mixture and other exactness parameterizations require their own calibration. Use theorem-matched asymptotics, bootstrap, or simulation for the fitted design. A favorable finite result remains branch-, population-, endpoint-, and measurement-conditional.

11.7.4 Latent atom versus narrow continuous distribution

Compare the atom-mixture model with continuous alternatives of varying tail and cusp behavior. An observed spike at a rounded zero may be quantization. A declining variance may be scale compression. A near-zero continuous distribution can mimic an atom under noise.

Use multiple resolutions, unrounded raw data, replicate channels, posterior predictive checks, and held-out proper scores. Report the smallest continuous scale that remains compatible rather than forcing a binary verdict when the data cannot discriminate.

11.7.4.1 Shrinking-band intercept test

For the latent mixture

$$Z \sim \pi_0\delta_0 + (1-\pi_0)F_c, \qquad F_c(\{0\}) = 0,$$

assume the continuous component has density f_c continuous at zero. Then as $k \downarrow 0$,

$$\mathbb{P}(|Z| \le k) = \pi_0 + 2(1-\pi_0)f_c(0)k + o(k).$$

The exact-atom branch therefore predicts a nonzero intercept in the small-band probability after observation error, rounding, and censoring are removed; a purely continuous near-zero law predicts zero intercept. Estimate the curve over multiple preregistered resolutions using raw unrounded data and conditionally independent or otherwise identified replicate channels. Fit cusp, spike-and-slab, heavy-tail, zero-inflated rounding, and scale-mixture alternatives, and calibrate finite-sample bias by simulation.

This is a progressive-resolution test, not an extrapolation license. The smallest defensible k is limited by instrument resolution and deconvolution stability. Report an uncertainty set for the intercept and the continuous slope, plus sensitivity to correlated errors and endpoint-dependent observation. Replication across independent measurement channels is more informative than merely adding finer bins to the same rounded instrument.

11.7.4.2 Exposure-scaling anti-concentration challenge

Let $S_n := \sum_{k=1}^{n} X_k$ be a fixed-horizon no-law benchmark with independent identically distributed nonlattice increments satisfying $\mathbb{E}[X_k] = 0$, $\mathrm{Var}(X_k) = \sigma^2 > 0$, and the conditions of a local limit theorem. For fixed $K > 0$,

$$\sqrt{n}\, P(|S_n| \le K) \to \frac{2K}{\sigma\sqrt{2\pi}}.$$

On an admissible aperiodic-lattice subsequence, $P(S_n = 0) = O(n^{-1/2})$; with nonzero drift under a Cramér condition, fixed-band probability has the corresponding exponential-order decay. Thus an ordinary additive no-law process does not preserve a constant exact-zero mass or fixed-band closure rate as effective exposure grows, and $S_n/n \to 0$ establishes only average neutrality, not terminal closure. A prospectively specified challenge bins complete streams by independently estimated effective exposure, fits the appropriate null scaling after the same endpoint and observation kernels, and compares it with the PTVC and declared route families. Bounded episodes may enter only as a separately labeled local-route validation branch. Endpoint selection, state-coupled stopping, dependence, heavy tails, and exposure heterogeneity remain explicit rivals (Shepp, 1964; Stone, 1965; Dembo & Zeitouni, 1998).

11.7.5 Measurement quality and the value of exactness evidence

Consider a local Gaussian benchmark:

$$P_0 = N(0, \sigma_\varepsilon^2), \qquad P_1 = N(0, \sigma_\varepsilon^2 + \sigma_Z^2),$$

and let $r = \sigma_Z^2 / \sigma_\varepsilon^2$. The per-observation Kullback–Leibler divergence is

$$D_{\mathrm{KL}}(P_0 \| P_1) = \frac{1}{2}[\log(1+r) - \frac{r}{1+r}] = \frac{r^2}{4} + O(r^3).$$

To accumulate expected log evidence Λ in this local benchmark requires approximately

$$n \approx \frac{4\Lambda\sigma_\varepsilon^4}{\sigma_Z^4}.$$

Thus evidence distinguishing a tiny latent variance from zero grows at fourth order in the ratio of latent standard deviation to measurement-error standard deviation. Better measurement can be more valuable than merely adding participants. The calculation is a planning benchmark, not a universal sample-size formula.

11.7.6 Bayesian evidence and prior sensitivity

A Bayes factor

$$\mathrm{BF}_{01} = \frac{p(D \mid M_0)}{p(D \mid M_1)}.$$

can compare an exact model and a continuous alternative, but evidence depends on the alternative's prior scale and mass near zero. The Jeffreys–Lindley phenomenon shows that significance and Bayesian evidence can diverge as sample size and prior dispersion change (Lindley, 1957). Report a preregistered sensitivity grid over plausible prior scales and near-zero shapes. Rouder et al. (2009) provide a canonical Bayesian testing framework, but PTVC requires bespoke latent and observation models rather than a default t test.

Bayes factors should be accompanied by predictive checks, prior predictive simulation, and calibration in synthetic data. A diffuse alternative should not be chosen merely because it makes the point null look favorable.

11.7.7 Proper predictive scoring

Models should predict the observed law under equal data, preprocessing, tuning resources, and acquisition conditions. Use strictly proper scores such as log score, Brier score, or continuous ranked probability score so a model is rewarded for honest probabilistic forecasts (Gneiting & Raftery, 2007). Compare calibration and sharpness, not only point error.

Latent-dynamics candidates and rivals share the frozen observation architecture, preprocessing, missingness/retention process, and information budget. Only predeclared observation-artifact rivals may vary the named kernel component under the same latent information, tuning budget, and score; these comparisons are reported separately.

11.7.8 Sequential and multiverse control

Long-running cohorts invite repeated looks, adaptive sampling, and analytic flexibility. Use confidence sequences or e-processes for anytime-valid monitoring where appropriate (Howard et al., 2021; Ramdas et al., 2022). Freeze the confirmatory specification before accessing decisive endpoints. Use a blinded multiverse to identify defensible analytic choices, then promote a bounded subset to preregistration. Specification-curve and multiverse analyses are sensitivity tools, not a license to select the most favorable result (Steegen et al., 2016; Simonsohn et al., 2020).

11.7.9 Robustness and false-support tournament

Every primary statistic should be tested on stress-test simulations. Generate data with:

- zero-inflated rounding but no latent atom;
- terminal missingness depending on unobserved distress;
- proxy substitution near death;
- floor and ceiling compression;
- random endpoints;
- conditioned sampling;
- response shift;
- peak–end memory;
- adaptive sampling triggered by extreme values;
- cross-person cancellation with individual violations;
- analyst-selected reference classes.

A statistic that frequently endorses PTVC under these generators is not confirmatory, regardless of its elegance.

Executable exactness program. Preserve all almost-sure and integrability qualifiers. Use one frozen observation, preprocessing, missingness/retention, and information architecture for candidates and rivals; only preregistered observation-artifact rivals may vary one named kernel under equal budgets. Freeze resolutions, replicate channels, quantization/censoring generators, stopping rules, and the unresolved-at-maximum-resolution rule. In synthetic CUT runs, score additivity and the shared residual separately. Report power, family-wise false support, No adequate estimate, and open-set operating characteristics.

11.8 Ordered implementation roadmap

The implementation sequence is organized by scientific promotion gates.

Where feasible, the confirmatory prerequisite and route studies should use a Stage 1 Registered Report or an equivalent outcome-blind review process so that the scientific question, hypotheses, methods, analysis plan, and publication decision are evaluated before results are known (Chambers & Tzavella, 2022).

When a PTVC prerequisite or route study is randomized, its protocol should additionally satisfy SPIRIT 2025 and its report should satisfy CONSORT 2025, with the registered estimand, intercurrent-event strategy, missingness plan, harms reporting, and adverse-result rules kept concordant across protocol and publication (Chan et al., 2025; Hopewell et al., 2025).

1. **Measurement gate.** Validate sign, origin, exposure, scalar/bipolar representation, additivity, and method-diverse channel bridges on held-out sites and states.
2. **Recovery gate.** Release synthetic generators, fixed seeds, schemas, scoring code, and tests for PTVC-band, atom/diffuse, OU/homeostatic, conditioned-bridge, endpoint-selection, missingness/observation-floor, heavy-tail, and open-set cases. Execution releases include exact code, dependency lock, output hashes, and documented clean-run results.
3. **Finite-class discriminator gate.** Run the completion-potential rotation and the factorial signed-dose local episode-complementarity study with common calibration, rivals, multiplicity control, and abstention rules.
4. **Endpoint gate.** Complete blinded endpoint-adjudication reliability and only then begin prospective endpoint-aligned observation under the controlling safety gate in Section 9.4.
5. **Transport/retirement gate.** Test environment, cohort, modality, endpoint-cause, and resolution transport; retire or narrow branches at preregistered thresholds. Every stage yields reusable measurement, causal, endpoint, simulation, and open-science infrastructure even if PTVC is not favored.

Declarations

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Competing interests. Stephen Takowsky originated PTVC, founded the Law of Fairness Project, and owns or controls rights in related publications and project materials.
Data, code, and materials. This Formal Model supplies the project's expanded definitions, formal derivations, protocols, formula provenance, and integrated research architecture. Empirical studies arising from this framework will release materials under their protocol-fixed ethics, privacy, and version-control plans. Artificial intelligence was used in the creation of this work.
Ethics statement. No human or animal participants or participant data were involved in the preparation of this Formal Model.
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Appendix A: Methods, Architectures, and Frontier Theorem Targets

Appendix A unifies the operational method bank and the frontier mechanism bank while preserving their different maturity levels. Inclusion means that an entry is concrete enough to affect an estimand, theorem, simulation, discriminator, or protocol; it does not imply equal empirical support. Part I entries are operational methods or near-term candidate architectures; Part II alone uses the frontier status-code legend.

Table 4. Appendix A Status Codes

Status	Meaning
[IR]	Identity or representation result; not an empirical evidence vote
[CM]	Candidate mechanism with a declared object and consequence
[CP]	Conditional proposition under displayed assumptions
[TT]	Open theorem target stated for proof or counterexample
[ST]	Simulation or generator-recovery target
[ED]	Empirical discriminator

Part I: Operational Methods and Near-Term Candidate Architectures

A1.1 Hidden-state control architecture

Optimization convention. An infimum or supremum is the primary object. An optimizer is introduced only after sufficient existence assumptions have been stated; otherwise the protocol uses an explicitly defined ε-optimal object.

Let the generic hidden state X_t^{hid} summarize reserve, affective momentum, biological load, environment, memory, and social context:

$$\mathrm{d}X_t^{\mathrm{hid}} = f\big(X_t^{\mathrm{hid}}, a_t, t\big)\,\mathrm{d}t + G\big(X_t^{\mathrm{hid}}, t\big)\,\mathrm{d}W_t.$$

Terminal neutrality becomes a boundary or viability condition on admissible hidden-state paths. The research value is that interventions can target identifiable state transitions rather than treating ledger changes as isolated observations.

Under partial observation, define the filtered belief state

$$b_t := \mathcal{L}\Big(\big(X_t^{\mathrm{hid}}, L_t^{\mathrm{phen}}\big) \mid \mathcal{H}_{i,t}^{\mathrm{pre}}\Big).$$

For a fixed belief b, finite band K, and tolerance ε, define

$$\mathcal{S}_{K,\varepsilon}(t) = \{b\colon \sup_{\pi\in\Pi_{\mathrm{adm}}(t,b)} P_b^{\pi}(|L(T^-)| \le K) \ge 1-\varepsilon\}.$$

For a nonempty partially identified belief set $\mathcal{B}$, define robust lower and upper solvency values

$$\begin{aligned}\underline{v}_K(t,\mathcal{B}) \quad &:= \sup_{\pi\in\Pi_{\mathrm{adm}}(t,\mathcal{B})} \inf_{b\in\mathcal{B}} P_b^{\pi}(|L(T^-)| \le K),\\ \overline{v}_K(t,\mathcal{B}) \quad &:= \sup_{\pi\in\Pi_{\mathrm{adm}}(t,\mathcal{B})} \sup_{b\in\mathcal{B}} P_b^{\pi}(|L(T^-)| \le K).\end{aligned}$$

Robust solvency requires the lower value to be at least $1-\varepsilon$; an upper value below $1-\varepsilon$ is adverse; the intermediate case is unresolved. Policy probability mass, conditioning, and attainment or ε-optimality assumptions are stated locally.

Action admissibility is evaluated under the intervention-indexed continuation law and the declared information set. Because b_t is estimated or partially identified, protocols compare this route with prognosis-only, generic POMDP, and viability rivals and never expose future endpoint labels or latent ledger truth to the controller.

A1.2 Allostatic reserve and scalar-envelope testing

Reserve may be a multiscale envelope over energetic, neural, immune, endocrine, cognitive, and social resources:

$$U_t = U\left(E_t^{\mathrm{eng}}, E_t^{\mathrm{neural}}, E_t^{\mathrm{immune}}, E_t^{\mathrm{endo}}, E_t^{\mathrm{cog}}, E_t^{\mathrm{social}}\right)$$

The route prediction is not that every component declines monotonically, but that the declared envelope of attainable future stream-owned contributions contracts near the endpoint. The envelope, route state, component bridges, metric, and transport tolerance must be operationally defined and validated before a proxy is exported across sites or populations.

A1.3 Opponent-process and adaptation dynamics

A minimal opponent architecture writes valence as immediate drive minus delayed opponent state:

$$F_t = R_t - O_t, \mathrm{d}O_t = \kappa(R_t - O_t)\mathrm{d}t + \xi \mathrm{d}W_t.$$

This can generate rebound, tolerance, withdrawal, and return toward baseline. Its role is both mechanism neighbor and rival: if opponent dynamics explain terminal patterns without a neutral-origin boundary, the PTVC route loses ground.

A1.4 Active-inference and expected-free-energy bridge

Use $\mathcal{G}_t^{\mathrm{EfE}}$ for expected free energy to avoid collision with the treatment or care process $G_i(t)$ and the felt-valence contribution $F_i(t)$. Any bridge from expected free energy to phenomenal valence requires independent calibration (Joffily & Coricelli, 2013). For a declared policy class Π_i define the infimum and an ε-optimal policy by

$$\mathcal{G}_{i,t}^{\inf} := \inf_{\pi \in \Pi_i} \mathbb{E}[\mathcal{G}_t^{\mathrm{EfE}} \mid \pi], \qquad \mathbb{E}[\mathcal{G}_t^{\mathrm{EfE}} \mid \pi_{i,\varepsilon}] \le \mathcal{G}_{i,t}^{\inf} + \varepsilon.$$

A PTVC-compatible active-inference route would predict that policies preserving viability also reduce terminal ledger dispersion under independent origin and endpoint rules. A viability-only account remains a rival unless it predicts the same terminal-neutrality structure.

A1.5 Filtered horizon and model-specific marginal constraint sensitivity

Let $\mathcal{W}_i(x,h)$ be the minimal expected constraint-violation cost-to-go, in registered cost units, attainable over remaining horizon h from state x under the declared admissible policy and model class. Where differentiable and where greater remaining time cannot worsen the attainable minimum, $\mathcal{W}_i$ is weakly nonincreasing in h and

$$\lambda_i^H(x,h) := -\partial_h \mathcal{W}_i(x,h) \ge 0.$$

If a value-to-go convention is used instead, the sign is reversed and the symbol is renamed; finite-horizon and infinite-limit statements retain their separate regularity conditions.

A1.6 Causal-measurement graph

A confirmatory protocol must provide a time-indexed causal graph or structural causal model with every node locally defined. At minimum distinguish latent clinical or contextual state $\Xi_i(t)$, phenomenal rate $F_i(t)$, reserve $U_i(t)$, treatment or care exposure $G_i(t)$, medication $M_i(t)$, current endpoint-predicate state $E_i^{\mathrm{end}}(t)$, reporting capacity $Q_i^{\mathrm{rep}}(t)$, channel-availability vector $\mathbf{O}_i^{\mathrm{avail}}(t) := \left(O_{i,c}^{\mathrm{avail}}(t)\right)_{c\in\mathcal{C}_i^{\mathrm{obs}}}$, with a separately defined attempt process when unattempted differs from failed or unavailable, and postcommitment retention $R_i^{\mathrm{ret}}(t)$.

A minimal schematic is

$$\{\Xi_i(t), G_i(t), M_i(t)\} \to \{F_i(t), U_i(t), E_i^{\mathrm{end}}(t), Q_i^{\mathrm{rep}}(t)\}.$$

$$\{F_i(t), U_i(t), E_i^{\mathrm{end}}(t), Q_i^{\mathrm{rep}}(t), G_i(t), M_i(t)\} \to \{\mathbf{O}_i^{\mathrm{avail}}(t), R_i^{\mathrm{ret}}(t)\}.$$

$$O_{i,c}^{\mathrm{avail}}(t) \to O_{i,c}^{\mathrm{obs}}(t), \qquad c \in \mathcal{C}_i^{\mathrm{obs}}.$$

A completed future endpoint may enter a retrospective selection graph, but it may not be used as a prospective parent of earlier state, reporting, or retention variables unless the model explicitly abandons forward causal semantics. The protocol graph must expose colliders, mediators, shared causes, endpoint leakage, reserve-proxy leakage, treatment-induced observation, and forbidden conditioning paths before analysis.

Causal-semantic typing. Each protocol declares either: (i) a dynamic structural-causal branch with explicit temporal ordering or time-unrolled nodes, structural equations, exogenous variables, intervention semantics, and treatment of contemporaneous relations; or (ii) a multivariate counting- or marked-point-process branch whose edges are defined through predictable intensities or compensators and local independence. Ordinary DAG separation is used only on the fully specified acyclic branch; the process branch uses its corresponding local-independence separation rule. The arrow displays are schematic dependency maps unless one of these semantics is supplied, and no causal direction, collider status, or conditional independence is inferred from them alone (Didelez, 2008; Pearl, 2009).

A1.7 Positive controls and calibration tasks

A measurement system should recover known perturbations before interpreting terminal signatures. If a protocol cannot detect a validated nonterminal affective perturbation of known sign and approximate magnitude, it cannot interpret terminal compression as PTVC-route-informative.

A1.8 Neutral-origin and gauge-identification program

For a neutral landmark state s with independently elicited or experimentally supported neutral semantics, let $\chi_{i,\eta}(s)$ denote the calibrated response assigned to that state under protocol η. Gauge identification requires that transformations preserving admissible landmark relations also preserve sign and approximate magnitude; it cannot be selected after terminal fit.

$$\chi_{i,\eta}(s_0) = 0, \chi_{i,\eta}(s_1) > 0, \chi_{i,\eta}(s_2) < 0.$$

The research task is to identify stable sign and origin anchors across contexts, not to recenter outcomes after observing terminal data. If the neutral landmark moves with culture, illness, sedation, disability state, or response shift, the report becomes context-specific or partially identified.

Bidirectional cross-sign calibration. Within each participant, independently calibrated positive and negative stimuli are paired and titrated to experienced indifference. The inferred exchange ratio must transport across at least two modality pairs within a preregistered tolerance; failure narrows or downgrades origin and scalarity status rather than being repaired by terminal recentering.

If the retained transformation group still permits independent positive/negative rescaling or context-dependent nonlinear remapping, scalar additive PTVC is unidentified. The protocol must remain bipolar, ordinal, partially ordered, or return No adequate estimate; a normalized ratio cannot substitute for within-stream cross-sign calibration.

A1.9 Replicate-channel deconvolution of a terminal atom

Use the declared terminal latent object:

$$O_i^{(c)} = Z_i^\star + \varepsilon_i^{(c)}, \qquad c = 1, \dots, k,$$

With calibrated unit loadings, a conditional two-replicate Kotlarski branch requires the same tiered package as §6.5: square-integrable latent and error terms, conditional mutual independence of the latent target and replicate errors given X (or the theorem-matched weaker package), zero conditional error means, a location normalization, nonvanishing relevant characteristic functions, and transported loadings/error laws. Analyst separation and method diversity are design safeguards, not stochastic-independence assumptions (Kotlarski, 1967; Fan, 1991; Kato et al., 2021).

A1.10 Product-factorization nonuniqueness

A product form is not uniquely identifying. For every strictly positive measurable gauge $\chi(t)$,

$$L(t) = U(t)Y(t) = \big(U(t)\chi(t)\big)\big(Y(t)/\chi(t)\big), \qquad \chi(t) > 0.$$

The observed product is invariant under this transformation. Identification of a factor requires an external normalization, an independently measured route-state channel, or cross-equation restrictions; product algebra alone does not identify reserve.

A1.11 Terminal boundary-layer exponent classification

Boundary-exponent discipline follows §6.2. Use the collision-free distance $d_i^{\mathrm{rc}}(t) = \tau_i^{\mathrm{rc}} - t$ and the same declared boundary-conditioning state $\Xi_i^\circ(t)$. Transfer of a fixed-reserve boundary exponent requires the conditional reserve law to remain tight and bounded away from zero near the boundary; otherwise use the random-reserve or non-power branch. For the hazard route, transport the declared boundary state, filtration, and clock together.

A1.12 Martingale-neutrality diagnostic in reserve coordinates

A normalized process can be tested for fair-game-like expectation neutrality under a declared generative measure:

$$\mathbb{E}\,[\Delta Y_{i,k} \mid \mathcal{H}_{i,t_k}^{\mathrm{pre}}] = 0.$$

For innovation-time increments, the diagnostic also examines serial dependence and variance ratios. Its value is comparative: it separates a reserve-normalized innovation coordinate from endpoint-conditioned drift, smoothing artifacts, denominator error, and selection-driven terminal concentration. It does not distinguish all bridges by itself, because a bridge transformed into its own innovation coordinates may also look martingale-like; the diagnostic must be interpreted with covariance, endpoint, reserve, observation, and recovery evidence.

A1.13 Entropic path-space steering

Finite-band entropic steering is defined formally in A2.12 and A2.18. This entry records only its operational use as a baseline-relative path-law comparison under a common observation and scoring architecture. Its large-deviation and conditioned-process context follows Dembo and Zeitouni (1998), Ethier and Kurtz (1986), and Chetrite and Touchette (2015).

Exact point pinning is singular in many continuous models; finite terminal bands allow tractable comparison across rivals.

A1.14 Critical-transition and regime-switching rivals

The regime-switching rival is discrete time: use S_k, X_k, and S_{k-1} throughout this branch. A continuous-time alternative requires a separately declared generator and observation law.

Critical-transition rivals test whether terminal concentration is ordinary regime change rather than PTVC-specific dynamics. Let S_k be the latent regime and X_k the measured or latent state. A hidden-regime branch may use

$$\mathbb{P}(S_k = j \mid S_{k-1} = r) = p_{rj}, \qquad X_k \mid S_k = j \sim F_j.$$

Critical-transition alternatives may predict rising autocorrelation, variance, skewness, or recovery time near a transition. The protocol fixes the state representation, transition model, window, smoothing, and critical-transition statistics before outcome inspection and compares them against bridge and reserve-route predictions (Kuehn, 2011; Rabiner, 1989; Wichers et al., 2016; Scheffer et al., 2009).

A1.15 Natural reserve perturbations and target-trial architecture

Define eligibility, time zero t, treatment strategies and versions, any grace period, intercurrent-event strategy, follow-up horizon τ, measured outcome, causal contrast, and the preassignment history available at time zero. Target-trial emulation follows Hernán and Robins (2016) and related longitudinal causal methods. Estimation may use marginal structural or targeted-learning methods when their identification conditions are met (Robins et al., 2000; van der Laan & Rose, 2011).

A counterfactual reserve-perturbation estimand can be written as

$$\begin{aligned}\Delta_{U,i}(\tau;t) := \mathbb{E}[\{&M_{\eta,i}^{(1)}(t+\tau) - M_{\eta,i}^{(1)}(t)\} \\ &-\{M_{\eta,i}^{(0)}(t+\tau) - M_{\eta,i}^{(0)}(t)\} \mid \mathcal{H}_{i,t}^{\mathrm{pre}}].\end{aligned}$$

Here the displayed object is the potential measured estimand under interventions 1 and 0, not its estimator. Identification and estimation require the assignment, consistency, exchangeability or randomization, positivity, censoring, endpoint, observation, missingness, treatment, and interference assumptions declared by the target trial. This architecture is useful only when the perturbation affects reserve before ledger contraction and is not merely a proxy for endpoint proximity, treatment selection, or observation intensity (Hernán & Robins, 2016; Hernán & Robins, 2020).

A lead–lag diagnostic can test temporal ordering. The protocol preregisters the dependence unit, covariance estimator, window, lag grid $\ell = 1, \dots, L$, irregular-time alignment, covariate history, logarithm floor, and multiplicity rule. Define $\log_{\delta_{\log}}(x) := \log(x \vee \delta_{\log})$ for a frozen $\delta_{\log} > 0$ and fit

$$\Delta \log_{\delta_{\log}} \hat{V}_{L,i,\eta}(t) = \alpha + \sum_{\ell=1}^{L} \beta_\ell \, \Delta \log_{\delta_{\log}} \hat{U}_{i,\eta}(t-\ell) + \sum_{\ell=0}^{L} \gamma_\ell^\top X_i(t-\ell) + \varepsilon_i(t).$$

For fixed weights $w_{\ell,\eta}$, define $\hat{B}_\eta := \sum_{\ell=1}^{L} w_{\ell,\eta} \, \hat{\beta}_\ell$. Construct the simultaneous region $\mathcal{C}_{B,\eta}$ from the declared full covariance estimator and multiplicity allocation. The route-ordering screen passes only when $\mathcal{C}_{B,\eta}$ lies inside the preregistered directional region and reverse-lag and placebo specifications do not recover an equally strong or earlier signal. A reserve route is weakened if ledger contraction consistently precedes the reserve signal, if the lead–lag relation disappears after leakage control, or if endpoint-cadence artifacts explain the timing.

Use the generative conditional dispersion from §6.2, or an explicitly identified process/innovation variance over a prospectively declared comparable reference class, as the primary lead–lag outcome:

$$V_L^{\text{gen}}(u,x,t) := \operatorname{Var}\{L_i^{\text{rc}}(t) \mid U_i(t) = u, \Xi_i^\circ(t) = x\}$$

The filtered posterior variance $\hat{V}_{L,i,\eta}^{\text{post}}(t) := \widehat{\operatorname{Var}}\left\{L_i^{\text{phen}}(t) \mid \mathcal{O}_{i,t}^{\text{obs}}\right\}$ is a recoverability and precision diagnostic. A smaller $\hat{V}^{\text{post}}$ may reflect improved information or calibration even when the underlying ledger has not contracted. Holding observation information, kernel precision, and filter calibration fixed or explicitly decomposed is necessary to rule out observation-only explanations, but it is not sufficient to establish ontic ledger contraction without an independently validated latent-state and measurement bridge. Simulation must show that observation-only precision changes do not pass the reserve-ordering screen.

A1.16 Network interference

Social and clinical systems may violate no-interference assumptions. Let $A_i^{\text{net}}(t)$ denote network exposure and G the declared interaction graph. A schematic model is (Hudgens & Halloran, 2008).

$$\mathrm{d}X_i(t) = b\,(X_i(t), A_i^{\text{net}}(t), G, t)\mathrm{d}t + \Sigma_i(t)\mathrm{d}W_i(t).$$

The estimand, exposure mapping, spillover radius, clustering, and interference sensitivity analysis must be declared. Network effects can create apparent compensation, convergence, or route-state dependence without a per-stream terminal law.

Network exposure may alter a stream's future owned contribution, but it never permits one stream's residual to cancel or offset another; spillovers and cross-stream transfer remain distinct hypotheses.

A1.17 Vector-valence geometry

Let $\vec{v}_i$ be a finite $\mathbb{R}^d$-valued measure absolutely continuous with respect to μ_i^{phen}, with density $\vec{f}_i$. Define

$$\vec{v}_i(B) := \int_B \vec{f}_i\,(x)\,\mu_i^{\text{phen}}(dx) \in \mathbb{R}^d.$$

For a preregistered dual vector w in the admissible dual set $\mathcal{W}$, define the projected ledger

$$L_i^{(w)}(t) := \langle w, \vec{v}_i(\Omega_i^{\leq}(t))\rangle.$$

Let $|\vec{v}_i|$ denote the vector-measure variation induced by the declared norm $\|\cdot\|$ on $\mathbb{R}^d$. Require $|\vec{v}_i|(\Omega_i) < \infty$ and state the dual norm used to bound $|L_i^{(w)}|$. A vector branch must preregister the scalar projection, componentwise closure, ordered cone, or partial-order rule; w may not be chosen because it makes terminal closure hold.

A1.18 Cross-scale reserve renormalization

Reserve may be scale-dependent. Let the route state at scale ℓ and the prospectively fixed scale map be represented by the objects in the following display. A decision-bearing cross-scale claim requires a declared metric and tolerance:

$$d_{\mathcal{U}}\left(\mathcal{R}_{\ell\to\ell+1}\left(U_{i,\ell}\right), U_{i,\ell+1}\right) \le \varepsilon_\ell.$$

The domain and codomain of $\mathcal{R}_{\ell\to\ell+1}$ are declared whenever they differ by scale. The scale map, environments, transported prediction, metric, tolerance, and failure rule are fixed before analysis. Without those objects the entry is exploratory rather than confirmatory.

A1.19 Dynamic measurement invariance and response shift

A terminal context can alter the meaning of a response. Dynamic invariance can be represented by time-varying loadings (Millsap, 2011):

$$O_t = \Lambda(t)\xi_t + \varepsilon_t.$$

If $\Lambda(t)$ changes near the endpoint, an apparent ledger movement may be a measurement shift. Invariance, response-shift, and differential-channel-functioning tests belong in the observation model.

The protocol states which loadings, thresholds, intercepts, residual variances, or response mappings may vary. Absent a validated partial-invariance or alignment model, failure of metric invariance invalidates comparisons of latent-unit relations such as associations and slopes; failure of scalar invariance invalidates comparisons of latent means and mean changes. Response shift must be modeled or the claim narrowed (Sprangers & Schwartz, 1999).

A1.20 Retrospective terminal-cohort reflection screen

First executable retrospective screen. Preregister parallel HRS, ELSA, and SOEP reanalyses with interview-wave end, discharge, device retirement, shifted windows, and pseudo-terminal windows as endpoint-negative controls. These studies evaluate measured well-being proxies and endpoint-selection alternatives and serve as low-cost screens for prospective ledger studies (Sonnega et al., 2014; Steptoe et al., 2013; Wagner et al., 2007).

Let $t_{0i} := T_{i,\eta}^{\mathrm{adj}} - \Delta$. The primary ANCOVA or latent-state screen is

$$M_{\eta,i}\left(T_{i,\eta}^{\mathrm{adj}}-\right) = \beta_0 + \beta_1 M_{\eta,i}(t_{0i}) + \beta_D D_i^{\theta}(t_{0i}) + \gamma^\top X_i(t_{0i}) + \varepsilon_{i,\eta}.$$

Fit this model with measurement-error correction, duration adjustment, cohort/design weights, and negative-control windows. A state-space implementation may replace both noisy levels by latent states under a validated observation model. The raw change-score regression is sensitivity-only because baseline measurement error mechanically couples change to baseline; its null distribution must be calibrated by simulation under no contraction. A favorable reflection pattern is a low-cost screen for prospectively testable dynamics, not a route-specific or latent-terminal verdict.

A1.21 Adaptive observation and robust protocol design

Observation schedules may adapt to burden only through a prospectively fixed predictable delay kernel. Let $\Delta T_{i,k+1} := T_{i,k+1} - T_{i,k}$. When simultaneous visits are forbidden,

$$\mathbb{P}\left(\Delta T_{i,k+1} \in B \mid \mathcal{H}^{\mathrm{pre}}_{i,T_{i,k}}\right) = Q_\eta\left(B \mid \mathcal{H}^{\mathrm{pre}}_{i,T_{i,k}}\right), \quad B \in \mathcal{B}\big((0,\infty)\big).$$

The full prospective history contains every clinical, burden, protocol, and observed-data input actually used for scheduling and contains no future endpoint information. The kernel, positivity region, missed-visit treatment, and stopping rule are preregistered. Rigorous design selects schedules that distinguish reserve, bridge, endpoint-selection, and observation-floor rivals without current-study held-out outcomes.

A1.22 Quantization, censoring, and multi-resolution coarse-graining

Choose and freeze one order. In the primary instrument model, latent signal plus pre-quantization error is quantized and post-quantization recording error is added:

$$O = Q_\Delta\big(Z + \varepsilon_{\mathrm{pre}}\big) + \varepsilon_{\mathrm{post}}.$$

The alternative order $Q_\Delta(Z) + \varepsilon$ is a separately scored sensitivity model; the two are not interchanged after inspection.

A1.23 Intrinsic-time and perturbation-recovery architecture

The logarithmic identity assumes $U_i > 0$ and $\mathrm{d}U_i = -\kappa_i^U U_i\, \mathrm{d}t$. Stochastic or jump reserves require their full semimartingale decomposition.

Calendar time may be the wrong coordinate. Distinguish the reserve-depletion, normalized-state innovation, and route-coordinate quadratic-variation clocks:

$$\tau_i^U(t) := \int_{S_i^\theta}^t \kappa_i^U(s)\, ds = \log \frac{U_i(S_i^\theta)}{U_i(t)}, \qquad \tau_i^Y(t) := \int_{S_i^\theta}^t \sigma_i^2(s)\, ds.$$

$$A_i^{\mathrm{QV,rc}}(t) := \int_{S_i^\theta}^t c_{\mathrm{LU}}^2\, U_i^2(s)\sigma_i^2(s)\, ds.$$

These are respectively the reserve-depletion clock, normalized-state innovation clock, and route-coordinate quadratic-variation clock. They have different dimensions and information requirements and are not interchangeable without an explicit empirical bridge.

A1.24 Information-gain/cost sequencing heuristic

Let $\mathcal{A}^{\mathrm{des}}$ be the admissible design set. For each design a, define information gain I_a under the declared predictive law and cost C_a in compatible units. Let $\lambda \geq 0$, $\varepsilon \geq 0$, and assume the displayed suprema are finite on the branch used.

$$J_\lambda^{\mathrm{sup}} = \sup_{a \in \mathcal{A}^{\mathrm{des}}} (I_a - \lambda C_a), \qquad a_{\lambda,\varepsilon} \in \{a : I_a - \lambda C_a \geq J_\lambda^{\mathrm{sup}} - \varepsilon\}.$$

For budget B,

$$\mathcal{A}_B := \{a \in \mathcal{A}^{\mathrm{des}} : C_a \leq B\},$$
$$I_B^{\mathrm{sup}} := \sup_{a \in \mathcal{A}_B} I_a,$$
$$a_{B,\varepsilon} \in \{a \in \mathcal{A}_B : I_a \geq I_B^{\mathrm{sup}} - \varepsilon\}.$$

Use an optimizer only after attainment is established. If $\mathcal{A}_B = \emptyset$ or the relevant value is undefined, return No adequate estimate.

A1.25 Invariant prediction across reference classes

Let $\mathcal{Y}$ be a common standard-Borel outcome space; let $d_{\mathcal{P}}$ be a preregistered measurable metric/divergence on $\mathcal{P}(\mathcal{Y})$ (bounded-Lipschitz for weak transport, or total variation only on a dominated branch); and let $\mu_{e,e'}$ be the declared overlap law. For regular conditional kernels $K^{(e,f)}(x,\cdot)$, define

$$\Delta_{e,e'}^{(f)} = \operatorname{ess}\sup_{x\sim\mu_{e,e'}} d_{\mathcal{P}}\left(K^{(e,f)}(x,\cdot), K^{(e',f)}(x,\cdot)\right).$$

or its registered $\mu_{e,e'}$-integral analogue. Overlap failure is a separate No adequate estimate state. Use sup, not max, unless the environment set is finite and attainment is established. The cross-environment program is related to invariant prediction and invariant risk minimization, while PTVC retains its own preregistered bridge and support requirements (Arjovsky et al., 2019; Peters et al., 2016).

A1.26 Orthogonal nonterminal calibration battery

Before any combined score is formed, fix $s_m \in \{-1,+1\}$ prospectively for every directional task using independent calibration. A zero/tie is unfavorable unless the preregistered tolerance says otherwise; if direction cannot be fixed without the tested outcome, return No adequate estimate. Define each task set, feature map, directional or null-control score, aggregation rule, tolerance, and failure disposition. Empty task sets, undefined features, or unavailable controls yield No adequate estimate rather than a favorable default.

$$D_m^{\mathrm{dir}} = \mathbf{1}\{\mathcal{I}_m \neq \emptyset,\ \inf_{\delta\in\mathcal{I}_m} s_m\delta \geq \delta_{\min,m}\}.$$

$$D_m^{\mathrm{null}} := \mathbf{1}_{\{\mathcal{I}_m\neq\emptyset,\ \mathcal{I}_m\subseteq[-K_{0,m},K_{0,m}]\}}.$$

Failure to reject a null-control effect is not successful null calibration; the complete compatibility set must lie inside the preregistered null band. Direction, magnitude, latency, and channel-specific recovery remain separate calibration requirements. A system that fails nonterminal calibration cannot interpret terminal patterns as PTVC-route-informative, even if the terminal plots look orderly.

A1.27 Endpoint-negative-control and placebo-window architecture

Endpoint-specific claims should be tested against control endpoints and placebo windows. Let

$$T^{\mathrm{ctrl}} = \left\{T^{\mathrm{shift}}, T^{\mathrm{admin}}, T^{\mathrm{discharge}}, T^{\mathrm{device}}, T^{\mathrm{random}}\right\}.$$

If terminal concentration, reserve-squared slopes, or final-window contraction appear equally at control endpoints, the result is an endpoint-normalization artifact or a broader terminal-decline phenomenon. A PTVC-route-informative signature should localize to the endpoint-separated stream endpoint and survive rival recovery.

A1.28 Stochastic-reserve stress test

Let reserve and normalized state be joint semimartingales on $t \geq S_i^{\theta}$, with positivity-preserving reserve coefficients and one boundary time $T_i^U := \inf\{t \geq S_i^{\theta} : U_i(t) \leq 0\}$. One generic interior model, with predictable $\rho_i(t) \in [-1,1]$, is

$$\mathrm{d}U_i(t) = b_U(t,U_i)\mathrm{d}t + \sigma_U(t,U_i)\mathrm{d}W_i^U(t) + \mathrm{d}J_i^U(t),$$

$$\mathrm{d}Y_i(t) = b_Y(t,Y_i)\mathrm{d}t + \sigma_Y(t,Y_i)\mathrm{d}W_i^Y(t) + \mathrm{d}J_i^Y(t), \mathrm{d}[W_i^U, W_i^Y]_t = \rho_i(t)\mathrm{d}t.$$

$$\begin{aligned}\mathrm{d}[U_i, Y_i]_t^c &= \sigma_U(t, U_i)\sigma_Y(t, Y_i)\rho_i(t)\,\mathrm{d}t,\\ \sum_{s\le t} \Delta\, U_i(s)\Delta Y_i(s) &\quad \text{records common-jump covariation.}\end{aligned}$$

Reflection, absorption, killing, overshoot, and continuation at the boundary must be declared. Product dynamics inherit the continuous bracket and common-jump terms; no independent-shock assumption is implicit.

A1.29 Experience-versus-memory observation rival

Measured affect may track memory reconstruction rather than lived experience (Kahneman & Riis, 2005; Redelmeier & Kahneman, 1996). A rival model separates online experience E_t, latent memory state C_t^{mem}, and measured report M_t:

$$\begin{aligned}\mathrm{d}C_t^{\text{mem}} &= r(C_t^{\text{mem}}, E_t, X_t^{\text{ctx}}, t)\,\mathrm{d}t + \zeta\,\mathrm{d}W_t,\\ M_t &\sim K^{\text{report}}\big(\cdot \mid C_t^{\text{mem}}, W_t^{\text{nuis}}\big).\end{aligned}$$

A terminal report may converge because memory narratives converge, not because latent felt valence closes.

Measured-memory branch. For a prospectively declared episode $W = [a_W, b_W)$ and validated real-time rating $R_{i,W}(t)$, define

$$I_{i,W} := \int_{a_W}^{b_W} R_{i,W}\,(t)\,\mathrm{d}t,$$

$$P_{i,W}^{+} := \sup_{t\in W}[R_{i,W}(t)]_{+},$$

$$P_{i,W}^{-} := \sup_{t\in W}[-R_{i,W}(t)]_{+}.$$

$$E_{i,W} := R_{i,W}(b_W -), D_{i,W} := b_W - a_W,$$

$$M_{i,W}^{\text{ret}} = \beta_0 + \beta_{\text{int}} I_{i,W} + \beta_D D_{i,W} + \beta_+ P_{i,W}^{+} + \beta_- P_{i,W}^{-} + \beta_{\text{end}} E_{i,W} + \varepsilon_{i,W}.$$

Preregister the episode window, sampling correction, peak definition, end-window rule, missingness handling, nonlinear alternatives, and held-out comparison. This is an observation or memory rival: it may invalidate retrospective-report evidence for an additive ledger but does not by itself refute countable additivity at the latent level (Fredrickson & Kahneman, 1993; Redelmeier & Kahneman, 1996).

Because the integral, duration, peaks, and end value are generated from the same sampled series, the confirmatory analysis uses a held-out model tournament or an orthogonalized feature construction, matches observation count, cadence, and measurement noise across models, and propagates generated-feature uncertainty. A nominal peak or end coefficient is not an independent evidence vote.

A1.30 Endpoint-clock, reserve-clock, and observation-clock separation

Distinguish $T_i^{\star}$ (truth endpoint), $T_{i,\eta}^{\text{cand}}$ (candidate endpoint), $T_{i,\eta}^{\text{adj}}$ (adjudicated protocol endpoint), τ_i^{rc} (route boundary), T_i^{obs} (observation cessation), and $A_i^{\text{QV,rc}}(t)$ (route-coordinate quadratic-variation clock). Each clock has a declared domain, common origin S_i^{θ}, monotonicity rule, and inverse convention where applicable.

A1.31 Structural identifiability and observability

The scientific functional $q(\vartheta)$ is structurally identifiable under the declared observation process when observationally equivalent parameter values agree on that functional (Villaverde, 2019):

$$P^{\text{obs}}_{\eta,\vartheta_1} = P^{\text{obs}}_{\eta,\vartheta_2} \Rightarrow q(\vartheta_1) = q(\vartheta_2).$$

Full parameter identifiability is stronger: it requires observational equivalence to imply $\vartheta_1 = \vartheta_2$. One high-likelihood fit is not identifiability. For confirmatory use, the state-space model must state whether the target is full parameter identification, state-coordinate observability, route classification, or identification of a named scientific functional, and must demonstrate the corresponding property under the declared observation process and planned sampling.

A1.32 Multiverse-to-preregistration shrinkage

Exploratory analyses may map a multiverse of preprocessing, endpoint, reserve, and observation choices. Confirmatory analysis should shrink that multiverse to a single preregistered specification or a declared model ensemble. The transition from exploration to confirmation is itself a research object. Multiverse analysis and specification-curve principles are cited at the design claim (Steegen et al., 2016; Simonsohn et al., 2020).

A1.33 Exploratory graded-unity sensitivity branch

Status: exploratory sensitivity branch. Let $w_i(t) \in [0,1]$ be a dimensionless graded-unity or exposure weight and $F_i(t)$ a signed physical-time contribution rate before weighting. No current consciousness index is assumed to possess ratio-scale phenomenal meaning or cross-substrate transportability.

$$L_i^{(\text{gw})}(t) := \int_{S_i^\theta}^{t} w_i\,(s) F_i(s)\,\mathrm{d}s.$$

Report a preregistered sensitivity family $w_{i,\kappa}$, including binary-gate, threshold, monotone-link, and bounded-misspecification branches. The weight family, calibration data, direction, bounds, and threshold are fixed without terminal-fit optimization. If the terminal verdict changes materially across scientifically defensible weights, return No adequate estimate and record 'unity-weight-sensitive' as a sensitivity flag. This branch is simulation or sensitivity analysis unless an independent graded-unity measurement bridge is validated.

A1.34 Endpoint hazard and killing-process architecture

Let $g^{\text{adj}}_{i,\eta}\left(t, \mathcal{H}^{\text{adj}}_{i,t}\right) \in \{0,1\}$ be the prospectively fixed adjudication predicate, including dwell, return, signal-quality, competing-event, and review rules. Define

$$T^{\text{adj}}_{i,\eta} := \inf\{t \geq S_i^\theta : g^{\text{adj}}_{i,\eta}\left(t, \mathcal{H}^{\text{adj}}_{i,t}\right) = 1\}, \qquad \inf\emptyset := \infty,$$

and only then define

$$N^{\text{adj}}_{i,\eta}(t) := \mathbf{1}\{T^{\text{adj}}_{i,\eta} \leq t\}.$$

The predicate is primitive; the counting process and endpoint are derived, eliminating the circular construction.

Each process has a declared right-continuous filtration, mark/cause rule, censoring convention, and compensator. For cause e, $\alpha_i^e(t \mid \mathcal{H}_{i,t-})$ is the cause-specific hazard and $\lambda_i^e(t) := R_i^{\mathrm{risk}}(t)\alpha_i^e(t \mid \mathcal{H}_{i,t-})$ the predictable intensity, with $d\Lambda_i^e(t) = \lambda_i^e(t)dt$ on the absolutely continuous branch. A Cox-type hazard may be

$$\alpha_i^e(t \mid \mathcal{H}_{i,t-}) = \alpha_0^e(t)\exp\{\beta_X^\top X_i(t-) + \beta_U U_i(t-) + \beta_A^\top \mathbf{O}_i^{\mathrm{avail}}(t-) + \beta_G^\top G_i(t-)\}.$$

Competing events are handled by declared cause-specific or subdistribution estimands, never by silently reclassifying them. A model containing a ledger-derived predictor is an endpoint-selection rival or sensitivity model, not an independent confirmatory endpoint predicate. The hazard construction follows Cox (1972) and Andersen et al. (1993).

A1.35 Path-signature and multiresolution shape discriminators

Path signatures provide a systematic family of order-sensitive multiscale features for continuous or embedded discrete trajectories (Chevyrev & Kormilitzin, 2016; Fermanian, 2021). The protocol must fix the interpolation, time or lead-lag augmentation, base-point treatment, truncation depth, normalization, and observation window outside terminal fit.

A signature discriminator is route-relevant only when the same frozen feature map predicts held-out terminal, perturbation, or cross-resolution behavior better than equally treated ordinary rivals. Parameterization invariance, tree-like equivalence, and translation invariance are properties to be declared or deliberately broken, not silently treated as psychophysical invariances.

A1.36 Nonequilibrium probability currents and steady-state routes

Let $p_t(x)$ be a normalized state density and $J_t(x)$ its probability current. The continuity equation is

$$\partial_t p_t(x) = -\nabla \cdot J_t(x).$$

$$\partial_t p_t(x) = 0, \qquad \nabla \cdot J(x) = 0, \qquad J \not\equiv 0 \text{ permitted.}$$

A steady-state special case additionally requires a time-invariant density and divergence-free current, together with normalization and declared boundary conditions; a nonequilibrium steady state may retain nonzero current.

Finite-variation branch. With F_t and ρ_t independently measured or constrained,

$$F_t = -\frac{d}{dt}\Phi(X_t) + \rho_t$$

is used only when $\Phi(X)$ has absolutely continuous finite variation. It is not a residual definition of ρ_t.

Semimartingale branch. For diffusion state X_t,

$$d\Phi(X_t) = \mathcal{L}\Phi(X_t)\,dt + \nabla\Phi(X_t)^\top \sigma(X_t)\,dW_t,$$

with the registered Itô–Lévy jump correction on jump paths. Martingale and jump terms are not absorbed into ρ by definition. A nonzero stationary probability current, $j_{\mathrm{ss}} \neq 0$, rejects detailed balance for that fitted state description but does not by itself reject a boundary law. The nonequilibrium comparators draw on Schnakenberg (1976), Crooks (1999), Jarzynski (1997), and England (2013).

A1.37 Anytime-valid evidence processes for sequential testing

Under the simple null P_0 whose conditional predictive density is q_{rival}, with common support and frozen prequential densities,

$$E_k := \prod_{r=1}^{k} \frac{q_{\mathrm{route}}\left(D_r \mid \mathcal{H}_{r-1}^{\mathrm{pre}}\right)}{q_{\mathrm{rival}}\left(D_r \mid \mathcal{H}_{r-1}^{\mathrm{pre}}\right)}$$

satisfies $\mathbb{E}_{P_0}\left(E_k \mid \mathcal{H}_{k-1}^{\mathrm{pre}}\right) = E_{k-1}$. For a composite null $\mathcal{M}_0$, anytime validity is claimed only for a construction proved to satisfy $\mathbb{E}_P\left(E_k \mid \mathcal{H}_{k-1}^{\mathrm{pre}}\right) \le E_{k-1}$ for every $P \in \mathcal{M}_0$, such as a theorem-matched safe mixture or split/prequential method. Outcome-adaptive refitting is prohibited unless incorporated in that proof.

A1.38 Shape-constrained latent trajectory bounds

Let $\mathcal{S}_{\mathrm{adm}}$ be the prospectively declared shape class, let $D_{i,\eta}^{\mathrm{obs}}$ denote the observed data object, and let $\mathcal{C}_{i,\eta}$ collect the declared observation-kernel, endpoint, missingness, calibration, representation, route-state, and numerical-validity constraints. Define

$$K_{i,\eta}^{\mathrm{traj}}\left(D_{i,\eta}^{\mathrm{obs}}; \mathcal{C}_{i,\eta}\right) := \left\{L : L \text{ is compatible with } D_{i,\eta}^{\mathrm{obs}} \text{ under } \mathcal{C}_{i,\eta}\right\},$$

$$I_{i,\eta,T}^{\mathrm{shape}} := \left\{L\left(T_{i,\eta}^{\mathrm{adj}} -\right) : L \in K_{i,\eta}^{\mathrm{traj}}\left(D_{i,\eta}^{\mathrm{obs}}; \mathcal{C}_{i,\eta}\right) \cap \mathcal{S}_{\mathrm{adm}}\right\}.$$

A validly computed empty intersection is a model-incompatibility result, distinct from No adequate estimate. No adequate estimate is reserved for prerequisites below their registered validity or information thresholds, undefined optimization, or numerical invalidity. Shape constraints, tolerances, endpoint objects, and the validity rule are fixed prospectively, and uncertainty in every defining constraint is propagated to the terminal set.

A1.39 Marked shock-and-repair process

Let $N_i^{\mathrm{evt}}(\mathrm{d}s, \mathrm{d}m)$ be a marked random measure on the fixed carrier $(0,\infty) \times \mathcal{M}_i$ with predictable compensator $\nu_i^{\mathrm{evt}}(\mathrm{d}s, \mathrm{d}m)$ and predictable amplitude $A_i^{\mathrm{evt}}(s,m)$ in the declared filtration. Stream entry is encoded by the predictable indicator $\mathbf{1}_{\{S_i^\theta < s\}}$. Assume the following local pathwise integrability condition, or state a prospectively fixed truncation or localization convention that yields a finite-valued ledger:

$$\int_{(0,T]\times\mathcal{M}_i} \mathbf{1}_{\{S_i^\theta < s\}} \left|A_i^{\mathrm{evt}}(s,m)\right| N_i^{\mathrm{evt}}(\mathrm{d}s, \mathrm{d}m) < \infty \quad \text{a.s. for every finite } T.$$

Under that declared domain condition, the uncompensated event ledger is

$$L_i^{\mathrm{evt}}(t) = L_i^{\mathrm{evt}}\left(S_i^\theta -\right) + \int_{(0,t]\times\mathcal{M}_i} \mathbf{1}_{\{S_i^\theta < s\}} A_i^{\mathrm{evt}}(s,m) N_i^{\mathrm{evt}}(\mathrm{d}s, \mathrm{d}m).$$

For the square-integrable compensated branch require predictable κ_i and, on every finite horizon T,

$$\mathbb{E} \int_{(0,T]\times\mathcal{M}_i} \mathbf{1}_{\{S_i^\theta < s\}} \parallel \kappa_i(s,m) \parallel^2 \ \nu_i^{\mathrm{evt}}(\mathrm{d}s, \mathrm{d}m) < \infty.$$

Define $\widetilde{N}_i^{\mathrm{evt}} := N_i^{\mathrm{evt}} - \nu_i^{\mathrm{evt}}$. Under the displayed square-integrability condition,

$$M_i^\kappa(t) = \int_{(0,t]\times\mathcal{M}_i} \mathbf{1}_{\{S_i^\theta < s\}} \kappa_i(s,m) \, \widetilde{N}_i^{\mathrm{evt}}(\mathrm{d}s, \mathrm{d}m).$$

The compensated event term is a square-integrable martingale. If only local integrability is available, label and localize it as a local martingale; infinite-activity branches declare their truncation convention.

This architecture distinguishes shocks, ordinary adaptation, repair, and persistent residuals without treating observed event catalogues as latent ledgers.

A1.40 Terminal-band predictive scoring and small-band asymptotics

For each $K > 0$, define

$$D_{\mathrm{eq},i,\eta}(K) := \mathbf{1}\{\mathcal{J}_{i,\eta,T}^{\mathrm{meas}} \neq \emptyset,\ \mathcal{J}_{i,\eta,T}^{\mathrm{meas}} \subseteq [-K, K]\},$$

This subsection is the declared symmetric identity/positive-linear special case, with $\mathcal{A}_{i,\eta,K}^{\mathrm{meas}} = [-K, K]$. The general decision uses the possibly asymmetric or disconnected acceptance set $\mathcal{A}_{i,\eta}^{\mathrm{meas}}$ defined in §2.3.

$$A_{i,K,t,h}^{\mathrm{obs}} = \{T_{i,\eta}^{\mathrm{adj}} \in (t, t+h],\ D_{\mathrm{eq},i,\eta}(K) = 1\}, \qquad Y_{i,K,t,h}^{\mathrm{obs}} = \mathbf{1}_{A_{i,K,t,h}^{\mathrm{obs}}}.$$

If $K_1 \leq K_2$, then $A^{\mathrm{obs}}(K_1) \subseteq A^{\mathrm{obs}}(K_2)$. Under one fixed compatibility object, the decreasing-band intersection is the observed exact-singleton event; it is promoted to a latent atom only after the measurement bridge and deconvolution gates pass.

Secondary protocol-level categorical score. Let $\mathcal{Y}_{\mathrm{cat}}$ contain six mutually exclusive outcomes: equivalent, registered off-band, indeterminate overlap, valid-empty/model-incompatible, NAE, and no endpoint. For model M, let $p_M(c)$ be the conditional probability assigned to outcome c given the frozen prospective information field; the probabilities are nonnegative and sum to one.

$$S_{\log}(M; Y_{\mathrm{cat}}) := -\sum_{c \in \mathcal{Y}_{\mathrm{cat}}} \mathbf{1}_{\{Y_{\mathrm{cat}}=c\}} \log p_M(c).$$

If a model assigns probability zero to the realized category, the proper logarithmic score is $+\infty$; a finite numerical floor is reported only as a separately labeled sensitivity score.

Primary hierarchical score. Score endpoint occurrence $E \in \{0,1\}$; procedure validity $V \in \{0,1\} \mid E = 1$; and compatibility status $B \in \{\mathrm{EQ}, \mathrm{OFF}, \mathrm{OVERLAP}, \emptyset\} \mid E = V = 1$, where OFF is the registered off-band category and $\emptyset$ is a valid model-incompatibility result. Report the conditional log-score components and their sum. NAE corresponds to $V = 0$; predicting NAE or no endpoint alone earns no scientific credit.

'Registered off-band' means incompatible with the band under the registered procedure at its declared confidence level; it does not assert certainty beyond that procedure. The binary joint-event score based on $\left(Y_i, K, t, h^{\mathrm{obs}}\right)$ is secondary and is reported with separate endpoint and availability scores.

A1.41 Joint longitudinal-endpoint-observation architecture

Every joint record is indexed by protocol endpoint status: no endpoint, candidate crossing, estimated endpoint, confirmed under protocol, revised endpoint, or No adequate estimate. Status-specific observation and retention laws are not pooled without an explicit bridge.

Let $X_i(t)$ be the longitudinal latent state, $U_i(t)$ a route coordinate where applicable, $G_i(t)$ treatment or care context, and $W_i^{\mathrm{nuis}}(t)$ observation-relevant nuisance state. A schematic longitudinal component is

$$dX_i(t) = b_X\big(X_i(t), U_i(t), G_i(t)\big)\, dt + \Sigma_X\big(X_i(t), U_i(t)\big)\, dW_i(t).$$

For a cause-specific endpoint class, a predictable hazard branch may be written

$$\lambda_i^e(t) = R_i^{\mathrm{risk}}(t) \alpha_T^e \left(X_i(t-), U_i(t-), G_i(t-), \mathbf{O}_i^{\mathrm{avail}}(t-)\right).$$

Retention and observation availability are separate stochastic processes:

$$\mathbb{P}\big(R_i^{\mathrm{ret}}(t) = 1 \mid \mathcal{H}_{i,t-}^{\mathrm{joint}}\big) = \pi_R\big(\mathcal{H}_{i,t-}^{\mathrm{joint}}\big).$$

$$\mathbb{P}\big(O_{i,c}^{\mathrm{avail}}(t) = 1 \mid \mathcal{H}_{i,t-}^{\mathrm{joint}}, R_i^{\mathrm{ret}}(t) = 1\big) = \pi_{O,c}\big(\mathcal{H}_{i,t-}^{\mathrm{joint}}\big).$$

For channel c at time t_k, conditional on $O_{i,c}^{\mathrm{avail}}(t_k) = 1$,

$$O_{i,k}^{(c)} \sim \mathcal{K}_{\eta,c,t_k}^{\mathrm{obs}}\big(\cdot \mid X_i(t_k), V_{i,c}^{\mathrm{lat}}(t_k), W_i^{\mathrm{nuis}}(t_k), \mathcal{O}_{i,t_k-}^{\mathrm{obs}}\big).$$

The latent channel target $V_{i,c}^{\mathrm{lat}}$ is prospectively declared; the cumulative ledger is one possible target, not the default target of every report, sensor, scan, behavior, or clinical channel. Dynamic, endpoint, retention, availability, observation, and missingness assumptions are stated jointly (Rizopoulos, 2012).

A1.42 Bridge theorem target between phenomenal integral and reserve product

The Appendix A bridge obeys the Section 1.1 pre-origin left-limit convention: any origin atom is owned exactly once, and neither a route-specific reset nor a one-sided atom assignment is admissible.

A mechanism claim distinguishes four bridge strengths.

Exact path bridge:

$$L_i^{\mathrm{phen}}(t) = L_i^{\mathrm{rc}}(t), S_i^{\theta} \le t < T_i^{\star},$$

$$L_i^{\mathrm{phen}}\big(S_i^{\theta}-\big) = L_i^{\mathrm{rc}}\big(S_i^{\theta}-\big).$$

If an owned origin atom exists, the two bridge paths must match that jump exactly once:

$$\Delta L_i^{\mathrm{phen}}\big(S_i^{\theta}\big) = \Delta L_i^{\mathrm{rc}}\big(S_i^{\theta}\big).$$

Exact terminal bridge:

$$L_i^{\mathrm{phen}}(T_i^{\star}-) = L_i^{\mathrm{rc}}(T_i^{\star}-).$$

Approximate latent bridge. Let $B_i := L_i^{\mathrm{phen}} - L_i^{\mathrm{rc}}$ and suppose

$$\| B_i \|_{\mathcal{B}} \le \varepsilon_B, \quad |B_i(T_i^{\star}-)| \le C_B \| B_i \|_{\mathcal{B}}, \quad \left|L_i^{\mathrm{phen}}(T_i^{\star}-)\right| \le |L_i^{\mathrm{rc}}(T_i^{\star}-)| + C_B\varepsilon_B.$$

Moreover, $|L_i^{\mathrm{rc}}(T_i^{\star}-)| + C_B\varepsilon_B \le K$ implies $L_i^{\mathrm{phen}}(T_i^{\star}-) \in [-K, K]$. This establishes finite-band latent compatibility. Exact PTVC transfers only when both route and bridge terminal residuals equal zero in the declared truth mode.

Protocol bridge. A measured route proxy can support only a measured-estimand claim unless the observation, endpoint, representation, calibration, and measured-to-latent bridges are separately defended. The bridge class, norm, tolerance, calibration procedure, and held-out failure rule are preregistered.

A1.43 Absorbing-boundary versus conditioned-bridge discriminator

A central ambiguity is whether terminal concentration is generated forward by an intrinsic boundary mechanism or by conditioning on a selected endpoint. An intrinsic-boundary model may be written

$$\mathrm{d}X_t = b(X_t, U_t, t)\,\mathrm{d}t + s(X_t, U_t, t)\,\mathrm{d}W_t, \qquad \tau_0 := \inf\{t: X_t \in \mathcal{N}_0\}.$$

For a positive-probability terminal or hitting event B, define $h_B(t,x) := P_0(B \mid X_t = x)$ and, on the interior domain where it is used, require $h_B > 0$, $h_B \in C^{1,2}$, and the theorem-matched backward equation, boundary/terminal data, existence, and nonexplosion assumptions. For a point endpoint or set with zero transition probability, do not write an ordinary conditional probability; use a positive transition density $h_y(t,x) := p_0(T-t,x,y)$ or prove convergence of conditioned laws for a declared shrinking-neighborhood sequence $\mathcal{N}_{0,n} \downarrow \mathcal{N}_0$.

$$h_{\text{hit}}(t,x) := P_0(\tau_0 \le T \mid X_t = x)$$

Under the applicable positive h on its declared interior domain, a state-based Doob transform is

$$\mathrm{d}X_t = [b(X_t,U_t,t) + a(X_t,U_t,t)\nabla_x \log h(t,X_t)]\mathrm{d}t + s(X_t,U_t,t)\mathrm{d}W_t, \quad a = ss^\top.$$

Compare joint path and endpoint likelihood, flux into $\mathcal{N}_0$, endpoint-time distribution, normalized volatility, perturbation response, out-of-window prediction, and generator recovery. If the conditioned process performs as well under the same observation law, the intrinsic-boundary interpretation is Model-ambiguous.

A1.44 Remaining-sum kernel and dependence-budget spectroscopy

Let $\ell := L_i(t)$ be fixed by the prospective prefix information $\mathcal{H}_{i,t}^{\text{pre}}$. Let p_i be a one-step suffix density and $p_{i,n}$ the corresponding n-fold convolution on the conditionally i.i.d. unit-weight branch. When $p_{i,n}(-\ell) > 0$, define

$$q_{i,n}(x \mid \ell) := \frac{p_i(x)p_{i,n-1}(-\ell - x)}{p_{i,n}(-\ell)}.$$

This closed form applies only to conditionally i.i.d. unit-weight increments. For deterministic or prefix-measurable weights define $Y_{i,j} = w_{i,j}\Delta_{i,j}$. If the $Y_{i,j}$ are conditionally independent with densities $f_{i,j}$, replace the unit-weight convolution by the convolution of the relevant $f_{i,j}$, including scaling. For dependent increments use the full conditional joint-density ratio. The covariance identity below belongs to the separately declared weighted closure law.

If finite conditional second moments hold, the weights are prefix-measurable, and exact weighted suffix closure satisfies $\sum_{j=1}^{n} w_{i,j}\,\Delta_{i,j} = -\ell$ almost surely conditionally on the prefix, then

$$0 = \sum_{j=1}^{n} w_{i,j}^2 \operatorname{Var}\big(\Delta_{i,j} \mid \mathcal{H}_{i,t}^{\text{pre}}\big) + 2 \sum_{1 \le j < k \le n} w_{i,j}\, w_{i,k} \operatorname{Cov}\big(\Delta_{i,j}, \Delta_{i,k} \mid \mathcal{H}_{i,t}^{\text{pre}}\big).$$

Unconditionally across varying prefixes, the suffix-sum variance is generally $\operatorname{Var}(\ell)$ rather than zero. This yields a nonterminal dependence test only under the declared observation and rival models.

A1.45 Prospectively frozen model ensemble and stacking verdict

Let $b = 1, \ldots, B$ index preregistered dependence blocks (participant, site, family, endpoint window, apparatus, or time block as appropriate). Every preprocessing, imputation, representation, endpoint model, candidate model, and weight is fitted within the training portion that excludes block b. With leave-one-block-out predictive density $p_k(y_b \mid y_{-b})$ (Vehtari et al., 2017; Yao et al., 2018),

$$J_{\text{stack}}(w) = \sum_{b=1}^{B} \log\Big[\sum_{k=1}^{K} w_k\, p_k(y_b \mid y_{-b})\Big], \qquad J_{\text{stack}}^{\text{sup}} = \sup_{w \in \Delta_K} J_{\text{stack}}(w).$$

Use an optimizer only after attainment; otherwise report an ε-optimal vector. Site- or endpoint-held-out evaluation is required for transport claims. Stable route weight identifies a model worth further testing, not a terminal verdict; weight spread across adequate rivals yields Model-ambiguous.

Stacking weights are predictive mixture coefficients, not posterior probabilities that a mechanism is true. Report the full ε-optimal weight set or a stability envelope when the optimum is nonunique or nearly flat.

A1.46 Measured implementation of the exogenous-endpoint occupation screen

Let $V(t) \in \{0,1\}$ indicate that all registered prerequisites yield a valid compatibility-set computation, let $\mathcal{J}(t-)$ be that set when $V(t) = 1$, and let $\mathcal{A}^{\text{sci}}$ be the registered measured acceptance set. Define five mutually exclusive states: equivalent, indeterminate overlap, off-band, valid-empty/model-incompatible, and No adequate estimate.

$$
\begin{aligned}
J^{\text{eq}}(t) \quad &:= \mathbf{1}\{V(t) = 1,\ \mathcal{J}(t-) \neq \emptyset,\ \mathcal{J}(t-) \subseteq \mathcal{A}^{\text{sci}}\},\\
J^{\text{ind}}(t) \quad &:= \mathbf{1}\{V(t) = 1,\ \mathcal{J}(t-) \neq \emptyset,\ \mathcal{J}(t-) \cap \mathcal{A}^{\text{sci}} \neq \emptyset,\ \mathcal{J}(t-) \nsubseteq \mathcal{A}^{\text{sci}}\}.\\
J^{\text{off}}(t) \quad &:= \mathbf{1}\{V(t) = 1,\ \mathcal{J}(t-) \neq \emptyset,\ \mathcal{J}(t-) \cap \mathcal{A}^{\text{sci}} = \emptyset\},\\
J^{\emptyset}(t) \quad &:= \mathbf{1}\{V(t) = 1,\ \mathcal{J}(t-) = \emptyset\}, \qquad J^{\text{NAE}}(t) := \mathbf{1}\{V(t) = 0\}.
\end{aligned}
$$

On the fixed carrier $(0, \tau]$, require the left-limit state $J^c(t-)$ to be predictable (or replace it by its predictable projection $\text{Pred}[J^c](t)$) and integrate $\mathbf{1}_{\{S_i^\theta < t\}} J^c(t-)$ against the same estimated class-e compensator to obtain $\widehat{\omega}_{i,\eta,e}^c(\tau)$ for $c \in \{\text{eq}, \text{ind}, \text{off}, \emptyset, \text{NAE}\}$. The five states partition at-risk compensator mass; none disappears from the denominator.

A1.47 Causal-locus calibration and modular mechanism benchmark

Object. A preregistered graph separates preassignment history H_0, assignment A, precommitment eligibility E_0^{prot}, commitment record I_C, private policy seed R_C^{priv}, commitment information $\mathcal{F}_C$, committed action C, structural action availability $\mathcal{A}_C^{\text{do}}$, physical transition X', target/randomizer Z^{rnd}, route mark J^{route}, endpoint object E^{prot}, postcommitment retention R^{ret}, observation availability $\mathbf{O}^{\text{avail}}$, and observed data O^{obs}. The benchmark combines conditional calibration, leakage budgeting, action-invariance tests, denominator flow, selection and garbling controls, redundant randomizer technologies, replay against unused committed target streams, common-pipeline generator recovery, open-set abstention, and worst-case false-support calibration.

Use. This is a mature-method wrapper for safe mechanism localization, not a new PTVC mechanism. It cross-references the existing causal-measurement graph, target-trial, endpoint-negative-control, adaptive observation, joint longitudinal-endpoint-observation, and registered model-ensemble entries. It keeps detailed experimental portfolios in Appendix B rather than duplicating them in Appendix A.

Part II: Frontier Exactness Mechanisms and Theorem Targets

Part II control. Every item is a frontier formal or empirical target and may carry one or more codes from Table 4. Only [TT] denotes an open theorem target; no proof is claimed beyond explicitly labeled [CP] results or cited theorems whose assumptions are shown to match. Each entry must expose assumptions, any established conditional result, the open extension where applicable, a countermodel or retirement condition, and the required specialist domain before promotion.

Part II develops frontier mechanisms at an earlier stage of maturity than the core Formal Model but already concrete enough to guide theorem construction, simulation, rigorous countermodeling, and low-burden empirical design. Each entry states a formal object, distinctive consequence, nearest adequate rival, outcome criterion, and promotion or retirement path.

Every frontier route remains subject to the common observation-garbling, posterior-uncertainty, missingness, and observation-floor controls in Section 7 and Appendix B.

Status convention. Each Part II entry is classified as an identity or representation result, a conditional proposition under displayed assumptions, a mechanism hypothesis, an empirical discriminator, a simulation target, or an open theorem target. An open theorem target becomes an established result only when a proof is supplied or a cited theorem matches its assumptions.

A2.1 Branch-indexed PTVC-admissibility law with separate selector

Status: [IR/CP/TT] branch-indexed representation, conditional normalized-kernel construction, and open measurable-extension target.

Let $(\mathsf{H}_0, \Sigma_{\mathsf{H}_0})$, $\quad P_0^{\mathrm{pre}} \in \mathcal{P}(\mathsf{H}_0)$ be the declared measurable prefix carrier and its probability law.

For prefix h and branch b, define $\Gamma_b(h) := \{\gamma : (h, \gamma, b) \text{ is admissible and } h \oplus \gamma \in \mathsf{H}\}$ and $\mathcal{F}^b_{\mathrm{PTVC}}(h) := \{\gamma \in \Gamma_b(h) : h \oplus \gamma \in \mathcal{F}^b_{\mathrm{PTVC}}\}$. Require the branch selector correspondence $\mathcal{I}_{\mathrm{sel}}(h)$, the continuation correspondence $\Gamma_b(h)$, their graphs and sections, and concatenation to be measurable.

$$\begin{aligned} S(\mathrm{d}b \mid h) &\quad \text{is supported on } \mathcal{I}_{\mathrm{sel}}(h), \\ Q^b(\mathrm{d}\gamma \mid h) &\quad \text{is supported on } \Gamma_b(h), \\ Q^b\big(\mathcal{F}^b_{\mathrm{PTVC}}(h) \mid h\big) &\quad = 1. \end{aligned}$$

$$Q^{\mathrm{sel}}(\mathrm{d}h, \mathrm{d}b, \mathrm{d}\gamma) = P_0^{\mathrm{pre}}(\mathrm{d}h) S(\mathrm{d}b \mid h) Q^b(\mathrm{d}\gamma \mid h).$$

The selector and branch-specific continuation kernels are normalized probability kernels on their declared measurable carriers.

All kernels are normalized. The complete-history law is the push-forward of Q^{sel} under $(h, b, \gamma) \mapsto h \oplus \gamma$; iterated integration proves normalization.

A2.2 Relative-coboundary and Hodge-completion law

Status: [CM/TT] candidate potential mechanism and theorem target.

On an augmented psychophysical state graph treated as a one-dimensional cell complex, let ω_s be a signed experiential 1-form. The graph Hodge decomposition is (Jiang et al., 2011)

$$\omega_s = \mathrm{d}\Phi_s + h_s.$$

A pure-potential branch has $h_s = 0$. A coexact component requires a higher-dimensional oriented cell or simplicial complex with declared inner products and boundary conditions; it is absent on a graph treated strictly as one-dimensional.

$$C_s[\gamma] = \int_\gamma \omega_s = \Phi_s(z_T) - \Phi_s(z_S).$$

If stream boundaries share the same potential level, closure follows by identity. Distinctive implication: held-out loop integrals satisfy $\oint_\gamma \omega_s = 0$ on recombinable domains. Nearest rival: ordinary state-value learning or adaptation. Failure condition: stable nonzero loop periods after complete-state specification defeat the pure-potential branch and require residual storage or global context.

A2.3 Protected stream charge or continuation register

Status: [CM/CP/TT] candidate mechanism, conditional proposition, and open zero-level derivation target. A continuation-charge route requires a separately identified continuation term R_i^{cont} and

$$Q_i^{\mathrm{ch}}(t) = L_i^{\mathrm{phen}}(t) + R_i^{\mathrm{cont}}(t), \qquad dQ_i^{\mathrm{ch}}(t) = 0.$$

The exact conditional result is

$$dQ_i^{\mathrm{ch}} = 0,\ Q_i^{\mathrm{ch}}\left(S_i^{\theta} -\right) = 0,\ R_i^{\mathrm{cont}}(T_i^{\star} -) = 0 \Rightarrow L_i^{\mathrm{phen}}(T_i^{\star} -) = 0.$$

If an owned origin atom belongs to the stream, require $\Delta Q_i^{\mathrm{ch}}\left(S_i^{\theta}\right) = 0$ so that the origin contribution is carried once rather than omitted or duplicated.

The conserved zero level follows from an independent symmetry, boundary condition, or theorem. That derivation and a phenomenal bridge promote conservation from an initial-charge identity to a PTVC-relevant mechanism.

A2.4 PTVC-completion committors and boundary geometry

Status: [CM/ED] candidate mechanism and empirical discriminator. Let $X_i^{\mathrm{comp}}(t) = x$, $\ell := L_i^{\mathrm{phen}}(t)$, and $Q_{h_t}^{a,g}$ be the prospectively declared continuation law conditional on the observed prefix under structural action a and frozen continuation regime g, on endpoint branch $T_i^{\star}(a,g)$. The history-indexed law may be reduced to the corresponding state-indexed continuation law only under a separately declared state-sufficiency condition; otherwise the history index is retained. For finite band $[-K, K]$, define

$$\mathcal{F}_{\mathrm{comp},K}^{a,g}(h_t) := \{\gamma\colon\ \ell + \Delta L_{t:T_i^{\star}(a,g)}(\gamma) \in [-K, K]\}.$$

The action-specific and state-level committors are

$$p_{\eta,h_t,g}^{\mathrm{comp}}(a) := Q_{\eta,h_t}^{a,g}\left(F_{\eta,\mathrm{comp}}^{a,g}(h_t)\right), \qquad 0 \le p_{\eta,h_t,g}^{\mathrm{comp}}(a) \le 1.$$

$$h_{g,K}(x, \ell, t) := Q^{g}\left(\ell + \Delta L_{t:T_i^{\star}(g)} \in [-K, K] \mid X_i^{\mathrm{comp}}(t) = x, L_i^{\mathrm{phen}}(t) = \ell\right).$$

The committor asks for the conditional probability that the declared suffix satisfies the registered completion event from the current state. The state, endpoint branch, baseline law, suffix space, and event are fixed independently and transported across actions, horizons, tasks, and controls (E & Vanden-Eijnden, 2006).

A2.5 Joint terminal reachability and constrained reach-avoid problem

Status: [CP/TT]. Proposition 2 supplies the controlling scalar directional reachability condition. This entry adds only the joint residual-solvency set and constrained reach-avoid extension for several streams or coupled resources.

For multiple active streams, define the current residual vector and the nonempty attainable joint future-contribution set only from ownership-preserving, nonduplicating continuations with stream-coordinate provenance. Cross-stream transfer or cancellation is prohibited unless a separately named junction or transfer theory is being tested. Define

$$\mathbf{L}(t) \in \mathbb{R}^m, \qquad \emptyset \neq \mathcal{A}_{\text{fut}}^{\text{joint}}(t,z) \subseteq \mathbb{R}^m.$$

$$-\mathbf{L}(t) \in \mathcal{A}_{\text{fut}}^{\text{joint}}(t,z).$$

Equivalently, define the residual-solvency set

$$\mathcal{S}_{\text{res}}^{\text{joint}}(t,z) := \{\ell \in \mathbb{R}^m : -\ell \in \mathcal{A}_{\text{fut}}^{\text{joint}}(t,z)\}.$$

Then exact joint completion requires

$$\mathbf{L}(t) \in \mathcal{S}_{\text{res}}^{\text{joint}}(t,z).$$

The scalar directional reachability screen is Proposition 2. This entry adds joint terminal reachability and subset-bottleneck constraints when future resources or opportunities are shared across streams. Scientific value: impossible prefixes become explicit and individual completion is separated from joint solvency. Nearest rival: ordinary allostasis, resource depletion, or constrained control. Failure condition: a realized prefix leaves the declared residual-solvency set and still terminates neutrally inside the claimed reference class.

Let $A \subset \mathbb{R}^m$ be the closed convex attainable joint future-contribution set and define its support function by $h_A(a) := \sup_{x \in A} a^\top x$. Then

$$-\mathbf{L}(t) \in A \Leftrightarrow a^\top[-\mathbf{L}(t)] \leq h_A(a) \quad \text{for every } a \in \mathbb{R}^m$$

For any stream subset J, $a = \pm \mathbf{1}_J$ yields the signed subset-bottleneck inequalities. For nonconvex $\mathcal{A}$, these inequalities are necessary only for its convex hull; exact membership remains the sufficiency target.

A2.6 Random-endpoint no-go and expected/unexpected endpoint fork

Status: [CP/ED] theorem-backed conditional-result and empirical-discriminator fork. Proposition 1 and its approximate-dependence Corollary 1.1 provide the controlling random-endpoint results; the preceding conditional zero-atom screen supplies a distinct at-risk diagnostic. This entry distinguishes expected state-coupled boundaries, unexpected exogenous endpoints, administrative endpoint coding, and endpoint-conditioned bridge rivals.

The route must specify whether endpoint timing is coupled to the latent completion coordinate, whether the endpoint is a stopping time in the declared filtration, whether any endpoint-coincident phenomenal contribution is included, and which continuation or routing alternatives remain admissible. A merely administrative endpoint cannot inherit a truth-level closure claim.

A2.7 Endpoint-support and compensator law

Status: [CM/TT]. Let $X_i^{\text{comp}}(t-)$ be a predictable route-specific completion coordinate. The route is PTVC-relevant only through an independently defended bridge to the phenomenal ledger.

Exact branch. Define the route and phenomenal singleton sets

$$\mathcal{N}_0^{\text{rc}} := \{0\}, \qquad \mathcal{C}_0^{\text{phen}} := \{0\}, \qquad X_i^{\text{comp}}(t-) = 0 \Leftrightarrow L_i^{\text{phen}}(t-) = 0.$$

$$\mathbf{1}_{\{X_i^{\text{comp}}(t-) \notin \mathcal{N}_0^{\text{rc}}\}} \, \mathrm{d}\Lambda_i^e(t) = 0 \quad \text{a.s. on the stopping-time branch,}$$

$$\mathbf{1}_{\{X_i^{\text{comp}}(t-) \notin \mathcal{N}_0^{\text{rc}}\}} \, \mathrm{d}\Pi_i^{e,p}(t) = 0 \quad \text{a.s. on the non-stopping branch.}$$

Finite-band branch. Calibrate route and phenomenal half-widths separately in their declared units:

$$\mathcal{N}_{K_{\text{rc}}}^{\text{rc}} = [-K_{\text{rc}}, K_{\text{rc}}], \qquad \mathcal{C}_{K_{\text{phen}}}^{\text{phen}} = [-K_{\text{phen}}, K_{\text{phen}}].$$

A bidirectional equivalence between membership in these bands is asserted only after independent bridge validation; otherwise use the declared one-way implication or compatibility relation. Endpoint support is stated in the route coordinate as

$$\mathbf{1}_{\{X_i^{\mathrm{comp}}(t-)\notin\mathcal{N}_{K_{\mathrm{rc}}}^{\mathrm{rc}}\}}\,\mathrm{d}\Lambda_i^{e}(t)=0\quad\text{or}\quad\mathbf{1}_{\{X_i^{\mathrm{comp}}(t-)\notin\mathcal{N}_{K_{\mathrm{rc}}}^{\mathrm{rc}}\}}\,\mathrm{d}\Pi_i^{e,p}(t)=0.$$

Stopping-time and non-stopping branches are reported separately. These route-specific support laws do not replace Proposition 3 and inherit its filtration, predictability, endpoint-class, projection, and bridge requirements.

A2.8 Stream junction algebra

Status: [IR/CP/TT] ownership-preserving representation, conditional nonduplication law, and measurable-composition theorem target.

For each elapsed event token e, apply exactly one prospectively typed measurable ownership allocation across the augmented output account:

$$w_{s',e}(\omega)\ge 0,\qquad \sum_{s'\in\mathcal{S}_{\mathrm{out}}^{+}} w_{s',e}(\omega)=1\quad\text{a.s.}$$

Here $\mathcal{S}_{\mathrm{out}}^{+}$ is the disjoint union of the outgoing streams with the retained, sink, and unresolved accounts. The map $e\mapsto w(s',e)$ is measurable, and no token is duplicated across accounts or streams.

Every unresolved account preserves exposure, positive variation, and negative variation as separate typed coordinates. Opposite-sign cancellation within the unresolved account is prohibited; bounds propagate each coordinate separately or return Model-ambiguous/No adequate estimate.

$$J_c\circ P_\pi^{\mathrm{in}}=P_\pi^{\mathrm{out}}\circ J_c.$$

Sequential composition and nonduplication remain required. No stream's residual may cancel or offset another stream's residual.

A2.9 Physical marginal transparency and no-survival-oracle constraint

Status: [CM] operational-transparency mechanism requirement.

A global PTVC-admissibility law must preserve ordinary physical predictions available to a modeled decision process or observer without terminal-outcome access. Let $\mathcal{I}_t^{\mathrm{avail}}$ denote exactly the information prospectively available to that process. For observed physical events A in the declared admissible preterminal sigma-field,

$$Q\big(A\mid\mathcal{I}_t^{\mathrm{avail}}\big)=P_0\big(A\mid\mathcal{I}_t^{\mathrm{avail}}\big).$$

For each declared exogenous randomizer Z, the route states whether $Q\{Z\in B\mid\mathrm{do}(\mathrm{A}_t=a),\mathcal{F}_{\mathrm{pre}}\}=P_0\{Z\in B\mid\mathcal{F}_{\mathrm{pre}}\}$ for every registered action c and measurable target set B, or it prospectively specifies and tests the deviation.

A2.10 Scalarization holonomy and cross-sign calibration

Status: [TT] scalarization-holonomy theorem target.

$$T_{\beta\leftarrow\alpha}\colon\mathcal{V}_\alpha\to\mathcal{V}_\beta,\qquad \mathrm{Hol}_{\alpha_1}(\gamma):=T_{\alpha_1\leftarrow\alpha_k}\circ\cdots\circ T_{\alpha_2\leftarrow\alpha_1}\colon\mathcal{V}_{\alpha_1}\to\mathcal{V}_{\alpha_1}.$$

A global scalarization claim requires identity holonomy or a preregistered tolerance in a declared metric:

$$\mathrm{d}_G\,(\mathrm{Hol}(\gamma), I) \le \varepsilon_{\mathrm{hol}}.$$

Failure indicates path-dependent calibration, unresolved chart dependence, or an omitted state variable; it prevents a unique global scalar PTVC claim without a further quotient or ordered-structure construction.

A2.11 Quantized or error-corrected residual code

Status: [CM/TT]. Let G be an ambient abelian group, $\Lambda_0 \le G$ a subgroup, and $\mathcal{R}_{\mathrm{code}} := G/\Lambda_0$ the quotient residual space equipped with a representative-invariant metric. A quantized route may impose

$$[r_s(T_s^\star -)] = [0] \quad \text{in } \mathcal{R}_{\mathrm{code}}.$$

Modular neutrality is not automatically phenomenal zero. A PTVC-relevant code requires an independently fixed decoder

$$d: \mathcal{R}_{\mathrm{code}} \to \mathcal{R}_{\mathrm{phen}}, \quad d([0]) = 0_{\mathrm{phen}}, \quad d^{-1}\big(\{0_{\mathrm{phen}}\}\big) = \{[0]\}.$$

Accumulation, routing, copying, restoration, fission, fusion, and junction maps must commute with encoding and decoding. Without this bridge the proposal is a quantized-equivalence or error-corrected residual theory, not exact PTVC.

A2.12 Entropic path-space steering and Schrödinger bridge frontier

Status: [CM/TT]. Define a branch-indexed dynamically admissible path-law class and keep its path-space steering cost distinct from the unrestricted event-level information projection in A2.18. (Chen et al., 2021; Föllmer, 1985; Léonard, 2014)

Let the branch-specific baseline path law, dynamically admissible law class, and terminal-band event be prospectively declared. For $\alpha \in [0,1]$, define the path-law steering cost by

$$J_{\mathrm{path}}^{(b)}(K,\alpha) := \inf_{\substack{Q \in \mathcal{Q}_{\mathrm{dyn}}^{(b)} \\ Q\left(A_K^{(b)}\right) \ge 1-\alpha}} D_{\mathrm{KL}}\left(Q \,\middle\|\, P_0^{(b)}\right).$$

The path-law cost is bounded below by the unrestricted event-level information projection in A2.18 whenever the dynamically admissible law class is a subset of the absolutely continuous comparison class used there. If the classes coincide, the distinction collapses. Introduce an optimizer only after attainment is established; otherwise report an ε-optimal law. Exact singleton pinning is a limiting or singular-support branch, not the default finite-cost object.

Empty feasible-class convention:

$$\inf \emptyset := +\infty.$$

A2.13 Topological, gauge, and cocycle exactness

Status: [CP/TT] conditional relative-cohomology result and open PTVC bridge target.

Let $\mathcal{X}$ be a smooth paracompact manifold and $\mathcal{B}_s \subset \mathcal{X}$ a closed embedded submanifold (or the explicitly declared boundary submanifold), and let ι denote inclusion. Define the relative de Rham complex by

$$\Omega^\bullet(\mathcal{X}, \mathcal{B}_s) = \{\omega : \iota^\star \omega = 0\}.$$

Require $\mathrm{d}\omega_s = 0$ before defining its relative cohomology class, and impose the independent obstruction-vanishing condition

$$[\omega_s] = 0 \quad \text{in } H^1_{\mathrm{dR}}(\mathcal{X}, \mathcal{B}_s; \mathbb{R}), \quad \omega_s = \mathrm{d}\Phi_s, \quad \Phi_s|_{\mathcal{B}_s} = 0.$$

Only then does there exist a relative potential Φ_s with $\omega_s = \mathrm{d}\Phi_s$ and $\Phi_s|_{\mathcal{B}_s} = 0$ after normalizing one common boundary value; declared relative-cycle periods then vanish. This relative-cohomology formulation follows standard differential-form and de Rham foundations (Bott & Tu, 1982).

A2.14 Frontier substrate routes

Status: [ST]. Quantum, time-symmetric, topological, holographic, computational, or other substrate proposals enter this appendix when they specify a carrier, a map to the declared phenomenal object, a distinctive consequence, a nearest adequate rival, a failure condition, and a promotion or retirement path. These frameworks are formally distinct candidate-route families. Their PTVC relevance becomes empirical when a validated phenomenal bridge and a distinctive held-out prediction support the protocol-fixed route; target-level adjudication proceeds through the full measurement, endpoint, transport, and rival-comparison architecture (Nielsen & Chuang, 2010; Shalizi & Crutchfield, 2001; Wharton & Argaman, 2020).

Promotion requires a theorem or quantitative prediction unavailable to ordinary classical, biological, computational, selection, or observation models under the same data and scoring rules.

A2.15 Open countermodel tournament and promotion map

Status: [ST] red-team simulation and promotion framework.

Beyond its separately reported status code, every mechanism is evaluated by the same five substantive fields: formal object; observable implication; closest adequate rival; disconfirming outcome; and promotion or retirement threshold.

Every Appendix A, Part II route should face a countermodel tournament. A route is promoted when it supplies all five elements:

object + consequence + rival + failure condition + recovery path.

Near-term frontier tests include loop integrals, common-completion tests, reachability-cone violations, endpoint-type discrimination, junction algebra, reserve-orthogonal perturbations, and stress-test simulators. Medium-term bridge hypotheses include PTVC-completion committors, protected charges, endpoint-support laws, and scalarization holonomy. Substrate proposals advance into the empirical program as soon as they produce a bridge theorem or safe distinctive discriminator.

A2.16 Characteristic-function constraint spectroscopy

Status: [CP/TT] conditional characteristic-function theorem target.

Let the residual vector have declared dimension m, mean vector μ, covariance matrix Σ, and row/column orientation. The characteristic-function identity is the distribution-free restriction. When finite second moments exist, the displayed mean-and-covariance restrictions are equivalent to exact additive closure almost surely; without finite second moments, use the characteristic-function identity.

For an additive-increment vector $\Delta \in \mathbb{R}^m$ and a declared constraint matrix $A \in \mathbb{R}^{r \times m}$, exact additive closure $A\Delta = 0$ almost surely is equivalent to the vector characteristic-function identity below (Lukacs, 1970).

$$\varphi_{A\Delta}(u) := \mathbb{E}[e^{iu^\top A\Delta}], \quad u \in \mathbb{R}^r.$$

The almost-sure constraint is equivalent to

$$A\Delta = 0 \text{ a.s.} \quad \Leftrightarrow \quad \varphi_{A\Delta}(u) = 1 \text{ for every } u \in \mathbb{R}^r.$$

When finite second moments exist, this is equivalent to

$$A\,\mathbb{E}[\Delta] = 0, \quad A\,\mathrm{Cov}(\Delta)A^\top = 0.$$

With an independent, calibrated additive measurement-error vector $\varepsilon \in \mathbb{R}^r$,

$$\varphi_{\mathrm{obs}}(u) = \varphi_{A\Delta}(u)\,\varphi_\varepsilon(u), \qquad u \in \mathbb{R}^r.$$

For r=1 and $A\Delta = Z_i^\star$, the theorem reduces to the scalar residual characteristic-function condition. Deconvolution is available only where $\varphi_\varepsilon(u) \neq 0$.

Distinctive implication: the rank and geometry of declared additive constraints can be tested without Gaussianity. Nearest rival: ordinary correlation, smoothing, or a fitted partition. Failure condition: ridge directions vanish under held-out partitions, calibrated error, or unconstrained rivals.

Limitation: this identity is algebraically equivalent to the declared additive constraint. It is not independent evidence when the partition, matrix, and increment vector are constructed from the same fitted ledger. Empirical use requires independently fixed partitions or routing matrices, held-out frequency ranges, competing constraint geometries, and a calibrated error model.

A2.17 Generator-ratio and cycle-affinity exactness

Status: [TT] generator-transform theorem target.

On a finite conservative irreducible chain, every positive Q_0-harmonic function is constant and the transform below is the identity. A nontrivial branch therefore requires a reducible or transient structure with a nonconstant harmonic function, a killed or sub-Markov generator, a positive eigenfunction with the matching diagonal shift, or a space-time harmonic function. On the inherited support, define

$$q_{xy}^h = q_{xy}^0 \frac{h(y)}{h(x)}, \quad x \neq y, \qquad q_{xx}^h = -\sum_{y \neq x} q_{xy}^h.$$

An eigenfunction transform uses its matching diagonal shift. A nonharmonic completion function belongs to a separately specified Feynman–Kac or killed-process branch with its potential or killing term. Every branch states whether the transformed operator is conservative, sub-Markov, time-inhomogeneous, or conditioned at a boundary.

Along a closed directed cycle C, the product ratio telescopes:

$$\prod_{(x,y)\in C} \frac{q_{xy}^h}{q_{xy}^0} = 1.$$

Thus a state-based Doob potential transform preserves cycle affinity on the inherited support at a fixed time slice. A time-dependent or history-dependent transform is a different class and must include its additional temporal or path terms; the fixed-slice telescoping identity may not be transferred to it without proof (Doob, 1957; Chetrite & Touchette, 2015).

A2.18 Finite-band steering lower bound and exact-pinning obstruction

Status: [CP/TT]. Let $Z^{(b)}$ be the terminal residual on branch b and define

$$A_K^{(b)} := \{|Z^{(b)}| \leq K\}, \qquad p_K^{(b)} := P_0^{(b)}\left(A_K^{(b)}\right).$$

For $p, q, \alpha \in [0,1]$, binary relative entropy uses the extended-real convention

$$d_{\mathrm{bin}}(q\|p) := q\log\frac{q}{p} + (1-q)\log\frac{1-q}{1-p} \qquad (0 < p < 1),$$

$$d_{\mathrm{bin}}(q\|0) := \begin{cases} 0, & q = 0, \\ +\infty, & q > 0, \end{cases} \qquad d_{\mathrm{bin}}(q\|1) := \begin{cases} 0, & q = 1, \\ +\infty, & q < 1. \end{cases}$$

The event-level value is

$$J_K^{(b)}(\alpha) := \begin{cases} 0, & p_K^{(b)} \ge 1-\alpha, \\ d_{\mathrm{bin}}\left(1-\alpha\|p_K^{(b)}\right), & 0 < p_K^{(b)} < 1-\alpha, \\ +\infty, & p_K^{(b)} = 0 \text{ and } \alpha < 1. \end{cases}$$

For the unrestricted absolutely continuous law class, this is the exact information projection:

$$\inf_{\substack{Q \ll P_0^{(b)} \\ Q\left(A_K^{(b)}\right) \ge 1-\alpha}} D_{\mathrm{KL}}\left(Q\|P_0^{(b)}\right) = J_K^{(b)}(\alpha).$$

Map each path to $\mathbf{1}_{A_K^{(b)}}$. In the binary divergence use $0\log 0 := 0$ and understand boundary values by lower-semicontinuous extension; the separately displayed $p = 0$ and $p = 1$ cases remain controlling.

[P] Proof. Map each path to $\mathbf{1}_{A_K^{(b)}}$ and apply KL data processing. Minimizing the resulting binary divergence over $q \ge 1-\alpha$ gives the displayed cases; when finite, a density constant on $A_K^{(b)}$ and its complement attains the Bernoulli tilt (Csiszár, 1975). For a restricted dynamic or intervention class, the infimum is at least $J_K^{(b)}(\alpha)$ and may be unattainable.

A2.19 Spectator factorization and completion-scarcity additivity

Status: [CP/ST] selector-separability condition and simulation target.

For a declared event A with completion probability $h_A(z)$, define logarithmic completion scarcity by

$$\mathcal{S}_A(z) := \begin{cases} -\log h_A(z), & h_A(z) > 0, \\ +\infty, & h_A(z) = 0. \end{cases}$$

Factorization of h implies additivity of this logarithmic scarcity. Any other proper score requires its own additivity theorem.

Keep $Q_{\mathcal{S}_1 \cup \mathcal{S}_2} = Q_{\mathcal{S}_1} \otimes Q_{\mathcal{S}_2}$ on the declared independent product branch. For coordinate completion events $A \subseteq \Omega_A$ and $B \subseteq \Omega_B$, define

$$C := (A \times \Omega_B) \cap (\Omega_A \times B) = A \times B.$$

Then

$$h_{A\times B}(z_A, z_B) = h_A(z_A)h_B(z_B), \qquad -\log h_{A\times B} = -\log h_A - \log h_B.$$

The union is reserved for stream labels; simultaneous completion is the intersection/product event.

Failure may indicate selector inconsistency, stream-count bias, unit-refinement bias, incorrect baseline factorization, hidden spectator dependence, omitted common cause, shared completion scarcity, or a genuine nonlocal coupling. Those alternatives require equal-rival treatment before the route is rejected.

A2.20 Dual-potential alignment

Status: [ED] local alignment diagnostic. Let x belong to one declared state space $\mathcal{X}$, let a(x) be a positive semidefinite local diffusion matrix, and define the local seminorm

$$\| v \|_{a(x)} := \sqrt{v^\top a(x) v}.$$

Let $B_L(x)$ be an independently oriented nonnegative ledger-burden coordinate and define completion scarcity by $\mathcal{S}_{\text{comp}}(x) := -\log h(x)$ on $h(x) > 0$. Where both local seminorms are nonzero, define

$$A_{\text{repair}}(x) := \frac{\nabla B_L(x)^\top a(x) \nabla \mathcal{S}_{\text{comp}}(x)}{\| \nabla B_L(x) \|_{a(x)} \| \nabla \mathcal{S}_{\text{comp}}(x) \|_{a(x)}}.$$

For a Doob-type completion drift $b^F - b^0 = -a\nabla\mathcal{S}_{\text{comp}}$,

$$\nabla B_L^\top (b^F - b^0) = -A_{\text{repair}} \| \nabla B_L \|_a \| \nabla \mathcal{S}_{\text{comp}} \|_a.$$

Positive alignment therefore means that the induced drift locally reduces the declared burden. If either denominator seminorm vanishes, report No adequate estimate. Alignment is a local diagnostic, not route identification.

A2.21 Random-time completion geometry

Status: [TT] random-time theorem target.

If the true stream endpoint $T_i^\star$ is not a stopping time in Proposition 3's completed right-continuous endpoint filtration $\left(\mathcal{H}_{i,t}^e\right)_{t\geq 0}$, endpoint analysis may require random-time tools (Nikeghbali, 2006). Under a declared law Q and the usual conditional-expectation regularity, define the càdlàg Azéma survival supermartingale

$$Z_t^{A,Q} := Q\left(T_i^\star > t \mid \mathcal{H}_{i,t}^e\right).$$

Frontier theorem target: relate $Z_t^{A,Q}$, the endpoint projection or compensator, and completion state to identify off-boundary risk without future labels.

A2.22 Completion-information lower bound

Status: [CP/TT] conditional information-budget obstruction. Let L denote the remaining signed residual that a local mechanism would need to cancel at a declared decision time, let X collect currently available physical and biological state, let Z denote the controller's additional information, and let $\hat{R} = g(X, Z)$ be the future compensation selected by that controller. Define the reconstruction $\tilde{L} = -\hat{R}$. Let $D(X)$ be a measurable pointwise distortion budget and suppose

$$\begin{aligned} \mathbb{E}[\left(L - \tilde{L}\right)^2 \mid X] &= \mathbb{E}[\left(L + \hat{R}\right)^2 \mid X] \leq D(X) \quad \text{a.s.}, \\ D_{\text{avg}} &:= \mathbb{E}[D(X)]. \end{aligned}$$

Conditional data processing and rate–distortion theory then imply

$$I(L; Z \mid X) \geq I\left(L; \tilde{L} \mid X\right) \geq R_{L|X}^{\text{pt}}[D(\cdot)] \geq R_{L|X}\left(D_{\text{avg}}\right).$$

where, with X available to both encoder and decoder, define

$$\begin{aligned} R^{\mathrm{pt}}_{L|X}[D(\cdot)] \quad &:= \inf_{P_{\tilde{L}|L,X}:\ \mathbb{E}[(L-\tilde{L})^2|X]\le D(X)\ \text{a.s.}} I(L;\tilde{L}\mid X), \\ R_{L|X}(D_{\mathrm{avg}}) \quad &:= \inf_{P_{\tilde{L}|L,X}:\ \mathbb{E}[(L-\tilde{L})^2]\le D_{\mathrm{avg}}} I(L;\tilde{L}\mid X). \end{aligned}$$

These are, respectively, the profile-constrained and average-distortion conditional rate–distortion functions under squared error (Cover & Thomas, 2006). On a conditionally Gaussian branch, $L \mid X = x \sim \mathcal{N}(m(x), \sigma_L^2(x))$, the pointwise distortion profile yields

$$I(L;Z \mid X) \ge \frac{1}{2}\mathbb{E}[\log^{+}\left(\frac{\sigma_L^2(X)}{D(X)}\right)], \quad \log^{+}(u) = \max\{0, \log u\}.$$

Under the displayed conditional rate–distortion premises, this establishes [CP] INF-1 (conditional information lower bound); any general dynamic or controller extension remains [TT].

Use natural logarithms unless another base is stated. In the conditional Gaussian specialization, adopt the extended-real integrand

$$\psi(\sigma_L^2, D) := \begin{cases} +\infty, & D = 0 < \sigma_L^2, \\ 0, & \sigma_L^2 = 0, \\ \dfrac{1}{2}\log(\sigma_L^2/D), & 0 < D < \sigma_L^2, \\ 0, & D \ge \sigma_L^2. \end{cases}$$

Require measurability of ψ and integrability of $\mathbb{E}\psi(\sigma_L^2(X), D(X))$. On the displayed Gaussian branch, if $D_n(X) \downarrow 0$ and $\sigma_L^2(X) \ge c > 0$ on a positive-probability set, the lower bound diverges there. Other residual classes require a matching conditional rate–distortion law.

Empirical use. Bound prospectively available target information; estimate conditional distortion under a frozen observation model; and compare it with a simulated or analytic frontier. Performance beyond the bound localizes leakage, misspecified conditional variance, selection, observation coupling, or a nonlocal route. Table 4 governs every status, and mixed labels retain all component requirements.

Appendix B: Separation, Information Constraints, and Mechanism Tests

Causal-locus chain.

$$\begin{gathered}\text{assignment} \rightarrow \text{eligibility} \rightarrow \text{information} \rightarrow \text{policy} \\ \rightarrow \mathcal{A}^{\mathrm{do}}(\mathcal{H}_t, \omega) \rightarrow \text{transition/randomizer/outcome} \\ \rightarrow \text{endpoint/routing} \rightarrow \text{retention} \rightarrow \text{observation.}\end{gathered}$$

The factorization is chain-rule bookkeeping; causal identification requires separately declared randomization, exchangeability, consistency, positivity/support, exclusion, and measurement premises. Calibration conditions on measurable realized support; task-additivity equations apply only to the additive measured branch.

Appendix B turns information, policy, transition, selection, endpoint, routing, and observation mechanisms into an integrated localization program. Its results become PTVC-relevant when joined to the target-law and measured-to-latent bridge gates. Appendix C supplies a separate conditional philosophical and mathematical interpretation.

B.1 Scientific role and claim ceiling

No randomizer or maze result is a PTVC test unless a preregistered PTVC route entails a quantitative deviation from equally informed calibration, learning, planning, and selection rivals. Until that condition is met, these tasks are null-calibration and mechanism-localization benchmarks whose outcomes map to causal loci and controlled verdicts.

The Formal Model defines PTVC, the value object, stream and endpoint requirements, measured-to-latent separation, route consequences, rivals, and evidential gates. Appendix A, Part I contains operationally grounded methods and near-term routes. Appendix A, Part II contains governed frontier mechanisms. Appendix B supplies an applied grammar for decision, randomization, selection, routing, and mechanism-recovery studies.

B.2 Representation gate and causal-locus architecture

B.2.1 Representation gate

Representation is logically prior to causal localization. Before testing a mechanism, the protocol must declare the state or residual proxy, sign, neutral origin, scalarization, phenomenal-exposure rule, attainable set, stream-routing rule, endpoint branch, and uncertainty set. A representation trained on the tested outcome, endpoint, eligibility event, or randomizer label is not an independent predictor. A psychophysical assignment selector, if proposed, is a bridge-level architecture and must be specified before behavioral failure, not introduced afterward.

Candidate completion-compatible mental-access route. A registered mental-content route must distinguish preconscious candidate generation, psychophysical assignment to qualifying phenomenal access, attention or maintenance, endorsement, intention, and overt action. On a probability-one hard-filter branch, a measurable access kernel on a standard-Borel candidate space assigns probability one to a prospectively defined measurable completion-compatible set under the declared information set, endpoint/routing branch, and frozen continuation regime. A separate topological-support claim requires a declared topology and its own support or closedness premises. Any immediate stream-owned phenomenal contribution assigned by the registered representation and exposure rule is included before the suffix is evaluated. The set and kernel are fixed independently of realized terminal outcomes. The B.4.3 averaging identity may be instantiated at this earlier locus only after the candidate space, access kernel, candidate-specific complete-continuation law, measurable completion event, information set, endpoint/routing branch, and frozen continuation regime have been declared; its action interpretation does not transfer automatically. On a prefix-local branch, once content enters qualifying phenomenal awareness, such a contribution belongs to the accrued prefix and is not retroactively erased by suppression, reinterpretation, nonendorsement, or nonaction. A complete-history restriction is the separate global locus of §1.5 and A2.1 and implies neither a local online controller nor literal prior simulation. Any empirical or modal promotion requires measurable-selection, partial-observation, time-consistency, bridge, rival-recovery, and retirement conditions. The screened set and route are protocol-defined candidate structures for operationalizing PTVC's terminal relation; target and route remain separately adjudicated. Thought occurrence remains distinct from endorsement, intention, action, moral approval, conscious ledger computation, divine agency, and metaphysical free will.

B.2.2 Diagnostic variables and baseline factorization

Protocol chronology for B-FAC-1. This baseline template orders eligibility and precommitment information, committed action, physical transition, outcome randomization or assignment Z^{rnd}, routing and endpoint adjudication, retention, availability, and observation. The factorization is protocol-specific: if Z^{rnd} is assigned or revealed before commitment or physical transition, if eligibility occurs at another stage, or if retention or observation timing changes, the protocol must refactor the joint law according to its preregistered DAG and may not reuse B-FAC-1 unchanged.

Define the temporally ordered diagnostic tuple

$$\mathsf{D} := \left(H_0, A, E_0^{\mathrm{prot}}, I_C, R_C^{\mathrm{priv}}, C, X', Z^{\mathrm{rnd}}, J^{\mathrm{route}}, E^{\mathrm{prot}}, R^{\mathrm{ret}}, \mathbf{O}^{\mathrm{avail}}, O^{\mathrm{obs}}\right),$$

where I_C is the observed commitment record and

$$\mathcal{F}_C := \sigma\left(H_0, A, E_0^{\mathrm{prot}}, I_C, R_C^{\mathrm{priv}}\right)$$

is the commitment-information sigma-field, including any private policy seed. A chronological baseline factorization (B-FAC-1) is

$$\begin{aligned}
P_0(\mathrm{d}\mathsf{D}) = {} & P_0(\mathrm{d}H_0)P_0(\mathrm{d}A \mid H_0)P_0\left(\mathrm{d}E_0^{\mathrm{prot}} \mid H_0, A\right) \\
& \times P_0\left(\mathrm{d}I_C, \mathrm{d}R_C^{\mathrm{priv}} \mid H_0, A, E_0^{\mathrm{prot}}\right)P_0(\mathrm{d}C \mid \mathcal{F}_C) \\
& \times P_0\left(\mathrm{d}X' \mid H_0, A, E_0^{\mathrm{prot}}, I_C, R_C^{\mathrm{priv}}, C\right) \\
& \times P_0\left(\mathrm{d}Z^{\mathrm{rnd}} \mid H_0, A, E_0^{\mathrm{prot}}, I_C, R_C^{\mathrm{priv}}, C, X'\right) \\
& \times P_0\left(\mathrm{d}J^{\mathrm{route}}, \mathrm{d}E^{\mathrm{prot}} \mid \begin{matrix} H_0, A, E_0^{\mathrm{prot}}, I_C, R_C^{\mathrm{priv}}, C, \\ X', Z^{\mathrm{rnd}} \end{matrix}\right) \\
& \times P_0\left(\mathrm{d}R^{\mathrm{ret}} \mid \begin{matrix} H_0, A, E_0^{\mathrm{prot}}, I_C, R_C^{\mathrm{priv}}, C, X', Z^{\mathrm{rnd}}, \\ J^{\mathrm{route}}, E^{\mathrm{prot}} \end{matrix}\right) \\
& \times P_0\left(\mathrm{d}\mathbf{O}^{\mathrm{avail}} \mid \begin{matrix} H_0, A, E_0^{\mathrm{prot}}, I_C, R_C^{\mathrm{priv}}, C, X', Z^{\mathrm{rnd}}, \\ J^{\mathrm{route}}, E^{\mathrm{prot}}, R^{\mathrm{ret}} \end{matrix}\right) \\
& \times P_0\left(\mathrm{d}O^{\mathrm{obs}} \mid \begin{matrix} H_0, A, E_0^{\mathrm{prot}}, I_C, R_C^{\mathrm{priv}}, C, X', Z^{\mathrm{rnd}}, \\ J^{\mathrm{route}}, E^{\mathrm{prot}}, R^{\mathrm{ret}}, \mathbf{O}^{\mathrm{avail}} \end{matrix}\right).
\end{aligned}$$

Every mechanism claim names the earliest factor it changes and which factors remain invariant.

Table 5. Appendix B Diagnostic Variables

Symbol	Object	Primary question
H_0	Preassignment history	What physical, phenomenal, measured, and contextual state existed before assignment?
A	Assignment	How were treatment, horizon, challenge, or future conditions assigned?
E_0^{prot}	Precommitment eligibility	Who reached the decision point before commitment and later selection?
I_C	Commitment record	What observed cues, measurements, and records were available at commitment?
R_C^{priv}	Private policy seed	What private random seed, if any, was used by the policy?
$\mathcal{F}_C$	Commitment sigma-field	What information was mathematically available when the action was committed?
C	Committed action	What option, stopping decision, search policy, or intervention was selected?
X'	Physical transition	How did the action alter the environment, state, or feasible future?
Z^{rnd}	Outcome assignment	Which consequence, label, mapping, or random target was assigned?
J^{route}	Route mark	Did the stream continue, pause, split, fuse, copy, or become unresolved?
E^{prot}	Endpoint object	Which protocol or truth-level endpoint event was adjudicated?
R^{ret}	Postcommitment retention	Who remained recorded, retained, analyzable, or included?
$\mathbf{O}^{\text{avail}}$	Joint observation-availability vector	Which channels were available at each scheduled or triggered observation time?
O^{obs}	Observed data	Complete multichannel observed record, including explicit missing or unavailable symbols consistent with the joint availability vector.

B.2.3 Ten local loci

Table 6. Appendix B Causal Loci

Locus	Object altered	Typical alternatives
1. Assignment	$P(A \mid H_0)$	Allocation dependence or randomization failure.
2. Precommitment eligibility	$\mathbb{P}\left(E_0^{\text{prot}} \mid H_0, A\right)$	Access, cancellation, illness, or failure to reach commitment.
3. Information	$P\left(I_C, R_C^{\text{priv}} \mid H_0, A, E_0^{\text{prot}}\right)$	Valid cues, leakage, active sensing, learning, memory, or apparatus asymmetry.
4. Policy	$P(C \mid \mathcal{F}_C)$	Attention, effort, persistence, risk, search, stopping, or soft reweighting.
5. Structural action availability	Structural domain $\mathcal{A}^{\text{do}}$	Hard pruning, unavailability, safety constraints, or declared admissibility.
6. Transition	$P(X' \mid \text{pa}_{X'})$	Action-sensitive environment, timing, or opportunity change.
7. Outcome/randomizer	$P_0\left(Z^{\text{rnd}} \mid \text{pa}_{Z^{\text{rnd}}}\right)$	Miscalibration, mapping error, hidden cause, or device defect.
8. Endpoint / routing	$P_0\left(J^{\text{route}}, E^{\text{prot}} \mid \text{pa}_{JE}\right)$	Endpoint support, continuation, fission, fusion, copying, or censoring.
9. Postcommitment retention	$P(R^{\text{ret}} \mid \text{pa}_R)$	Recording, analyzability, device survival, censoring, or selection.
10. Observation	$P_0\left(\mathbf{O}^{\text{avail}}, O^{\text{obs}} \mid \text{pa}_O\right)$	Missingness, clipping, timing error, coding, adjudication, or shutdown.

Table notation. In the Object altered column, pa with a target subscript abbreviates only that target's prospectively named predecessor set in the registered chronology; it does not authorize a data-selected or unrestricted parent set. The nonconflation rule in §8.5 governs every row: better cue use is not randomizer dependence; a changed transition is not a changed randomizer; differential eligibility is not a policy effect; endpoint reclassification is not latent closure; and observation loss is not ontic contraction.

B.2.4 Earliest divergence, mechanism equivalence, and global architectures

Let $\mathcal{A}_C^{\mathrm{do}}(\omega)$ be a nonempty $\mathcal{F}_C$-measurable random action set with measurable graph. Policies satisfy $\pi\big(\mathcal{A}_C^{\mathrm{do}}(\omega) \mid \mathcal{F}_C\big)(\omega) = 1$ a.s.; measurable selectors are required.

Fix a measurable success map $s\colon \mathcal{A} \times \mathcal{Z} \to \{0,1\}$ and define $q(a,\omega) := \mathbb{E}[s(a,Z) \mid \mathcal{F}_C](\omega)$. Then

$$\operatorname*{ess\,sup}_{\pi \in \Pi(\mathcal{F}_C)} \int_{\mathcal{A}_C^{\mathrm{do}}(\omega)} q\,(a,\omega)\, \pi(\mathrm{d}a \mid \mathcal{F}_C)(\omega).$$

The special case $s(a,z) = \mathbf{1}\{a = z\}$ is used only when actions and successful labels share a declared measurable space. If action changes the target law, use the intervention-indexed q^a. On general spaces, the essential supremum is over admissible measurable policies and structurally unavailable actions are excluded by $\mathcal{A}_C^{\mathrm{do}}$.

For each declared intervention a, define the observable-data map on standard Borel spaces

$$D_\eta^{\mathrm{obs},a}\colon (\Omega,\mathcal{F}) \to \big(\mathcal{D}_\eta, \mathfrak{D}_\eta\big), \quad P_M^a\big(D_\eta^{\mathrm{obs},a} \in B\big), \quad B \in \mathfrak{D}_\eta.$$

All compared P_M^a live on this common domain or are pushed forward to the common observable space $\big(\mathcal{D}_\eta, \mathfrak{D}_\eta\big)$. Two mechanisms are observationally equivalent under the declared intervention family exactly when these probabilities agree for every $a \in \mathcal{A}_\eta^{\mathrm{do}}(h_t)$ and every $B \in \mathfrak{D}_\eta$. Structural action availability remains distinct from behavioral policy probability mass; equivalence establishes protocol-relative agreement, while ontic identity remains a separate scientific target.

B.3 Conditional calibration and information limits

B.3.1 Conditional calibration lemma

Let C be $\{1,\dots,m\}$-valued and measurable with respect to the full information $\mathcal{F}_C$ available at irrevocable commitment, including any private policy seed. Let Z be a $\{1,\dots,m\}$-valued successful label and $S := \mathbf{1}_{\{C=Z\}}$. Then

$$\mathbb{P}(S = 1 \mid \mathcal{F}_C) = \sum_{j=1}^{m} \mathbf{1}_{\{C=j\}}\, \mathbb{P}(Z = j \mid \mathcal{F}_C).$$

If the target is conditionally uniform,

$$\mathbb{P}(Z = j \mid \mathcal{F}_C) = \frac{1}{m} \quad (j = 1, \dots, m),$$

then every adapted deterministic or randomized policy satisfies

$$\mathbb{P}_\pi(C = Z \mid \mathcal{F}_C) = \sum_{j=1}^{m} \mathbf{1}_{\{C=j\}} \ \mathbb{P}(Z = j \mid \mathcal{F}_C) = \frac{1}{m} \quad \text{a.s.}$$

More generally, let $q_j(\omega) := \mathbb{P}(Z = j \mid \mathcal{F}_C)(\omega)$. On this matching-label branch, require $\mathcal{A}_C^{\mathrm{do}}(\omega) \subseteq \{1, \ldots, m\}$ almost surely. For the nonempty finite realized action-value set, the operational optimum is

$$\operatorname*{ess\,sup}_{\pi \in \Pi(\mathcal{F}_C)} \mathbb{P}_\pi(C = Z \mid \mathcal{F}_C)(\omega) = \max_{j \in \mathcal{A}_C^{\mathrm{do}}(\omega)} q_j(\omega).$$

This equality assumes that the finite available set is nonempty; on general spaces, use the essential supremum over admissible measurable policies. The full maximum is the all-actions-available special case.

Randomization cannot exceed this maximum. If action changes the target law, use the corresponding intervention-indexed comparison instead.

B.3.2 Approximate calibration and leakage budget

For probability distributions p and q on a finite label space, this appendix uses

$$\mathrm{TV}(p, q) := \frac{1}{2} \sum_j |p_j - q_j|.$$

All total-variation leakage bounds in Appendix B use this convention.

Let $q(\cdot \mid \mathcal{F}_C)$ be the conditional target distribution and Unif_m the uniform distribution. The maximum matching advantage is bounded by the remaining conditional nonuniformity:

$$\max_{1 \le j \le m} q_j - \frac{1}{m} \le \frac{1}{2} \sum_{j=1}^{m} \left| q_j - \frac{1}{m} \right| = \mathrm{TV}(q, \mathrm{Unif}_m).$$

Condition on $\mathcal{F}_C$, apply this inequality to the operational optimum, and take expectations to obtain the leakage bound below.

$$\mathbb{P}(C = Z) - \frac{1}{m} \le \mathbb{E}[\mathrm{TV}(q(\cdot \mid \mathcal{F}_C), \mathrm{Unif}_m)]$$

If the calibrated conditional target probabilities all remain within δ of $1/m$, then the matching probability is at most $1/m + \delta$. An excess inside the validated leakage budget is not anomalous. An excess outside it first rejects the declared information, calibration, timing, eligibility, or observation model; it does not uniquely identify PTVC.

B.3.3 General action invariance

For the realized measurable structural action set $\mathcal{A}_C^{\mathrm{do}}(\omega)$ and admissible policy kernel $\pi(\mathrm{d}a \mid \mathcal{F}_C)$,

$$\mathcal{L}^\pi(Y \mid \mathcal{F}_C)(\omega) = \int_{\mathcal{A}_C^{\mathrm{do}}(\omega)} K_a(\cdot \mid \mathcal{F}_C)(\omega) \, \pi(da \mid \mathcal{F}_C)(\omega).$$

If $K_a(\cdot \mid \mathcal{F}_C) = K(\cdot \mid \mathcal{F}_C)$ for π-almost every admissible a, the integral equals K. The discrete sum is the finite/countable special case. Marginal invariance additionally requires the distribution of $\mathcal{F}_C$ to be held fixed or standardized; search, abstention, stopping, structural-action-domain changes, policy-mass changes, or transition changes violate the premise.

B.3.4 Sequential calibration

Multiple attempts, adaptive search, retries, optional stopping, trial exclusion, and repeated inspection change the reference process. Under the calibrated null, the centered cumulative process is a martingale:

$$M_n := \sum_{t=1}^{n} (S_t - p_t)\,, p_t := \mathbb{P}(S_t = 1 \mid \mathcal{F}_{t-1}).$$

The centered martingale M_n is not itself an anytime-valid p-value or evidence process. Optional-stopping-valid inference requires a preregistered nonnegative test supermartingale, e-process, or confidence sequence satisfying A1.37.

B.4 Completion-dependent action mass and policy exactness

Write $h_t = (x, \ell, t)$. Let $\mathcal{A}_\eta^{\mathrm{do}}(h_t)$ be a nonempty measurable set-valued correspondence with a measurable graph on the declared standard-Borel action space, and let $\pi_t(\cdot \mid h_t, g)$ be a policy kernel satisfying $\pi_t\big(\mathcal{A}_\eta^{\mathrm{do}}(h_t) \mid h_t, g\big) = 1$. This is probability-one structural admissibility and does not by itself imply topological-support containment. A separately registered [SUPPORT] claim requires a declared topology and direct support or closedness premises. Policy behavior never defines structural availability. Define the action-conditioned completion set with the committed action visible unless the suffix convention explicitly includes it. Require the soft-policy normalizer to be strictly positive; finiteness is automatic because the completion probability lies in [0,1] and the baseline policy is a probability kernel. A zero normalizer makes the reweighted policy undefined and returns No adequate estimate.

B.4.1 Modal ladder

Let $\mathcal{A}_\eta^{\mathrm{do}}(h_t)$ be the structurally available intervention/action domain at prefix h_t. For each $a \in \mathcal{A}_\eta^{\mathrm{do}}(h_t)$ and one prospectively frozen continuation regime, let $\big(\Gamma_{\eta,h_t,g}, \mathcal{G}_{\eta,h_t,g}\big)$ be a common standard-Borel canonical suffix space. Under $\mathrm{do}(\mathsf{A}_t = a)$, let $Q_{\eta,h_t}^{a,g}$ be a probability kernel on that space and let $F_{\eta,\mathrm{comp}}^{a,g}(h_t)$ be the action- and regime-specific PTVC-completion event in its sigma-field.

Here $Q_{\eta,h_t}^{a,g}$ contains one frozen continuation regime. If several regimes remain live, construct and report $Q_{\eta,h_t}^{a,g}$, $F_{\eta,\mathrm{comp}}^{a,g}$, and $p_{\eta,h_t,g}^{\mathrm{comp}}(a)$ branchwise; never optimize silently over g. Define

$$\mathcal{A}_{\eta,F}^{\exists}(h_t; g) := \{a \in \mathcal{A}_\eta^{\mathrm{do}}(h_t) \colon F_{\eta,\mathrm{comp}}^{a,g}(h_t) \neq \emptyset\}.$$

$$\mathcal{A}_{\eta,F}^{+}(h_t; g) := \{a \in \mathcal{A}_\eta^{\mathrm{do}}(h_t) \colon Q_{\eta,h_t}^{a,g}\left(F_{\eta,\mathrm{comp}}^{a,g}(h_t)\right) > 0\}.$$

$$\mathcal{A}_{\eta,F}^{1}(h_t; g) := \{a \in \mathcal{A}_\eta^{\mathrm{do}}(h_t) \colon Q_{\eta,h_t}^{a,g}\left(F_{\eta,\mathrm{comp}}^{a,g}(h_t)\right) = 1\}.$$

Structural availability, completion-compatible action sets, and behavioral policy probability mass are different objects. An action may be physically available yet receive zero policy probability, or receive positive policy mass while having zero PTVC-completion probability.

B.4.2 Hard exclusion, soft weighting, and anti-tautology gate

For each available action $a \in \mathcal{A}_\eta^{\mathrm{do}}(h_t)$ under the same frozen continuation regime, define the action-specific completion probability

$$p^{\mathrm{comp}}_{\eta,h_t,g}(a) := Q^{a,g}_{\eta,h_t}\left(F^{a,g}_{\eta,\mathrm{comp}}(h_t)\right), \qquad 0 \le p^{\mathrm{comp}}_{\eta,h_t,g}(a) \le 1.$$

A hard action-mass exclusion route assigns zero policy probability outside a declared completion-compatible action set. On a finite or countable action domain, a normalized soft policy for β > 0 is

$$\pi_{\beta,\eta}(a \mid h_t, g) := \frac{\pi_{0,\eta}(a \mid h_t, g)\left[p^{\mathrm{comp}}_{\eta,h_t,g}(a)\right]^{\beta}}{\sum_{a' \in \mathcal{A}^{\mathrm{do}}_{\eta}(h_t)} \pi_{0,\eta}(a' \mid h_t, g)\left[p^{\mathrm{comp}}_{\eta,h_t,g}(a')\right]^{\beta}}.$$

On a general measurable action space, require the action-indexed completion probability to be measurable and use the same regime-indexed baseline probability kernel $\pi_{0,\eta}(\mathrm{d}a \mid h_t, g)$:

$$\pi_{\beta,\eta}(\mathrm{d}a \mid h_t, g) := \frac{\left[p^{\mathrm{comp}}_{\eta,h_t,g}(a)\right]^{\beta}\pi_{0,\eta}(\mathrm{d}a \mid h_t, g)}{\int_{\mathcal{A}^{\mathrm{do}}_{\eta}(h_t)}\left[p^{\mathrm{comp}}_{\eta,h_t,g}(a')\right]^{\beta}\pi_{0,\eta}(\mathrm{d}a' \mid h_t, g)}.$$

For $\beta = 0$, use the regime-indexed identity $\pi_{\beta,\eta}(\mathrm{d}a \mid h_t, g) := \pi_{0,\eta}(\mathrm{d}a \mid h_t, g) \quad$ for $\beta = 0$. This identity is defined branchwise under the same frozen continuation regime. Require the normalizer to be strictly positive; finiteness is automatic because the completion probability lies in [0,1] and the baseline policy is a probability kernel. The baseline policy, continuation law, completion event, endpoint rule, and state representation are fixed independently of the behavior later explained.

B.4.3 Policy-mixture exactness under a frozen action-first regime

Fix protocol η, a reached prefix h_t, and a frozen downstream regime g. Measurability conditions: the action space is standard Borel; the structural action correspondence has a measurable graph; the mapping from history and action to the continuation law is a probability kernel on the common canonical suffix space; the completion event is jointly measurable in history, action, and continuation; and the action-specific completion probability is measurable in the action. The frozen regime includes every declared downstream policy, physical transition, endpoint, routing, selection, and measurement-reactivity component that causally affects the continuation. The truth-level completion event remains defined independently of the observation kernel used to estimate it.

$$\pi_t\left(\mathcal{A}^{\mathrm{do}}_{\eta}(h_t) \mid h_t, g\right) = 1, \qquad \pi_{0,\eta}\left(\mathcal{A}^{\mathrm{do}}_{\eta}(h_t) \mid h_t, g\right) = 1,$$
$$P^{\pi}_{\eta,h_t,g}(da, d\gamma) = \pi_t(da \mid h_t, g)\, Q^{a,g}_{\eta,h_t}(d\gamma).$$

$$\tilde{F}_{\eta,\mathrm{comp}}(h_t; g) := \{(a,\gamma): a \in \mathcal{A}^{\mathrm{do}}_{\eta}(h_t),\ \gamma \in F^{a,g}_{\eta,\mathrm{comp}}(h_t)\} \in \mathcal{B}(\mathcal{A}) \otimes \mathcal{G}_{\eta,h_t,g}.$$

Define

$$p^{\mathrm{comp}}_{\eta,h_t,g}(a) := Q^{a,g}_{\eta,h_t}\left(F^{a,g}_{\eta,\mathrm{comp}}(h_t)\right), \qquad 0 \le p^{\mathrm{comp}}_{\eta,h_t,g}(a) \le 1.$$

$$P^{\pi}_{\eta,h_t,g}\left(\tilde{F}_{\eta,\mathrm{comp}}(h_t; g)\right) = \int_{\mathcal{A}^{\mathrm{do}}_{\eta}(h_t)} p^{\mathrm{comp}}_{\eta,h_t,g}(a)\,\pi_t(da \mid h_t, g).$$

$$P^{\pi}_{\eta,h_t,g}\left(\tilde{F}_{\eta,\mathrm{comp}}(h_t; g)\right) = 1 \Leftrightarrow p^{\mathrm{comp}}_{\eta,h_t,g}(a) = 1 \quad \pi_t(\cdot \mid h_t, g)\text{-almost everywhere.}$$

$$\mathcal{A}^{1}_{\eta,F}(h_t; g) := \{a \in \mathcal{A}^{\mathrm{do}}_{\eta}(h_t): p^{\mathrm{comp}}_{\eta,h_t,g}(a) = 1\}, \qquad \pi_t\left(\mathcal{A}^{1}_{\eta,F}(h_t; g) \mid h_t, g\right) = 1.$$

[P] Proof. The mixture identity follows from the registered disintegration. Equality to one is equivalent to

$$\int \left(1 - p^{\mathrm{comp}}_{\eta,h_t,g}(a)\right)\pi_t(da \mid h_t, g) = 0.$$

The integrand is nonnegative, so it vanishes almost everywhere under the policy; the converse is immediate.

Scope. This is a probability-mass statement under one registered factorization. Topological-support results and causal identification from observed choices are available through their corresponding registered extensions. Conscious ledger access, endpoint knowledge, computation of completion probabilities, and psychological motivation remain distinct hypotheses with their own tests. If every structurally available action already completes with probability one, the proposition is exact but behaviorally silent.

The mixture equivalence is an elementary conditional consequence of the registered action-first disintegration. Its contribution here is PTVC-specific typing, modality separation, dynamic extension, and identification consequences rather than a claim of a new general integration theorem.

B.4.4 Finite soft-guidance ceiling, attainability, and information

For finite β > 0, define

$$Z_{\beta,\eta}(h_t, g) := \int \left[p^{\mathrm{comp}}_{\eta,h_t,g}(a)\right]^{\beta} \pi_{0,\eta}(da \mid h_t, g), \qquad 0 < Z_{\beta,\eta}(h_t, g) \le 1.$$

$$\pi_{\beta,\eta}(da \mid h_t, g) := \frac{\left[p^{\mathrm{comp}}_{\eta,h_t,g}(a)\right]^{\beta} \pi_{0,\eta}(da \mid h_t, g)}{Z_{\beta,\eta}(h_t, g)}, \qquad \beta > 0.$$

$$P^{\pi_{\beta,\eta}}_{\eta,h_t,g}\left(\tilde{F}_{\eta,\mathrm{comp}}(h_t; g)\right) = \frac{Z_{\beta+1,\eta}(h_t, g)}{Z_{\beta,\eta}(h_t, g)}.$$

$$\begin{aligned} &P^{\pi_{\beta,\eta}}_{\eta,h_t,g}\left(\tilde{F}_{\eta,\mathrm{comp}}(h_t; g)\right) = 1 \\ &\Leftrightarrow \quad \pi_{0,\eta}\left(\{a: 0 < p^{\mathrm{comp}}_{\eta,h_t,g}(a) < 1\} \mid h_t, g\right) = 0 \\ &\qquad \text{and} \quad \pi_{0,\eta}\left(\{a: p^{\mathrm{comp}}_{\eta,h_t,g}(a) = 1\} \mid h_t, g\right) > 0. \end{aligned}$$

Positive power weighting removes actions with zero completion probability but retains baseline-positive actions with intermediate completion probability; any retained intermediate mass prevents exact probability-one completion. For example,

$$a \in (0,1), \quad p^{\mathrm{comp}}(a) = a, \quad \pi_0 = \mathrm{Uniform}(0,1) \quad \Rightarrow \quad P^{\pi_\beta}\left(F_{\mathrm{comp}}\right) = \frac{\beta+1}{\beta+2} < 1.$$

The values approach one as β increases, but no finite β attains one. Freeze a nonpolicy environment before optimizing over complete admissible policies. At each history under consideration, require the admissible policy class to be nonempty, and define

$$\mathcal{S}_{\mathrm{AS}}(\bar{g}) := \left\{h: \exists \pi \in \Pi_{\mathrm{adm}}(h, \bar{g}),\ P^{\pi,\bar{g}}\left(F_{\mathrm{comp}} \mid h\right) = 1\right\}.$$

$$\mathcal{S}_{\mathrm{LS}}(\bar{g}) := \left\{h: \sup_{\pi \in \Pi_{\mathrm{adm}}(h,\bar{g})} P^{\pi,\bar{g}}\left(F_{\mathrm{comp}} \mid h\right) = 1\right\}.$$

$$\mathcal{S}_{\mathrm{AS}}(\bar{g}) \subseteq \mathcal{S}_{\mathrm{LS}}(\bar{g}),$$

Almost-sure attainability requires an attaining policy; limit-sure attainability permits only a sequence approaching probability one. If the admissible policy class is empty, return RouteStatus: unavailable — empty admissible policy class rather than evaluating an almost-sure or limit-sure condition. Neither region is called a viability or completion kernel without controlled invariance, measurable selection, endpoint compatibility, and recursive time consistency. This follows established reachability terminology: almost-sure attainability requires one policy that reaches the target with probability one, whereas limit-sure attainability permits policies whose values approach one without an attaining policy (Chatterjee, Chmelík, & Davies, 2016; Asadi et al., 2025).

Under partial observation o with registered belief b_t and observation-adapted policy π_t,

$$P\big(F_{\text{comp}} \mid o\big) = \int \int p^{\text{comp}}_{\eta,h,g}(a)\, \pi_t(da \mid o)\, b_t(dh \mid o).$$

Posterior probability-one completion requires the action-specific probability to equal one for almost every live history–action pair. A literal robust guarantee over every compatible history uses

$$\mathcal{A}^1_{\text{rob}}(o;g) := \bigcap_{h \in \mathcal{H}(o)} \mathcal{A}^1_{\eta,F}(h;g).$$

A robust observation-adapted policy exists only where the robust completion-compatible action set is nonempty on the registered observation support and the corresponding multifunction admits a measurable selector satisfying structural admissibility. Failure of either condition yields robust-route unavailability rather than probability-one completion. If observationally indistinguishable histories require disjoint probability-one action sets, no policy measurable only with respect to that observation can guarantee both. Additional information, overlapping safe actions, or another causal locus is required.

Off-policy identification ceiling. When an observed policy assigns zero probability to putatively unsafe actions, observed behavior does not identify their counterfactual completion probabilities without additional positivity, structural, instrumental, experimental, or partial-identification assumptions. Hard exclusion therefore creates precisely the off-policy region whose safety claim is hardest to verify. Evidence about excluded actions must come from ethically bounded surrogate tasks, exogenous availability or incentive perturbations, natural experiments, structural restrictions, partial-identification bounds, or sealed generator-recovery studies. Harmful, illegal, endpoint-manipulating, or lifetime-risk interventions remain prohibited.

B.4.5 Sequential completion-value consistency

Fix registered decision times $t_0 < \cdots < t_n$, a frozen nonpolicy environment, and a nonempty admissible full-policy class. For k = 0,…,n define the conditional completion value $V_k\pi$; for k = 0,…,n−1 define the action value $q_k\pi$. The terminal quantity $V_n\pi$ is the conditional completion probability under the frozen post-t_n continuation. If t_n ends the declared suffix and the completion event is measurable at t_n, then $V_n\pi$ is its indicator. Let the registered transition kernel from t_k to t_{k+1} be K_k. Define

$$V^\pi_k(h) := P^{\pi,\bar{g}}\big(F_{\text{comp}} \mid H_{t_k} = h\big).$$

$$q^\pi_k(h,a) := \int V^\pi_{k+1}(h')\, K^{\bar{g}}_k(dh' \mid h,a).$$

$$V^\pi_k(h) = \int q^\pi_k(h,a)\, \pi_k(da \mid h).$$

$$P^{\pi,\bar{g}}\big(F_{\text{comp}}\big) = 1 \Rightarrow V^\pi_k\big(H_{t_k}\big) = 1 \quad P^{\pi,\bar{g}}\text{-a.s.}, \qquad q^\pi_k\big(H_{t_k},a\big) = 1 \quad \pi_k\big(da \mid H_{t_k}\big)\text{-a.e., } P^{\pi,\bar{g}}\text{-a.s.}$$

This tower-property result gives a coherent repeated-decision meaning to preserving a completing future. The displayed implications hold under the governing law at almost every reached registered history;

robust positive claims additionally require the completion-compatible correspondence to be nonempty and to admit an observation-measurable, time-consistent selector. The result does not imply monotonic movement of the ledger toward zero, conscious computation, or a psychological controller. A continuous-time Hamilton–Jacobi–Bellman, martingale, or viability formulation requires separately stated regularity, admissibility, and measurable-selection assumptions.

B.5 Experimental forks

Table 7. Appendix B Experimental Arms

Arm	Outcome timing	Information structure	Primary purpose
A. Weak-cue fixed target	Before commitment	Small valid cue	Attention, learning, active sensing, and candidate motivational constructs after independent construct validation.
B. Cue-null fixed target	Before commitment	No usable cue under calibrated observer model	Leakage, procedural regularity, and preexisting dependence.
C. Action-sensitive control	After action	Action genuinely changes probabilities	Validates causal learning and contingency detection.
D. Post-choice randomizer	After irrevocable commitment	No ordinary target information	Randomizer transparency, timing, selection, or joint-law dependence.
E. Random horizon	Fixed, concealed, or action-dependent	Known or uncertain future opportunity	Finite-horizon planning, reachability, and endpoint-conditioned selection.
F. Concealed / sham future assignment	Committed before preperiod	Unavailable to participants, handlers, systems, and analysts	Allocation leakage, eligibility, attrition, and administrative selection.
G. Observation garbling	Same latent task	Randomly masked or degraded channels	Tests whether information loss creates apparent contraction or certainty.

The proposed primary study would cross one benign policy manipulation with weak-cue, cue-null, action-sensitive, and post-choice modules. The same manipulation may alter behavior in all arms, but calibrated success should improve only where valid information or causal control exists under ordinary models.

B.6 Task contribution, selection, endpoint, and observation

B.6.1 The protocol is part of the trajectory

On an additive measured branch, protocol bookkeeping is per stream:

$$M_{\eta,i,\text{post}} = M_{\eta,i,\text{pre}} + J_{\eta,i,\text{task}} + J_{\eta,i,\text{later}} + J_{\eta,i,\text{endpoint}}.$$

$J_{\eta,\text{task}}$ includes reward, fear, frustration, uncertainty, effort, fatigue, relief, social evaluation, and measurement burden. $J_{\eta,\text{later}}$ includes later experiential and opportunity consequences. $J_{\eta,\text{endpoint}}$ exists only in an explicitly endpoint-inclusive branch. This is protocol-level bookkeeping, not a claim that the terms are identified. A corresponding latent identity must use L_i^{phen} and requires the declared measurement bridge.

B.6.2 Manipulation is not state identification

Assignment to an abundance, adversity, reward, stress, relief, or control condition does not itself establish a ledger sign or magnitude. A ledger interpretation requires an independently validated measured estimand, sign rule, neutral origin, scalarization, exposure rule, uncertainty model, and prospectively frozen mapping. Confirmatory residual estimates must be generated without using held-out test outcomes to set the condition's sign or magnitude.

B.6.3 Eligibility and collider selection

Let $R_1 := R^{\text{ret}}$ denote postcommitment retention and O^{avail} observation availability. Even when action C and target Z are independent marginally, conditioning on later selection variables can induce dependence:

$$C \perp\!\!\!\perp Z \quad \not\Rightarrow \quad C \perp\!\!\!\perp Z \mid \left(R_1 = 1,\ O^{\text{avail}} = 1\right).$$

Conditioning on retention and observation availability can induce dependence; it is not claimed that it must do so in every selection model. Assignment-level denominator flow and selection sensitivity are required before interpreting a retained-sample anomaly.

B.6.4 Endpoint ontology and no-survival-oracle requirement

Task absorption, study exit, device loss, biological crisis, apparent death, and the true endpoint of a qualifying stream are distinct. Every endpoint-dependent design must predeclare strict-pre-endpoint, endpoint-inclusive, extended-continuation, or no-endpoint status. A failed branch may not be repaired after outcome inspection by switching interpretations.

A model in which modeled decision makers or controllers can accurately observe a ledger, deliberately keep it off a terminal surface, and thereby suppress true endpoint hazard would permit survival or postselection arbitrage. An endpoint-support or global-selector route must state which premise fails and which indirect, risky, or noncontrollable observables remain testable. Opacity without a surviving prediction is not an explanation.

B.6.5 Observation and audit

Action timestamps, cue exposure, policy revisions, randomizer outputs, retries, eligibility, outcomes, routing, exclusions, and analytic transformations require redundant logging and independent audit. A result appearing only after missing records, timing exclusions, smoothing, endpoint-conditioned preprocessing, or outcome-dependent retention is an observation-locus result unless those operations were preregistered and recoverable under synthetic nulls.

B.7 Mechanism-by-intervention tournament

Table 8. Appendix B Mechanism Localization

Mechanism	Distinctive localization	Randomizer expectation	Primary defeat
Ordinary policy / motivation	Choice, effort, latency, or search changes with valid cues or incentives.	Transparent post-choice randomizer remains calibrated.	Excess remains after cue, planning, and calibration audit.
Enhanced information	Performance tracks independently measured cue information.	No effect after commitment when no information remains.	Effect survives cue-null and post-choice modules.
Hard action-mass exclusion	Prospectively declared action zeros and strong option preservation.	Cannot alter an otherwise transparent randomizer.	Pruned actions occur under validated availability and incentives.

Mechanism	Distinctive localization	Randomizer expectation	Primary defeat
Soft completion weighting	Graded policy shifts tied to one fixed completion potential.	No randomizer effect under transparency.	The fixed completion potential fails held-out transport.
Transition-law modification	Action changes environment or future opportunities.	Randomizer changes only if the action changes that module.	No predicted physical transition difference appears.
Outcome / randomizer dependence	Post-choice conditional calibration fails.	Randomizer is nontransparent by hypothesis.	Effect is technology-, retry-, timing-, or logging-specific.
Eligibility before commitment	Changes who reaches commitment before action.	May change the committed population without altering a within-unit randomizer.	Effect survives assignment/access-flow accounting and eligibility modeling.
Retention after commitment / analysis selection	Changes who is recorded, observable, retained, or analyzed after action.	Can create analyzed action-target dependence without changing the underlying randomizer.	The effect persists after validated denominator-flow or selection correction and complete-data recovery, and the selection model fails held-out recovery.
Endpoint / routing	Endpoint risk or stream routing is state-dependent.	No necessary within-task choice or randomizer effect.	Adequately adjudicated off-support endpoint or routing failure.
Observation artifact	Effect depends on channel quality, coding, smoothing, or retention.	Apparent excess or contraction follows observation loss.	The effect survives calibrated redundant channels, randomized-garbling correction, and independent pipeline replay, and the artifact model fails held-out recovery.
Global / psychophysical selector	Several loci or physical-to-phenomenal assignments are coordinated.	May preserve or alter declared randomizers.	Fails dynamic consistency, spectator, no-signaling, bridge, or held-out joint-law tests.

B.8 Safe experimental program

Randomized garbling-by-placebo-window experiment. At prospectively fixed ethical landmarks, randomize a noncritical channel to full-resolution or garbled observation and independently classify windows as truth-proxy endpoint or matched placebo windows. Hold risk-set standardization, treatment information, observation opportunities, and latent model family fixed. Route-level contraction predicts endpoint-localized latent contraction surviving garbling correction; an observation-artifact rival predicts stronger apparent measured contraction under degradation; a generic endpoint-normalization rival predicts similar behavior at placebo windows.

All studies below are low-burden mechanism-separation, branch-testing, or analysis-recovery studies. Successful results may promote the tested mechanism branch to the next registered evidence level. Whole-life inference additionally requires the independently validated complete-domain, endpoint, routing, observation, and measured-to-latent bridges and the applicable claim-ceiling gate.

Table 9. Appendix B Experiment Bank

Experiment	Core design	Primary value
1. Modular cue-policy-transition-randomizer task	Cross weak-cue, cue-null, action-sensitive, and post-choice modules under one benign policy manipulation.	Separates information and causal control from motivation.
2. Information-budget calibration	Estimate residual target information and compare observed advantage with the TV leakage budget.	Tests excess beyond demonstrably available information.
3. Post-commitment remapping and randomizer audit	After commitment, separately randomize option-label mapping and target; add redundant technologies and unused precommitted replay streams.	Separates option bias, mapping error, target defects, timing, and analysis artifacts.

Experiment	Core design	Primary value
4. Option-preservation and reachability	Match immediate utility, risk, delay, effort, uncertainty, and conventional option value while varying declared future attainable sets.	Tests hard action-mass exclusion, soft weighting, and ordinary planning.
5. Modal solvency simulation	Construct actions with existential, positive-probability, or probability-one PTVC completion.	Tests which admissibility claim a guidance rule implements.
6. Completion/interruption × signed-dose × random-horizon study	At repeated eligible times in a safe task, independently randomize a brief sign-opposed dose, genuine task completion versus matched interruption or a pseudo-endpoint, and a concealed horizon satisfying Corollary 1.1's registered density and dependence conditions; estimate immediate and later signed areas.	Separates generic adaptation and opponent response from completion-specific compensation, endpoint salience, horizon sensitivity, and selection; licenses only a finite-episode route result.
7. Matched-state collision and regeneration	Create different safe histories, match current state, then apply an identical suffix before and after candidate resets such as sleep or task reset.	Tests state sufficiency, hidden continuation state, and path dependence.
8. Concealed future and sham assignment	Commit an effective and unused harmless future assignment before a preperiod; conceal both from participants, systems, and analysts.	Audits leakage, scheduling, eligibility, and administrative selection.
9. Selected-sample randomizer trap	Generate action and target independently, then retain observations as a function of their match.	Mandatory negative control for collider-generated post-choice dependence.
10. Randomized observation garbling	Randomly mask channels while leaving the latent process unchanged.	Tests whether information loss creates spurious neutrality or certainty.
11. Spectator and modularity test	Insert, remove, duplicate, or refine an independent module while holding the focal process fixed.	Tests selector separability, stream-count bias, and normalization artifacts.
12. Multi-participant bottleneck game	Use harmless shared future capacities with a precomputed joint capacity polytope.	Tests joint terminal reachability and subset bottlenecks without cross-person ledger transfer.
13. Replay, rollback, copying, and merge simulation	Restore, duplicate, merge, or stop explicitly nonconscious modeled controllers after different histories.	Tests continuation-state and routing algebra.
14. Completion-field overidentification	Estimate one completion object from nonterminal transitions and require it to predict policy, reachability, perturbation, and cycle-ratio restrictions without refitting.	Tests whether one mechanism, rather than separate flexible functions, explains the cross-arm vector.

Completion-specific signed interaction. At an eligible decision time, let $A_t \in \{d, 0\}$ be the randomized signed-dose assignment, $J_t^{\text{cont}} \in \{1,0\}$ be the randomized task-continuation assignment, where one permits planned completion and zero assigns a matched interruption, I_t denote availability. For each assigned dose–continuation pair and the frozen post-decision policy, let $Y_{t,h}(a, c; \pi)$ denote the proximal signed-area potential outcome over the registered horizon; under consistency, the observed outcome is the ledger increment over that interval. With later assignments marginalized under the frozen policy π, define

$$\Delta_{\text{CSI}}(d, h) = \mathbb{E}[Y_{t,h}(d, 1; \pi) - Y_{t,h}(0,1; \pi) \mid I_t = 1]$$
$$-\mathbb{E}[Y_{t,h}(d, 0; \pi) - Y_{t,h}(0,0; \pi) \mid I_t = 1].$$

Under consistency, registered sequential randomization of the joint factorial assignment conditional on preassignment history, positivity for all four factorial cells within availability, the declared interference and carryover conditions, and the frozen observation and missingness model, randomized cell means identify this intention-to-treat proximal interaction. If actual completion rather than assigned continuation is analyzed, completion is a postassignment mediator and requires a separately declared compliance or mediation model with sensitivity analysis. A sign-structured interaction that transports across held-out horizons supports a completion-specific local mechanism beyond generic dose adaptation; it does not adjudicate complete-stream PTVC. Repeated decision times follow the micro-randomized-trial framework (Klasnja et al., 2015).

Recommended first empirical study. The modular cue-policy-transition-randomizer task is safe, repeatable, fully loggable, and directly tests the core information-versus-policy distinction.

Mandatory negative controls. The public benchmark should include the selected-sample trap, randomized observation garbling, a known technology-specific randomizer defect, and an open-set hybrid generator. A pipeline that misclassifies any of these is not ready for anomaly interpretation.

B.9 Natural and observational stress tests

Table 10. Appendix B Study Portfolio

Study	Design	Scientific use
Sudden versus expected natural endpoints	Prospective pre-endpoint data with cause-specific models and no alteration of care.	Pressures endpoint routes; requires strong value, endpoint, and routing bridges.
Clinical random-horizon variation	Naturally varying discharge, procedure, treatment, or scheduling horizons.	Tests horizon-sensitive policy without treating the horizon as a true endpoint.
Sleep and candidate regeneration	Mild histories followed by sleep or reset and next-day state matching.	Tests persistence, transformation, or reset of continuation-relevant state.
Natural assignment shocks	Schedules, notifications, lotteries, outages, or weather interruptions.	Validates leakage, selection, eligibility, and observation pipelines.
Existing panels and behavior data	Repeated affect or option-preservation data with explicit entry, origin, exposure, and measurement limits.	Feasibility or falsification screens only; no whole-life ledger claim.

No natural study may equate ordinary mood, life satisfaction, reward history, stress exposure, or treatment group with the cumulative phenomenal ledger without a validated bridge. No care, endpoint, suffering, or consent condition may be altered to increase informativeness.

B.10 Simulator benchmark and model recovery

Observation-mechanism rivals use preregistered model-indexed kernels; latent-dynamics rivals share the fixed kernel. All receive equal raw inputs, preprocessing, tuning budget, and observed-law scoring.

Before empirical interpretation, the complete analysis pipeline must pass a blinded generator-recovery study or an equally stringent analytic or formally verified validation appropriate to the declared model class. Its purpose is to determine whether one frozen pipeline can localize the generating mechanism and return model ambiguity when mechanisms are observationally or interventionally equivalent.

Table 11. Appendix B Generator Bank

ID	Generator	Distinctive signature
M0	Calibrated ordinary null	Behavior varies; matching remains calibrated.
M1	Motivated valid-cue use	Effects grow with valid information and vanish at cue nullity.
M2	Hidden cue or common cause	Preassigned advantage with apparatus, batch, trace, or context dependence.
M3	Hard action-mass exclusion	Prospectively predicted actions receiving zero policy mass.
M4	Soft completion weighting	Graded policy shifts tied to one fixed completion object.
M5	Transition-law modification	Environment changes after action while explicit randomizer stays calibrated.
M6-E	Eligibility before commitment	Who reaches commitment changes before action.

ID	Generator	Distinctive signature
M6-R	Retention after commitment / analysis selection	Recording, analyzability, or inclusion changes after action.
M7	Endpoint bridge or routing selection	Task-ending, terminal-pull, or routing patterns change without randomizer dependence.
M8	Action-randomizer dependence	Post-choice conditional calibration fails.
M9	Observation or pipeline artifact	Effect depends on response format, anchors or defaults, prompt cadence, repeated-assessment reactivity, logging, smoothing, clipping, missingness, retries, or exclusions.
M10	Flexible ordinary latent-state controller	Several signatures arise without PTVC structure.
M11	Global or psychophysical selector	Cross-locus dependence with declared consistency and bridge constraints.
M12	Open-set, hybrid, or challenging corruption	True generator is absent, spans loci, or exploits code, timing, custody, or analyst leakage.
M13	AS/LS attainment gap	Values approach one but no admissible policy attains probability one.
M14	Partial-information action conflict	Observationally identical states require disjoint probability-one completion-compatible actions.
M15	Composite action-plus-endpoint route	Action mass and cause-specific endpoint support jointly supply closure.
M16	All-actions-safe null	Every available action already completes; fixed-policy probability-one compatibility holds, but no differential action-route signal is predicted.

False-support certification uses the single family-wise event, threshold, upper bound, and tolerance defined in §9.3.

Candidate and rival models receive identical observation channels, preprocessing, representation budgets, endpoint rules, missingness handling, tuning opportunities, and proper scoring. Required outputs are a generator-by-model confusion matrix, calibration, held-out predictive score, open-set abstention rate, and worst-case false route-specific support. When adequate generators remain equivalent, the correct verdict is Model-ambiguous.

The most informative inference is a cross-arm fingerprint, not one success count. One architecture should jointly predict policy, confidence, latency, cue sensitivity, actions receiving zero policy mass, transition dynamics, target calibration, eligibility, routing, and observation. Diagnostics derived from one altered kernel are one evidence family and must be scored jointly rather than counted as independent votes.

B.11 Statistical, procedural, and interpretive standards

- Use conditional calibration rather than an assumed ideal binomial null as the primary randomizer target.
- Use hierarchical or design-based methods when trials share participants, apparatus, sessions, sites, laboratories, randomizer batches, or analysis pipelines.
- Propagate cue-nullity, target-calibration, state-estimation, endpoint, eligibility, and observation uncertainty rather than treating them as fixed.
- Freeze the primary and ordered secondary estimands, stopping rule, retries, exclusions, multiplicity control, and No adequate estimate gates before reveal.
- Use append-only synchronized logs, replay protection, seed or entropy custody, blinded cleaning, and complete reporting of failures and abandoned branches.
- Use held-out proper predictive scores, cross-arm transport, and recovery calibration rather than significance alone.

- Require replication with different code, personnel, laboratory, analyst team, and randomizer technology before interpreting a post-choice anomaly.
- Report No adequate estimate when representation, cue, commitment, randomizer, eligibility, endpoint, or observation integrity cannot be established.

B.11.1 Claim ceiling

Table 12. Appendix B Evidence Levels

Level	Permissible conclusion
C0	The thought experiment or simulator is internally coherent.
C1	The pipeline recovers declared generators and controls false route-specific support.
C2	A safe study identifies an effect at a declared causal locus.
C3	One mechanism class predicts a transported cross-arm pattern better than rivals receiving equal treatment.
C4	A mechanism package connects to a measured estimand via an independently validated bridge to the declared PTVC target and survives false-support calibration.
T2	On the separate theory axis, an independently successful physical–psychophysical theory entails closure in a declared class.

A post-choice anomaly without PTVC-specific state dependence is an anomalous-correlation result, not route-specific support. A policy effect is evidence about policy. A selection effect is evidence about selection. A simulator result is evidence about software and recoverability.

B.11.2 Route promotion and retirement rules

- A policy-only route fails if it predicts above-calibration success in an audited action-invariant task.
- A hard action-mass exclusion route fails if prospectively inadmissible actions occur under adequate incentives and validated availability.
- A soft completion route fails if one independently fixed completion object does not transport across actions, tasks, horizons, perturbations, and cycle restrictions.
- A randomizer-dependence route fails if the effect is technology-, retry-, timing-, logging-, or selection-specific.
- A nontransparent selector fails if it permits signaling or survival arbitrage, violates intervention composition, or changes an independent spectator marginal without a declared coupling.
- A continuation-state route fails if it adds no held-out prediction beyond ordinary state or disappears under leakage controls.
- Any rescue requiring post hoc changes in representation, information, endpoint, stream, origin, scalarization, action set, or hidden-state dimension is a new model and inherits new recovery obligations.

B.12 Concealed-target completion benchmark and welfare-prioritized translation

Status and scope. This benchmark targets only a predeclared action-guidance or endpoint-selector subclass in which a completion constraint changes choice or transition probabilities. Descriptive PTVC, the boundary statement that complete streams terminate balanced, does not by itself imply concealed-route accuracy. A simultaneous confidence region for the selector advantage wholly contained in a preregistered practically-null region can reject the registered minimum-effect selector subclass without deciding every PTVC branch; failure to detect an advantage alone is inconclusive. The primary implementation is in silico and explicitly nonphenomenal; Section 11 gives the only permissible animal translation.

Calibrated null. Let Z be a route target assigned by an audited external randomizer, C the selected route, and $\mathcal{F}_C$ the information available when commitment becomes irreversible. If Z is conditionally uniform given $\mathcal{F}_C$, every $\mathcal{F}_C$-measurable policy, including private randomization represented inside that information field, has success probability exactly 1/m. A reproducible excess beyond the registered sampling-error boundary therefore implies conditional dependence through cue leakage, randomization failure, a hidden common cause, selection, or an additional physical channel. It is never evidence for a "non-informational" channel; statistical dependence carries information in the relevant sense.

$$[\mathbb{P}(Z = j \mid \mathcal{F}_C) = \frac{1}{m} \text{ for every } j] \Rightarrow \mathbb{P}(C = Z \mid \mathcal{F}_C) = \frac{1}{m}.$$

Primary simulator tournament. The blinded concealed-target generator bank contains: CT0, the calibrated null; CT1, ordinary perceptible leakage; CT2, hidden common-cause dependence; CT3, endpoint-conditioned or selected trajectories; and CT4, an explicitly programmed completion controller. The benchmark varies residual sign and magnitude, remaining capacity, route count m, precommitment target mapping, observation garbling, and intervention availability. A separate transition branch remaps the target after commitment only while the physical outcome remains genuinely alterable. One frozen pipeline must recover the generator or return Model-ambiguous/open-set inadequate on held-out seeds.

All generator, route-count, residual-sign, residual-magnitude, garbling, and intervention cells belong to one preregistered family. Family-wise false support is controlled through §9.3's frozen multivariate decision map; sequential monitoring uses an anytime-valid construction proved for the declared composite null.

PTVC-specific fingerprints. An action-selector route earns promotion only through the joint pattern: sign-appropriate directional sorting; monotonic dependence on independently estimated residual magnitude; transport across route counts and precommitment apparatus mappings; survival of leakage audits; and compatibility with the completion potential estimated from separate transition or action data. Postcommitment remapping is scored only on the separately registered transition branch. Generic disturbance can change behavior but does not prospectively predict which residual sign, route mapping, and held-out module will carry the effect.

Precommitted falsifiers. The registered minimum-effect selector route fails if the simultaneous confidence region for its advantage lies wholly inside the preregistered practically-null region and the registered minimum-effect alternative lies outside that region; if an effect is undirected or nonmonotone in the independently estimated residual; if it follows handler, batch, odor, geometry, timing, or apparatus; if target remapping eliminates it; or if equal-flexibility ordinary channel models win held-out scoring. Increasing m estimates the information capacity of any surviving channel. Exact PTVC does not imply infinite implementation capacity, so finite-m breakdown localizes capacity rather than automatically deciding the boundary law.

Animal-study boundary. Lethal, induced-terminal, deprivation, untreated-pain, withheld-rescue, or altered-veterinary-endpoint selector paradigms are outside this research program. Section 9.4 is controlling; §11.6.12 supplies the protocol-specific replacement, welfare, care, and stopping gates. The permitted translation uses reversible nonterminal choices and observational natural- or humane-endpoint follow-up under ordinary or enriched care.

Comparative measurement basis. Rat judgment bias can track affect-related state; positive facial changes, 50-kHz vocalizations, grimace, behavior, sleep, and activity provide complementary channels (Harding et al., 2004; Finlayson et al., 2016; Sotocinal et al., 2011). None is accepted as a stand-alone valence meter: rewarding morphine, for example, did not consistently increase 50-kHz calls (Wright et al., 2012). The protocol therefore freezes a bipolar, multimethod latent model and validates each channel against method-diverse outcomes before any ledger verdict.

Interpretation. In-silico recovery validates the discriminator and analysis pipeline, not PTVC in nature. A gate-valid off-band endpoint in the prospective animal cohort rejects the registered measured animal projection; terminal within-band compatibility plus transported fingerprints motivates stronger bridges but does not by itself establish strict lifetime phenomenal closure. This asymmetric decision rule makes the experiment informative without inflating what animal proxies can show.

Proof locations. B.2.4, B.3, and B.6 derive the conditional-calibration, leakage, action-invariance, selection, and mechanism-equivalence results under their displayed assumptions.

Animal enrollment follows only after replacement, proxy, simulation, welfare, governance, and analysis-plan gates pass. Power uses registered pilot or external variance and clustering assumptions, not placeholders. Before enrollment, fix applicable ARRIVE/PREPARE guidance, cluster and attrition structure, humane stopping, care noninterference, and the local-mechanism claim ceiling.

Appendix C: Interpretive Scope and Conditional Canonicality of Intrastream Terminal Balance

C.1 Philosophical coordinate

Interpretive scope. PTVC concerns the terminal within-stream relation and leaves path monotonicity and the number of intermediate zero crossings open. Urges, aversions, addictions, fears, preferences, felt certainty, libertarian free will, determinism, and the phenomenology of choice remain separate research questions, to be addressed through independently specified models and evidence rather than treated as PTVC premises or personal guidance.

PTVC selects one explicitly delimited coordinate among broader theories of equality, justice, welfare, and priority (Dworkin, 1981a, 1981b; Rawls, 1999; Scanlon, 1998; Sen, 1980; Temkin, 1993; Parfit, 1997):

Table 13. Appendix C Scope Matrix

Dimension	PTVC coordinate	Nearby alternatives
Bearer	Each maximal qualifying conscious stream, separately routed under the declared ownership and routing rule.	Organism, legal person, episode, population, species
Currency	Declared signed phenomenal value	Resources, capabilities, rights, preference satisfaction, meaning, plural goods
Temporal scope	Complete strict-pre-endpoint stream evaluated through $L_i^{\text{phen}}(T_i^{\star}-)$; endpoint-inclusive and extended-continuation variants are separate.	Momentary or episode-level evaluation; birth-to-date prefixes; endpoint-inclusive totals; extended-continuation scopes.
Relation	Positive-negative equality within the same stream	Common residual, common ratio, sufficiency, priority, suffering cap
Transfer	No cross-stream offset	Population or ensemble compensation
Gross magnitude	Finite on the finite signed-measure branch, but neither equalized across streams nor otherwise upper-bounded by PTVC.	Equalized gross magnitude across streams; bounded total suffering; capped peak intensity; unbounded accumulation.
Modal scope	Declared separately	Actual history, probability one, all policies, all interventions, compositional closure

The most precise philosophical label is universal intrastream terminal valence balance. This is a defined dimension of possible fairness rather than a complete theory of justice, welfare, rights, or flourishing.

C.2 Additive mathematics: equal balance, unequal gross valence variation

On the finite scalar signed-measure branch, write the Jordan decomposition and complete-stream magnitudes as

$$\nu_i = \nu_i^+ - \nu_i^-, V_i^+ = \nu_i^+(\Omega_i), V_i^- = \nu_i^-(\Omega_i).$$

The complete-stream residual and gross scalar valence variation are

$$C_i[\Gamma] = V_i^+ - V_i^-, V_i^{\text{gross}} = V_i^+ + V_i^-.$$

Where $V_i^- > 0$, define the secondary pole ratio

$$\rho_i^{\text{bal}} := \frac{V_i^+}{V_i^-}, V_i^- > 0.$$

The literal additive PTVC condition is

$$C_i[\Gamma] = 0 \quad \Leftrightarrow \quad V_i^+ = V_i^- < \infty.$$

Under PTVC, the pole ratio equals one wherever it is defined. Equality remains primary because a zero-zero stream satisfies PTVC while the ratio is undefined. PTVC imposes no cross-stream equality of gross magnitude. On the bipolar representation, primitive component measures, overlap, and gross totals are separately identified; when the finite components induce the same net measure on the same event domain, Lemma 1 gives the same terminal residual.

C.3 Conditional canonicality

The following propositions establish PTVC's conditional canonicality under their displayed common-unit, scalarity, nondegeneracy, reference-class, and transformation assumptions. Empirical evaluation then asks whether those assumptions and the resulting terminal relation hold in the declared world-domain.

Equal moral standing motivates a family of candidate fairness formalisms. The present philosophical case is developed under a declared bearer, currency, scope, no-transfer rule, finite cross-sign unit, transformation family, and modal branch. Within that family, the four propositions identify conditions under which zero or one-to-one balance has a specified invariance or canonicality property. Broader theories of fairness, phenomenal value, and nature remain open for comparison.

[CP] Proposition C.1: streamwise-unit invariance. Consider the common-residual family

$$C_i[\Gamma] = \delta_{\text{res}} \quad \text{for every qualifying stream } i.$$

[P] Proof. If each stream's cardinal unit may be independently rescaled by $a_i > 0$ while the same numerical residual is required to remain valid, then

$$a_i \delta_{\text{res}} = \delta_{\text{res}} \quad \text{for every } a_i > 0.$$

This equation has the unique solution

$$\delta_{\text{res}} = 0.$$

Zero is therefore the unique common residual invariant under independent positive streamwise unit changes.

This uniqueness conclusion assumes that the same numerical residual survives independent streamwise positive unit changes. If a genuinely fixed common cross-stream cardinal unit is independently established, a nonzero common residual remains a coherent comparator. The empirical program therefore tests the invariant common-residual family alongside that fixed-unit alternative.

[CP] Proposition C.2: reciprocal-polarity fixed point. In the proportional family

$$V_i^+ = \rho V_i^-, \quad \rho > 0.$$

[P] Proof. Exchanging the positive and negative poles maps ρ to ρ^{-1}. The unique positive fixed point is

$$\rho = 1.$$

One-to-one balance is thus canonical inside the polarity-exchange-invariant proportional family. Physical, psychophysical, or moral symmetry of the poles remains an additional premise.

[CP] Proposition C.3: valence conjugation. Suppose the admissible class is closed under an involution J satisfying

$$C_i[J\Gamma] = -C_i[\Gamma],$$

and suppose every admissible history has the common residual δ_{res}. Then $\delta_{\text{res}} = 0$.

[P] Proof. Closure under J gives $\delta_{\text{res}} = -\delta_{\text{res}}$, hence $\delta_{\text{res}} = 0$.

[CP] Proposition C.4: ratio-residual intersection. Suppose one common ratio and one common residual satisfy $V_i^+ = \rho V_i^-$ and $C_i[\Gamma] = \delta_{\text{res}}$ throughout a reference class containing two nondegenerate streams with distinct finite negative magnitudes.

[P] Proof. For each stream, $(\rho - 1)V_i^- = \delta_{\text{res}}$. Distinct finite negative magnitudes force $\rho = 1$ and then $\delta_{\text{res}} = 0$. PTVC is therefore the unique intersection of the common-ratio and common-residual families in this reference class; under the common-ratio premise, the two streams also have distinct gross magnitudes.

C.4 Philosophical significance and complementary dimensions

PTVC has several distinctive strengths:

- Identity blindness after bearer and routing qualification: the same relation applies throughout the declared reference class. This does not make genotype, species, morphology, legal identity, or software label a bearer criterion by itself.
- Separateness of streams: a population mean cannot compensate an unbalanced stream.
- Diversity without equalized lives: duration, intensity, narrative, opportunity, and gross valence variation may differ.
- Whole-trajectory discipline: the hypothesis directs attention to stream origin, endpoint, routing, and unobserved experience rather than inferring a complete-stream verdict from a visible prefix.
- Truth-condition economy: once the representation is fixed, $C_i[\Gamma] = 0$ is a compact universal boundary predicate.

PTVC contributes a precisely defined balance coordinate within a broader humane and moral landscape. Gross negative magnitude, peak suffering, duration, voluntariness, rights, agency, restitution, temporal order, narrative shape, and the distribution of value through time remain complementary dimensions (Broome, 2004; Velleman, 1991). Their importance motivates separate gross-load, suffering-priority, temporal-shape, rights-based, and plural-value hypotheses rather than an expansion of PTVC by definition.

A zero signed residual leaves gross magnitude, temporal order, agency, rights, narrative structure, and the fact of experience intact; it therefore neither assigns the welfare value of nonexistence nor equates a conscious life with matter that never lived.

C.5 Conceptual rivals and interpretive boundaries

Expected-neutrality uses the prospectively declared ex-ante sigma-field $\mathcal{G}_{i,0}^{\text{ex}}$ and the exclusions defined in §8.6; this appendix adds only the philosophical comparison.

The following rivals remain live:

$$H_\rho^{(b)}: \quad V_i^{+,(b)} = \rho V_i^{-,(b)}, \qquad \rho > 0,$$

$$H_{\delta_{\text{res}}}^{(b)}: \quad C_i^{(b)}[\Gamma] = \delta_{\text{res}},$$

$$H_E^{(b)}: \quad \mathbb{E}_{Q^{(b)}}\left[C_i^{(b)}[\Gamma] \mid \mathcal{G}_{i,0}^{\text{ex}}\right] = 0,$$

$$H_g^{(b)}: \quad V_i^{\mathrm{gross},(b)} = g, \qquad g \in [0,\infty).$$

A numerical common-gross equality across streams additionally requires a common cross-stream unit; it is not invariant under independent streamwise rescaling.

Additional rivals include suffering caps, weighted or lexicographic negative priority, temporal-shape laws, nonhedonic currencies, and ordinary no-law processes.

PTVC admits multiple philosophical interpretations that enrich its significance, while its empirical evidence level remains governed by the same transparent scientific tests.

C.6 Ethical and practical synthesis

PTVC's ethical interpretation is clear: it specifies a terminal intrastream balance relation without asserting debt, punishment, desert, sin, payment, or causal payback. This distinction also separates PTVC from just-world cognition, the documented tendency to perceive outcomes as deserved (Lerner, 1980; Hafer & Bègue, 2005).

The hypothesis is evaluated through the preregistered representation, endpoint, measurement, rival, and recovery gates of Sections 7–9, and it never licenses withholding relief or treating suffering as necessary compensation.

The practical rule is direct: reduce preventable suffering, protect consent, provide rescue and comfort, and preserve ordinary ethical and clinical duties. Those actions remain part of the realized trajectory and are fully compatible with the research program.

The philosophical contribution of PTVC is precise and constructive: it formulates a candidate universal intrastream terminal-balance relation, derives conditional uniqueness results, and organizes a disciplined family of nearby alternatives. The scientific framework turns that structure into theorem targets, measurement programs, mechanism tests, and empirical comparisons that can progressively test finite-class implications and narrow viable mechanisms.

C.7 Constraint-conditional footprint heuristic

A constraint-conditional thought experiment may be used without making the candidate constraint a premise: if terminal balance were a substrate-general boundary constraint, what observable architecture should systems satisfying it exhibit, and which signatures would differ under a no-law alternative? The heuristic generates discriminators and organizes the observations that would matter. Evidentiary weight arises only from prospectively specified, held-out empirical performance under the model's existing scientific gates.

Table 14. Constraint-Conditional Differential Footprint

Candidate footprint	Why PTVC may require it	No-law expectation	Discriminating test
Residual incremental value and residual compression sufficiency	Completion depends on what remains to be balanced, not only the present state	Current state, recent history, and opportunity suffice	Held-out state-only versus state-plus-residual loss for incremental value, and residual-only versus registered richer-history loss for compression sufficiency, each with its own practical margin and rival bank.
One completion potential governs several modules	A genuine boundary constraint should shape transitions, actions, and admissible endpoints coherently	Separate flexible kernels fit each module without common geometry	Cross-equation overidentification and rotation of training/held-out modules

Candidate footprint	Why PTVC may require it	No-law expectation	Discriminating test
Robustness to unexpected endpoints	A global path or endpoint law need not rely on a local forecast of death	Horizon-aware local controllers lose calibration as endpoint surprise rises	Prospective anticipated-versus-unexpected endpoint cohorts
Coherence across resolution	Exact or banded closure is a path property rather than a rounding artifact	Apparent zero changes with bin width, censoring, or instrument floor	Replicate-channel deconvolution and shrinking-band intercept tests
Substrate and culture transport	A substrate-general candidate law predicts transport beyond one language, religion, or adult justice concept	Effects track belief, demand, species-specific ecology, or instrument	Cross-cultural, indirect, prelinguistic, and comparative protocols
Intervention-compatible compensation	Perturbations alter the amount remaining and should change later reachable structure	Rebound follows ordinary adaptation kernels only	Randomized signed-dose response surfaces across horizon and reserve
Cut-point past–future complementarity	A completed zero residual fixes future signed remainder as the opposite of accrued past area	State recovery and generic adaptation need not cancel path area across arbitrary cuts	Random-cut, independent-channel errors-in-variables test of slope -1 plus a separately estimated residual $D_i(\tau) = P_i(\tau) + R_i(\tau) = 0$; impose $\beta = 1$ before interpreting the raw intercept as a mean-residual target.
Feasible completion-information budget	A local controller must obtain enough information to reconstruct the residual it cancels	Ordinary controllers remain bounded by available cues and state	Rate–distortion frontier with leakage audit and hidden-channel alternatives

The proposed program uses redundant, dependency-audited routes to the same formal object to test whether any recoverable boundary signature persists under noise, missingness, heterogeneity, and multiple endpoint causes. Because ordinary physiology also uses redundancy, the discriminator is whether diverse mechanisms share PTVC's completion geometry rather than merely preserving local viability.

Physics offers conditioned paths, coboundaries, large deviations, steering, killing processes, and first-passage boundaries. PTVC posits a terminal boundary constraint, not conservation of energy or a known charge. A physics-facing branch must identify a phenomenal carrier on which CUT-1, the completion potential, and INF-1 transport across substrates while nonequilibrium and integral-feedback rivals fail under equal scoring. Such a finding would support a previously unrecognized substrate-general boundary regularity without subsuming rights, agency, suffering priority, narrative value, or social justice.

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About the Book Creator

Stephen Takowsky has spent his life studying worlds: how rules shape them, how people move within them, and how strategy changes what becomes possible. He is a world-champion Civilization player; the Guinness World Records-recognized captain of Team Liquid's pioneering Civilization team; a former community manager whose work included Terraria, Ghostrunner, WUCHANG: Fallen Feathers, and other titles; the creator of the United Servers of Discord Alliance; an author and community builder; and the founder of the Law of Fairness research program. His path has crossed philosophy, finance, esports, online communities, and science, yet one question has remained remarkably consistent: what hidden structure governs a complex system, and what future becomes possible once that structure is understood?

At Morgan Stanley in Beverly Hills, Takowsky completed Tier 1 of the firm's Financial Advisor Associate Program while building a client practice and contributing strategy essays to the firm's internal Advisor Insights platform. In a field where abstractions carried immediate consequences for trust, money, and people's futures, he learned to identify a system's decisive variables, distinguish appearance from underlying structure, and turn complex ideas into strategies people could use.

After leaving Morgan Stanley, Takowsky joined Team Liquid. Following aXiomatic's September 2016 acquisition of a controlling interest, he was named captain of Team Liquid's newly created Civilization team. Competing as "MrGameTheory," he reached number one on the Civilization Revolution leaderboard and in the international Civilization IV and Civilization V leagues, while setting records in outnumbered Civilization IV ladder formats. Across years of competition, Civilization became a self-directed education in efficiency, long horizons, and consequential choice: see the whole system before committing to a move, think in centuries rather than turns, anticipate intelligent opposition, and keep searching after every obvious path fails.

Competition soon became community leadership. While representing Team Liquid, Takowsky helped organize mobile gaming communities across titles including Clash Royale and Arena of Valor. Team Liquid credited him with a major role in establishing a casual Clash Royale network of more than 3,000 players. The work required developing leaders, aligning incentives across large networks, preserving trust under pressure, and turning independent decisions into coordinated action. Those lessons became part of the practical foundation of the work that followed.

At 505 Games, Takowsky served as Senior Digital Content & Engagement Manager across several major titles; his work included worldwide community strategy and engagement for Terraria, a seventy-million-copy phenomenon and one of the best-selling games ever made. He founded the United Servers of Discord Alliance, bringing together leaders from communities built around Roblox, Star Wars, Pokémon Go, Warframe, League of Legends, Grand Theft Auto, Animal Crossing, Terraria, and other properties. The alliance gives leaders a forum to exchange strategy, coordinate events, and create opportunities for their communities to discover and support one another. He also served as a site producer and community manager at ZAM Network, which operated the TF2Outpost marketplace. Together, these roles taught him how trust, conflict resolution, incentives, communication, and fair rules shape growing communities.

Takowsky began developing the Law of Fairness more than twenty years ago while earning bachelor's degrees in Religious Studies and Philosophy/Law and Society at the University of California, Riverside. His 2009 book, *Of Grandeur*, gave the idea its first public form. He then brought the idea into sustained dialogue with academics across disciplines, inviting rigorous counterexamples, references, technical critique, and stronger formulations. That process clarified the claim, strengthened its architecture, and helped transform a solitary inquiry into a research program built for serious scrutiny. The present work carries the imprint of every scholar and community member who read, replied, challenged an assumption, suggested a source, or made time for an unconventional question; their generosity continues to shape the project today.

The technical formulation is Per-Stream Terminal Valence Closure, a precise and falsifiable boundary hypothesis with defined objects, serious rivals, explicit decision conditions, ethical safeguards, and a staged experimental program. Its animating conviction is simple: humanity's largest questions deserve its strongest methods. The Formal Model specifies what must be represented, measured, and compared; how evidence should update the target and its rivals; where uncertainty requires restraint; and how equal-rule comparisons can distinguish among explanations. Takowsky's role is to make the hypothesis explicit enough that researchers can test, refine, narrow, or reject it through evidence.

The ten-book Law of Fairness series carries that inquiry across ten domains. It moves from the formal hypothesis and its scientific testing to a mature vision of utopia and the responsible transition toward it, then into ordinary life and spiritual meaning, and onward to government, civic formation, constitutional design, and the material foundations of lasting peace. Together, the books ask not only whether the Law is true, but what humanity might understand, how people might live, and what institutions we might build if we approached fairness with greater honesty, discipline, and care.

The project has been independent and self-funded to this stage. Its future is collaborative. Takowsky welcomes researchers, universities, laboratories, funders, technologists, students, critics, writers, artists, community builders, and volunteers prepared to bring rigor, imagination, and courage to one of humanity's oldest questions. The next breakthrough may begin with a theorem, an instrument, a dataset, a counterexample, or an experiment no one has yet imagined. The Law of Fairness began as one person's question. Its next chapter belongs to everyone willing to help discover what is true and build what becomes possible.

Explore the project and its official communities at LawOfFairness.com. Research, scholarly, media, funding, volunteer, and collaboration inquiries: team@lawoffairness.com.